MATH2831/2931
LINEAR MODELS

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Contents

1 Introduction

1.1 Some Basic Terminology

In this course we will be concerned with building statistical models which describe the dependence of some variable of interest (called the response) on a number of other variables (called predictors).

An example will help to illustrate the idea. The Bureau of Meteorology in Sydney issues daily maximum temperature forecasts at many locations around the city. One of the locations at which a temperature forecast is issued is Sydney Airport, and forecasters in the Bureau make use of a statistical model to help them in deciding on the forecast.

Forecasters have available to them the predictions of a deterministic physical model (based on the equations of motion for fluids) which gives predicted values of certain meteorological variables such as temperature, wind speed and direction and pressure. However, the physical model may contain biases, and for this reason the Bureau uses a statistical model to relate the observed maximum temperature at Sydney airport (the response) to the set of variables predicted by the physical model (the predictors). The use of the statistical model to modify the output of the physical model gives a more accurate forecast, and allows quantification of uncertainty about predictions of the response.

In situations where the predictors are under the control of an experimenter the response is sometimes called the dependent variable and the predictors are called independent variables. In this course we will not be concerned with modelling the independent variables as random quantities, but assume that they are controlled or at least accurately measured.

1.2 The Model Building Process

Before we proceed to a discussion of specific models, it is helpful to consider some of the reasons we might have for building a statistical model. We build statistical models in order to make decisions. For instance, in a medical study we might administer several different kinds of treatment to a collection of patients with a certain condition, and we want to decide which treatment is most effective. Or we may wish to predict the future level of a share price index based on information available to us now. Or we may wish to compare different descriptions of a set of data based on competing scientific theories: we might ask ourselves which model (and hence which theory) do the data support?

Qualitatively, we can write a statistical model for the response variable in the form

$$\text{Response} = \text{Signal} + \text{Noise}$$

"Signal" here denotes systematic variation in the response, which for us will be variation which can be explained in terms of the predictors and a small number of unknown parameters to be estimated from the data. The other component of the model, the "Noise" term, captures residual variation unexplained in the signal: typically there will be some unknown parameters to be estimated in this component of the model as well.

A good statistical model provides a reduction of a possibly large and complex data set to a description in terms of a small number of parameters. When building a model a good statistician will keep in mind that the the reduction of the data obtained must be a useful one for making the decisions which motivated the building of the model. Finding or selecting a good statistical model is a complex iterative process which is difficult to formally describe. Usually statisticians cycle between stages of tentative model formulation, estimation and model criticism.

It is often an objective of a statistical analysis to describe as much of the variation in the response as possible systematically (that is, in the "Signal"). When we achieve this objective we say that the model fits the data well. However, as we have mentioned, parsimony of the model (that is, having a model with a small number of parameters) is also desirable in order to provide a reduction of the data which is useful for human decision making and for making predictions of future response values. So we must manage a trade off between simplicity of a statistical model and goodness of fit. How we should manage this trade off often depends on the use to which the model will be put.

We have talked about the process of model building in general terms. We will now look in detail at the *general linear model* which is the subject of this course. We discuss perhaps the simplest example of the general linear model first (the simple linear regression model) and illustrate the processes of estimation, model selection and model criticism in this situation. Once we have done this, the rest of the course will be concerned with describing the same processes in the general case.

2 The Simple Linear Regression Model

As we have mentioned, perhaps the simplest example of a linear model is the simple linear regression model, which we now describe. In the simple linear regression model, the conditional distribution of the response is allowed to depend on a single predictor. Write $y_1, ..., y_n$ for response variable, and write $x_1, ..., x_n$ for the corresponding values of the predictor.

The fundamental assumption of the simple linear regression model is that, for (unknown) parameters β_0 and β_1 taking values in $\mathbb{R}$, the response variables are of the form

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \tag{1}$$

where ε_i, $i = 1, ..., n$ denotes a collection of uncorrelated error terms with mean zero and variance σ^2. The formula (1) illustrates the decomposition mentioned in the introduction of the response into signal and noise: here $\beta_0 + \beta_1 x_i$ is the signal or systematic component of the model, and the terms ε_i describe the noise or residual variation unexplained by the signal.

With the assumptions we have made so far, we will be able to derive estimators for the unknown parameters β_0, β_1 and σ^2 and examine some of their properties. However, for deriving confidence intervals and performing hypothesis tests we will also need to assume that the errors ε_i are normally distributed. The assumption of the normality of errors also allows us to make stronger conclusions about the optimality of estimators for the unknown parameters.

2.1 Estimation of parameters by least squares

Good estimators b_0 and b_1 of the parameters β_0 and β_1 will be estimators which minimize the variation unexplained in the response by the systematic component of the fitted model.

One would like to fit the data to a linear model, given by $\beta_0 + \beta_1 x_i$. Then one goodness of fit criterion which might be minimized to obtain good estimators of β_0 and β_1 is

$$\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_i))^2. \tag{2}$$

where β_0 and β_1 are parameters which we minimize over. If b_0 and b_1 are the values which minimize (2) with respect to β_0 and β_1, then b_0 and b_1 minimize in some overall sense the deviations of the responses y_i from the fitted values $\widehat{y_i}$.

Of course it is possible to define global measures of the discrepancy of the fitted values from the responses which are different to (2). For instance, we could consider estimators obtained by minimizing

$$\sum_{i=1}^{n} |y_i - (\beta_0 + \beta_1 x_i)| \tag{3}$$

with respect to β_0 and β_1. However, in the case of normally distributed errors, there are good reasons for considering the criterion (2), which is also useful due to its computational tractability.

The estimators b_0 and b_1 minimizing (2) are called the *least squares estimators* of β_0 and β_1. We derive expressions for these estimators now. Differentiating (2) with respect to β_0 and β_1 gives

$$\begin{aligned}
\frac{\partial}{\partial \beta_0} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 &= -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) \\
\frac{\partial}{\partial \beta_1} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 &= -2 \sum_{i=1}^{n} x_i (y_i - \beta_0 - \beta_1 x_i).
\end{aligned}$$

Setting these to zero, we obtain for b_0, b_1 that

$$\begin{aligned}
-2 \left(\sum_{i=1}^{n} y_i - n b_0 - \left(\sum_{i=1}^{n} x_i \right) b_1 \right) &= 0 \\
-2 \left(\sum_{i=1}^{n} x_i y_i - \left(\sum_{i=1}^{n} x_i \right) b_0 - \left(\sum_{i=1}^{n} x_i^2 \right) b_1 \right) &= 0.
\end{aligned}$$

Hence upon rearranging we have that the least squares estimators b_0 and b_1 satisfy

$$n b_0 + \left(\sum_{i=1}^{n} x_i \right) b_1 = \sum_{i=1}^{n} y_i \qquad (4)$$

$$\left(\sum_{i=1}^{n} x_i \right) b_0 + \left(\sum_{i=1}^{n} x_i^2 \right) b_1 = \sum_{i=1}^{n} x_i y_i. \qquad (5)$$

These are linear equations in b_0 and b_1 (called the normal equations) which are easily solved. Dividing (4) by n and solving for b_0 gives

$$b_0 = \bar{y} - b_1 \bar{x}. \qquad (6)$$

Subsituting (6) into (5) gives

$$\left(\sum_{i=1}^{n} x_i \right) (\bar{y} - b_1 \bar{x}) + \left(\sum_{i=1}^{n} x_i^2 \right) b_1 = \sum_{i=1}^{n} x_i y_i$$

or

$$b_1 \left(\sum_{i=1}^{n} x_i^2 - \bar{x} \sum_{i=1}^{n} x_i \right) = \sum_{i=1}^{n} x_i y_i - \bar{y} \sum_{i=1}^{n} x_i.$$

Solving for b_1,

$$b_1 = \frac{\sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}.$$

Table 2.1: Zinc levels in plants and lake sediment samples

Concentration of Zinc in sediment	Concentration of Zinc in plant
37.5	15.9
72.5	42.7
85.2	85.7
76.5	52.6
64.5	49.1
86.8	59.0
90.8	53.6
105.8	77.8
85.8	63.2
87.9	62.3
53.9	22.7
102.3	66.1
90.7	47.4
86.0	59.4
79.0	50.9

We write S_{xy} for the numerator in (7) and S_{xx} for the denominator so that $b_1 = S_{xy}/S_{xx}$. We can then compute b_0 from (6). An example will help to reinforce some of the concepts we have discussed so far.

Example: zinc concentrations in plants

The following example is described in *Understanding Data: Principles and Practice of Statistics*, David Griffiths, W. Douglas Stirling and K. Laurence Weldon, Wiley, 1998.

The data in the table below were collected from fifteen lakes in central Ontario. The purpose of collecting the data was to assess how zinc levels in a certain plant in the lake (the response) were related to zinc levels in the lake sediment (the predictor). By describing and modelling this relationship we can predict zinc levels in the plant from an analysis of lake sediment. A scatterplot of these data is shown in Figure 2.1, and superimposed on the plot is the fitted least squares regression line. To compute the fitted regression line, write $y_1, ..., y_{15}$ for the fifteen response values and $x_1, ..., x_{15}$ for the corresponding fifteen values of the

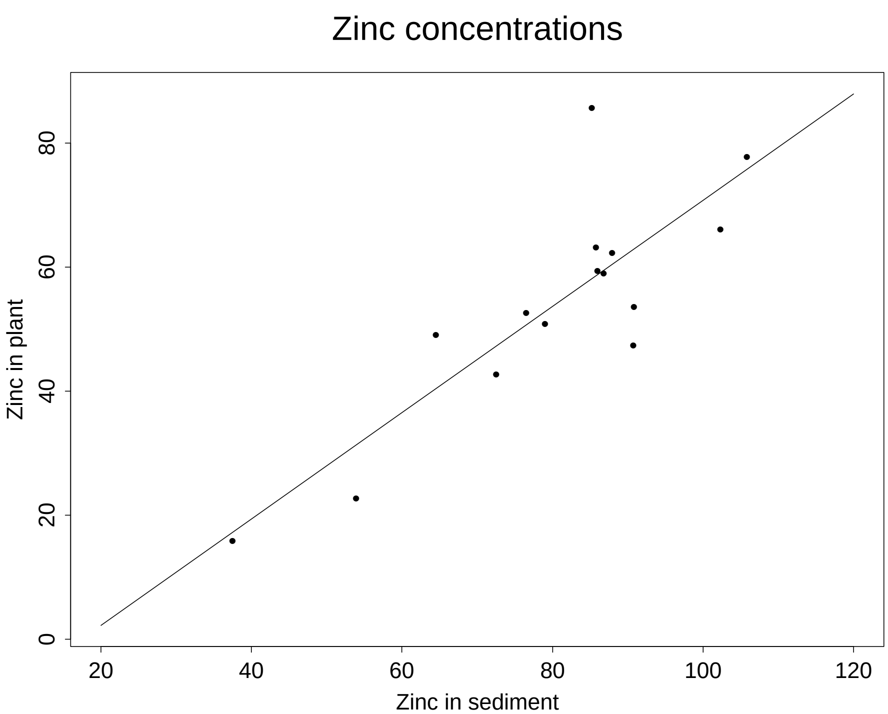

Figure 2.1: Scatter plot showing Zinc concentrations in plants and lake sediment samples.

predictor and observe that

$$\begin{aligned}
\sum_{i=1}^{15} y_i &= 808.4 \\
\sum_{i=1}^{15} x_i &= 1205.2 \\
\sum_{i=1}^{15} x_i^2 &= 101228.6 \\
\sum_{i=1}^{15} x_i y_i &= 68716.78
\end{aligned}$$

so that

$$\begin{aligned}
S_{xy} &= 68716.78 - \frac{(808.4).(1205.2)}{15} \\
&= 3764.5
\end{aligned}$$

and

$$\begin{aligned}
S_{xx} &= 101228.6 - \frac{1205.2^2}{15} \\
&= 4394.8.
\end{aligned}$$

Hence the least squares estimate of the slope coefficient is $3764.5/4394.8 = 0.857$ and the least squares estimate of the intercept is $808.4/15 - 0.857 \times 1205.2/15 = -14.9$. Hence the fitted least squares regression line (the systematic component of the fitted model) is

$$-14.9 + 0.857x.$$

2.2 Properties of Least Squares Estimators of slope and intercept

We now discuss some properties of the least squares estimators of β_0 and β_1. The properties of the estimators derived in this section hold regardless of whether or not the errors in the simple linear regression model are normally distributed.

To begin, we ask the question: how do we assess the quality of estimators of parameters in a statistical model? For us, a good estimator is one that performs well on average with repeated use. Suppose our data arise from an experiment which can be repeated. We can imagine computing estimates from our data for the same experiment over many repetitions, and examining the variablility of the estimates obtained. We would want the average of the estimates to be equal to the true parameter value (no systematic bias), and we want the variation of the estimates about the true parameter value to be as small as possible (small variance).

We would like to compute the mean and variance of the least squares estimators of the slope and intercept in the simple linear regression model as a way of evaluating the performance of these estimators.

Some Notation

In the rest of the note, for convenience, we write y_i to represent both the i-th response variable (see model assumption (1)) and its realization (a data point). Recall that the least squares estimator b_1 for β_1 is given by

$$b_1 = \frac{S_{xy}}{S_{xx}} \tag{7}$$

where

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n}$$

and

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}.$$

It is a very easy exercise to derive the following alternative expressions for S_{xy} and S_{xx}:

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x}) y_i$$

and

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x}) x_i.$$

Also,

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

and

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

In the following, we compute the expectation and variance of b_0 and b_1, which are **random** under the model assumption (1).

Expected value of b_1

Consider the expected value of b_1. We have

$$\begin{aligned}\mathbb{E}(b_1) &= \mathbb{E}\left(\frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{S_{xx}}\right) \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})\mathbb{E}(y_i)}{S_{xx}} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})(\beta_0 + \beta_1 x_i)}{S_{xx}}.\end{aligned}$$

Using the fact that

$$\begin{aligned}\sum_{i=1}^n (x_i - \bar{x}) &= \sum_{i=1}^n x_i - n\bar{x} \\ &= n\bar{x} - n\bar{x} \\ &= 0,\end{aligned}$$

we have

$$\begin{aligned}\mathbb{E}(b_1) &= \beta_1 \frac{\sum_{i=1}^n (x_i - \bar{x})x_i}{S_{xx}} \\ &= \beta_1.\end{aligned}$$

So if we were to use the estimator b_1 repeatedly for estimating β_1 in a certain experiment, on average the estimator gives the true parameter (no systematic bias, we say that b_1 is *unbiased* for β_1).

Expected value of b_0

Recall that

$$b_0 = \bar{y} - b_1\bar{x}.$$

Now,

$$\begin{aligned}\mathbb{E}(\bar{y}) &= \frac{\sum_{i=1}^n \mathbb{E}(y_i)}{n} \\ &= \frac{\sum_{i=1}^n (\beta_0 + \beta_1 x_i)}{n} \\ &= \frac{n\beta_0 + \beta_1 \sum_{i=1}^n x_i}{n} \\ &= \beta_0 + \beta_1\bar{x}.\end{aligned}$$

Hence since b_1 is unbiased for β_1,

$$\begin{aligned}\mathbb{E}(b_0) &= \mathbb{E}(\bar{y}) - \mathbb{E}(b_1)\bar{x} \\ &= \beta_0 + \beta_1\bar{x} - \beta_1\bar{x} \\ &= \beta_0.\end{aligned}$$

So b_0 is unbiased for β_0.

Variance of b_0 and b_1

For an estimator (such as b_0 or b_1) to be useful, we must be able to assess how reliable it is. Following our previous discussion, it would be nice to know how variable the estimators b_0 and b_1 are in repetitions of an experiment. We give expresssions for the variances of b_0 and b_1. These expressions are a special case of a later more general result and are given without proof.

$$\mathrm{Var}(b_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)$$

$$\mathrm{Var}(b_1) = \frac{\sigma^2}{S_{xx}}$$

$$\mathrm{Cov}(b_0, b_1) = -\frac{\sigma^2\bar{x}}{S_{xx}}.$$

Example: market model of stock returns

The following example is from Keller, Warrack and Bartel, “Statistics for Management and Economics: a Systematic Approach (Second Edition)”, Wadsworth, Belmont, 1990, p. 725.

A well known model in finance, called the *market model*, assumes that the monthly rate of return on a stock (R) is linearly related to the monthly rate of return on the overall stock market (R_m). The mathematical description of the model is

$$R = \beta_0 + \beta_1 R_m + \varepsilon$$

where the error term ε is assumed to satisfy the requirements of the linear regression model. For practical purposes, R_m is taken to be the monthly rate of return on some major stock market index, such as the New York Stock Exchange (NYSE) Composite Index.

The coefficient β_1, called the stock’s *beta coefficient*, measures how sensitive the stock’s rate of return is to changes in the level of the overall market. For example, if $\beta_1 > 1$ ($\beta_1 < 1$), the stock's rate of return is more (less) sensitive to changes in the level of the overall market than is the average stock. The monthly rates of return to Host International Inc. stock and to the overall market (as approximated by the NYSE composite index) over a 5-year period are shown in the table below.

Month	Host International	Market	Month	Host International	Market
Jan 1975	26.7	13.5	Jul 1977	−4.2	−1.5
Feb 1975	7.0	6.1	Aug 1977	2.2	−1.4
Mar 1975	15.9	2.9	Sep 1977	3.1	0.1
Apr 1975	18.6	4.7	Oct 1977	6.4	−3.9
May 1975	−6.0	5.5	Nov 1977	11.0	4.2
Jun 1975	−4.2	5.2	Dec 1977	−1.0	0.5
Jul 1975	4.0	−6.4	Jan 1978	−2.7	−5.7
Aug 1975	−5.2	−2.0	Feb 1978	10.4	−1.2
Sep 1975	−1.7	−3.6	Mar 1978	10.2	3.2
Oct 1975	28.2	6.1	Apr 1978	15.6	8.3
Nov 1975	17.6	3.1	May 1978	10.2	3.2
Dec 1975	1.6	−1.0	Jun 1978	1.2	−1.3
Jan 1976	17.6	12.5	Jul 1978	22.3	5.7
Feb 1976	−13.4	0.1	Aug 1978	3.1	3.8
Mar 1976	−12.1	3.0	Sep 1978	−13.6	−0.6
Apr 1976	−6.2	−1.1	Oct 1978	−28.8	−10.2
May 1976	−12.1	3.0	Nov 1978	19.0	3.1
Jun 1976	−2.5	4.7	Dec 1978	−2.1	1.6
Jul 1976	−9.3	0.7	Jan 1979	−7.8	4.7
Aug 1976	0.0	0.0	Feb 1979	−10.1	−2.9
Sep 1976	−1.5	2.6	Mar 1979	11.4	6.2
Oct 1976	−5.3	−2.1	Apr 1979	−5.5	0.7
Nov 1976	9.7	0.5	May 1979	−6.6	−1.5
Dec 1976	18.7	5.8	Jun 1979	19.6	4.5
Jan 1977	−10.7	−4.0	Jul 1979	3.7	1.5
Feb 1977	−8.4	−1.6	Aug 1979	7.9	6.3
Mar 1977	6.3	−1.1	Sep 1979	−3.2	0.0
Apr 1977	1.2	0.4	Oct 1979	−10.4	−6.9
May 1977	2.5	−1.2	Nov 1979	−12.4	6.0
Jun 1977	14.3	5.3	Dec 1979	−4.2	2.3

Figure 2.2 shows a scatterplot of these data, together with a fitted least squares regression line. As usual writing y for the response (Host International

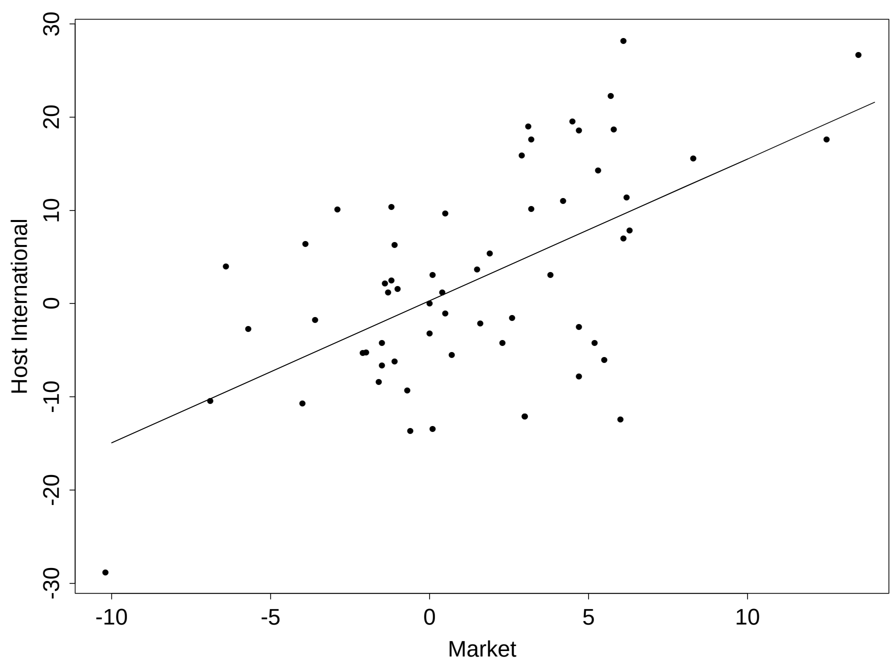

Figure 2.2: Scatter plot of Host International returns versus overall market returns with fitted least squares regression line.

return) and x for the predictor (overall market return) we have in this case that

$$\sum_{i=1}^{n} x_i = 90.4 \qquad \sum_{i=1}^{n} x_i^2 = 1254.1$$

$$\sum_{i=1}^{n} y_i = 136.0 \qquad \sum_{i=1}^{n} y_i^2 = 8158.9$$

$$\sum_{i=1}^{n} x_i y_i = 1992.8.$$

Hence the estimated beta coefficient for the stock (the least squares estimate of the slope in the linear regression) is

$$b_1 = \frac{S_{xy}}{S_{xx}}$$

and we calculate S_{xy} and S_{xx} as

$$\begin{aligned} S_{xy} &= \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} \\ &= 1992.8 - \frac{(90.4).(136.0)}{60} \\ &= 1787.89 \end{aligned}$$

and

$$\begin{aligned} S_{xx} &= \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} \\ &= 1254.1 - \frac{90.4^2}{60} \\ &= 1117.90 \end{aligned}$$

giving

$$b_1 = 1787.89/1117.90 = 1.60.$$

So it seems as though the rate of return on Host International stock is more sensitive to the level of the overall market than is the average stock. The least squares estimate b_0 of β_0 here is $\bar{y} - b_1\bar{x} = 2.2667 - (1.60)(1.5067) = -0.14$.

We have previously shown that b_0 and b_1 are unbiased, and derived expressions for the variance of b_0 and b_1. How reliable is the estimate of the beta coefficient in this example? Our expressions for the variance of b_0 and b_1 contain the parameter σ^2, which is unknown. We need to estimate this if we are to obtain estimated values for the variances of b_0 and b_1.

2.3 Estimation of the Error Variance

So far we have described least squares estimation of the parameters β_0 and β_1 in the simple linear regression model, but have not considered estimation of the error variance σ^2. If β_0 and β_1 were known, a reasonable estimator might be

$$\frac{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2}{n}$$

(since $\varepsilon_i = y_i - \beta_0 - \beta_1 x_i$ and so the above expression is a sample average of the squared ε_i which is an unbiased estimator of $\sigma^2 = \mathbb{E}(\varepsilon^2)$).

Since β_0 and β_1 are not known, we might suggest plugging the least squares estimators of β_0 and β_1 into the above expression.

In particular, consider the estimator

$$\sigma^{*2} = \frac{\sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2}{n}.$$

What are the properties of this estimator? In particular, is it unbiased for σ^2 (does it give the right value for σ^2 on average with repeated use)? It can be shown (proof is a special case of later results) that

$$\mathbb{E}(\sigma^{*2}) = \frac{n-2}{n}\sigma^2$$

which suggests using the modified estimator

$$\widehat{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2}{n-2}$$

which is unbiased for σ^2.

A useful alternative computational formula for $\widehat{\sigma}^2$ is

$$\widehat{\sigma}^2 = \frac{S_{yy} - b_1^2 S_{xx}}{n-2}$$

where

$$S_{yy} = \sum_{i=1}^n y_i^2 - \frac{\left(\sum_{i=1}^n y_i\right)^2}{n}.$$

We can "plug in" the estimate $\widehat{\sigma}^2$ for σ^2 in our expressions for the variances of b_0 and b_1 to obtain estimated variances for these quantities.

Example: market model of stock returns

We can estimate the error variance as

$$\widehat{\sigma}^2 = \frac{S_{yy} - b_1^2 S_{xx}}{n-2}$$

where

$$\begin{aligned} S_{yy} &= \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} \\ &= 8158.9 - \frac{136.0^2}{60} \\ &= 7850.63 \end{aligned}$$

so that

$$\begin{aligned} \widehat{\sigma^2} &= \frac{7850.63 - 1.60^2 1117.90}{58} \\ &= 86.01. \end{aligned}$$

Hence $\widehat{\sigma} = 9.27$.

By "plugging in" this $\widehat{\sigma}$ value in the expressions for $\text{Var}(b_0)$ and $\text{Var}(b_1)$ that we obtained before, we get that the estimated variance of b_0 is 1.61 (estimate standard deviation of 1.27) and the estimated variance of b_1 is 0.077 (estimated standard deviation of 0.28).

2.4 Maximum Likelihood Estimators

In MATH2801 you have learnt about maximum likelihood estimators, which often have certain optimality properties in large samples. In later discussion we will assume that the error terms in the simple linear regression model are normally distributed, so it is of interest to ask what the maximum likelihood estimators of the slope, intercept, and error variance are under normal assumptions.

First we review the concept of maximum likelihood estimation. Suppose that we have a discrete n-dimensional random vector Y with a distribution depending on a p-dimensional parameter vector θ. Now suppose that we observe a realization y of Y and that we wish to estimate θ. Since Y is discrete, we can consider the probability that $Y = y$ for different values of θ, and this probability $\mathbb{P}(Y = y|\theta)$ considered as a function of θ for the fixed observed y is called the *likelihood function*.

The idea of maximum likelihood estimation is the following: we should choose the parameter θ so that the observed value y of Y has the highest probability of occurrence. That is, the maximum likelihood estimator is chosen to be the maximizer of the likelihood function. In the case where Y is a continuous random vector, the likelihood is the density function of Y considered as a function of θ for the fixed observed y, and again the maximum likelihood estimator maximizes the likelihood function.

What are the maximum likelihood estimators of β_0, β_1 and σ^2 in the simple linear regression model if we assume that the errors are normal? The density function of the ith response is

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \beta_0 - \beta_1 x_i)^2\right)$$

and since the responses are now assumed to be independent we can write down the likelihood function as

$$\begin{aligned} L(y; \beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \beta_0 - \beta_1 x_i)^2\right) \\ &= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2\right). \end{aligned}$$

We must maximize this function with respect to β_0, β_1 and σ^2 to obtain the maximum likelihood estimators. Note that maximizing the likelihood function is equivalent to maximizing the logarithm of the likelihood function (since log is an increasing function and hence if L_1 and L_2 are values of the likelihood function for two different values of the parameters and $L_1 < L_2$ then $\log(L_1) < \log(L_2)$). If we write $l(y; \beta_0, \beta_1, \sigma^2)$ for $\log L(y; \beta_0, \beta_1, \sigma^2)$ we have

$$\begin{aligned} l(y; \beta_0, \beta_1, \sigma^2) &= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2 \\ &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2 \end{aligned}$$

Clearly regardless of the value of σ^2, $l(y; \beta_0, \beta_1, \sigma^2)$ is maximized with respect to β_0 and β_1 by choosing β_0 and β_1 to minimize

$$\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2.$$

That is, the maximum likelihood estimators of β_0 and β_1 are just the least squares estimators. It remains to find the maximum likelihood estimator of σ^2. Differentiating the log-likelihood with respect to σ^2 gives

$$\frac{\partial}{\partial\sigma^2}\left(-\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2\right) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2.$$

Setting this to zero, writing $\widehat{\sigma}^2$ for the maximum likelihood estimator of σ^2 and substituting in the maximum likelihood (least squares) estimators for β_0 and β_1 we get

$$\frac{n}{2\widehat{\sigma}^2} = \frac{1}{2\widehat{\sigma}^4}\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2$$

which gives

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2.$$

So the maximum likelihood estimator of σ^2 is the biased estimator that we considered in our last lecture.

2.5 Attributing variation to different sources

In this section we will prove a fundamental identity for the simple linear regression model which we will later generalize when considering the general linear model. The identity we derive describes a partition of the total variation into terms representing variation explained in different parts of the model (signal and noise).

Writing $\widehat{y}_i = b_0 + b_1 x_i$ for the ith fitted value, the identity we derive is

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\widehat{y}_i - \bar{y})^2 + \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2. \tag{8}$$

On the left hand side of (8) we have a term which we can think of as being the total variation in the data – note that this term is the numerator of the expression we would use for computing the sample variance of the y's. On the right hand side we have a sum of two terms: the first term represents the variation of the predictions about the sample mean of the y's (we can think of this as the variation explained by the fit) and the second term represents variation of the responses from the predictions (we can think of this as variation unexplained by the fit).

We now derive the identity above. We have

$$\begin{aligned}\sum_{i=1}^{n}(y_i - \bar{y})^2 &= \sum_{i=1}^{n}(y_i - \widehat{y}_i + \widehat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 + \sum_{i=1}^{n}(\widehat{y}_i - \bar{y})^2 + 2\sum_{i=1}^{n}(y_i - \widehat{y}_i)(\widehat{y}_i - \bar{y})\end{aligned}$$

so we only need to show that

$$\sum_{i=1}^{n}(y_i - \widehat{y}_i)(\widehat{y}_i - \bar{y}) = 0$$

or

$$\sum_{i=1}^{n}(y_i - \widehat{y}_i)\widehat{y}_i - \bar{y}\sum_{i=1}^{n}(y_i - \widehat{y}_i) = 0. \tag{9}$$

Now, we have that

$$\begin{aligned}\sum_{i=1}^{n}(y_i-\widehat{y}_i) &= \sum_{i=1}^{n}(y_i-b_0-b_1x_i)\\ &= \sum_{i=1}^{n}y_i-nb_0-b_1\sum_{i=1}^{n}x_i. \qquad (10)\end{aligned}$$

Recalling that the normal equations solved to find b_0 and b_1 are

$$nb_0+\left(\sum_{i=1}^{n}x_i\right)b_1 = \sum_{i=1}^{n}y_i \qquad (11)$$

and

$$\left(\sum_{i=1}^{n}x_i\right)b_0+\left(\sum_{i=1}^{n}x_i^2\right)b_1 = \sum_{i=1}^{n}x_iy_i. \qquad (12)$$

it follows from (11) and (10) that

$$\sum_{i=1}^{n}(y_i-\widehat{y}_i)=0$$

and hence to show (9) we just have to prove that

$$\sum_{i=1}^{n}(y_i-\widehat{y}_i)\widehat{y}_i=0.$$

Now,

$$\begin{aligned}\sum_{i=1}^{n}(y_i-\widehat{y}_i)\widehat{y}_i &= \sum_{i=1}^{n}(y_i-\widehat{y}_i)(b_0+b_1x_i)\\ &= b_1\sum_{i=1}^{n}(y_i-\widehat{y}_i)x_i\\ &= b_1\left(\sum_{i=1}^{n}x_iy_i-\sum_{i=1}^{n}(b_0+b_1x_i)x_i\right)\\ &= b_1\left(\sum_{i=1}^{n}x_iy_i-b_0\sum_{i=1}^{n}x_i-b_1\sum_{i=1}^{n}x_i^2\right)\\ &= 0\end{aligned}$$

from (12) as required.

Coefficient of determination

The identity (8) motivates one way of measuring the goodness of fit of a statistical model. We introduce the notation

$$\begin{aligned} SS_{total} &= \sum_{i=1}^{n}(y_i - \bar{y})^2 \\ SS_{reg} &= \sum_{i=1}^{n}(\widehat{y}_i - \bar{y})^2 \end{aligned}$$

and

$$SS_{res} = \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$$

so that

$$SS_{total} = SS_{reg} + SS_{res}. \quad (13)$$

We call SS_{total} the total sum of squares, SS_{reg} the regression sum of squares and SS_{res} the residual sum of squares: as we have mentioned, SS_{total}, SS_{reg} and SS_{res} represent the total variation, variation explained by the fit, and residual variation respectively. One measure for describing the fit of a linear regression model is the *coefficient of determination*,

$$R^2 = \frac{SS_{reg}}{SS_{total}}$$

which from (13) can also be written as

$$R^2 = 1 - \frac{SS_{res}}{SS_{total}}.$$

Clearly R^2 is always non-negative and less than or equal to one: it will be one when $SS_{reg} = SS_{total}$, which occurs when $SS_{res} = 0$ (that is, when the fitted values are all equal to the responses). So R^2 is a measure of the quality of the fit, with values close to one indicating a good fit.

R^2 needs to be used with some care for assessing the quality of linear statistical models however. When we discuss the general linear model where we consider multiple predictors, we will see that adding a new predictor cannot decrease R^2, even if the predictor is actually unrelated to the response. So according to this criterion the best model is always the most complex one.

2.6 Distribution theory of estimators in simple linear regression

In this section we describe some basic distribution theory for b_0, b_1 and $\widehat{\sigma}^2$ which enables us to construct confidence intervals and hypothesis tests for β_0 and β_1 in the simple linear regression model.

The χ^2, t and F distributions

We start by reviewing some results from MATH2801. Recall that if $Z_1, ..., Z_n$ are independent standard normal random variables, $Z_1, ..., Z_n \sim N(0,1)$ and if $X = \sum_{i=1}^n Z_i^2$, then X has a chi-squared distribution with n degrees of freedom, and we write $X \sim \chi^2_n$.

Recall also that if $X \sim \chi^2_m$ and $Y \sim \chi^2_n$ and if X and Y are independent, then we say that the ratio

$$F = \frac{X/m}{Y/n}$$

has an F distribution with m and n degrees of freedom, and we write $F \sim F_{m,n}$.

Finally, recall that if Z is a standard normal random variable $Z \sim N(0,1)$, and if X is a chi-squared random variable with n degrees of freedom independent of Z, and if

$$T = \frac{Z}{\sqrt{X/n}}$$

then we say that T has a t distribution with n degrees of freedom, and we write $T \sim t_n$. Observe also that T^2 has an F distribution with 1 and n degrees of freedom.

Distribution theory for b_0, b_1 and $\widehat{\sigma}^2$

We now state some results about the distributions of b_0 and b_1. It is easy to show that b_0 and b_1 are normally distributed (since linear combinations of independent normal random variables are normal, and b_0 and b_1 are linear combinations of the responses). Since b_0 and b_1 are normal, their distributions are determined by their means and variances. From our previous results, we have

$$b_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

and

$$b_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right).$$

For making inferences about β_0 and β_1, we will also need to know something about the distribution of $\widehat{\sigma}^2$. It can be shown that

$$\frac{(n-2)\widehat{\sigma}^2}{\sigma^2}$$

has a chi-squared distribution with $n-2$ degrees of freedom, and that this variate is independent of b_0 and b_1.

Some important results for inference

These results allow us to state the distributions of some important statistics. Consider

$$\frac{(b_0-\beta_0)}{\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}}}.$$

We can write this as

$$\left(\frac{b_0-\beta_0}{\sigma\sqrt{\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}}}\right)\Big/\sqrt{\frac{\frac{(n-2)\widehat{\sigma}^2}{\sigma^2}}{n-2}}$$

which takes the form of a standard normal random variable (b_0 minus its mean divided by its standard deviation) divided by the square root of an independent χ^2_{n-2} random variable ($(n-2)\widehat{\sigma}^2/\sigma^2$) divided by its degrees of freedom. Hence the above statistic has a t distribution with $n-2$ degrees of freedom,

$$\frac{(b_0-\beta_0)}{\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}}} \sim t_{n-2}.$$

Similar reasoning shows that

$$\frac{(b_1-\beta_1)}{\frac{\widehat{\sigma}}{\sqrt{S_{xx}}}}$$

has a t distribution with $n-2$ degrees of freedom. These statistics are useful for creating confidence intervals for β_0 and β_1, as we now describe.

2.7 Confidence intervals for $\boldsymbol{\beta_0}$ and $\boldsymbol{\beta_1}$

We review the basic idea of a confidence interval before describing how to construct confidence intervals for the coefficients in the simple linear regression model.

Confidence intervals

Suppose we have a random vector $Y = (Y_1, ..., Y_n)$ with a distribution depending on an unknown parameter θ. A confidence interval for θ is a rule for computing an interval of plausible values for θ based on the data. Our rule involves functions of the data $L(Y)$ and $U(Y)$ which specify the lower and upper limits of the interval $(L(Y), U(Y))$ of plausible values.

We can talk about the properties of confidence intervals in terms of their behaviour in repetitions of an experiment. Suppose we repeat some experiment many times, and that we apply the rule $(L(Y), U(Y))$ to the data for each experiment in order to specify a range of plausible values for θ. In general, in some experiments the interval $(L(Y), U(Y))$ will contain θ, and in some experiments it will not. If with repeated use our interval covers θ a proportion $1-\alpha$ of the time, we say that $(L(Y), U(Y))$ is a $100(1-\alpha)\%$ confidence interval for θ.

Confidence intervals for $\boldsymbol{\beta_0}$ and $\boldsymbol{\beta_1}$

Suppose we wish to compute a confidence interval for β_1, the slope term in the simple linear regression model. If $t_{\alpha/2;n-2}$ denotes the upper $\alpha/2$ percentage point of the t distribution with $n-2$ degrees of freedom (that is, $t_{\alpha/2;n-2}$ is the value that bounds an area of $\alpha/2$ under the curve in the upper tail of a t_{n-2} density), then we can write

$$\mathbb{P}\left(-t_{\alpha/2,n-2} \leqslant \frac{(b_1-\beta_1)}{\frac{\widehat{\sigma}}{\sqrt{S_{xx}}}} \leqslant t_{\alpha/2,n-2}\right) \quad = \quad 1-\alpha.$$

Rearranging the inequality, we get

$$\mathbb{P}\left(b_1 - t_{\alpha/2;n-2}\frac{\widehat{\sigma}}{\sqrt{S_{xx}}} \leqslant \beta_1 \leqslant b_1 + t_{\alpha/2;n-2}\frac{\widehat{\sigma}}{\sqrt{S_{xx}}}\right) \quad = \quad 1-\alpha.$$

Hence a $100(1-\alpha)$ percentage confidence interval for β_1 is

$$\left(b_1 - t_{\alpha/2;n-2}\frac{\widehat{\sigma}}{\sqrt{S_{xx}}}, b_1 + t_{\alpha/2;n-2}\frac{\widehat{\sigma}}{\sqrt{S_{xx}}}\right).$$

Similar reasoning to the above shows that a $100(1-\alpha)$ percentage confidence interval for β_0 is

$$\left(b_0 - t_{\alpha/2;n-2}\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}}, b_0 + t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}}\right).$$

Example: zinc concentrations in plants

Consider once more the data on zinc concentrations described in the first lecture. The data consist of fifteen measurements $y_1, ..., y_{15}$ of zinc concentrations in a certain plant taken from fifteen different lakes, and fifteen measurements $x_1, ..., x_{15}$ of zinc concentrations from sediment samples in the lakes. Using the fact that

$$\sum_{i=1}^{15} y_i = 808.4 \qquad \sum_{i=1}^{15} x_i = 1205.2$$

$$\sum_{i=1}^{15} x_i^2 = 101228.6 \qquad \sum_{i=1}^{15} y_i^2 = 48130.92$$

$$\sum_{i=1}^{15} x_i y_i = 68716.78$$

we compute the least squares estimates b_0 and b_1 of β_0 and β_1 as -14.9 and 0.857 respectively. We now derive a 95% confidence interval for β_1.

The expression for a 95% confidence interval for β_1 is

$$\left(b_1 - t_{0.025,n-2}\frac{\widehat{\sigma}}{\sqrt{S_{xx}}}, b_1 + t_{0.025,n-2}\frac{\widehat{\sigma}}{\sqrt{S_{xx}}}\right).$$

Now, we have that

$$\widehat{\sigma}^2 = \frac{S_{yy} - b_1^2 S_{xx}}{n-2}$$

where as usual

$$S_{yy} = \sum_i y_i^2 - \frac{\left(\sum_i y_i\right)^2}{n}$$

and

$$S_{xx} = \sum_i x_i^2 - \frac{\left(\sum_i x_i\right)^2}{n}.$$

Substituting, we have $S_{yy} = 4563.54$ and $S_{xx} = 4394.80$ and hence

$$\begin{aligned}\widehat{\sigma}^2 &= \frac{4563.54 - 0.857^2 4394.80}{13} \\ &= 102.99\end{aligned}$$

or $\widehat{\sigma} = 10.14$. Furthermore, the upper 2.5 percentage point of a t-distribution with $n - 2 = 13$ degrees of freedom is $t_{0.025,13} = 2.16$ (this can be computed from tables or a computer package like R). Hence the confidence interval is

$$(0.857 - 2.16\frac{10.14}{\sqrt{4394.80}}, 0.857 + 2.16\frac{10.14}{\sqrt{4394.80}}) = (0.526, 1.188).$$

2.8 Hypothesis testing for $\boldsymbol{\beta_0}$ and $\boldsymbol{\beta_1}$

We review the basic ideas of hypothesis testing before discussing hypothesis tests for β_0 and β_1 in the simple linear regression model.

Hypothesis testing

Suppose we have a random vector $Y = (Y_1, ..., Y_n)$ with a distribution depending on a parameter θ. In hypothesis testing there are two competing claims about θ called the null hypothesis (written H_0) and the alternative hypothesis (written H_1) and we want to decide between them based on the data.

The null hypothesis H_0 is distinguished by the fact that we only want to reject H_0 if there is very strong evidence against it. An analogy is often made between hypothesis testing and a criminal trial. In a criminal trial, the state of the defendant is either "innocent" or "guilty" (think of the defendant's state as the parameter) and we want to decide which of these competing claims is true based on the evidence. The null hypothesis in this analogy would be "The defendant is innocent," since (in our justice system) we only want to convict someone if there is very strong evidence that they are guilty (beyond reasonable doubt). The probability of wrongfully convicting an innocent person must be controlled at some small level.

The way that hypothesis testing proceeds is by calculating a function of the data $t(Y)$ (called the test statistic) and then defining a set of values for the test statistic where H_0 will be rejected (the critical region). If $t(Y)$ does not lie in the critical region then H_0 is accepted. So we have some rule for deciding between H_0 and H_1 based on the data.

In hypothesis testing there are two kinds of errors which can be made. We can reject H_0 when it is true (this is called a type I error) or we can accept H_0 when it is false (this is called a type II error). Since we only want to reject H_0 if there is very strong evidence, we fix the probability of making a type I error at some small level, called the significance level of the test (usually denoted α). In the example of the criminal trial, we can think of the significance level as defining our standard of "reasonable doubt".

Subject to fixing the probability of a type I error at the level α, a good test is one that minimizes the probability of making a type II error.

Hypothesis testing for $\boldsymbol{\beta_1}$

We can use our results about the distributions of b_0, b_1 and $\widehat{\sigma}^2$ to test hypotheses about β_0 and β_1.

To test the hypothesis

$$H_0 : \beta_1 = \beta$$

versus the alternative

$$H_1 : \beta_1 \neq \beta$$

we use the test statistic

$$T = \frac{b_1 - \beta}{\widehat{\sigma}/\sqrt{S_{xx}}}$$

which has a t_{n-2} distribution under H_0.

The critical region for a test at significance level α is

$$T < -t_{\alpha/2;n-2} \text{ or } T > t_{\alpha/2;n-2}.$$

Clearly the probability of lying in this region (rejecting H_0) when H_0 is true is α (since under H_0 the test statistic T has a t_{n-2} distribution).

p-values

There is an alternative way of reporting the results of a hypothesis test that you may remember from MATH2801. Rather than adopting a certain significance level for a test, finding the critical region for that test and then seeing whether or not the test statistic lies in the critical region, we can find the value for the significance level such that the observed value of the test statistic lies "just on the boundary" of the critical region. This critical value for the signifiance level is called the p-value for the test. If the significance level is bigger than the p-value, then the test statistic lies in the critical region and H_0 is rejected, whereas if the significance level is less than the p-value than H_0 is accepted.

In the case of the test we have just described for β_1, if we observe $T = t$, we can compute the p-value as

$$p = \mathbb{P}(|T| \geqslant |t||\beta_1 = \beta).$$

and then we compare this probability with the significance level α for the test. If $p < \alpha$, then we reject H_0, whereas if $p \geqslant \alpha$ we fail to reject H_0. In computing the p-value, we are asking ourselves: what is the probability of getting a value as extreme as the one observed for the test statistic or more extreme given H_0 is true (where what is "extreme" is defined by the alternative)? If this probability is small, then we look to the alternative hypothesis as a possible explanation.

One-sided test for β_1

In the discussion above we have considered only the alternative hypothesis $\beta_1 \neq \beta$ (called a two sided alternative). We can also construct tests for $H_0 : \beta_1 = \beta$ against the one sided alternatives $H_1 : \beta_1 > \beta$ or $H_1 : \beta_1 < \beta$. For $H_1 : \beta_1 > \beta$ the critical region for a size α test is $T > t_{\alpha;n-2}$ and the p-value of the test is comptued as

$$p = \mathbb{P}(T \geqslant t|\beta_1 = \beta).$$

For $H_1 : \beta_1 < \beta$ the critical region is $T < -t_{\alpha;n-2}$ and the p-value of the test is

$$p = \mathbb{P}(T \leqslant t | \beta_1 = \beta).$$

Hypothesis testing for β_0

So far we have only dealt with the construction of tests for the coefficient β_1. We can also construct a test of the null hypothesis $H_0 : \beta_0 = \beta$ against one and two sided alternatives.

To test the hypothesis

$$H_0 : \beta_0 = \beta$$

versus the alternative

$$H_1 : \beta_0 \neq \beta$$

we use the test statistic

$$T = \frac{b_0 - \beta}{\widehat{\sigma}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}}$$

which has a t_{n-2} distribution under H_0. The critical region for a test at significance level α is

$$T < -t_{\alpha/2,n-2} \text{ or } T > t_{\alpha/2,n-2}$$

and the p-value of the test is

$$p = \mathbb{P}(|T| \geqslant |t| | \beta_0 = \beta)$$

where t is the observed value of the test statistic.

For the one-sided alternatives $H_1 : \beta_0 > \beta$ or $H_1 : \beta_0 < \beta$ we modify the critical region to $T > t_{\alpha,n-2}$ or to $T < -t_{\alpha,n-2}$ respectively. The p-value for the test with $H_1 : \beta_0 > \beta$ is

$$p = \mathbb{P}(T \geqslant t | \beta_0 = \beta).$$

The p-value for the test with $H_1 : \beta_0 < \beta$ is

$$p = \mathbb{P}(T \leqslant t | \beta_0 = \beta).$$

Example: zinc concentrations in plants

Continuing our example on zinc concentration in plants, suppose we want to test the null hypothesis

$$H_0 : \beta_1 = 0$$

versus

$$H_1 : \beta_1 \neq 0$$

at the 5% level. We illustrate the computation of the p-value for the test. The test statistic here is

$$\frac{b_1}{\widehat{\sigma}/\sqrt{S_{xx}}}$$

which we compute as

$$\frac{0.857}{10.14/\sqrt{4934.8}} = 5.59.$$

Under H_0, this is a realization of a t random variable with 13 degrees of freedom, so that if $T \sim t_{13}$ we can compute the p-value as

$$\begin{aligned} p &= \mathbb{P}(|T| \geqslant 5.59) \\ &= 2Pr(T \geqslant 5.59) \\ &= 8.69\mathtt{e} - 05 \end{aligned}$$

(from tables or R). So at the 5% level we reject H_0 since $p < 0.05$. That is, we believe that the zinc concentration in the sediment samples is a useful predictor of zinc concentration in the plants.

Instead of computing the p-value we could also have computed the critical region for the test. The critical region is

$$T < -t_{0.025;13} \text{ or } T > t_{0.025;13}$$

and from R or tables we have $t_{0.025;13} = 2.16$, so that the value of the test statistic (5.59) lies in the critical region, and H_0 is rejected.

2.9 The Analysis of Variance (ANOVA) table

It is often of particular interest to test the null hypothesis

$$H_0 : \beta_1 = 0$$

versus the alternative

$$H_1 : \beta_1 \neq 0.$$

When we test this hypothesis we are asking ourselves: is the predictor in the linear regression model at all useful for explaining variation in the response? As we have just seen, we can do this test using the test statistic

$$\frac{b_1}{\widehat{\sigma}/\sqrt{S_{xx}}}. \tag{14}$$

which has a t_{n-2} distribution under the null hypothesis.

We now derive an equivalent F statistic for the above test, and describe a way of showing the calculations involved using the *analysis of variance table*. Observe that

$$\begin{aligned} SS_{reg} &= \sum_{i=1}^{n}(\widehat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^{n}(b_0 + b_1 x_i - \bar{y})^2 \\ &= \sum_{i=1}^{n}(\bar{y} - b_1\bar{x} + b_1 x_i - \bar{y})^2 \\ &= \sum_{i=1}^{n} b_1^2(x_i - \bar{x})^2 \\ &= b_1^2 S_{xx}. \end{aligned} \tag{15}$$

Now, squaring the statistic (14) gives (under H_0) a random variable with an F distribution with 1 and $n-2$ degrees of freedom:

$$\frac{b_1^2 S_{xx}}{\widehat{\sigma}^2}.$$

Using (15) we can write this last statistic as

$$F = \frac{SS_{reg}/1}{SS_{res}/(n-2)}. \tag{16}$$

So we can write down an F statistic for testing for the usefulness of a predictor in a simple linear regression model, and this F statistic is defined from the terms in the fundamental partition of variation we have described previously. The critical region for the test based on (16) is $F > F_{\alpha;1,n-2}$, where $F_{\alpha;1,n-2}$ is the upper 100α percentage point of the $F_{1,n-2}$ distribution (so we reject $H_0 : \beta_1 = 0$ if the variation explained by the model is large compared to the residual variation).

The *analysis of variance table* presents for the simple linear regression model the decomposition of total variation into SS_{reg} and SS_{res} and shows the computation of the F statistic above. The analysis of variance table has the form shown below. The sum of squares column shows the partitioning of the total variation. The second column, the degrees of freedom column, shows the degrees of freedom parameters describing the null distribution of the statistic (16) (we will come to a deeper understanding of the degrees of freedom parameters when we talk about the general linear model). Entries in the mean square column are computed by dividing the corresponding entry in the sum of squares column by its degrees of freedom, and in the final column we have the F statistic which is used for testing for the usefulness of the predictor. We will generalize this analysis of variance

Source	Sum of Squares	Degrees of freedom	Mean Square	F
Regression	SS_{reg}	1	$MS_{reg} = SS_{reg}/1$	$MS_{reg}/\widehat{\sigma}^2$
Residual	SS_{res}	$n-2$	$MS_{res} = \frac{SS_{res}}{(n-2)} = \widehat{\sigma}^2$	
Total	SS_{total}	$n-1$		

table later when we talk about the general linear model. The analysis of variance table is a useful tool for displaying the sources of variation in data, and for organizing the calculations involved in certain tests of hypotheses. We conclude this section with two examples.

Example: market model of stock returns

We return to our example on the market model of stock returns (from Keller, Warrack and Bartel, "Statistics for Management and Economics: a Systematic Approach (Second Edition)", Wadsworth, Belmont, 1990, p. 725). Recall that in the market model the monthly rate of return on a stock (R) is linearly related to the monthly rate of return on the overall stock market (R_m). The mathematical description of the model is

$$R = \beta_0 + \beta_1 R_m + \varepsilon$$

where the error term ε is assumed to satisfy the requirements of the simple linear regression model. We are interested in the coefficient β_1, called the stock's *beta coefficient*, which measures how sensitive the stock's rate of return is to changes in the level of the overall market. If $\beta_1 > 1$ ($\beta_1 < 1$), the stock's rate of return is more (less) sensitive to changes in the level of the overall market than is the average stock. Our data consist of 5 years of monthly rates of return on Host International stock and rates of return on the NYSE composite index (measuring the rate of return on the overall stock market). When we considered this data set previously, we computed b_1 as 1.60, b_0 as -0.14 and $\widehat{\sigma}$ as 9.27. Also, $S_{xx} = 1117.90$.

Since there is uncertainty in our estimate of the beta coefficient it is of interest to compute a confidence interval for this coefficient, and to test the hypothesis that the beta coefficient is 1 (is there any real evidence that the stock is more or less sensitive than average to the overall market level given the inherent variation in the data?) A $100(1-\alpha)$ percentage confidence interval for β_1 is

$$\left(b_1 - t_{\alpha/2,n-2} \frac{\widehat{\sigma}}{\sqrt{S_{xx}}}, b_1 + t_{\alpha/2,n-2} \frac{\widehat{\sigma}}{\sqrt{S_{xx}}} \right)$$

which gives a 95% confidence interval in this case of

$$\left(1.60 - 2.002\frac{9.27}{\sqrt{1117.90}}, 1.60 + 2.002\frac{9.27}{\sqrt{1117.90}}\right) = (1.04, 2.16).$$

We see that the 95% confidence interval does not contain 1, so there seems to be some support for the assertion that the stock is more sensitive than average to the overall market level. Looking at whether the confidence interval contains 1 is equivalent to performing a certain hypothesis test on β_1. We test

$$H_0 : \beta_1 = 1$$

versus

$$H_1 : \beta_1 \neq 1$$

using the test statistic

$$\frac{b_1 - 1}{\widehat{\sigma}/\sqrt{S_{xx}}}$$

which evaluates here to

$$\frac{1.60 - 1}{9.27/\sqrt{1117.90}} = 2.16.$$

So if $T \sim t_{58}$, the p-value for the test is

$$\begin{aligned} p &= \mathbb{P}(|T| \geqslant 2.16) \\ &= 2Pr(T \geqslant 2.16) \\ &= 0.0349 \end{aligned}$$

(from tables or R) so that we reject $H_0 : \beta_1 = 1$ at the 5% level. If we were to conduct a one-tailed test, $H_1 : \beta_1 > 1$ say, computation of the p-value would be altered to

$$\begin{aligned} p &= \mathbb{P}(T \geqslant 2.16) \\ &= 0.0175 \end{aligned}$$

and again we would reject $H_0 : \beta_1 = 1$ at the 5% level.

Example: Risk Assessment from Financial Reports

We describe a data set now that we will return to later when considering linear regression models with multiple predictors. The following example is described in Gail E. Farrelly, Kenneth R. Ferris and William R. Reichenstein, "Perceived Risk, Market Risk and Accounting-Determined Risk Measures", *Accounting Review* 60 (1985), pp. 287–88.

Investors are interested in assessing the riskiness of a company's common stock, as well as its expected rate of return. It is therefore desirable to potential investors that a company's financial reports provide information to help them assess the company's risk.

Farrelly, Ferric and Reichenstein conducted an investigation into the relationship between seven accounting-determined measures of risk and the average risk assessment of financial analysts. The seven accounting-determined measures of risk (all of which could be computed from a company's financial reports) and their definitions are as follows:

Dividend payout, x_1: (Cash dividends)/(Earnings)

Current ratio, x_2: (Current assets)/(Current liabilities)

Asset size, x_3: log(Total assets)

Asset growth, x_4: Average growth rate in asset size for the years 1977-1981

Leverage, x_5: (Total senior debt)/(Total assets)

Variability in earnings, x_6: Standard deviation of the price-earnings ratio for the years 1977-1981

Covariability in earnings, x_7: Strength of the relationship between a firm's price-earnings ratio and the average price-earnings ratio of the market overall

These seven measures were computed for 25 well-known stocks, based on data from the companies' annual reports from 1977-1981. These data were then sent to a random sample of 500 financial analysts, who "were requested to assess the risk of each of the 25 companies on a scale of 1 (low) to 9 (high), assuming that the stock was to be added to a diversified portfolio." The mean rating, $\boldsymbol{y}$, assigned by the 209 financial analysts who responded is recorded for each of the 25 stocks. This measure of the financial analysts' risk perception was taken to be a reasonable surrogate for the (market) risk of each stock.

Prediction of market risk based on a linear regression model with multiple predictors is a problem we will consider later. For the moment, we consider a simple linear regression model for predicting market risk based on asset size. A scatterplot of market risk versus asset size is shown in Figure 2.3. We do not describe in detail the computation of parameters or computation of test statistics here, but the fitted line is

$$\widehat{y}_i = 8.1433 - 0.4123x_i,$$

the estimated error standard deviation is 1.475, and the p-value for testing $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$ using the usual t-statistic is $p = 0.0211$. So it seems that asset size is a useful predictor of market risk. In later work on linear regression models with multiple predictors we consider the problem of which

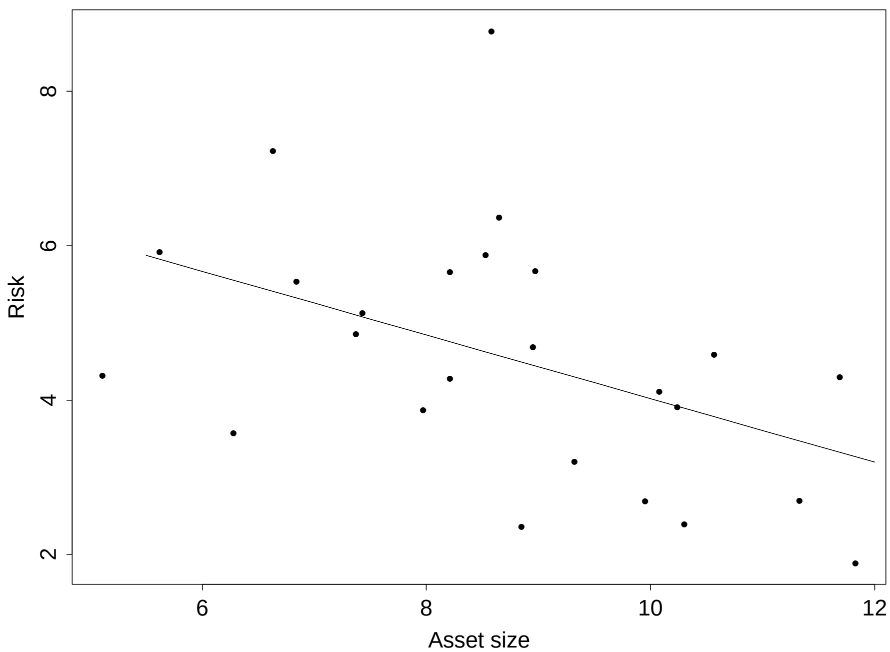

Figure 2.3: Scatter plot of assessment of market risk versus asset size with fitted least squares regression line.

predictors should be chosen for predicting market risk from amongst the seven predictors described above. Choosing subsets of predictors when there are many to choose from in explaining variation in the response is a much trickier problem than the one we have considered here, and further discussion is delayed until we have given a thorough treatment of the general linear model.

2.10 Prediction in the simple linear regression model

The purpose of building a simple linear regression model is often to predict a future response value when the value of the predictor (x_0 say) for that observation is known. Consider, for instance, the risk assessment data in the last subsection. For the companies in the study we know their asset size and an average rating of risk from 209 financial analysts. For a new company which has not been assessed by the analysts we can obtain the predictor (asset size) from company reports, and we can forecast risk using the fitted regression line.

Obviously we may also wish to quantify our uncertainty about predictions from a simple linear regression model. In this subsection we consider construction of confidence intervals on the mean response for a new observation, as well as construction of *prediction intervals* which cover the actual response for a new observation with specified probability.

Suppose that we have fitted a simple linear regression model to data consisting of responses $y_1, ..., y_n$ and predictors $x_1, ..., x_n$ and that we have obtained least squares estimates b_0 and b_1 of β_0 and β_1, and an estimate $\widehat{\sigma}^2$ of σ^2. How do we estimate the mean response for a future observation given a value x_0 for the predictor? Also, how do we estimate confidence in the estimated mean and in a future response when we know the value of the predictor?

An intuitively reasonable estimator of the mean response when the predictor is x_0 is of course simply

$$b_0 + b_1 x_0$$

which following standard notation we write as $\widehat{y}(x_0)$. The actual mean response when the predictor is x_0 is

$$\beta_0 + \beta_1 x_0$$

and a new observation y_0 at this predictor value can be written as

$$\beta_0 + \beta_1 x_0 + \varepsilon_0$$

where ε_0 is a normally distributed error term with mean zero and variance σ^2, independent of $y_1, ..., y_n$. A confidence interval for the mean response at x_0 will take into account the uncertainty in estimating β_0 and β_1, and a prediction interval for y_0 (which is a random interval containing y_0 with specified probability) will take into account both the uncertainty in estimating β_0 and β_1 and the variability of ε_0.

Confidence intervals for the mean response

First we give an expression for the variance of $\widehat{y}(x_0)$: it can be shown that

$$\mathrm{Var}(\widehat{y}(x_0)) = \sigma^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right).$$

In discussing the general linear model later we will show that b_0 and b_1 are jointly normally distributed, so that $\widehat{y}(x_0)$ (which is a linear combination of b_0 and b_1) is normally distributed. Hence since $\mathbb{E}(\widehat{y}(x_0)) = \mathbb{E}(b_0+b_1x_0) = \beta_0+\beta_1x_0$ we have

$$\widehat{y}(x_0) \sim N\left(\beta_0+\beta_1x_0, \sigma^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)\right)$$

or

$$\frac{\widehat{y}(x_0)-\beta_0-\beta_1x_0}{\sigma\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}} \sim N(0,1). \tag{17}$$

Also, we have as before that

$$\frac{(n-2)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2} \tag{18}$$

and that this variate is independent of b_0 and b_1 (and hence of $\widehat{y}(x_0)$). So dividing (17) by the square root of (18) divided by its degrees of freedom gives

$$\frac{\widehat{y}(x_0)-\beta_0-\beta_1x_0}{\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}} \sim t_{n-2}.$$

Hence if $t_{\alpha/2,n-2}$ is the upper $100\times\alpha/2$ percentage point of a t distribution with $n-2$ degrees of freedom, we have

$$\mathbb{P}\left(-t_{\alpha/2,n-2} \leqslant \frac{\widehat{y}(x_0)-\beta_0-\beta_1x_0}{\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}} \leqslant t_{\alpha/2,n-2}\right) = 1-\alpha$$

or

$$\mathbb{P}\left(\widehat{y}(x_0)-t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}} \leqslant \beta_0+\beta_1x_0 \leqslant \widehat{y}(x_0)+t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}\right) = 1-\alpha.$$

Hence a $100(1-\alpha)$ percentage confidence interval for the mean response $\beta_0+\beta_1x_0$ when the predictor is x_0 is

$$\left(\widehat{y}(x_0)-t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}, \widehat{y}(x_0)+t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}\right).$$

Prediction intervals

It remains to find a prediction interval for a new observation y_0 when the predictor value is x_0. Since y_0 and $\widehat{y}(x_0)$ are independent, we have that

$$\begin{aligned}\mathrm{Var}(y_0 - \widehat{y}(x_0)) &= \mathrm{Var}(y_0) + \mathrm{Var}(\widehat{y}(x_0)) \\ &= \sigma^2 + \sigma^2\left(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right) \\ &= \sigma^2\left(1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right).\end{aligned}$$

Also, $\mathbb{E}(y_0 - \widehat{y}(x_0)) = \beta_0 + \beta_1 x_0 - \beta_0 - \beta_1 x_0 = 0$. Now, y_0 is normally distributed, and so is $\widehat{y}(x_0)$, and y_0 and $\widehat{y}(x_0)$ are independent, so we can write

$$\frac{y_0 - \widehat{y}(x_0)}{\sigma\sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}} \sim N(0,1)$$

and of course as before

$$\frac{(n-2)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$$

independently of y_0. So similar reasoning to before gives

$$\frac{y_0 - \widehat{y}(x_0)}{\widehat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}} \sim t_{n-2}$$

and a $100(1-\alpha)\%$ prediction interval for y_0 is

$$\left(\widehat{y}(x_0) - t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}, \widehat{y}(x_0) + t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}\right).$$

The expression given above for the prediction interval is a very natural one. It tells us that our uncertainty of prediction is related to the residual error variance (estimated by $\widehat{\sigma}^2$), the sample size, and how far x_0 is away from the mean of the predictors $\bar{x}$. If x_0 is far away from where most of the observed predictors lie, then our uncertainty of prediction is increased. We must in any case be very careful about prediction for values of the predictor far away from any of the observed x_i: while linearity and the other assumptions of the simple linear regression model may seem to hold locally, these assumptions can break down when we are far from observed values of the predictors. Extrapolation is dangerous!

Example: risk assessment data

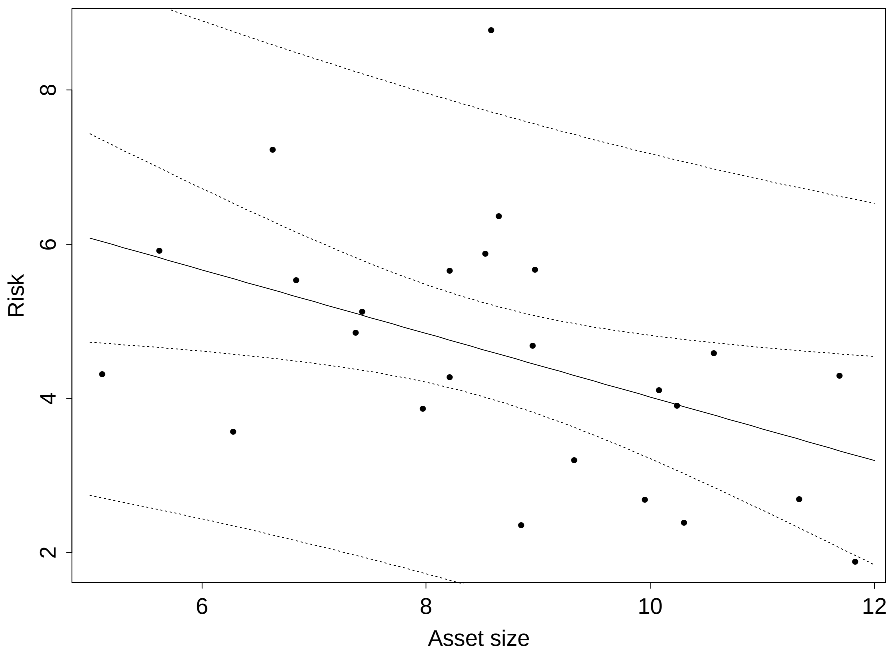

Figure 2.4: Scatter plot of assessment of market risk versus asset size with fitted least squares regression line and 95 percent confidence intervals for the mean (inner bands) and 95 percent prediction intervals (outer bands).

We conclude this section with an application to the risk assessment data. Recall that in the previous subsection we built a simple linear regression model to predict risk based on asset size. Figure 2.4 shows a scatterplot of these data, together with 95 percent confidence intervals for the mean over a range of values for the predictor (the inner bands) and prediction intervals over the same range of predictor values (the outer bands). Of course, the prediction intervals are wider than the confidence intervals for the mean, since they reflect both our uncertainty in estimating the conditional mean and the inherent variability of the errors.

It should be pointed out that the bands on the graph in Figure 2.4 show pointwise confidence and prediction intervals and this must be kept in mind when interpreting the graph. Consider, for instance, the inner bands, which give the upper and lower confidence limits for the mean response over the range of predictors in the graph. At each point we have 95 percent confidence that the mean response lies between the bounds at that point: however, are we 95 percent confident that the true mean response lies between the upper and lower limits over the whole range of the predictors in the graph? That is, if we repeated this experiment a large number of times, would the true mean response lie between all the upper and lower confidence limits (the confidence bands) simultaneously in 95 percent of the repetitions of the experiment? In general, the answer is no.

2.11 Simultaneous inference and Bonferroni adjustment

We now turn to a discussion of simultaneous inference, which we touched on at the end of the last subsection. The confidence intervals for the mean that we drew in Figure 2.4 were pointwise intervals, and it is not true that the true regression line lies between the bands with 95% confidence, even though for any point x_0 $\beta_0 + \beta_1 x_0$ lies between the bands at x_0 with 95% confidence.

Let's be more precise. Write $\mathbb{E}(x_0; \alpha)$ for the random interval

$$\left(\widehat{y}(x_0) - t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{n}}, \widehat{y}(x_0) + t_{\alpha/2,n-2}\widehat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{n}} \right).$$

We know that

$$\mathbb{P}(\beta_0 + \beta_1 x_0 \in \mathbb{E}(x_0; \alpha)) = 1 - \alpha$$

so that $\mathbb{E}(x_0; \alpha)$ defines a $100(1-\alpha)\%$ confidence interval.

Now suppose we consider two values for the predictor, $x_{0,1}$ and $x_{0,2}$ say. We want to specify a pair of intervals which will contain $\beta_0 + \beta_1 x_{0,1}$ and $\beta_0 + \beta_1 x_{0,2}$ simultaneously with $100(1-\alpha)\%$ confidence.

Pointwise $100(1-\alpha)\%$ confidence intervals do not guarantee joint coverage with confidence $100(1-\alpha)\%$. That is, it is not true in general that

$$\mathbb{P}(\beta_0 + \beta_1 x_{0,1} \in \mathbb{E}(x_{0,1}; \alpha) \text{ and } \beta_0 + \beta_1 x_{0,2} \in \mathbb{E}(x_{0,2}; \alpha)) = 1 - \alpha.$$

To achieve the desired joint confidence level we must adjust the confidence level for the pointwise intervals.

2.12 Bonferroni adjustment

Suppose that we have parameters $\theta_1, ..., \theta_k$ and that we have a method for constructing pointwise confidence intervals for each of these parameters. Then it can be shown that if we construct pointwise $100(1-\alpha/k)\%$ confidence intervals for $\theta_1, ..., \theta_k$, then these intervals will have joint coverage of at least $100(1-\alpha)\%$. This adjustment to the pointwise confidence levels is referred to as *Bonferroni adjustment*. (See your tutorial sheet for justification).

The ideas of simultaneous inference and Bonferroni adjustment can be applied in hypothesis testing as well. An analogy here is with the notion of double jeopardy in a criminal trial. This is the idea that a person should not be tried for the same crime twice. Why would this not be fair?

Suppose that someone is tried for the same crime twice. They are convicted either if they are convicted in the first trial or in the second trial. If the probability is α that the person is wrongfully convicted in one of the two trials, then in general the probability of wrongful conviction on either of the two trials will be bigger than α. (Running the same small risk twice will result in a bigger risk).

In statistical language, suppose that we have parameters $\theta_1, ..., \theta_k$ and that we have for each $i = 1, ..., k$ a way of testing the null hypothesis

$$H_0 : \theta_i \in \Theta_i$$

against the alternative

$$H_1 : \theta_i \in \Theta_i^c$$

where Θ_i is some appropriate subset of the parameter space and Θ_i^c is its complement (think of these tests as being the individual trials in our double jeopardy analogy with H_0 as "The defendant is innocent").

Suppose we want to test

$$H_0 : \theta_1 \in \Theta_1, ..., \theta_k \in \Theta_k$$

against

$$H_1 : \text{Not all } \theta_i \in \Theta_i,\ \ i = 1, ..., k$$

at significance level α. One approach to this test is to conduct individual tests on the coefficients θ_i and to reject the null hypothesis if any of the individual tests are rejected. But what significance level should be used for the individual tests to ensure significance level α for the joint test? It turns out that if we adjust the significance level for the individual tests to be α/k, then the joint test has significance level less than or equal to α. This is the Bonferroni adjustment to the significance level.

We should point out that the Bonferroni adjustment is conservative in the sense that the Bonferroni confidence intervals discussed above give a joint coverage which is usually greater than $100(1-\alpha)\%$, and the Bonferroni adjustment to significance levels in hypothesis testing results in a test with significance level less than α. Other methods for constructing joint confidence regions and tests are preferable when they are available: for linear models, alternative methods often are available (see later discussion for the general linear model).

2.13 Criticism of the model: residuals

At the beginning of the course we described the process of building a statistical model as an iterative procedure where the analyst cycles through stages of model formulation, estimation and criticism. We have described in detail for the simple linear regression model some of the tools which are used in model formulation and selection and estimation. In this section we take a brief look at one method for criticism of a model.

Suppose we have fitted a simple linear regression model to data consisting of responses $y_1, ..., y_n$ and predictors $x_1, ..., x_n$. Under the assumptions of the model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

where the ε_i, $i = 1, ..., n$, are uncorrelated zero mean errors with common variance σ^2. We can estimate β_0 and β_1 via the least squares estimators b_0 and b_1. The fitted values are

$$\widehat{y_i} = b_0 + b_1 x_i$$

for $i = 1, ..., n$. We define the residuals of the fitted model to be the differences between the responses and the fitted values,

$$e_i = y_i - \widehat{y_i}$$

$i = 1, ..., n$. If we can estimate β_0 and β_1 precisely using b_0 and b_1, then the residuals should be approximately the error terms ε_i and so we can use the residuals to check some of the assumptions made about the error terms in the simple linear regression model. As well as the residuals e_i (sometimes called the raw residuals) there are other kinds of residuals which may be more useful in some situations, but we will not discuss these at the moment.

What happens when the assumptions of the simple linear regression model are violated? It is common for statisticians to use plots of the residuals against the fitted values or predictor values to try to detect an incorrect specification of the systematic part of the model or to detect departures from the constancy of variance assumption for the errors. Since the error terms in the simple linear regression model are supposed to have zero mean, any systematic trend in the residuals as the fitted values or predictor values increase provides some evidence that the errors do not have zero mean and that the mean structure is not correctly specified. Similarly, increasing variability as the fitted values or predictor values increase can indicate that the residuals do not have a common variance σ^2. We can also use the residuals to check the assumption of normality of errors, although we do not discuss this further at the present time.

The usefulness of residual plots for model criticism is perhaps best shown via some examples. Figure 2.5 shows a scatterplot of some synthetic data together with a fitted linear regression model. Below is a plot of the residuals versus the fitted values: there is a clear trend in the mean level of the residuals as the fitted values increase, suggesting that perhaps a model which is nonlinear in the predictor would be more appropriate than the model considered here. This example shows how beneficial residual plots can be for detecting departures from the assumed mean structure for the model.

We give another synthetic example which shows the usefulness of the residuals for detection of violations of the assumption of constancy of variance of the errors. It often happens in data sets from many areas of science, social science and economics that the variation in the response increases as the mean response increases. Figure 2.6 shows a scatterplot for a synthetic data set which exhibits this kind of behaviour, together with a fitted linear regression model. Below it is the corresponding residual plot: the increasing trend in the variance of the residuals is obvious.

We will say much more about residuals later in this course.

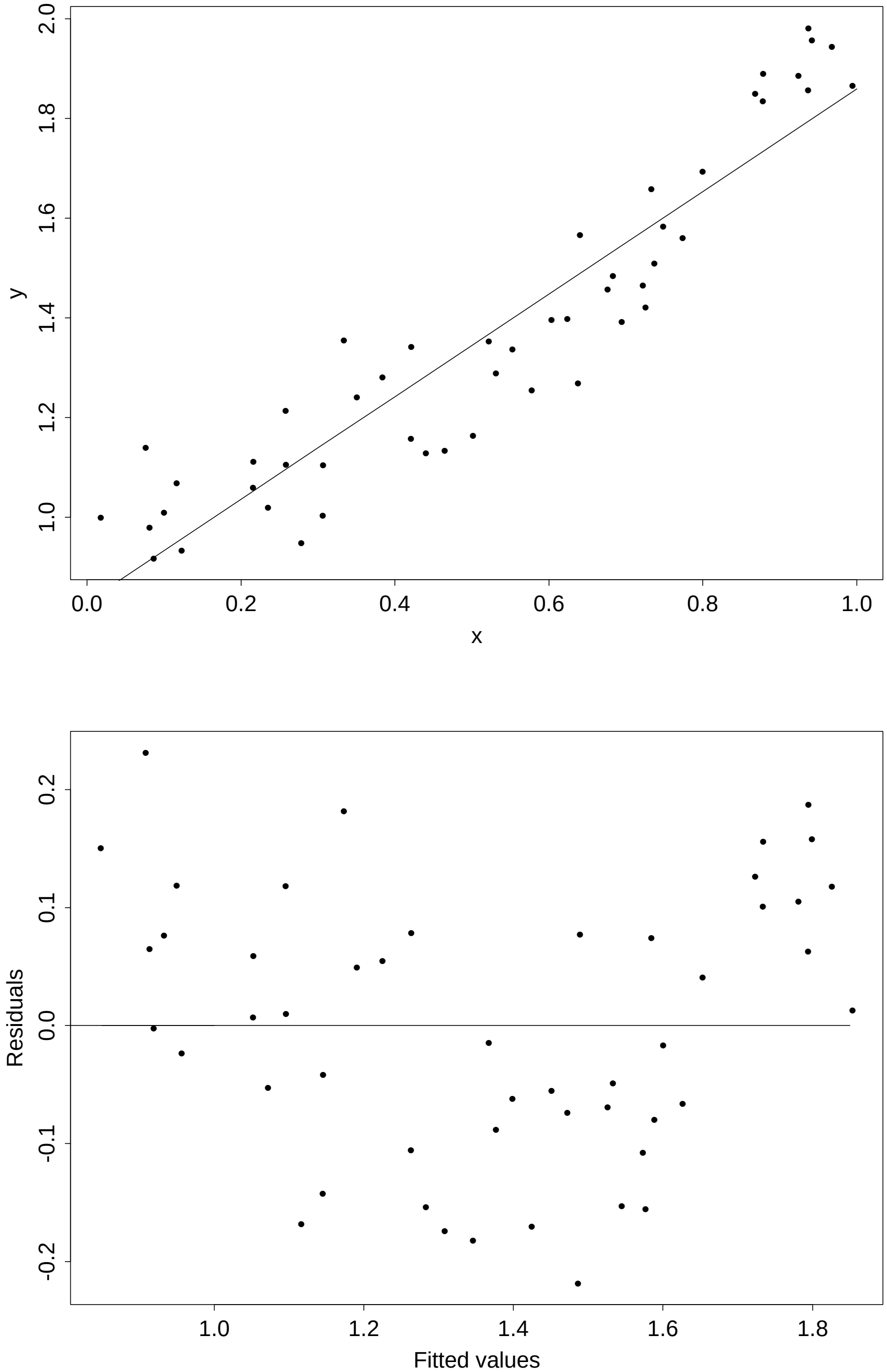

Figure 2.5: Scatter plot of synthetic data set with fitted linear regression (top) and plot of residuals versus fitted values.

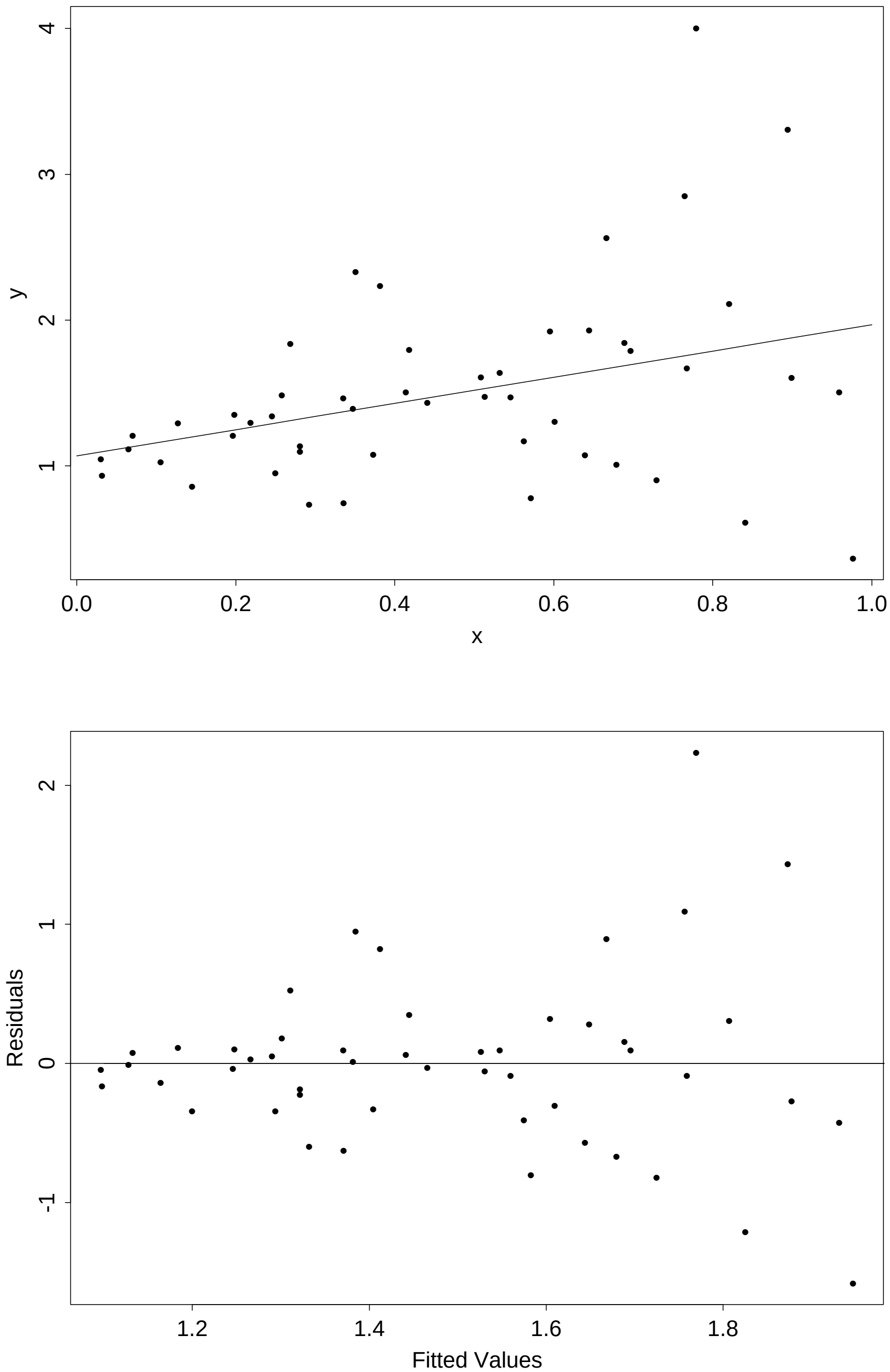

Figure 2.6: Scatter plot of synthetic data set with fitted linear regression (top) and plot of residuals versus fitted values.

3 The general linear model

In previous sections we considered the process of model building for the simple linear regression model. We attempted to explain variation in the response in terms of variation of a single predictor. In many practical problems, however, there are multiple predictors which may be useful for explaining variation in the response, and we can extend the simple linear regression model to the so-called *general linear model* to deal with this more complicated situation.

The rest of this course is concerned with developing tools for model selection, estimation and model criticism for the general linear model, just as we did for the simple linear regression model.

3.1 Formulation of the general linear model

Extending our previous notation, we write y_i $i = 1, ..., n$ for the response variable, but now corresponding to each y_i we have values of k predictor variables $x_{i1}, ..., x_{ik}$. In the *general linear model* we assume that

$$y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_k x_{ik} + \varepsilon_i$$

where $\beta_0, ..., \beta_k$ are unknown parameters and ε_i, $i = 1, ..., n$ are a collection of uncorrelated errors with zero mean and common variance σ^2. As for the simple linear regression model, we will also sometimes make the assumption that the errors are normally distributed. This assumption is needed for constructing confidence intervals and prediction intervals and for hypothesis testing.

Example: Risk Assessment Data

We discussed earlier the problem of forecasting the risk of a company's common stock based on accounting-determined measures of risk which can be obtained from a company's financial reports. The measures of risk considered were as follows:

Dividend payout, x_1: (Cash dividends)/(Earnings)

Current ratio, x_2: (Current assets)/(Current liabilities)

Asset size, x_3: log(Total assets)

Asset growth, x_4: Average growth rate in asset size for the years 1977- 1981

Leverage, x_5: (Total senior debt)/(Total assets)

Variability in earnings, x_6: Standard deviation of the price-earnings ratio for the years 1977-1981

Covariability in earnings, x_7: Strength of the relationship between a firm's price-earnings ratio and the average price-earnings ratio of the market overall

These measures of risk were determined for 25 companies, and sent to 500 randomly chosen financial analysts, 209 of whom responded to a request to rank the risk of each of the 25 companies on a scale of 1 (low) to 9 (high), assuming that the stock was to be added to a diversified portfolio. The average of the ratings ($\mathbf{y}$) for the 209 analysts was then computed for each of the 25 stocks.

We are interested in predicting risk (the response $\mathbf{y}$) in terms of the accounting-determined measures of risk $\mathbf{x}_1 - \mathbf{x}_7$. One way of approaching this prediction problem is to build a linear model for the response in terms of the multiple predictors $\mathbf{x}_1 - \mathbf{x}_7$. Specifically, we write

$$y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_7 x_{i7} + \varepsilon_i$$

where y_i is the risk for the ith company, $x_{i1}, ..., x_{i7}$ are the accounting-determined measures of risk for the ith company, and ε_i, $i = 1, ..., 25$ are error terms which are assumed uncorrelated with a common variance.

Some of the problems we face in using this model to predict risk are the same as problems we have faced for the simple linear regression model, but some are more complicated. We need to estimate the parameters $\beta_0 - \beta_7$, and the error variance σ^2. We also need to decide what subset of the predictors is most useful for predicting the risk (and deciding this question can be much more complicated than deciding whether a single predictor is useful or not in a simple linear regression, particularly if the various predictors considered contain much the same kind of information). We must also develop tools for checking model assumptions, and once we have a model we are happy with we must be able to use it for making predictions and for describing our uncertainty about those predictions. These problems and others are considered in the remainder of the course.

It is important to note that when we talk about a linear model we mean a model which is linear in the parameters. It is perfectly valid to consider new predictors constructed from non-linear transformations of the original predictors in the linear model, provided that the new predictors enter into our model linearly in the parameters. For instance, consider the scatterplot of Figure 3.7. It is evident from this scatterplot that a simple linear regression model for the response y with the predictor x would not be appropriate. The mean response seems to change non-linearly with x. However, one possible model for these data might be

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \varepsilon_i.$$

That is, we can use the general linear model with three predictors (the original predictor x together with x^2 and x^3) as a model for the variation in the response.

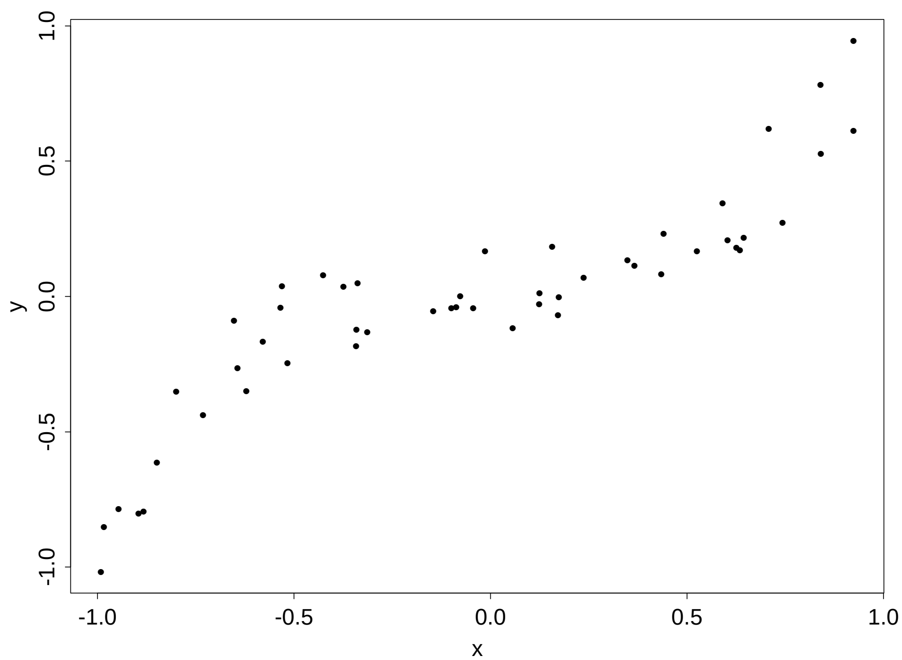

Figure 3.7: Scatter plot for synthetic data set showing non-linear relationship between mean response and predictor.

This model is still linear in the parameters, and hence it is a linear model by our definition. We can describe quite complicated relationships between the mean response and a predictor using polynomials, and we may wish to develop inferential techniques for deciding on an appropriate order for a polynomial approximation (we have used a cubic polynomial here). We will develop hypothesis tests for the linear model that can provide one approach to problems of model selection such as this. There are other flexible approximation techniques for non-linear relationships between the mean response and a predictor which perform better than polynomial regression for many purposes.

As another example we consider a synthetic data set in which there are two predictors of the response. Below is a meshplot showing the responses, the values of the predictors, and the true mean response as a function of the predictors. I

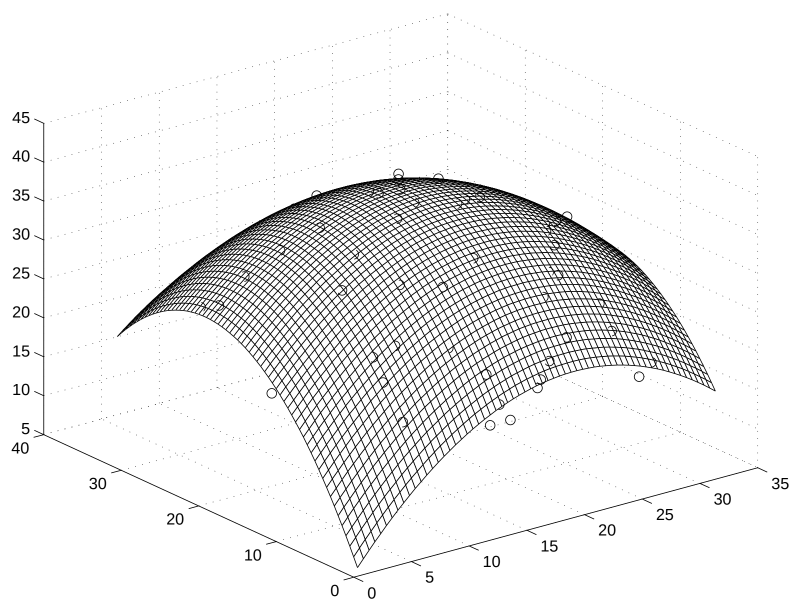

Figure 3.8: Scatter plot for synthetic data set showing non-linear relationship between mean response and two predictors. The circles show the response values, and the surface shows the true mean response used in the simulation.

simulated this data set from the model

$$y_i = 1 + 2x_{i1} + 2.5x_{i2} - 0.02x_{i1}x_{i2} - 0.05x_{i1}^2 - 0.06x_{i2}^2 + \varepsilon_i$$

$i = 1, ..., 50$ where $\mathrm{Var}(\varepsilon_i) = 1.0$ here and the ε_i were taken to be normally distributed. This data set also comes from a linear model, as the relationship of the mean response to the predictors in the above model is linear in the parameters (the coefficients in the above bivariate polynomial would be unknown for a real data set and would need to be estimated from the data). By now you should realize that the linear model is a very flexible tool for modelling complicated relationships between a response variable and a set of predictors.

Before we go on we formulate the linear model in matrix notation: this matrix notation will be used throughout the rest of the course, and the use of matrix language is indispensible in describing the theory of linear models. Write y for the vector of responses, $y = (y_1, ..., y_n)^\top$ where $a^\top$ denotes the transpose of the vector a here (so y is a column vector). Also, write β for the vector of the mean parameters in the linear model, $\beta = (\beta_0, ..., \beta_k)^\top$. and letting $p = k + 1$ (so that p is the length of the vector β), write X for the $n \times p$ matrix whose 1st column is a column of 1's and whose $(i + 1)$th column is the vector $(x_{1i}, ..., x_{ni})^\top$ for $i = 1, ..., k$. Finally write $\varepsilon = (\varepsilon_1, ..., \varepsilon_n)$ for the vector of errors in the linear model. Then we can write the general linear model in matrix notation as

$$\begin{bmatrix} y_1 \\ \\ \vdots \\ \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1k} \\ & & & \\ \vdots & & \vdots & \\ & & & \\ 1 & x_{n1} & \dots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \\ \vdots \\ \\ \varepsilon_n \end{bmatrix}$$

or

$$y = X\beta + \varepsilon.$$

3.2 Least squares estimation of mean parameters

In the simple linear regression model we obtained least squares estimators of the parameters β_0 and β_1 in the systematic part of the model by minimizing the sum of the squared residuals. In this subsection we develop a similar least squares method of estimation for the parameters β in the general linear model.

The least squares criterion

In the general linear model, we have

$$y = X\beta + \varepsilon$$

where y is an $n \times 1$ vector of responses, X is an $n \times p$ matrix and ε is an $n \times 1$ vector of zero mean errors, uncorrelated with a common variance σ^2. The systematic component of the model here is given by the term $X\beta$. If β is known, then $X\beta$ is the vector of predicted values for the responses. As for the simple linear regression model, we wish to define a goodness of fit criterion which reflects the discrepancy between the responses y and predictions $X\beta$: when β is unknown this criterion can then be minimized to obtain an estimator of β.

A natural extension of the criterion considered for the simple linear regression model is to minimize the sum of squares of the differences between the responses and predictions (that is, the sum of the squared elements of the vector $y - X\beta$). Note that for any $n \times 1$ vector x, $x^\top x$ is the scalar value $\sum_{i=1}^n x_i^2$, so that we can write this sum of squared differences as

$$(y - X\beta)^\top (y - X\beta). \tag{19}$$

We will refer to (19) as the *least squares criterion*. Of course, it reduces to the criterion we considered in the previous topic for the case of simple linear regression.

We will call the vector b which minimizes (19) with respect to β the *least squares estimator* of β. We derive this estimator now. First we express (19) in a more convenient form, before giving some results about differentiation with respect to vectors. We will then differentiate (19) to get a set of equations which can be solved to obtain the least squares estimator of β.

An alternative expression for the least squares criterion

First we recall a result about matrix transposes: for two matrices A and B which have dimensions so that A and B can be multiplied together, the transpose of AB, $(AB)^\top$ is equal to $B^\top A^\top$. If you don't know this result, or have forgotten it, then you should prove it as an (easy) exercise. Of course, using the above result, if A, B and C are three matrices such that the product ABC is well defined, then it is easy to show that $(ABC)^\top = C^\top B^\top A^\top$, and a similar result holds for a product of four or more matrices.

Now, observe that (19) can be written

$$\begin{aligned}
(y - X\beta)^\top (y - X\beta) &= y^\top (y - X\beta) - (X\beta)^\top (y - X\beta) \\
&= y^\top y - y^\top X\beta - \beta^\top X^\top y + \beta^\top X^\top X\beta \\
&= y^\top y - (X^\top y)^\top \beta - \beta^\top X^\top y + \beta^\top X^\top X\beta.
\end{aligned}$$

Since $\beta^\top X^\top y$ is a scalar, we know that the transpose of a scalar is itself, and so

$$\beta^\top X^\top y = (X^\top y)^\top \beta.$$

Hence the least squares criterion is

$$(y - X\beta)^\top (y - X\beta) = y^\top y - 2(X^\top y)^\top \beta + \beta^\top X^\top X\beta. \tag{20}$$

We must take the partial derivatives of the above expression with respect to the elements of the vector β, and set these to zero to obtain the least squares estimator of β. We do this now, but first we state some results which will help us about differentiating vector expressions.

Differentiating Vector Expressions

For a real valued function $f(x)$ defined for an argument $x = (x_1, ..., x_p)$ which is a p-dimensional vector of real numbers, we define the derivative of f with respect to x as the vector

$$\frac{\partial f(x)}{\partial x} = \left(\frac{\partial f(x)}{\partial x_1}, ..., \frac{\partial f(x)}{\partial x_p} \right)^\top .$$

With this notation, in order to find the least squares estimator we need to find the value b such that the function

$$\frac{\partial}{\partial \beta}(y - X\beta)^\top (y - X\beta)$$

is zero at $\beta = b$. To differentiate the least squares criterion with respect to β, the following result is useful.

Lemma: Let a be a fixed p-dimensional vector of real numbers, and let A be a fixed $p \times p$ matrix of real numbers. If x is also a p-dimensional vector of real numbers, and if $f(x)$ is the function

$$f(x) = a^\top x$$

then

$$\frac{\partial f(x)}{\partial x} = a.$$

Also, if

$$f(x) = x^\top Ax$$

then

$$\frac{\partial f(x)}{\partial x} = Ax + A^\top x.$$

Proof: An easy exercise.

The least squares estimator of $\boldsymbol{\beta}$

Using the above result to differentiate the least squares criterion, we have

$$\begin{aligned} \frac{\partial}{\partial \beta}(y - X\beta)^\top (y - X\beta) &= \frac{\partial}{\partial \beta}\left(y^\top y - 2(X^\top y)^\top \beta + \beta^\top X^\top X\beta \right) \\ &= -2X^\top y + X^\top X\beta + (X^\top X)^\top \beta \\ &= -2X^\top y + 2X^\top X\beta. \end{aligned}$$

Hence the least squares estimator b of β satisfies

$$-2X^\top y + 2X^\top Xb = 0$$

from which we have

$$(X^\top X)b \;=\; X^\top y. \tag{21}$$

The set of equations (21) is referred to as the *normal equations*. Now, $X^\top X$ is a $p \times p$ matrix, and if this matrix is invertible we can write the least squares estimator as

$$b = (X^\top X)^{-1} X^\top y.$$

In this course we will deal mainly with the so-called *full rank model*, for which $X^\top X$ is invertible so that the normal equations have a unique solution. We explain precisely what we mean by the full rank linear model now, but first we need to revise some more concepts from linear algebra.

Linear Independence and Rank

First we describe the notion of linear independence. If $u_1, ..., u_p$ are a set of vectors, we say that they are *linearly dependent* if there is some collection of constants $a_1, ..., a_p$ not all zero such that

$$a_1 u_1 + ... + a_p u_p = 0.$$

That is, the vectors are linearly dependent if we can express one of them as a linear combination of the others. If the vectors $u_1, ..., u_p$ are not linearly dependent, then they are said to be linearly independent.

The idea of linear independence is needed in defining the *rank* of a matrix. Suppose U is an $n \times p$ matrix with columns $u_1, ..., u_p$ so that

$$U = [u_1 \cdots u_p].$$

Then we define the rank of U to be the maximal number of linearly independent vectors in the set $u_1, ..., u_p$. Hence the rank of U is at most p. Now consider the design matrix X in the general linear model. This is a matrix with p columns and n rows, and we assume in what follows that the number of observations n is greater than or equal to the number of mean parameters in the model p. A linear model is said to be *full rank* if the rank of X is equal to p.

We now state some of the properties of the rank of a matrix. The following result is from Raymond H. Myers and Janet S. Milton, "A First Course in the Theory of Linear Statistical Models," PWS-KENT, Boston, 1991, p. 26. In the statement of the Lemma, U is an $n \times p$ matrix with $n \geqslant p$. One implication of the results below is that if X is the design matrix in a full rank linear model, then $X^\top X$ is invertible and hence the normal equations have a unique solution. We have used the notation rank(U) here for the rank of a matrix U.

Lemma:

1. If rank$(U) = p$, then rank$(U) = $ rank$(U^\top) = $ rank$(U^\top U) = p$.
2. Suppose $n = p$ so that U is a $p \times p$ matrix. Then U is nonsingular (that is, U has an inverse) if and only if rank$(U) = p$.
3. If P is a nonsingular $n \times n$ matrix, and Q is a nonsingular $p \times p$ matrix, then rank$(U) = $ rank$(PU) = $ rank(UQ).
4. The rank of a diagonal matrix is equal to the number of nonzero columns in the matrix.
5. If V is a matrix with dimensions so that UV is well defined, then the rank of UV is less than or equal to the rank of U and less than or equal to the rank of V.

We end this subsection with two examples. In the first, we give the least squares estimates of parameters in a multiple linear regression model for the risk assessment data. In the second example, we show for the simple linear regression model that the expression we have just derived in matrix notation for the least squares estimator is equivalent to the expressions we gave in our previous work for b_0 and b_1, the least squares estimators of β_0 and β_1.

Example: risk assessment data

For the risk assessment data, the response y_i was risk assessment, and there were seven accounting determined measures of risk (predictors) $x_{i1}, ..., x_{i7}$. The model is

$$y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_7 x_{i7} + \varepsilon_i.$$

The result of fitting the parameters $\beta_0, ..., \beta_7$ using the least squares criterion is given next.

```
Regression Analysis

The regression equation is
Mean Risk Assessment = 2.19 + 0.443 Dividend Payout + 0.865 Current Ratio
           - 0.247 Asset Size + 1.96 Asset Growth + 3.59 Leverage
           + 0.135 Variability Earnings + 1.05 Covariability Earnings

Predictor        Coef        StDev          T        P
Constant        2.191        1.870       1.17    0.258
Dividend       0.4426       0.1417       3.12    0.006
Current        0.8645       0.3733       2.32    0.033
Asset Si      -0.2472       0.1316      -1.88    0.078
Asset Gr        1.963        6.253       0.31    0.757
```

```
Leverage        3.592       1.436       2.50    0.023
Variabil      0.13459     0.05978       2.25    0.038
Covariab       1.0450      0.9380       1.11    0.281

S = 0.9816      R-Sq = 74.2%     R-Sq(adj) = 63.5%
```

Example: the simple linear regression model

In the simple linear regression model, $y = (y_1, ..., y_n)^\top$ and X is an $n \times 2$ matrix,

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}.$$

Hence we have that

$$X^\top X = \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix}$$

and

$$(X^\top X)^{-1} = \frac{1}{nS_{xx}} \begin{bmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{bmatrix}.$$

Also,

$$X^\top y = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{bmatrix}.$$

Hence

$$\begin{aligned} b &= (X^\top X)^{-1}(X^\top y) \\ &= \frac{1}{nS_{xx}} \begin{bmatrix} \sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i - \sum_{i=1}^n x_i \sum_{i=1}^n x_i y_i \\ n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i \end{bmatrix} \end{aligned} \tag{22}$$

Using our previous notation, we recognize the second element of (22) as $(nS_{xy})/(nS_{xx}) = S_{xy}/S_{xx}$, which was the expression we derived for the least squares estimator of the slope when studying the simple linear regression model. Our previous expression for the least squares estimator of the intercept was

$$b_0 = \bar{y} - b_1 \bar{x}$$

which we must show is equal to the second element of (22). Now,

$$\begin{aligned}
b_0 &= \bar{y} - b_1\bar{x} \\
&= \bar{y} - \frac{S_{xy}}{S_{xx}}\bar{x} \\
&= \frac{\bar{y}S_{xx} - \bar{x}S_{xy}}{S_{xx}} \\
&= \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}\right) - \left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n}\right)}{nS_{xx}} \\
&= \frac{\left(\sum_{i=1}^n x_i^2\right)\left(\sum_{i=1}^n y_i\right) - \frac{\left(\sum_{i=1}^n x_i\right)^2\left(\sum_{i=1}^n y_i\right)}{n} - \left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n x_i y_i\right) + \frac{\left(\sum_{i=1}^n x_i\right)^2\left(\sum_{i=1}^n y_i\right)}{n}}{nS_{xx}} \\
&= \frac{\left(\sum_{i=1}^n x_i^2\right)\left(\sum_{i=1}^n y_i\right) - \left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n x_i y_i\right)}{nS_{xx}}
\end{aligned}$$

as required.

3.3 Properties of least squares estimator of mean parameters

We now describe some of the properties of the least squares estimator b of β. In our previous development of the simple linear regression model, we showed that b_0 and b_1 were unbiased estimators of β_0 and β_1. We also gave expressions for the variance of b_0 and b_1 and the covariance $\mathrm{Cov}(b_0, b_1)$ of b_0 and b_1. In this subsection we develop similar results for the least squares estimator b of β in the general linear model. We will then discuss various optimality properties of the estimator b.

Expectations of random vectors

Before we discuss properties of b, however, we need to give some results about expectations for random vectors. If $Y = (Y_1, ..., Y_k)^\top$ is a random vector, and if $\mathbb{E}(Y_i) = \mu_i$, then

$$\mathbb{E}(Y) = \mu = (\mu_1, ..., \mu_k)^\top.$$

We state the following rules for expectations from Raymond H. Myers and Janet S. Milton, "A First Course in the Theory of Linear Statistical Models," PWS-KENT, Boston, 1991, p. 52.

Lemma:

i) If a is a $k \times 1$ vector of constants then $\mathbb{E}(a) = a$.

ii) If a is a $k \times 1$ vector of constants, and Y is a $k \times 1$ random vector with $\mathbb{E}(Y) = \mu$, then $\mathbb{E}(a^\top Y) = a^\top \mu$.

iii) If A is an $n \times k$ matrix, and Y is a $k \times 1$ random vector with $\mathbb{E}(Y) = \mu$, then $\mathbb{E}(AY) = A\mu$.

Proof:

i) is obvious from the corresponding property for expectations of random variables.

ii) $a^\top Y = \sum_{i=1}^k a_i Y_i$ and hence

$$\begin{aligned}\mathbb{E}(a^\top Y) &= \mathbb{E}(\sum_{i=1}^k a_i Y_i) \\ &= \sum_{i=1}^k a_i \mathbb{E}(Y_i) \\ &= \sum_{i=1}^k a_i \mu_i \\ &= a^\top \mu\end{aligned}$$

iii) $(AY)_i = \sum_{m=1}^k A_{im} Y_m$ and so

$$\begin{aligned}\mathbb{E}((AY)_i) &= \mathbb{E}(\sum_{m=1}^k A_{im} Y_m) \\ &= \sum_{m=1}^k A_{im} \mathbb{E}(Y_m) \\ &= \sum_{m=1}^k A_{im} \mu_m \\ &= (A\mu)_i.\end{aligned}$$

Therefore

$$\mathbb{E}(AY) = A\mu.$$

Covariance matrices

We also need to discuss the idea of a *covariance matrix* (sometimes called a variance-covariance matrix) of a random vector. If $Y = (Y_1, ..., Y_k)^\top$ is a random vector with $\mathbb{E}(Y) = \mu$, then the covariance matrix of Y is the $k \times k$ matrix whose (i, j)th entry is $\mathrm{Cov}(Y_i, Y_j) = \mathbb{E}((Y_i - \mu_i)(Y_j - \mu_j))$. If we define the expected value of a random matrix to be the matrix whose (i, j)th entry is just the expectation of the (i, j)th entry of the random matrix, and if we use the notation $\mathrm{Var}(Y)$ for the covariance matrix of Y, we can write

$$\mathrm{Var}(Y) = \mathbb{E}((Y - \mu)(Y - \mu)^\top)$$

since

$$(Y - \mu)(Y - \mu)^\top$$

is the $k \times k$ matrix whose (i, j)th element is $(Y_i - \mu_i)(Y_j - \mu_j)$. Note that the ith diagonal element of $\mathrm{Var}(Y)$ is simply the variance of Y_i, $i = 1, ..., n$. Also, $\mathrm{Var}(Y)$ is symmetric, since $\mathrm{Cov}(Y_i, Y_j) = \mathrm{Cov}(Y_j, Y_i)$. We can now state the following result (Raymond H. Myers and Janet S. Milton, "A First Course in the Theory of Linear Statistical Models," PWS-KENT, Boston, 1991, p. 54).

Lemma:

i) Let Y be a $k \times 1$ random vector with $\mathrm{Var}(Y) = V$. If a is a $k \times 1$ vector of real numbers, then

$$\mathrm{Var}(a^\top Y) = a^\top V a.$$

ii) Let Y be a $k \times 1$ random vector with $\mathrm{Var}(Y) = V$. Let A be a $k \times k$ matrix. If $Z = AY$, then

$$\mathrm{Var}(Z) = AVA^\top.$$

Proof:

i)

$$
\begin{aligned}
\mathrm{Var}(a^\top Y) &= \mathrm{Var}(\sum_{i=1}^k a_i Y_i) \\
&= \mathbb{E}\left(\left(\sum_{i=1}^k a_i Y_i - \sum_{i=1}^k a_i \mu_i\right)^2\right) \\
&= \mathbb{E}\left(\left(\sum_{i=1}^k a_i (Y_i - \mu_i)\right)^2\right) \\
&= \mathbb{E}\left(\sum_{i=1}^k \sum_{j=1}^k a_i a_j (Y_i - \mu_i)(Y_j - \mu_j)\right) \\
&= \sum_{i=1}^k \sum_{j=1}^k a_i a_j \mathbb{E}((Y_i - \mu_i)(Y_j - \mu_j)) \\
&= \sum_{i=1}^k \sum_{j=1}^k a_i a_j V_{ij} \\
&= a^\top V a.
\end{aligned}
$$

ii)

$$\begin{aligned}
\mathrm{Cov}(Z_i, Z_j) &= \mathrm{Cov}\left(\sum_{q=1}^{k} A_{iq}Y_q, \sum_{r=1}^{k} A_{jr}Y_r\right) \\
&= \mathbb{E}\left(\left(\sum_{q=1}^{k} A_{iq}Y_q - \sum_{q=1}^{k} A_{iq}\mu_q\right)\left(\sum_{r=1}^{k} A_{jr}Y_r - \sum_{r=1}^{k} A_{jr}\mu_r\right)\right) \\
&= \mathbb{E}\left(\left(\sum_{q=1}^{k} A_{iq}(Y_q - \mu_q)\right)\left(\sum_{r=1}^{k} A_{jr}(Y_r - \mu_r)\right)\right) \\
&= \mathbb{E}\left(\sum_{q=1}^{k}\sum_{r=1}^{k} A_{iq}A_{jr}(Y_r - \mu_r)(Y_q - \mu_q)\right) \\
&= \sum_{q=1}^{k}\sum_{r=1}^{k} A_{iq}A_{jr}V_{qr} \\
&= \sum_{q=1}^{k} A_{iq}\sum_{r=1}^{k} V_{qr}A_{jr} \\
&= \sum_{q=1}^{k} A_{iq}(VA^\top)_{qj} \\
&= (AVA^\top)_{ij}
\end{aligned}$$

So

$$\mathrm{Var}(Z) = AVA^\top.$$

Mean and covariance of b

Using the above result, we can prove the following.

Theorem:

In the full rank linear model, the least squares estimator $b = (X^\top X)^{-1}X^\top y$ is unbiased,

$$\mathbb{E}(b) = \beta$$

with covariance matrix

$$\mathrm{Var}(b) = \sigma^2(X^\top X)^{-1}.$$

Proof:

From the above lemma, we have

$$\begin{aligned}\mathbb{E}(b) &= \mathbb{E}((X^\top X)^{-1}X^\top y) \\ &= (X^\top X)^{-1}X^\top X\beta \\ &= \beta\end{aligned}$$

so that b is unbiased. Next we consider $\text{Var}(b)$. Also from the above lemma, and noting that $\text{Var}(y) = \sigma^2 I$, where I is the identity matrix, we have

$$\begin{aligned}\text{Var}(b) &= \sigma^2(X^\top X)^{-1}X^\top((X^\top X)^{-1}X^\top)^\top \\ &= \sigma^2(X^\top X)^{-1}X^\top X(X^\top X)^{-1}\end{aligned}$$

where we have used the fact that for a nonsingular matrix A, $(A^\top)^{-1} = (A^{-1})^\top$. Hence

$$\text{Var}(b) = \sigma^2(X^\top X)^{-1}$$

The Gauss-Markov theorem

The ability to compute the covariance matrix of the estimator b provides us with a way of comparing the properties of this estimator with alternative estimators. The least squares estimator of β is an example of a *linear estimator* which is an estimator of the form Ay for some $p \times n$ matrix A (setting $A = (X^\top X)^{-1}X^\top$ gives the least squares estimator). The *Gauss-Markov Theorem* states that if b^* is any unbiased linear estimator of β, and if b is the least squares estimator, then $\text{Var}(b_i) \leqslant \text{Var}(b_i^*)$, $i = 1, ..., p$. We say that b is the best linear unbiased estimator (BLUE) of β. We prove this result now.

Theorem:

The best linear unbiased estimator of β in the full rank linear model is the least squares estimator b.

Proof:

Suppose that b^* is an arbitrary linear unbiased estimator of β. We can write this estimator in the form

$$b^* = ((X^\top X)^{-1}X^\top + B)y$$

for some $p \times n$ matrix B (note that $B = 0$ just gives the least squares estimator).

Using the rules we have stated about means of random vectors,

$$\begin{aligned}\mathbb{E}(b^*) &= ((X^\top X)^{-1}X^\top + B)X\beta \\ &= \beta + BX\beta\end{aligned}$$

Now, we know that b^* is unbiased, so for the above to hold for arbitrary β we must have $BX = 0$. Now consider $\mathrm{Var}(b^*)$. We have that

$$\begin{aligned}\mathrm{Var}(b^*) &= \mathrm{Var}\left(\left((X^\top X)^{-1}X^\top + B\right)y\right) \\ &= \sigma^2((X^\top X)^{-1}X^\top + B)((X^\top X)^{-1}X^\top + B)^\top \\ &= \sigma^2((X^\top X)^{-1}X^\top + B)(X(X^\top X)^{-1} + B^\top) \\ &= \sigma^2((X^\top X)^{-1}X^\top X(X^\top X)^{-1} + (X^\top X)^{-1}X^\top B^\top + BX(X^\top X)^{-1} + BB^\top).\end{aligned}$$

But $BX = 0$, and so the third term on the right in the above expression is zero: also, the second term is zero, as $(X^\top B^\top) = (BX)^\top = 0$. Hence

$$\mathrm{Var}(b^*) = \sigma^2\left((X^\top X)^{-1} + BB^\top\right)$$

Note that the diagonal elements of $\mathrm{Var}(b^*)$ (which are the variances of the elements of b^*) are just the variances of the least squares estimators plus the diagonal elements of $BB^\top$. But the ith diagonal element of $BB^\top$ is

$$(BB^\top)_{ii} = \sum_{j=1}^{n} B_{ij}^2,$$

a positive quantity. So the least squares estimator is BLUE.

3.4 Maximum likelihood estimation of mean parameters

In the case of the simple linear regression model, we were able to show that the least squares estimates of β_0 and β_1 were also the maximum likelihood estimates if normality of the errors in the model was assumed. A similar result holds for the least squares estimator of β in the general linear model: least squares corresponds to maximum likelihood under an assumption of normality.

Maximum likelihood estimation of $\boldsymbol{\beta}$

Under normal assumptions, we can write down the likelihood in the case of the general linear model as

$$\begin{aligned}L(\beta, \sigma^2) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\left(y_i - (X\beta)_i\right)^2\right) \\ &= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - (X\beta)_i)^2\right).\end{aligned}$$

But

$$\sum_{i=1}^{n}(y_i - (X\beta)_i)^2 \quad = \quad (y - X\beta)^\top (y - X\beta)$$

So the likelihood becomes

$$L(\beta, \sigma^2) \quad = \quad (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)\right).$$

Hence the log-likelihood $l(\beta, \sigma^2)$ is

$$\begin{aligned} l(\beta, \sigma^2) &= \log L(\beta, \sigma^2) \\ &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta). \end{aligned} \tag{23}$$

Maximizing the likelihood is equivalent to maximizing the log-likelihood, and clearly the log-likelihood is maximized with respect to β here regardless of the value of σ^2 by minimizing $(y-X\beta)^\top(y-X\beta)$ with respect to β. So the maximum likelihood estimator of β is also the least squares estimator.

Maximum likelihood estimation of σ^2

We also find the maximum likelihood estimator for σ^2. Differentiating (23) with respect to σ^2, we get

$$\frac{\partial}{\partial \sigma^2} l(\beta, \sigma^2) \quad = \quad -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(y - X\beta)^\top (y - X\beta).$$

Writing b for the least squares estimator of β, and σ^{*2} for the maximum likelihood estimator of σ^2, we have

$$\frac{n}{2\sigma^{*2}} \quad = \quad \frac{1}{2\sigma^{*4}}(y - Xb)^\top (y - Xb)$$

so that

$$\sigma^{*2} \quad = \quad \frac{1}{n}(y - Xb)^\top (y - Xb).$$

Note that this expression agrees with the expression we obtained before for the simple linear regression model.

3.5 Estimation of the error variance

We have just shown that the maximum likelihood estimator of the error variance σ^2 is

$$\frac{1}{n}(y - Xb)^\top (y - Xb).$$

We showed for the case of the simple linear regression model that this estimator was biased. What is $\mathbb{E}(\sigma^{*2})$ in the general linear model?

Bias of the maximum likelihood estimator

Observing that $b = (X^\top X)^{-1}X^\top y$, we have that

$$\begin{aligned}\mathbb{E}(\sigma^{*2}) &= \frac{1}{n}\mathbb{E}\left((y - X(X^\top X)^{-1}X^\top y)^\top (y - X(X^\top X)^{-1}X^\top y)\right) \\ &= \frac{1}{n}\mathbb{E}(y^\top (I - X(X^\top X)^{-1}X^\top)(I - X(X^\top X)^{-1}X^\top)y).\end{aligned}$$

I claim that

$$(I - X(X^\top X)^{-1}X^\top)(I - X(X^\top X)^{-1}X^\top) = (I - X(X^\top X)^{-1}X^\top). \tag{24}$$

Matrices like $I - X(X^\top X)^{-1}X^\top$ which are unchanged when multiplied by themselves have a special name: they are called *idempotent* matrices, and they play an important role in the theory of linear models.

To see (24), observe that

$$\begin{aligned}(I - X(X^\top X)^{-1}X^\top)(I - X(X^\top X)^{-1}X^\top) &= I - X(X^\top X)^{-1}X^\top - X(X^\top X)^{-1}X^\top \\ &\quad + X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top \\ &= I - 2X(X^\top X)^{-1}X^\top + X(X^\top X)^{-1}X^\top \\ &= I - X(X^\top X)^{-1}X^\top.\end{aligned}$$

Hence we have that

$$\mathbb{E}(\sigma^{*2}) = \frac{1}{n}\mathbb{E}\left(y^\top (I - X(X^\top X)^{-1}X^\top)y\right). \tag{25}$$

To compute (25), we first give a general result about computing expectations which have this general form. Before we can state this result, however, we need to define the *trace* of a matrix, and to give some results about matrix traces. For a $k \times k$ matrix X, the trace of X, written $\mathrm{tr}(X)$ is simply the sum of the diagonal elements of X,

$$\mathrm{tr}(X) = \sum_{i=1}^{k} X_{ii}.$$

The following properties of the trace are easy to prove (see Raymond H. Myers and Janet S. Milton, “A First Course in the Theory of Linear Statistical Models,” PWS-KENT, Boston, 1991, p. 26.)

Theorem:

i) Let c be a real number. Then $\mathrm{tr}(cX) = c.\mathrm{tr}(X)$.

ii) $\mathrm{tr}(X+Y) = \mathrm{tr}(X) + \mathrm{tr}(Y)$

iii) If X is an $n \times p$ matrix, and Y is a $p \times n$ matrix, then $\mathrm{tr}(XY) = \mathrm{tr}(YX)$.

Means of quadratic forms of a random vector

We are now in a position to prove the result we need in order to find the mean of σ^{*2}. We are interested in *quadratic forms* of a random vector, which are expressions of the form $y^\top Ay$ for a random vector y and matrix A.

Lemma:

Let y be a $k \times 1$ random vector with $\mathbb{E}(y) = \mu$ and $\mathrm{Var}(y) = V$. Let A be a $k \times k$ matrix of real numbers. Then

$$\mathbb{E}(y^\top Ay) = \mathrm{tr}(AV) + \mu^\top A\mu.$$

Proof:

See your tutorial sheet.

Unbiased estimation of $\boldsymbol{\sigma^2}$

Applying the above lemma to compute $\mathbb{E}(\sigma^{*2})$ and observing that $\mathrm{Cov}(y) = \sigma^2 I$, we have

$$\begin{aligned}
\mathbb{E}(\sigma^{*2}) &= \frac{1}{n}\mathbb{E}(y^\top(I - X(X^\top X)^{-1}X^\top)y) \\
&= \frac{1}{n}\left(\sigma^2\mathrm{tr}(I - X(X^\top X)^{-1}X^\top) + (X\beta)^\top(I - X(X^\top X)^{-1}X^\top)(X\beta)\right) \\
&= \frac{1}{n}\left(\sigma^2\mathrm{tr}(I) - \sigma^2\mathrm{tr}(X(X^\top X)^{-1}X^\top) + \beta^\top X^\top(I - X(X^\top X)^{-1}X^\top)X\beta\right) \\
&= \frac{1}{n}\left(\sigma^2 n - \sigma^2\mathrm{tr}(X^\top X(X^\top X)^{-1}) + \beta^\top X^\top X\beta - \beta^\top X^\top X(X^\top X)^{-1}X^\top X\beta\right)
\end{aligned}$$

where we have used the fact that $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ for matrices A and B for which these products are defined. Observe that $(X^\top X)^{-1}(X^\top X)$ is the $p \times p$ identity matrix, so that

$$\mathbb{E}(\sigma^{*2}) = \frac{n-p}{n}\sigma^2.$$

So the maximum likelihood estimator is biased, and the bias depends on the number of mean parameters p. The above expressions suggest estimation of σ^2 by

$$\widehat{\sigma}^2 = \frac{1}{n-p}(y - Xb)^\top (y - Xb)$$

which will be unbiased.

We conclude this section with an example where we illustrate the computation of the least squares estimates of β, the estimation of σ^2, and the computation of estimated standard errors for the least squares estimators.

Example: Business failures and key economic indicators

This data set is from Gerald Keller, Brian Warrack and Henry Bartel, "Statistics for Management and Economics: a Systematic Approach," Wadsworth, Belmont, California, 1990, p. 804.

Many people have attempted to get an overall picture of business failures and bankruptcies – both by studying the incidence of business failure on a national or industry-wide scale and by analyzing individual firms one by one – with varying degrees of success. In the latter instance, promising results have been achieved by using various accounting and financial ratios to predict bankruptcies.

In looking at the overall picture, some analysts have been tempted by dramatic swings in the number of business failures from year to year to try to relate them to overall economic conditions. The annual percentage change in certain key economic indicators is reproduced in the table below.

In this example we fit a linear model (multiple regression model) to these data, using percentage change in business failures as the response and percentage change in the economic indicators shown in the table as predictors. For the moment we do not discuss model selection or criticism of a model, but simply fit this linear model. The coefficient estimates obtained from a common statistical package are shown below.

```
Coefficients:
                          Value Std. Error  t value Pr(>|t|)
          (Intercept) -59.4138  47.4629    -1.2518   0.2572
Real.domestic.product   2.9335   3.1052     0.9447   0.3813
   Wages.and.Salaries   3.7996   2.8988     1.3108   0.2379
    Unemployment.Rate   8.6019   5.7039     1.5081   0.1823
         Retail.sales  -5.5136   3.2695    -1.6864   0.1427
       Housing.Starts  -0.2079   0.5414    -0.3840   0.7142
              C.P.I.    1.7576   5.0222     0.3500   0.7383

Residual standard error: 15.47 on 6 degrees of freedom
Multiple R-Squared: 0.5537
```

Year	Business failures	Real domestic product	Wages and salaries	Unemployment rate	Retail sales	Housing starts	C.P.I.
1968	0.3	3.5	10.9	3.8	6.8	22.0	3.6
1969	-5.1	5.5	8.7	4.5	6.0	20.0	4.0
1970	24.3	6.2	12.1	4.4	6.9	6.9	4.6
1971	4.0	2.2	8.4	5.7	2.3	-9.5	3.3
1972	1.2	5.9	9.9	6.2	8.9	22.6	2.9
1973	-4.8	5.6	11.3	6.2	11.3	7.0	4.8
1974	-4.9	7.8	15.8	5.5	12.6	7.4	7.6
1975	5.1	4.7	19.4	5.3	16.9	17.3	10.8
1976	-4.9	0.7	16.4	6.9	14.5	4.2	10.8
1977	40.0	5.3	15.4	7.1	10.8	-7.3	7.5
1978	40.4	2.8	10.0	8.1	8.4	-10.1	8.0
1979	2.7	3.5	8.8	8.4	11.7	-7.3	9.0
1980	15.8	3.2	11.0	7.5	11.9	-13.4	8.8

It is instructive to compute the matrix $(X^\top X)^{-1}$ for this example, to multiply by the estimated residual variance 15.47^2 and then to take the square roots of the diagonal elements of this matrix to see whether they correspond to the standard errors quoted in the above table. We have

$$X^\top X = \begin{bmatrix} 13.0 & 56.90 & 158.10 & 79.60 & 129.00 & 25.20 & 85.70 \\ 56.9 & 293.79 & 702.71 & 335.52 & 568.08 & 225.37 & 353.78 \\ 158.1 & 702.71 & 2067.73 & 962.77 & 1693.85 & 145.45 & 1118.59 \\ 79.6 & 335.52 & 962.77 & 512.20 & 812.72 & 15.86 & 551.68 \\ 129.0 & 568.08 & 1693.85 & 812.72 & 1462.36 & 28.82 & 960.96 \\ 25.2 & 225.37 & 145.45 & 15.86 & 28.82 & 2341.46 & -145.68 \\ 85.7 & 353.78 & 1118.59 & 551.68 & 960.96 & -145.68 & 662.19 \end{bmatrix}$$

From this, we have $\widehat{\sigma}^2(X^\top X)^{-1}$ is

$$\begin{bmatrix} 2251.9 & -46.619 & -101.23 & -237.93 & 81.109 & -13.972 & -18.089 \\ -46.619 & 9.6388 & -1.0828 & 0.89594 & -4.3519 & 0.20687 & 8.3275 \\ -101.23 & -1.0828 & 8.4000 & 11.622 & -4.5641 & 0.43326 & -3.4736 \\ -237.93 & 0.89594 & 11.622 & 32.522 & -6.6688 & 1.2118 & -6.4694 \\ 81.109 & -4.3519 & -4.5641 & -6.6688 & 10.686 & -0.91736 & -10.615 \\ -13.972 & 0.20687 & 0.43326 & 1.2118 & -0.91737 & 0.29297 & 1.3521 \\ -18.089 & 8.3275 & -3.4736 & -6.4694 & -10.615 & 1.3521 & 25.213 \end{bmatrix}$$

Taking the square root of the diagonal elements of this matrix gives the values

$$(47.454, 3.1046, 2.8982, 5.7028, 3.2689, 0.54127, 5.0213)$$

which after allowing for rounding error are the entries given in the table.

3.6 Interval estimation in the general linear model

For the simple linear regression model, we were able to construct confidence intervals for the parameters of the model. In this subsection we do the same thing for the general linear model.

The multivariate normal distribution

To construct our confidence intervals we will first need to give some results about the distribution of the least squares estimator b of β and the distribution of our estimator $\widehat{\sigma}^2$ of the residual error variance.

To state the distribution of b we need to introduce the *multivariate normal distribution*. You are of course already familiar with the univariate normal distribution. An $n \times 1$ random vector Y is said to have a multivariate normal distribution with mean vector μ and covariance matrix Σ (and we write this as $Y \sim N(\mu, \Sigma)$) if it has the density

$$f_Y(y; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(y-\mu)^\top \Sigma^{-1}(y-\mu)\right),$$

where $|\Sigma|$ denotes the determinant of the matrix Σ. You will study some of the properties of the multivariate normal distribution in later statistics courses. It can be shown that the $n \times 1$ vector μ actually is the mean of a random vector with the above density. Similarly, if Y is a random vector with the above density, $\mathrm{Var}(Y) = \Sigma$. The mean vector μ and covariance matrix Σ completely characterize a multivariate normal distribution. Note that if $Y = (Y_1, ..., Y_n)$ is a vector of univariate independent normal random variables with $\mathbb{E}(Y_i) = \mu_i$ and $\mathrm{Var}(Y_i) = \sigma^2$, then Y is a multivariate normal random vector according to the above definition with mean μ and covariance matrix $\sigma^2 I$, where I is the $n \times n$ identity matrix here. In order to state the distribution of b, we need the following result.

Lemma:

Let Y be an $n \times 1$ multivariate normal random vector with mean μ and covariance matrix Σ,

$$Y \sim N(\mu, \Sigma).$$

Let A be a $p \times n$ matrix. Then $Z = AY$ is a multivariate normal random vector with mean $A\mu$ and covariance matrix $A\Sigma A^\top$,

$$Z \sim N(A\mu, A\Sigma A^\top).$$

We can now state the distribution of the least squares estimator b of β.

Distribution of b

Theorem:

In the full rank linear model, the least squares estimator b of β has a multivariate normal distribution with mean β and covariance matrix $\sigma^2(X^\top X)^{-1}$,

$$b \sim N(\beta, \sigma^2(X^\top X)^{-1}).$$

Proof:

We have already proven that the mean and covariance matrix of b are given by $\mathbb{E}(b) = \beta$ and $\mathrm{Var}(y) = \sigma^2(X^\top X)^{-1}$. The fact that b is multivariate normal follows from multivariate normality of y, the fact that b is a linear transformation of y,

$$b = (X^\top X)^{-1}X^\top y,$$

and the above lemma.

In an analogous way to the case of the simple linear regression model, it is true that

$$\frac{(n-p)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p}$$

and that b and $\widehat{\sigma}^2$ are independent

Interval estimation

We can now write down test statistics which can be used for constructing confidence intervals for the parameters β_j.

Write c_{jj} $j = 0, ..., k$ for the diagonal elements of $(X^\top X)^{-1}$. Then we know that

$$\frac{b_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

Also

$$\frac{(n-p)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p}$$

and $(n-p)\widehat{\sigma}^2/\sigma^2$ is independent of b_j. So

$$\left(\frac{b_j-\beta_j}{\sigma\sqrt{c_{jj}}}\right)\Big/\sqrt{\frac{\frac{(n-p)\widehat{\sigma}^2}{\sigma^2}}{n-p}} \sim t_{n-p}.$$

But the above statistic simplifies to

$$\frac{b_j-\beta_j}{\widehat{\sigma}\sqrt{c_{jj}}}.$$

We can use the above statistic to derive a confidence interval for β_j. Writing $t_{\alpha/2,n-p}$ for the upper $100\alpha/2$ percentage point of a t distribution with $n-p$ degrees of freedom, we have that

$$\mathbb{P}(-t_{\alpha/2,n-p} \leqslant \frac{b_j-\beta_j}{\widehat{\sigma}\sqrt{c_{jj}}} \leqslant t_{\alpha/2,n-p}) \;=\; 1-\alpha.$$

Rearranging the above inequality, we get

$$\mathbb{P}(-t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}} \leqslant b_j-\beta_j \leqslant t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}}) \;=\; 1-\alpha$$

or

$$\mathbb{P}(-b_j-t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}} \leqslant -\beta_j \leqslant -b_j+t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}}) \;=\; 1-\alpha$$

which gives

$$\mathbb{P}(b_j-t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}} \leqslant \beta_j \leqslant b_j+t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}}) \;=\; 1-\alpha.$$

Hence a $100(1-\alpha)$ percentage confidence interval for β_j is

$$(b_j-t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}}, b_j+t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{c_{jj}}).$$

Example: risk assessment data

For the risk assessment data, fitting a multiple linear regression model with mean risk assessment as the response and the accounting determined measures of risk as predictors gives the following.

```
Regression Analysis

The regression equation is
Mean Risk Assessment = 2.19 + 0.443 Dividend Payout + 0.865 Current Ratio
           - 0.247 Asset Size + 1.96 Asset Growth + 3.59 Leverage
           + 0.135 Variability Earnings + 1.05 Covariability Earnings
```

Predictor	Coef	StDev	T	P
Constant	2.191	1.870	1.17	0.258
Dividend	0.4426	0.1417	3.12	0.006
Current	0.8645	0.3733	2.32	0.033
Asset Si	-0.2472	0.1316	-1.88	0.078
Asset Gr	1.963	6.253	0.31	0.757
Leverage	3.592	1.436	2.50	0.023
Variabil	0.13459	0.05978	2.25	0.038
Covariab	1.0450	0.9380	1.11	0.281

```
S = 0.9816      R-Sq = 74.2%     R-Sq(adj) = 63.5%
```

We can use the above table of estimated coefficients and standard errors to construct confidence intervals for coefficients. The estimated standard errors of the estimators b_j, $\widehat{\sigma}\sqrt{c_{jj}}$, are listed in the table in the 'StDev' column. Noting that the upper 2.5 percentage point of a t distribution with $n - p = 17$ degrees of freedom is approximately 2.1098, a 95 percent confidence interval for β_1, the coefficient for Dividend, can be computed as

$$(0.4426 - (2.1098)(0.1417), 0.4426 + (2.1098)(0.1417)) = (0.1436, 0.7416)$$

and similarly for the other coefficients.

3.7 Hypothesis testing for coefficients

We can also use the above result about $(b_j - \beta_j)/(\widehat{\sigma}\sqrt{c_{jj}})$ to develop a test for the null hypothesis $\beta_j = \gamma$ against the alternatives $\beta_j \neq \gamma$, $\beta_j < \gamma$ or $\beta_j > \gamma$ in the presence of the other terms in the model. This last qualification "in the presence of the other terms in the model" is an important one: the conclusion may depend on what predictors are already included.

Hypothesis testing for β_j

To test

$$H_0 : \beta_j = \gamma$$

against the alternative

$$H_1 : \beta_j \neq \gamma$$

we use the test statistic

$$T = \frac{b_j - \gamma}{\widehat{\sigma}\sqrt{c_{jj}}}$$

which has a t_{n-p} distribution under H_0.

The critical region for the test (with significance level α) is

$$T < -t_{\alpha/2;n-p} \text{ or } T > t_{\alpha/2;n-p}.$$

We can also conduct the test by computing a p-value. If we write t for the observed value of T, then the p-value for the test is computed as

$$p = \mathbb{P}(|T| \geqslant |t| | \beta_j = \gamma)$$

(where of course under the null hypothesis $\beta_j = \gamma$ we have $T \sim t_{n-p}$). So we just compute the probability that a t_{n-p} random variable is as extreme as our observed statistic or more extreme: if p is smaller than the significance level for the test, we reject H_0 in favour of the alternative $\beta_j \neq \gamma$.

One-sided alternatives

For a one-sided test, we must modify the critical region and the definition of the p-value.

For

$$H_1 : \beta_j > \gamma$$

the critical region is

$$T > t_{\alpha;n-p}$$

and we compute the p-value as

$$p = \mathbb{P}(T \geqslant t | \beta_j = \gamma).$$

For the alternative

$$H_1 : \beta_j < \gamma$$

the critical region is

$$T < -t_{\alpha;n-p}$$

and we compute the p-value as

$$p = \mathbb{P}(T \leqslant t).$$

We conclude this section by looking once again at the risk assessment data.

Example: risk assessment data

In the column labelled 'T' here we have the values of the t-statistics described above for testing whether each element of β is zero in the presence of the other terms in the model: that is, for each parameter β_j, $j = 0, ..., k$ the statistics

$$\frac{b_j}{\widehat{\sigma}\sqrt{c_{jj}}}$$

are listed (and they are realizations of t_{n-p} random variables under $H_0 : \beta_j = 0$).

Also listed are the p-values for testing $H_0 : \beta_j = 0$ against the two-sided alternative $H_1 : \beta_j \neq 0$. A small p-value indicates rejection of the null hypothesis. A large p-value indicates that the given predictor seems to make no contribution to explaining the variation in the response in the presence of the other terms.

The t-tests we have just described are sometimes called *partial t-tests*. We note from the above p-values that asset growth and covariability earnings do not seem to make a contribution to explaining the variation in the presence of the other terms in the model. A word of caution here: partial t-tests can be hard to interpret. We certainly can't conclude from a partial t-test that a predictor is unrelated to the response, or that it should not be included in a good model for the response. For instance, it may happen that if two predictors carry much the same information about the response then a good model for the response should contain one of the two predictors, but that any model which contains both predictors results in a non-significant p-value for the partial t-tests for both predictors. Upon deleting one of the predictors from a model containing both, however, we might then find that the p-value for the partial t-test for the remaining variable is significant! Note that there is also a possible problem of simultaneous inference here: it could be argued that we should make an adjustment to the significance level used in the partial t-tests when we examine a table of p-values like the one given above.

3.8 Confidence intervals for the mean and prediction intervals.

In the case of simple linear regression in the previous topic we developed confidence intervals for the conditional mean and prediction intervals for a given value of the predictor. In this subsection we do this for the general linear model.

Suppose that for each value y_i of the response, $i = 1, .., n$, we have values of k predictor variables $x_{i1}, ..., x_{ik}$ which are used in a linear model for the response. If we write x_i for the vector $x_i = (1, x_{i1}, ..., x_{ik})^\top$ then under the assumptions of the general linear model

$$y_i = x_i^\top \beta + \varepsilon_i$$

where $\beta = (\beta_0, ..., \beta_k)^\top$ is a vector of unknown parameters and the ε_i are independent normal errors with zero mean and common variance σ^2.

Now consider a new response value y_* for which the values of the predictor variables are $x_{*1}, ..., x_{*k}$. If we write x_* for the vector $x_* = (1, x_{*1}, ..., x_{*k})^\top$, then the conditional mean of y_* given the predictor values is $x_*^\top \beta$. Also, we can write

$$y_* = x_*^\top \beta + \varepsilon_*$$

where ε_* is a zero mean normal random variable with variance σ^2 independent of ε_i, $i = 1, ..., n$. We wish to develop a confidence interval for the conditional mean

$x_*^\top \beta$ and a prediction interval for y_*.

Confidence interval for the mean

First we develop a test statistic which can be used for constructing a confidence interval for the conditional mean. Writing b for the least squares estimator of β, we consider $x_*^\top b$. We have that

$$\mathbb{E}(x_*^\top b) = x_*^\top \beta$$

and

$$\begin{aligned} \mathrm{Var}(x_*^\top b) &= x_*^\top \mathrm{Var}(b) x_* \\ &= \sigma^2 x_*^\top (X^\top X)^{-1} x_* \end{aligned}$$

where X is the design matrix for the fitted linear model. Furthermore, $x_*^\top b$ is normal (as a linear combination of a multivariate normal random vector). So we have

$$x_*^\top b \sim N(x_*^\top \beta, \sigma^2 x_*^\top (X^\top X)^{-1} x_*)$$

or

$$\frac{x_*^\top b - x_*^\top \beta}{\sigma \sqrt{x_*^\top (X^\top X)^{-1} x_*}} \sim N(0,1).$$

It can also be shown that $x_*^\top b$ and $(n-p)\widehat{\sigma}^2/\sigma^2$ are independent (try to prove this as an exercise).

Hence

$$\frac{x_*^\top b - x_*^\top \beta}{\sigma \sqrt{x_*^\top (X^\top X)^{-1} x_*}} \Big/ \sqrt{\frac{\frac{(n-p)\widehat{\sigma}^2}{\sigma^2}}{n-p}}$$

has a t distribution with $n-p$ degrees of freedom (since as we showed in the last lecture $(n-p)\widehat{\sigma}^2/\sigma^2$ has a χ^2_{n-p} distribution). But the above statistic simplifies to

$$\frac{x_*^\top b - x_*^\top \beta}{\widehat{\sigma} \sqrt{x_*^\top (X^\top X)^{-1} x_*}}. \tag{26}$$

From the result that (26) has a t distribution with $n-p$ degrees of freedom we can derive a confidence interval in the usual way: we can show that

$$\begin{aligned} \mathbb{P}(x_*^\top b - t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{x_*^\top (X^\top X)^{-1} x_*} \leqslant x_*^\top \beta \leqslant \\ x_*^\top b + t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{x_*^\top (X^\top X)^{-1} x_*}) = 1-\alpha \end{aligned}$$

which gives us our confidence interval for $x_*^\top \beta$.

Prediction intervals

It remains to derive a prediction interval for y_*. Write $\widehat{y}_*$ for the value

$$\widehat{y}_* = x_*^\top b$$

and consider $y_* - \widehat{y}_*$. We have that

$$y_* - \widehat{y}_* = x_*^\top \beta + \varepsilon_* - x_*^\top b$$

Now, clearly $\mathbb{E}(y_* - \widehat{y}_*) = 0$. Furthermore, ε_* and b are independent, and we have just shown that

$$\mathrm{Var}(x_*^\top b) = \sigma^2 x_*^\top (X^\top X)^{-1} x_*$$

so that

$$\begin{aligned}
\mathrm{Var}(y_* - \widehat{y}_*) &= \mathrm{Var}(x_*^\top b) + \mathrm{Var}(\varepsilon_*) \\
&= \sigma^2 x_*^\top (X^\top X)^{-1} x_* + \sigma^2 \\
&= \sigma^2 (1 + x_*^\top (X^\top X)^{-1} x_*).
\end{aligned}$$

It can also be shown that $y_* - \widehat{y}_*$ and $(n-p)\widehat{\sigma}^2/\sigma^2$ are independent, and so

$$\frac{y_* - \widehat{y}_*}{\sigma\sqrt{1 + x_*^\top (X^\top X)^{-1} x_*}} \Big/ \sqrt{\frac{\frac{(n-p)\widehat{\sigma}^2}{\sigma^2}}{n-p}} \sim t_{n-p}.$$

Simplifying, we have

$$\frac{y_* - \widehat{y}_*}{\widehat{\sigma}\sqrt{1 + x_*^\top (X^\top X)^{-1} x_*}} \sim t_{n-p}.$$

So the usual manipulations lead to

$$\begin{aligned}
\mathbb{P}(\widehat{y}_* - t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{1 + x_*^\top (X^\top X)^{-1} x_*} \leqslant y_* \leqslant \\
\widehat{y}_* + t_{\alpha/2,n-p}\widehat{\sigma}\sqrt{1 + x_*^\top (X^\top X)^{-1} x_*}) = 1 - \alpha.
\end{aligned}$$

which gives us the desired $100(1-\alpha)$ percentage prediction interval for y_*.

Example: risk assessment data

To illustrate the computation of confidence intervals for the mean and prediction intervals we consider again the risk assessment data. Suppose we are interested in predicting the market risk of the stock of a company with a divident payout of 0.5, current ratio of 1.0, asset size of 10.0, asset growth of 0.1, leverage of 0.3, variability earnings of 2.0 and covariability earnings of 0.6. We can compute a 95 percent confidence interval for the conditional mean of a company with these accounting determined risk measures, as well as a prediction interval for our company of interest using software.

The fitted value (estimated market risk) here is 2.975, the 95 percent confidence interval for the mean is

$$(2.138, 3.811)$$

and the 95 percent prediction interval is

$$(0.741, 5.208).$$

3.9 Joint confidence regions for the coefficients

Earlier we discussed the use of Bonferroni adjustment to obtain confidence regions with a guaranteed level of coverage by adjusting the coverage of pointwise confidence intervals for parameters. In the general linear model, adjusting the pointwise coverage of confidence intervals for $\beta_0, ..., \beta_k$ to be $1 - \alpha/(k+1)$ guarantees joint coverage for these intervals of at least $1 - \alpha$. However, in the case of the general linear model there is an alternative way of deriving confidence regions which have exact coverage of $1 - \alpha$ for the parameter vector β. We discuss this method now.

It can be shown that the statistic

$$\frac{(b-\beta)^\top X^\top X(b-\beta)}{p\widehat{\sigma}^2}$$

has an F distribution with p and $n-p$ degrees of freedom. Knowing this allows us to develop a joint confidence region for the coefficients. In particular, if $F_{\alpha;p,n-p}$ denotes the upper 100α percentage point of an F distribution with p and $n-p$ degrees of freedom, we have that

$$\mathbb{P}((b-\beta)^\top X^\top X(b-\beta)/p\widehat{\sigma}^2 \leqslant F_{\alpha;p,n-p}) = 1-\alpha.$$

Hence a $100(1-\alpha)$ percentage confidence region for β consists of all points β such that

$$(b-\beta)^\top X^\top X(b-\beta)/p\widehat{\sigma}^2 \leqslant F_{\alpha;p,n-p}.$$

3.10 Decomposing variation in the full rank linear model

For the simple linear regression model we proved a fundamental identity which partitioned total variation into variation explained by the model and residual variation. More precisely, we showed that

$$\sum_{i=1}^n (y_i - \bar{y})^2 \;=\; \sum_{i=1}^n (\widehat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \widehat{y}_i)^2. \tag{27}$$

where y_i, $i = 1, ..., n$ were the responses, $\bar{y}$ is the mean of the responses, and $\widehat{y}_i$ is the ith fitted value. We expressed this identity in the notation

$$SS_{total} = SS_{reg} + SS_{res}$$

where SS_{total} is called the total sum of squares (the sum of squared deviations of the responses about their mean), SS_{reg} is called the regression sum of squares (the sum of the squared deviations of the fitted values about their mean, which is $\bar{y}$) and SS_{res} is called the residual sum of squares (the sum of the squared deviations of the fitted values from the responses).

The identity (27) holds in the general linear model, where of course the fitted values are defined as the elements of the vector Xb where X is the design matrix and b is the least squares estimator of β. I won't prove this identity, but MATH2931 students will have to do this in their next assignment.

The ANOVA table

In the case of simple linear regression, the partition (27) of total variation was presented in an *analysis of variance* table. This table was also a convenient way of organizing calculations for hypothesis testing. We will cover some theory about hypothesis testing later, but for now we will look at the output of fitting a regression model, and try to understand the analysis of variance table and some of the other information. Below is the result of fitting a multiple regression model for the risk assessment data discussed in previous lectures (where mean risk assessment is the response, and the seven accounting determined measures of risk are predictors). See previous notes for a description of the data.

```
Regression Analysis

The regression equation is
Mean Risk Assessment = 2.19 + 0.443 Dividend Payout + 0.865 Current Ratio
           - 0.247 Asset Size + 1.96 Asset Growth + 3.59 Leverage
           + 0.135 Variability Earnings + 1.05 Covariability Earnings

Predictor        Coef       StDev          T        P
Constant        2.191       1.870       1.17    0.258
Dividend       0.4426      0.1417       3.12    0.006
Current        0.8645      0.3733       2.32    0.033
Asset Si      -0.2472      0.1316      -1.88    0.078
Asset Gr        1.963       6.253       0.31    0.757
Leverage        3.592       1.436       2.50    0.023
Variabil      0.13459     0.05978       2.25    0.038
Covariab       1.0450      0.9380       1.11    0.281
```

```
S = 0.9816      R-Sq = 74.2%     R-Sq(adj) = 63.5%

Analysis of Variance

Source            DF         SS          MS        F       P
Regression         7    47.0282      6.7183     6.97   0.001
Residual Error    17    16.3808      0.9636
Total             24    63.4090

Source       DF     Seq SS
Dividend      1    18.4253
Current       1     5.6024
Asset Si      1    10.1251
Asset Gr      1     1.6456
Leverage      1     6.0841
Variabil      1     3.9497
Covariab      1     1.1961
```

By now you should understand how the coefficients in the fitted model are computed, and how their standard errors are estimated. You should also understand the computation of the partial t statistics and associated p-values listed in the table of coefficient estimates, and their use for testing for the significance of a predictor in the multiple linear regression model in the presence of the other terms.

Below the table of coefficient estimates is listed the estimated standard deviation of the errors $S = 0.9816$ (this was $\widehat{\sigma}$ in our notation). Also listed is R^2, which takes the value 0.742 here (or 74.2 percent). For the simple linear regression model, we defined R^2 in terms of our fundamental partition of variation as

$$R^2 = \frac{SS_{reg}}{SS_{total}} \tag{28}$$

and the definition of R^2 is the same in the general linear model. Beside the quoted value for R^2 is a value for the so-called *adjusted* R^2, which we write as $\bar{R}^2$. We have $\bar{R}^2 = 0.635$ here (or 63.5 percent). To define $\bar{R}^2$, rewrite R^2 as

$$R^2 = 1 - \frac{SS_{res}}{SS_{total}} \tag{29}$$

(the equivalence of the two expressions for R^2 comes from replacing SS_{reg} in the numerator of (28) by the equivalent expression $SS_{total} - SS_{res}$, which comes from (27)). We define $\bar{R}^2$ by replacing SS_{res} in (29) by $\widehat{\sigma}^2$ (which is $SS_{res}/(n-p)$) and replacing SS_{total} by $SS_{total}/(n-1)$. $SS_{total}/(n-1)$ is in fact the estimated error variance for a linear model in which we fit only an intercept term. Hence we can write

$$\bar{R}^2 = 1 - \frac{(n-1)SS_{res}}{(n-p)SS_{total}} \tag{30}$$

or

$$\bar{R}^2 = 1 - \frac{\widehat{\sigma}^2(n-1)}{SS_{total}}. \tag{31}$$

In terms of R^2, we can write

$$\bar{R}^2 = 1 - \frac{n-1}{n-p}(1-R^2).$$

What is the reason for the introduction of $\bar{R}^2$? R^2 is an easily interpreted measure of fit of a linear model: it is simply the proportion of total variation explained by the model. However, one might be tempted to use R^2 as a basis for comparing models with different numbers of parameters. R^2 is generally not helpful for this purpose: the reason for this is that if we add a new predictor to a linear model, the residual sum of squares decreases, and R^2 will increase. Hence if we attempt to select a subset of good predictors from a set of possible predictors using the value of R^2, the best model according to this criterion will be the model containing all the predictors, even if many of the predictors are in fact unrelated to the response.

$\bar{R}^2$, on the other hand, does not necessarily increase as new predictors are added to a model. If we look at (31), we see that $\bar{R}^2$ increases as $\widehat{\sigma}^2$ decreases. So if we were to rank different models using $\bar{R}^2$ as a crude model selection device, it would be equivalent to ranking the models based on $\widehat{\sigma}^2$. Does $\widehat{\sigma}^2$ necessarily decrease as new predictors are added to the model, and hence must $\bar{R}^2$ increase? The answer is no.

Recall that

$$\widehat{\sigma}^2 = \frac{(y-Xb)^\top(y-Xb)}{n-p}.$$

Now, consider two models in which one model contains a subset of the predictors included in the other. Certainly for the larger model, the numerator in the above expression is smaller, but the denominator would also be smaller, as the number of parameters p is larger for the more complicated model. Hence any reduction in the residual sum of squares must be large enough to overcome the reduction in the denominator, a reduction which increases as the number of parameters in the model increases. $\bar{R}^2$ is consequently sometimes useful as a crude device for comparison of different models. In future lectures we will say much more about the process of model selection, and will develop other criteria for comparing models.

Sequential sums of squares

We note that the column labelled ‘SS’ in the analysis of variance table lists the components of the decomposition (27). SS stands for sum of squares here, and this column shows the partition of variation for a multiple linear regression model in exactly the same way as it did for the simple linear regression model.

It will be helpful to introduce some notation. Suppose we have k predictors $x_1, ..., x_k$ in a linear model for a response y. There are $p = k+1$ parameters in the vector $\beta = (\beta_0, ...\beta_k)^\top$ in the linear model with the predictors $x_1, ..., x_k$. Partition the vector β into two parts $\beta = ({\beta^{(1)}}^\top, {\beta^{(2)}}^\top)^\top$ where $\beta^{(1)}$ is an $r \times 1$ subvector say, and $\beta^{(2)}$ is a $(p-r) \times 1$ subvector. We write $R(\beta^{(2)}|\beta^{(1)})$ for the increase in SS_{reg} in (27) when the predictors corresponding to the parameters $\beta^{(2)}$ are added to a model involving the parameters $\beta^{(1)}$. Think of $R(\beta^{(2)}|\beta^{(1)})$ as the variation explained by the term involving $\beta^{(2)}$ in the presence of the term involving $\beta^{(1)}$. We define $R(\beta_1, ..., \beta_k|\beta_0)$ to be SS_{reg}.

The sequential sums of squares shown below the analysis of variance table are simply the values

$$\begin{aligned}
&R(\beta_1|\beta_0) \\
&R(\beta_2|\beta_0, \beta_1) \\
&R(\beta_3|\beta_0, \beta_1, \beta_2) \\
&\vdots \\
&R(\beta_k|\beta_0, ..., \beta_{k-1})
\end{aligned}$$

Note that these contributions add up to $R(\beta_1, ...\beta_k|\beta_0)$. For instance, consider the case where $k = 2$. $R(\beta_2|\beta_0, \beta_1)$ is the increase in the regression sum of squares when β_2 is added to the model involving β_1, so that by definition

$$R(\beta_1, \beta_2|\beta_0) = R(\beta_1|\beta_0) + R(\beta_2|\beta_0, \beta_1).$$

The sequential sums of squares are useful when we have first ordered the variables in our model in a meaningful way (based on the underlying science or context). They tell us about how much a term contributes to explaining variation given all the previous terms in the table (but ignoring the terms which come after it).

3.11 Hypothesis testing in the general linear model

In the case of the simple linear regression model, we discussed the use of a t test and an equivalent F test for examining the usefulness of the predictor. For the general linear model we have already discussed the use of partial t tests for deciding whether a predictor is needed in the presence of the other terms. For a two-sided alternative there is an F test corresponding to this partial t test (where the test statistic is just the square of the partial t statistic). These F tests are usually called partial F tests.

While partial t or F tests can be useful, in the case of the general linear model there are more complicated hypotheses that may be considered. For example, we may ask: are any of the predictors helpful for explaining variation in the response? Or is the model including all the predictors no better than the one containing just the intercept? There is an F statistic for testing this hypothesis, and this F statistic is usually displayed in the analysis of variance table when we fit a multiple linear regression model.

The ANOVA table

The ANOVA table for a typical multiple regression model with k predictors and where $p = k + 1$ is shown below. Here as usual we write SS_{total} for the total sum of squares, SS_{reg} for the regression sum of squares and SS_{res} for the residual sum of squares. Writing $\beta = (\beta_0, \beta_1, ..., \beta_k)^\top$ for the parameters in the general

Source	Degrees of freedom	Sum of squares	Mean square	F	P
Regression	$p-1$	SS_{reg}	$MS_{reg} = \frac{SS_{reg}}{(p-1)}$	$F = \frac{MS_{reg}}{MS_{res}}$	$\mathbb{P}(F_{p-1,n-p} \geqslant F)$
Residual	$n-p$	SS_{res}	$MS_{res} = \frac{SS_{res}}{(n-p)}$		
Total	$n-1$	SS_{total}			

linear model, where β_0 is an intercept term and $\beta_1, ..., \beta_k$ are the coefficients corresponding to the k predictors, it can be shown that the quantity MS_{reg}/MS_{res} has an F distribution with $p-1$ and $n-p$ degrees of freedom when $\beta_1 = ... = \beta_k = 0$. Hence we can use the statistic MS_{reg}/MS_{res} to test the null hypothesis

$$H_0 : \beta_1 = ... = \beta_k = 0$$

versus the alternative

$$H_1 : \text{Not all } \beta_j = 0,\ j = 1, ..., k.$$

Writing $F = MS_{reg}/MS_{res}$, and writing F^* for a random variable with $p-1$ and $n-p$ degrees of freedom, we compute the p-value for the test as

$$\mathbb{P}(F^* \geqslant F).$$

An alternative to the use of p-values is to find the critical region: for a test with significance level α, the critical region is $F > F_{\alpha;p-1,n-p}$.

Example: Canadian business failures

We illustrate the test for the overall significance of the model using the Canadian business failures data that we have discussed previously. These data describe the annual percentage change in business failures and in the level of certain key economic indicators. If we fit a multiple linear regression model with percentage change in business failures as the response and percentage change in the economic indicators (real domestic product, wages and salaries, unemployment rate, retail sales, housing starts and C.P.I.) as predictors, we get the results shown below.

```
Regression analysis

The regression equation is
Business failures = - 59.4 + 2.93 Real domestic product
           + 3.80 Wages and Salaries + 8.60 Unemployment rate
           - 5.51 Retail sales - 0.208 Housing starts + 1.76 C.P.I.

Predictor        Coef       StDev          T        P
Constant       -59.41       47.46      -1.25    0.257
Real dom        2.934       3.105       0.94    0.381
Wages an        3.800       2.899       1.31    0.238
Unemploy        8.602       5.704       1.51    0.182
Retail s       -5.514       3.270      -1.69    0.143
Housing       -0.2079      0.5414      -0.38    0.714
C.P.I.          1.758       5.022       0.35    0.738

S = 15.47       R-Sq = 55.4%     R-Sq(adj) = 10.7%

Analysis of Variance

Source            DF          SS          MS         F        P
Regression         6      1782.3       297.1      1.24    0.400
Residual Error     6      1436.4       239.4
Total             12      3218.7

Source       DF      Seq SS
Real dom      1         8.3
Wages an      1         2.8
Unemploy      1       478.5
Retail s      1      1164.3
Housing       1        99.1
C.P.I.        1        29.3
```

We may have anticipated that we would not learn very much here by fitting a multiple linear regression model with six predictors and only thirteen observations. We see in the ANOVA table that the F statistic for testing overall model adequacy is 1.24, and the associated p-value is 0.4. So at the 5 percent level we accept the null hypothesis that the regression coefficients corresponding to the six predictors are all zero.

Example: risk assessment data

As a further example of testing for overall model adequacy consider the risk assessment data, which has been described in previous lectures. On pages 88 and 89 of your lecture notes I have given results when fitting a multiple linear regression model for these data with mean risk assessment as the response and the seven accounting determined measures of risk as predictors. It can be seen in the ANOVA table that the F statistic for testing overall model adequacy is 6.97, and the associated p-value

$$p = \mathbb{P}(F^* \geqslant 6.97)$$

where $F^* \sim F_{7,17}$ is approximately 0.001. So for the risk assessment data we reject the null hypothesis

$$H_0 : \beta_1 = ... = \beta_k = 0$$

in favour of the alternative

$$H_1 : \text{Not all } \beta_j = 0,\ j = 1, ..., k.$$

General hypothesis testing in the linear model

The F-test given in the ANOVA table illustrates a general method for comparison of linear models via hypothesis testing.

When we conduct a hypothesis we are always comparing two different models, and the null hypothesis imposes a restriction upon the parameters in the more general model which holds under the alternative hypothesis.

For instance, in the F-test in the ANOVA table we compare the model

$$y_i = \beta_0 + \varepsilon_i$$

(the model which holds under the null hypothesis) to the more general model

$$y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_k x_{ik} + \varepsilon_i.$$

In general, we might wish to compare models where only a subset of the predictors are removed under the null hypothesis. Reordering predictors so that $x_1, ..., x_r$ are the predictors to be included under both the null and alternative hypotheses, we might wish to compare the model

$$y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_r x_{ir} + \varepsilon_i$$

to the model

$$y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_r x_{ir} + .. + \beta_k x_{ik} + \varepsilon_i.$$

Here H_0 is $\beta_{r+1} = ... = \beta_k = 0$ and H_1 states that at least one of $\beta_{r+1}, ..., \beta_k$ is nonzero.

Write SS_{null} for the residual sum of squares for the model where H_0 holds, and write SS_{full} for the model which holds under H_1. The extra variation explained by the more general model is $SS_{null} - SS_{full}$. Is the amount of extra variation explained so large that we would favour the more complex model? We can develop a hypothesis test to address this question.

Writing Δp for the number of restrictions imposed by the null hypothesis on the most general model (in H_0 given above this is $k - r$, the number of extra parameters in the model under H_1) it can be shown that

$$F = \frac{(SS_{null} - SS_{full})/\Delta p}{MS_{res}}$$

has an F distribution with Δp and $n - p$ degrees of freedom under H_0 By comparing this test statistic with percentage points of an $F_{\Delta p, n-p}$ distribution, we can determine if the amount of extra variation explained by the more general model indicates that this model is decisively favoured. An example will help to illustrate the idea.

Example: risk assessment data

Suppose we feel that the most important predictor of mean risk assessment for the risk assessment data is dividend, and that we wish to see whether the other predictors in the model are needed in a multiple linear regression model in which dividend is included. To compute the test statistic for testing the hypothesis that the coefficients corresponding to current ratio, asset size, asset growth, leverage, variability earnings and covariability earnings are all zero, we need to compute the decrease in the residual sum of squares when we add these terms to a model involving dividend. This is obtained from the R output from the table of sequential sums of squares (see previous notes for the R output for the risk assessment data). Let $\beta^{(1)} = (\beta_0, \beta_1)^\top$ and $\beta^{(2)} = (\beta_2, ..., \beta_7)^\top$. Then we wish to test

$$H_0 : \beta^{(2)} = 0$$

against the alternative

$$H_1 : \text{Not all elements of } \beta^{(2)} \text{ are zero.}$$

Write $R(\beta^{(2)}|\beta^{(1)})$ for $SS_{null} - SS_{full}$ (the reduction in residual sum of squares when the $\beta^{(2)}$ term is added to the null model). We have

$$\begin{aligned} R(\beta^{(2)}|\beta^{(1)}) &= 5.6024 + 10.1251 + 1.6456 + 6.0841 + 3.9497 + 1.1961 \\ &= 28.6030. \end{aligned}$$

We need to divide this by $k - r = 6$, the number of extra parameters in the full model, and then divide by MS_{res} to get the value of our test statistic:

$$\begin{aligned} \frac{R(\beta^{(2)}|\beta^{(1)})/6}{MS_{res}} &= \frac{28.6030/6}{0.964} \\ &= 4.947. \end{aligned}$$

Hence if F^* is a $F_{6,17}$ random variable, we compute the p-value for the test as

$$p = \mathbb{P}(F^* \geqslant 4.947).$$

From tables or using R we obtain $p = 0.0042$ approximately. Hence we reject the null hypothesis $\beta^{(2)} = 0$ at the 5 percent level of significance.

Sequential F tests

In the last lecture we discussed the table of sequential sums of squares which appears in the computer output, and mentioned that the values which appear in this table represent the increases in the regression sum of squares as terms are added sequentially to the model. These sequential sums of squares can be used as the basis for a formal test of the hypothesis of the usefulness of each predictor in the model in the presence of the terms which appear before it but ignoring terms which come after it. As we mentioned last time, the table of sequential sums of squares gives the values

$$\begin{aligned} &R(\beta_1|\beta_0) \\ &R(\beta_2|\beta_0, \beta_1) \\ &R(\beta_3|\beta_0, \beta_1, \beta_2) \\ &\vdots \\ &R(\beta_k|\beta_0, ..., \beta_{k-1}). \end{aligned}$$

Consider the jth of these sequential sums of squares,

$$R(\beta_j|\beta_0, ..., \beta_{j-1})$$

and the model
$$y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_j x_{ij} + \varepsilon_i.$$
We can test the hypothesis
$$H_0 : \beta_j = 0$$
in this model against the alternative
$$H_1 : \beta_j \neq 0$$
by using the test statistic
$$F = \frac{R(\beta_j|\beta_0, ..., \beta_{j-1})}{MS_{res}}$$
which has an $F_{1,n-p}$ distribution assuming $H_0 : \beta_j = 0$. As usual if F^* is a random variable with an F distribution with 1 and $n-p$ degrees of freedom, then we compute the p-value for the test as
$$p = \mathbb{P}(F^* \geqslant F).$$
Alternatively, for a test with significance level α, the critical region is $F > F_{\alpha;1,n-p}$. The test we have just described is called a sequential F test. These sequential F tests will in general be hard to interpret unless we first carefully choose the ordering of the predictors in the model: a sequential F test examines the usefulness of a predictor in the presence of previous predictors but ignoring predictors which come after it.

Example: Risk assessment data

For the risk assessment data. In fitting the model, the predictors are ordered so that dividend is the first predictor, current ratio is next, then asset size, then asset growth, then leverage, then variability earnings and finally covariability earnings. Suppose we wish to test whether asset size is a useful predictor in the model which includes dividend and current ratio. The increase in the regression sum of squares when a term for the predictor asset size is added to the model involving dividend and current ratio is 10.1251. Writing β_0 for the intercept term, β_1 for the coefficient for dividend, β_2 for the coefficient for current ratio and β_3 for the coefficient for asset size, we want to test
$$H_0 : \beta_3 = 0$$
versus
$$H_1 : \beta_3 \neq 0$$
in the model involving the terms β_0, β_1 and β_2. The mean square error MS_{res} can be computed from the output by $(16.3808+1.1961+3.9497+6.0841+1.6456)/21 =$

$29.2563/21 = 1.3932$, and we know from the discussion above that the ratio $10.1251/1.3932 = 7.267514$ is a realization of an $F_{1,21}$ random variable under the null hypothesis. If $F^* \sim F_{1,21}$, then we compute the p-value for the sequential test as

$$p = \mathbb{P}(F^* \geqslant 7.267514)$$

which from tables or a computer package gives $p = 0.01353433$ approximately. So adding the term asset size to a model involving dividend and current ratio does help to explain variation in the response in a multiple linear regression model.

We have seen that in the analysis of variance table the "Degrees of freedom" column specifies the degrees of freedom which are appropriate for computing a p-value for an F test of overall model adequacy. We have not discussed the question, however, of where these degrees of freedom parameters come from, and of how we derive the distributions of the test statistics we have studied. The key theorem in proving the results we have stated about the distributions of test statistics is *Cochran's Theorem.* I will state this result for the MATH2931 students.

First we need a definition.

Definition:

Let y be an $n \times 1$ normal random vector with mean μ and variance I. Then $y^\top y$ follows the noncentral χ^2 distribution with n degrees of freedom and noncentrality parameter $\lambda = \frac{1}{2}\mu^\top \mu$. We write $y^\top y \sim \chi^2_{n,\lambda}$.

Theorem:

Suppose that A is an $n \times n$ idempotent matrix, and that y is an $n \times 1$ normal random vector with mean vector μ and covariance matrix $\sigma^2 I$, where I is the $n \times n$ identity matrix. Write Q for the quadratic form $Q = \frac{1}{\sigma^2} y^\top A y$, and suppose that with $Q_i = \frac{1}{\sigma^2} y^\top A_i y$ $i = 1, ..., k$ we have

$$Q = \sum_{i=1}^{k} Q_i.$$

If we write r_i for the rank of A_i and r for the rank of A then any one of the following conditions implies the other two.

1. $\sum_{i=1}^{k} r_i = r$

2. Each Q_i has a noncentral χ^2 distribution with r_i degrees of freedom and noncentrality parameter $\lambda_i = \frac{1}{2\sigma^2}\mu^\top A_i \mu$.

3. Each Q_i is independent of every other.

What we have called the total sum of squares SS_{total}, in the general linear model takes the form $y^\top Ay$ for an idempotent matrix A, where y here is the vector of responses. Also, the regression sum of squares and residual sum of squares can be written as quadratic forms in y. When we are deriving the test statistics above we are considering decompositions of a quadratic form into meaningful parts, and showing that the parts are independent and χ^2 distributed (which is what we need to do to prove that our test statistics, which involves ratios of quadratic forms, are F distributed). For a quadratic form $y^\top Ay$, the rank of the quadratic form is defined to be simply the rank of A. Each of the sums of squares in the analysis of variance table are quadratic forms, and the degrees of freedom column in the analysis of variance table specifies the ranks of those quadratic forms.

3.12 Multicollinearity

In a previous lecture where we developed a partial t test for the significance of a predictor in the general linear model in the presence of other predictors we mentioned that the interpretation of these tests is complicated when some of the predictors in the model contribute similar information. In this subsection we discuss this idea more formally.

In the course so far we have been dealing with the full rank linear model, in which the design or model matrix X has full rank. Recall that X has full rank if no column of X can be expressed as a linear combination of the remaining columns. When the predictors in our linear model contribute similar information, then it may happen that although X is of full rank, we can still nearly express one of the predictors as a linear combination of the remaining predictors: *multicollinearity* is the name given to this condition.

Effects of multicollinearity

Why is multicollinearity a potential problem, and what are its effects? If X does not have full rank, then there is no unique solution to the normal equations: we can get the same vector of fitted values by forming different linear combinations of the vectors of predictor values. When multicollinearity occurs in the full rank linear model, something close to this situation also occurs: quite different linear combinations of the predictor values can result in almost the same vector of fitted values. Hence for quite different β values the least squares criterion (which was just the sum of the squared differences between the observations and fitted values) may not change very much. Roughly speaking, there may be a large region in the space of the possible parameter values where we have much the same fit, and this makes it difficult to estimate β precisely.

Figure 3.9 illustrates the phenomenon of multicollinearity for an artificial data set involving two predictors. In the Figure, $\mathbf{y}$ is the response and $\mathbf{x}_1$ and $\mathbf{x}_2$ are the predictors. We can see that $\mathbf{x}_1$ and $\mathbf{x}_2$ are strongly linearly related, and the $(\mathbf{y}, \mathbf{x}_1, \mathbf{x}_2)$ points nearly lie on a line in three-dimensional space. Since there are an infinite number of planes which can pass through a given line in three-dimensional space, it's clear that when attempting to fit a plane through these points we could have two planes with very different coefficients for $\mathbf{x}_1$ and $\mathbf{x}_2$ resulting in almost the same quality of fit (residual sum of squares). Because $\mathbf{x}_1$ and $\mathbf{x}_2$ are so closely related, it is hard to separate out their effects. The coefficients will not be well estimated, and attempting to make predictions at new points which don't respect the dependencies in the original data could result in nonsensical answers.

The example above is sometimes referred to as the picket fence of multicollinearity. Think of the heights $\mathbf{y}$ as representing the tops of pickets at the $(\mathbf{x}_1, \mathbf{x}_2)$ points in a picket fence. Imagine trying to balance a table top on the pickets: the balance is very precarious in directions perpendicular to the line of pickets.

More formally, recall that the covariance matrix of the least squares estimator b of β in the general linear model was

$$\sigma^2(X^\top X)^{-1}.$$

If multicollinearity occurs, then $X^\top X$ is "nearly" singular, and this may cause the diagonal elements of $(X^\top X)^{-1}$ to be very large. Hence if multicollinearity occurs, estimation of the parameters β can be very poor, with standard errors of the components of b extremely large. It is important to note that we will not detect multicollinearity by looking at goodness of fit measures like R^2 or adjusted R^2, or by examining residuals. Even though estimates of the regression parameters are poor, the least squares estimator b of β by definition minimizes the sum of the squared residuals, and the fitted model involving b may produce a high R^2 value. The presence of multicollinearity can also have an adverse effect on prediction: predictions can be wild for vectors of predictor values which are not typical of the linear dependencies in the original data.

How do we detect multicollinearity when it occurs? There are many sophisticated diagnostics which can be used. However, we won't discuss these diagnostics. One simple measure which may be examined, however, is the correlation matrix for the predictor variables. If two predictor variables are closely linearly related, then the correlation coefficient for these two predictors will be close to 1, and multicollinearity occurs. Note, however, that multicollinearity can be more subtle than this: it is possible for all correlations between pairs of predictors to be

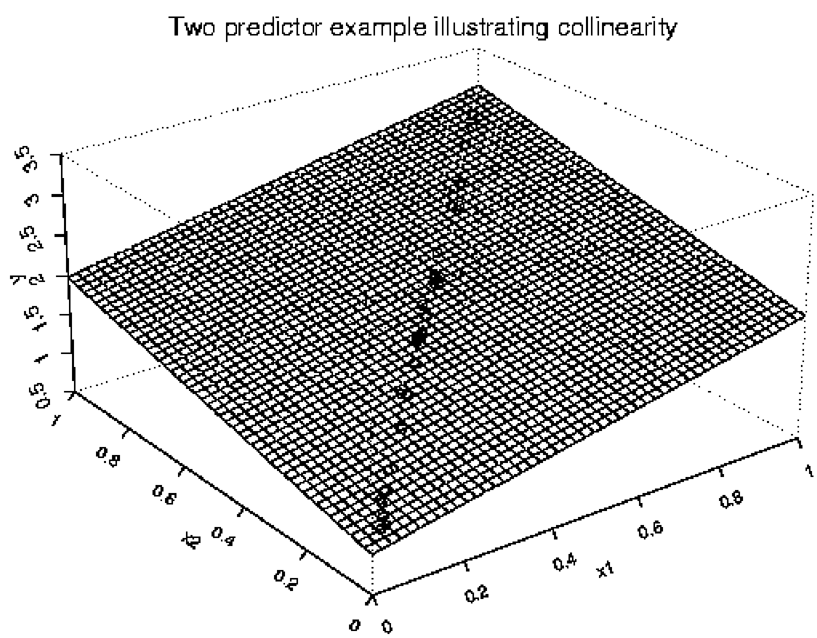

Figure 3.9: Example illustrating multicollinearity: on the plot, $\mathbf{y}$ is the response and $\mathbf{x}_1$ and $\mathbf{x}_2$ are the predictors. The points nearly lie on a line in three-dimensional space, and we can rotate the fitted plane and still get almost the same quality of fit (residual sum of squares).

small, but for one of the predictors to be capable of nearly being expressed as a linear combination of two or more of the remaining predictors. We now describe an example which illustrates some of the effects of multicollinearity.

Example: Hospital manpower data

The following data are from Raymond H. Myers, "Classical and Modern Regression with Applications (Second Edition)," Duxbury, Belmont, California, 1990, pp. 130–133. Data were collected from seventeen U.S. Naval hospitals at various sites around the world. The regressors are variables which are thought to predict workload (monthly man hours) at the hospitals. The variables are as follows:

$$\begin{aligned} y &= \text{Monthly man-hours} \\ x_1 &= \text{Average daily patient load} \\ x_2 &= \text{Monthly X-ray exposures} \\ x_3 &= \text{Monthly occupied bed days} \\ x_4 &= \text{Eligible population in the area / 1000} \\ x_5 &= \text{Average length of patients' stay in days.} \end{aligned}$$

It was desired to predict workload (i.e. the monthly man-hours y) based on the predictor variables x_1, x_2, x_3, x_4 and x_5. Fitting a multiple linear regression model gives the following output.

```
Regression Analysis

The regression equation is
y = 1963 - 15.9 x1 + 0.0559 x2 + 1.59 x3 - 4.22 x4 - 394 x5

Predictor        Coef       StDev          T        P
Constant         1963        1071       1.83    0.094
x1             -15.85       97.65      -0.16    0.874
x2            0.05593     0.02126       2.63    0.023
x3              1.590       3.092       0.51    0.617
x4             -4.219       7.177      -0.59    0.569
x5             -394.3       209.6      -1.88    0.087

S = 642.1       R-Sq = 99.1%     R-Sq(adj) = 98.7%

Analysis of Variance

Source            DF         SS          MS         F        P
Regression         5  490177488    98035498    237.79    0.000
Residual Error    11    4535052      412277
```

x_1	x_2	x_3	x_4	x_5	y
15.57	2463	472.9	18.0	4.45	566.5
44.02	2048	1339.8	9.5	6.92	696.8
20.42	3940	620.3	12.8	4.28	1033.2
18.74	6505	568.3	36.7	3.90	1603.6
49.20	5723	1497.6	35.7	5.50	1611.4
44.92	11520	1365.8	24.0	4.60	1613.3
55.48	5779	1687.0	43.3	5.62	1854.2
59.28	5969	1639.9	46.7	5.15	2160.6
94.39	8461	2872.3	78.7	6.18	2305.6
128.02	20106	3655.1	180.5	6.15	3503.9
96.00	13313	2912.0	60.9	5.88	3571.9
131.42	10771	3921.0	103.7	4.88	3741.4
127.21	15543	3865.7	126.8	5.50	4026.5
252.90	36194	7684.1	157.7	7.00	10343.8
409.20	34703	12446.3	169.4	10.78	11732.2
463.70	39204	14098.4	331.4	7.05	15414.9
510.22	86533	15524.0	371.6	6.35	18854.5

```
Total             16   494712540

Source        DF       Seq SS
x1             1    480612694
x2             1      7231656
x3             1       598469
x4             1       276098
x5             1      1458572
```

At first sight this seems like a satisfactory model: we have a very high R^2 value of 99.1 percent here. However, when we compute the correlation matrix of the predictors we find that the correlation between x_1 and x_3 is greater than 0.99.

The opposite extreme to collinearity is *orthogonality*. Write $\bar{x}_j$ for the mean of the values for the jth predictor,

$$\bar{x}_j = \frac{\sum_{i=1}^n x_{ij}}{n}.$$

Also, write s_j for the quantity

$$s_j = \sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

Now suppose we create a new set of predictor values by

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}.$$

Let Z be the $n \times k$ matrix with (i, j)th element equal to z_{ij}. If we fit a model using the predictors in Z instead of the predictors in the last k columns of X we do not really change the model: note that

$$\begin{aligned} \beta_i z_{ij} &= \beta_i \frac{x_{ij} - \bar{x}_j}{s_j} \\ &= -\beta_i \frac{\bar{x}_j}{s_j} + \frac{\beta_i}{s_i} x_{ij} \end{aligned}$$

which is a linear function of the original predictors (using the transformed predictors just changes the intercept and slope). Also note that the matrix $Z^\top Z$ is simply the correlation matrix of the predictors. You can easily verify that the (j, k)th element of this matrix is

$$\frac{\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_i (x_{ij} - \bar{x}_j)^2 \sum_i (x_{ik} - \bar{x}_k)^2}}$$

which is the sample correlation between the jth and kth predictors.

We say that the predictors are orthogonal if they are uncorrelated: that is, $Z^\top Z = I$ where I is the identity matrix. In the case of orthogonality, if we fit the model involving the transformed predictors, it is easy to see that the estimators of the predictor coefficients are independent (the covariance matrix of the predictor coefficients is $\sigma^2(Z^\top Z)^{-1} = \sigma^2 I$ and normality implies independence from this). This independence makes interpretation of coefficient estimates and of the partial t statistics much easier.

In general, if orthogonality doesn't hold and we fit a model using the standardized predictors Z, then the diagonal elements of

$$\sigma^2(Z^\top Z)^{-1}$$

will be larger than σ^2 (their value in the case of orthogonality). The diagonal elements of $(Z^\top Z)^{-1}$ represent the increase in the variances of the coefficients over the ideal case of orthogonality. These diagonal elements are called the variance inflation factors of the coefficients, and they are very useful for detecting multicollinearity. For the hospital manpower data the variance inflation factors are $(9597.57, 7.94, 8933.09, 23.29, 4.28)$. The first and third elements (corresponding to x_1 and x_3, which are highly correlated) are very large here, and the quality of estimation for the coefficients for these predictors is very poor.

More sophisticated methods for detecting multicollinearity and diagnosing what variables are involved are based on looking at the eigenvalues and eigenvectors of the correlation matrix $Z^\top Z$. (MATH2831 students can ignore the rest of this section if they wish). Roughly speaking, the eigenvectors corresponding to eigenvalues which are close to zero describe a near linear dependence among the columns of Z. We can write

$$Q^\top(Z^\top Z)Q = \Lambda$$

where Λ is a diagonal matrix of eigenvalues and Q is an orthogonal matrix of eigenvectors, $QQ^\top = I$. The eigenvalues of $Z^\top Z$ are always positive, and $Z^\top Z$ will be "nearly" singular (indicating multicollinearity) if some of the eigenvalues are close to zero. If an eigenvalue λ_j is small, we have for the eigenvector q_j (jth column of Q) that

$$q_j^\top(Z^\top Z)q_j = (Zq_j)^\top(Zq_j) \approx 0$$

The vector Zq_j (length n) is a linear combination of the columns of Z (the vectors of predictor values) and $(Zq_j)^\top(Zq_j)$ is the squared norm of this vector. This squared norm is close to zero only if the vector itself is nearly zero: the elements of the eigenvector q_j describe a near linear dependence among the predictors.

4 Model Selection

At the beginning of this course we discussed some of the reasons why we build statistical models. We build statistical models in order to make decisions, and in the process of selecting a model we must manage a trade off between complexity and goodness of fit in order to provide a reduction of the data that is useful for decision making.

In the context of the general linear model, the problem of model selection appears when we are deciding which predictor variables should be included for explaining variation in the response.

However, often data may have been collected for a large number of predictors, some of which might be unrelated to the response. We may wish to choose a simple model containing a subset of the predictors (or perhaps to choose a small collection of simple models containing subsets of the predictors) which offer a simpler explanation of the observed variation than a model incorporating all the predictors. Our interest may be in summarizing the data succinctly, in trying to determine which predictors are related to the response and which are not, or in prediction of future response values. Often there is no single best model.

Model selection for prediction

One of the most common reasons for building a statistical model is prediction, and we now discuss the issue of model selection when prediction is the goal. We begin by asking the question: is it harmful for prediction if we fit a statistical model which is more complicated than we really need? We will illustrate that fitting an unnecessarily complicated model can be harmful by considering the most elementary case, the simple linear regression model.

Suppose we have a data set consisting of responses $y_1, ..., y_n$ and corresponding predictor values $x_1, ..., x_n$, and suppose that the simple linear regression model holds,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

where ε_i $i = 1, ..., n$ is a collection of zero mean errors uncorrelated with common variance σ^2 say. We write M_0 for the model

$$y_i = \beta_0 + \varepsilon_i$$

in which the predictor x_i is excluded and write M_1 for the full model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

We consider fitting these two models to the data, and develop expressions for the expected squared prediction errors for the models for a new observation y_* when the predictor value is x_*. The expressions for expected squared prediction error will help to illustrate the trade off between complexity and goodness of fit that is involved in all model selection problems.

Prediction for a model with just an intercept

Consider first fitting the model M_0 which involves just an intercept. To estimate β_0, we must minimize

$$\sum_{i=1}^{n}(y_i - \beta_0)^2.$$

Differentiating this with respect to β_0, we get

$$-2\sum_{i=1}^{n}(y_i - \beta_0).$$

Hence the least squares estimator b_0 of β_0 satisfies

$$\sum_{i=1}^{n}(y_i - b_0) = 0$$

from which we have

$$\sum_{i=1}^{n} y_i = nb_0$$

or

$$b_0 = \bar{y}.$$

For the model M_0 then, our prediction of a new observation y_* from the fitted model is simply $\bar{y}$, the mean of the responses. We write $\widehat{y}^0(x_*)$ for the predicted value of y_* for model M_0 when the predictor is x_*.

Prediction for the full model

For the model M_1, write $\widehat{y}^1(x_*)$ for the predicted value of y_* when the predictor is x_*: that is, let

$$\widehat{y}^1(x_*) = b_0 + b_1 x_*$$

where b_0 and b_1 are the least squares estimators of β_0 and β_1 in the simple linear regression model that we developed at the beginning of the course. (At the risk of some confusion we have used the notation b_0 for the least squares estimator of the intercept in both models M_0 and M_1 even though the estimator of the intercept is different for the two models).

Comparing predictive performance

One way of measuring predictive performance of the models is to consider the expected values of the squared prediction errors. That is, we consider

$$\mathbb{E}((y_* - \widehat{y}^0(x_*))^2)$$

and

$$\mathbb{E}((y_* - \widehat{y}^1(x_*))^2).$$

Consider first $\mathbb{E}((y_* - \widehat{y}^0(x_*))^2)$. Recall that for a random variable Z, $\mathrm{Var}(Z) = \mathbb{E}(Z^2) - \mathbb{E}(Z)^2$ so that

$$\mathbb{E}(Z^2) = \mathrm{Var}(Z) + \mathbb{E}(Z)^2.$$

Applying this identity to $y_* - \widehat{y}^0(x_*)$, we have

$$\begin{aligned}
\mathbb{E}((y_* - \widehat{y}^0(x_*))^2) &= \mathrm{Var}(y_* - \widehat{y}^0(x_*)) + \mathbb{E}(y_* - \widehat{y}^0(x_*))^2 \\
&= \mathrm{Var}(y_*) + \mathrm{Var}(\widehat{y}^0(x_*)) + \mathbb{E}(y_* - \widehat{y}^0(x_*))^2 \\
&= \sigma^2 + \mathrm{Var}(\widehat{y}^0(x_*)) + \mathbb{E}(y_* - \widehat{y}^0(x_*))^2. \qquad (32)
\end{aligned}$$

By entirely similar reasoning, we can also show that

$$\mathbb{E}((y_* - \widehat{y}^1(x_*))^2) = \sigma^2 + \mathrm{Var}(\widehat{y}^1(x_*)) + \mathbb{E}(y_* - \widehat{y}^1(x_*))^2.$$

The terms in the above expressions are easily interpreted. The second term is just the variance of the prediction. The variance of prediction depends on the variance of our estimators of the model parameters, and we might expect that this is larger for the more complex model M_1 where there are more parameters to estimate. In fact, if either model M_0 or M_1 holds,

$$\mathrm{Var}(\widehat{y}^0(x^*)) = \frac{\sigma^2}{n}$$

and

$$\mathrm{Var}(\widehat{y}^1(x^*)) = \sigma^2\left(\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}\right).$$

So prediction variance is larger for the more complex model.

The third term in both expressions for expected squared prediction error is the square of the prediction bias. If the data conform to the model M_1, then we showed in our discussion of prediction for the general linear model that $\mathbb{E}(y_* - \widehat{y}^1(x_*)) = 0$ and so the squared bias term is zero for model M_1. However, if we fit the model M_0 when model M_1 holds with a nonzero β_1, then the bias term $\mathbb{E}(y_* - \widehat{y}^0(x_*))^2$ may be nonzero. If M_0 holds then the bias term will be zero. So the bias term is always smaller for the more complex model, and the variance term is always larger.

The key idea we are coming to here is the following: a complicated model has a higher variance of prediction than a simpler one, but a model which is too simple and ignores important predictors may make predictions which are systematically biased. Good model selection involves managing a trade off between the contributions of bias and variance to prediction error.

4.1 Out of sample prediction, PRESS residuals and the PRESS statistic

If we want to get some idea of the magnitude of prediction errors for a fitted linear model for new observations we cannot simply look at the residuals from the fitted model. The within sample prediction errors (residuals) will typically be smaller than so-called out of sample prediction errors (prediction errors for new observations which were not used in fitting the model). We have chosen the parameters in the fitted model by minimizing a measure of within sample prediction error, so that the true extent of typical discrepancies between fitted values and new observations may be underestimated by looking at the residuals for the fitted model.

We illustrate this phenomenon for the hospital manpower data introduced in the last lecture before discussing one measure which attempts to capture out of sample predictive performance and which may be used as a criterion for model selection.

Example: hospital manpower data

The following example of the difference between within sample and out of sample predictive performance is given by Myers (1990), "Classical and Modern Regression with Applications," Duxbury, Belmont, California, pp. 168–169. The example relates to the hospital manpower data we discussed in the last lecture. There are 17 observations here. If we fit a linear regression model with y as the response and x_1, x_2, x_3, x_4 and x_5 as predictors, then the residual for the last observation (within sample prediction error) is -466.27. Now, suppose we refit the model with the 17th observation deleted. Then we can produce a forecast of the response for the 17th observation based on this fitted model: when we do this, the prediction error is -3675.121, which is much larger (in absolute value) than the within sample prediction error of -466.27. Similarly, we can delete the other observations one at a time, and produce forecasts of each observation from a model not including that observation. For the fifteenth and sixteenth observations, for instance, this gives prediction errors of -2510.842 and 2242.496 respectively (and the residuals for the fifteenth and sixteenth observations when the model is fitted to all the data are -504.857 and 376.549 respectively, see Myers, p. 169). So we can see that in this example, prediction out of sample seems to be much poorer than prediction within sample.

In the example above, we tried to measure out of sample predictive performance by omitting one observation at a time, fitting a model without that observation and then using the fitted model to forecast the omitted observation. In this way we avoid the problem of using an observation both for fitting a model and assessment of the model. We write $\widehat{y}_{i,-i}$ for the forecast of the ith observation obtained by fitting a model to the data using all observations except the ith. The quantity

$$e_{i,-i} \quad = \quad y_i - \widehat{y}_{i,-i}$$

is called the ith PRESS residual. A global measure of goodness of fit based on the PRESS residuals is the PRESS statistic, which is the sum of the squared PRESS residuals:

$$\text{PRESS} \quad = \quad \sum_{i=1}^{n} e_{i,-i}^2.$$

Unlike the residual sum of squares, which the PRESS statisic resembles, the PRESS statistic does not necessarily decrease as we make the model more complex. The PRESS statistic can be computed and compared for different models, and the model with minimum PRESS chosen as one likely to have good out of sample predictive performance.

4.2 The hat matrix and computation of the PRESS residuals

It may appear at first sight that computation of the PRESS statistic is difficult. To compute the PRESS statistic, it seems that we must fit n different models where we delete one of the observations for each fit. If there are a large number of models to compare (large number of predictors) and a large number of observations, the computations quickly become unmanageable. As it turns out, it is not necessary to fit n separate regressions to compute the PRESS residuals. If we compute the residuals for the full fitted model, and a measure related to the influence of each observation on the fit (the so-called leverage of each observation) then the PRESS residuals and PRESS statistic can be computed.

Leverage and the hat matrix

In order to give the computational formula for the PRESS residuals in terms of ordinary residuals and leverages, we need to first define the leverages. The leverages are the diagonal elements of the so-called *hat matrix*,

$$H = X(X^\top X)^{-1}X^\top.$$

To interpret the hat matrix, just observe that the vector of fitted values is

$$\widehat{y} = Xb = X(X^\top X)^{-1}X^\top y = Hy.$$

So multiplying the vector of responses by H gives the fitted values. The ith diagonal element h_{ii} of H (ith leverage) multiplies y_i in determining $\widehat{y}_i$. So the leverage is some measure of the influence of y_i on the fit at the ith set of predictor values.

There is another interpretation of the leverages. If we write x_i for the vector of predictors corresponding to the ith observation of the response, $x_i = (1, x_{i1}, x_{i2}, .., x_{ik})^\top$, then we can write h_{ii} as

$$h_{ii} = x_i^\top (X^\top X)^{-1} x_i.$$

This can help us to interpret the leverages if we observe that

$$\begin{aligned} \mathrm{Var}(\widehat{y}(x_i)) &= \mathrm{Var}(x_i^\top b) \\ &= x_i^\top \mathrm{Var}(b) x_i \\ &= \sigma^2 x_i^\top (X^\top X)^{-1} x_i \\ &= \sigma^2 h_{ii} \end{aligned}$$

so that the ith leverage is apart from σ^2 simply the prediction variance for the fitted model at x_i. It can be shown that the leverage always lies between zero and one (the prediction variance is always nonnegative of course, and the variance of a prediction is never worse than the residual error variance at one of the observed x_i). Note that the leverage depends only on the vector of predictors (not on the response value): you should think of it as measuring the potential influence of an observation, with high leverage indicating that the vector of predictors is somehow extreme compared to the other predictor vectors.

PRESS residuals and leverage

Now that we have discussed the idea of leverage, we can describe computation of the PRESS residuals and the PRESS statistic. If e_i is the ordinary residual for the ith observation, h_{ii} is the ith leverage value, and if $e_{i,-i}$ is the ith PRESS residual, then

$$e_{i,-i} = \frac{e_i}{1 - h_{ii}}.$$

So to compute the PRESS residuals (and hence the PRESS statisic) all we need to do is compute the ordinary residuals and the diagonal of the hat matrix. If the leverage is large (so that the ith observation is an influential one in the fit) then the PRESS residual is made much larger in absolute value than the ordinary residual by the divisor of $1 - h_{ii}$ in the above equation. In effect, the PRESS residuals are obtained by increasing in absolute value the size of the ordinary residuals, with the amount of the increase related to how influential the observation may be in fitting the full model.

Example: capital asset pricing model

An example will help to make the concept of the leverage clear for the simple linear regression model. As we have mentioned, the ith leverage value is (apart from σ^2) the variance of our predictor of the response at x_i. For the simple linear regression model, we showed that the variance of our predictor of the conditional mean at x_0 was

$$\sigma^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)$$

so that the ith leverage value is

$$\frac{1}{n}+\frac{(x_i-\bar{x})^2}{S_{xx}}.$$

We see that the value of the leverage will be large if x_i is far away from the mean of the predictors $\bar{x}$: outlying points in the predictor space (predictor values distant from $\bar{x}$) have potentially high influence and a high leverage. Below is the result of fitting a simple linear regression model involving stock price as the response and TSE300 index as the predictor for the capital asset pricing model data discussed on Tutorial Sheet 5.

Also shown is a scatterplot of stock price versus TSE300 index.

```
Regression Analysis

The regression equation is
Stock Price = - 0.982 + 0.0101 TSE 300 Index

Predictor        Coef        StDev           T        P
Constant      -0.9817       0.8685       -1.13    0.272
TSE 300     0.0101020    0.0008176       12.36    0.000

S = 0.3176      R-Sq = 88.9%     R-Sq(adj) = 88.3%

Analysis of Variance

Source             DF          SS          MS         F         P
Regression          1      15.400      15.400    152.65     0.000
Residual Error     19       1.917       0.101
Total              20      17.317
```

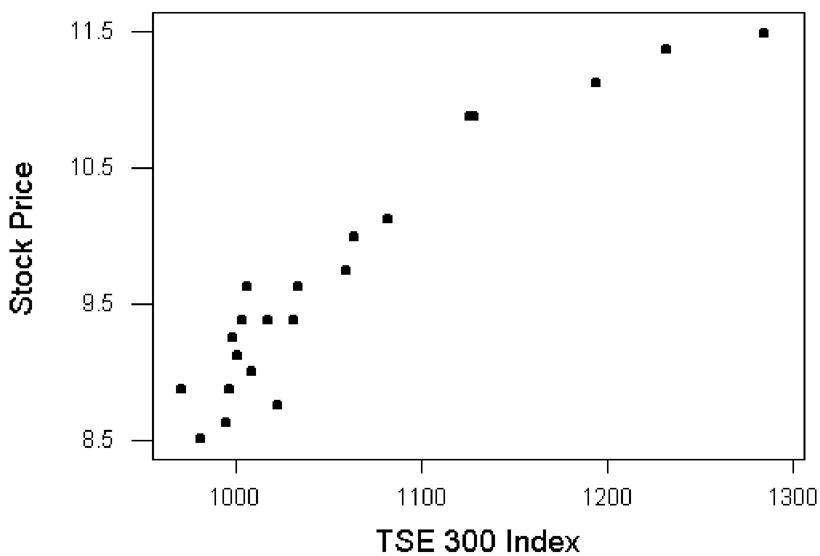

Figure 4.10: Scatter plot of stock price against TSE300 index.

The leverage values for the 21 observations are 0.073, 0.064, 0.057, 0.075, 0.088, 0.053, 0.052, 0.068, 0.070, 0.099, 0.059, 0.048, 0.072, 0.066, 0.048, 0.051, 0.080, 0.078, 0.168, 0.247, 0.386. We see that the largest leverage value corresponds to the last observation (which happens to be the rightmost predictor value in the scatter plot).

We now give an example illustrating the use of the PRESS statistic for model selection.

Example: data on cheddar cheese

In this example we will consider some data from a study on cheddar cheese in the La Trobe Valley, Victoria. For this data set, we have a subjective measure of cheese taste as the response (taste) and as predictors we have measures of concentration of acetic acid (acetic), concentration of hydrogen sulfide (H2S) and concentration of lactic acid (lactic). The table below shows the values of R^2, adjusted R^2 and the PRESS statistic for all possible models with 1, 2 and 3 predictors.

Model	R^2	$\bar{R}^2$	PRESS
H2S	0.571	0.556	3688.08
lactic	0.496	0.478	4375.64
acetic	0.302	0.277	6111.26
H2S, lactic	0.652	0.626	3135.44
H2S, acetic	0.582	0.551	3877.62
lactic, acetic	0.520	0.485	4535.47
H2S, lactic, acetic	0.652	0.612	3402.24

For this example, we see from the table that the best model in terms of the PRESS statistic is the model which contains Lactic and H2S. This model is also the best in terms of maximum adjusted R^2.

4.3 Cross validation

In this subsection we discuss the idea of cross validation, which is related to the ideas we developed in the definition of PRESS residuals and the PRESS statistic. In developing the PRESS residuals we left out one observation at a time and fitted a model to the remaining observations, and then predicted the observation which was omitted. The idea of cross validation is to split the data into two parts, a training sample (which is used to estimate parameters in candidate models) and a validation sample (which is used to assess predictive performance). Selection of a model is based on predictive performance in the validation sample. If we have $m+n$ observations (responses $y_1, .., y_{m+n}$ and corresponding vectors of predictors $x_1, ..., x_{m+n}$) and if the first m observations constitute the training sample and the last n constitute the validation sample, then we might compare different models according to the criterion

$$\sum_{i=m+1}^{m+n} (y_i - x_i^\top b)^2 \tag{33}$$

where b is the least squares estimator of β based on the first m observations. A common alternative is to compare different models based on

$$\sum_{i=m+1}^{m+n} |y_i - x_i^\top b|.$$

Once a model has been selected, we refit the selected model using the whole data set.

There are a number of difficulties with the idea of cross validation. Firstly, we must have a large enough number of observations so that we can split the data into two parts which are large enough to enable reliable estimation of parameters and reliable estimation of predictive performance. This may not always be the case. Also, there is the problem of how exactly to split the data (which observations should go in the training and validation samples, and what should m and n be?)

We give an example to illustrate the use of cross validation and some of the problems involved in its application.

Example: cross validation for data on cheddar cheese tastings

For the cheddar cheese tastings data, there are 30 observations which we split into two parts of 15 observations each (the first 15 observations in the sample constitute the training sample, and the last 15 constitute the validation sample). I have computed the criterion (33) using the validation sample after fitting each of the possible models involving the three predictors to the training sample. The results are shown in the table below. Also shown are the values of (33) when the second half of the data is used as the training sample and the first half is used as the validation sample. We can see from the table that the best model according to the cross validation criterion for both cases is the one involving H2S and Lactic. However, there are quite different predictor values in the first fifteen and last fifteen observations, and our ranking of the models is highly dependent on the choice of training set: for instance, the full model is the second best model

Model	Cross validation criterion (fitting to first half)	Cross validation criterion (fitting to second half)
H2S	2257.94	3312.33
Lactic	2510.53	3381.18
Acetic	5820.82	3423.51
H2S, Lactic	2139.56	2924.22
H2S, Acetic	9814.08	3339.40
Lactic, Acetic	5081.51	3473.03
H2S, Lactic, Acetic	8930.57	2959.95

when the second half of the data is used for the training set, but it is the second worst model when the first half of the data is used for the training set. How to split the data into two parts can be a difficult question in application of the idea of cross validation.

4.4 The conceptual predictive criterion (Mallow's C_p).

We describe another criterion for selection of a statistical model when prediction is the goal of a study. MATH2931 students will derive this criterion in Assignment 3. The idea of the criterion is to minimize an estimate of the quantity

$$\sum_{i=1}^{n} \frac{\mathrm{MSE}(\widehat{y}(x_i))}{\sigma^2} \tag{34}$$

where (writing $y(x_i)$ for the mean of response at x_i) $\mathrm{MSE}(\widehat{y}(x_i))$ is the mean squared error of prediction at x_i,

$$\mathrm{MSE}(\widehat{y}(x_i)) \quad = \quad \mathbb{E}((\widehat{y}(x_i) - y(x_i))^2)$$

Now, using the formula $\mathrm{Var}(Z) = \mathbb{E}(Z^2) - \mathbb{E}(Z)^2$, we have $\mathbb{E}(Z^2) = \mathrm{Var}(Z) + \mathbb{E}(Z)^2$ and hence

$$\mathrm{MSE}(\widehat{y}(x_i)) = \mathrm{Var}(\widehat{y}(x_i)) + \mathrm{Bias}(\widehat{y}(x_i))^2$$

where $\mathrm{Bias}(\widehat{y}(x_i)) = \mathbb{E}(\widehat{y}(x_i) - y(x_i))$. So (34) is equal to

$$\frac{\sum_{i=1}^{n} \mathrm{Var}(\widehat{y}(x_i))}{\sigma^2} + \frac{\sum_{i=1}^{n} \mathrm{Bias}(\widehat{y}(x_i))^2}{\sigma^2}.$$

It can be shown that $(n-p)(\widehat{\sigma}^2 - \sigma^2)$ is an unbiased estimator of

$$\sum_{i=1}^{n} \text{Bias}(\widehat{y}(x_i))^2$$

and that

$$\frac{\sum_{i=1}^{n} \text{Var}(\widehat{y}(x_i))}{\sigma^2} = p$$

so that

$$p + \frac{(n-p)(\widehat{\sigma}^2 - \sigma^2)}{\sigma^2}$$

is an estimator of (34). In practice σ^2 is not known, but an unbiased estimate of σ^2 (provided one of the models in the class of models being considered is correct) is the estimate $\widehat{\sigma}_F^2$ based on the full model: substituting into the expression above gives Mallow's C_p statistic,

$$C_p \quad = \quad p + \frac{(\widehat{\sigma}^2 - \widehat{\sigma}_F^2)(n-p)}{\widehat{\sigma}_F^2}.$$

Although we don't derive Mallow's C_p here, it is easy to see that by minimizing C_p we are managing a goodness of fit/complexity trade off. The first term p in the above expression grows with the complexity of the model, and the last term is a measure of goodness of fit (since if we have omitted important covariates the estimate $\widehat{\sigma}^2$ of σ^2 will be biased upwards, and hence $\widehat{\sigma}^2 - \widehat{\sigma}_F^2$ will tend to be positive.)

Example: cheddar cheese tastings

To illustrate the use of C_p, we consider once more the data on cheddar cheese tastings.

```
Best Subsets Regression


Response is taste


                                        A   L
                                        c   a
                                        e   c
                                        t H t
              Adj.                      i 2 i
Vars   R-Sq   R-Sq    C-p          s    c S c

   1   57.1   55.6    6.0     10.833      X
   1   49.6   47.8   11.6     11.745        X
   1   30.2   27.7   26.1     13.821    X
```

2	65.2	62.6	2.0	9.9424		X	X
2	58.2	55.1	7.2	10.890	X	X	
2	52.0	48.5	11.8	11.668	X		X
3	65.2	61.2	4.0	10.131	X	X	X

The best model according to the C_p criterion is again the one involving H2S and Lactic. All the various model selection criteria we have considered so far have led to the model involving H2S and Lactic as the best model. The choice of a best model (for predictive or other purposes) will often not be so clear, and it is always worthwhile to consider a number of model selection criteria in choosing a model.

4.5 Sequential variable selection procedures

The approaches we have developed so far to selection of a model are based on computing some criterion (Mallow's C_p or PRESS for instance) for all possible subset models and then choosing the model which optimizes the criterion. If the number of predictors under consideration is very large, then there is an enormous number of possible models, and computing any criterion for all possible models may be difficult.

Consider the situation where we have k predictors. Write $\gamma = (\gamma_1, ..., \gamma_k)$ for a $k \times 1$ vector where $\gamma_j = 1$ if the jth predictor is in the model and $\gamma_j = 0$ if the jth predictor is not in the model. Each distinct combination of the binary variables in γ corresponds to a different model. There are two possible values for γ_1 (zero or one), two possible values for γ_2, and so on. In general the number of different distinct values for the vector of binary variables γ is 2 multiplied by itself k times, or 2^k. For instance, consider the situation where there are just 2 predictors. γ has length two here, and its possible values are $(0,0)$, $(0,1)$, $(1,0)$ or $(1,1)$. So there are four possible models: the model containing no predictors, two possible models containing one predictor and the full model.

As k increases, 2^k quickly become very large indeed. With modern computing packages it is now possible to compute a model selection criterion like C_p for every possible model in most typical problems, but this was not always the case. To cope with the computational difficulties of model selection for large k so-called sequential methods of variable selection have been developed, and these methods are still sometimes used in large problems. The methods we will describe are also implemented in most computer packages, and are often used for this reason despite the associated statistical problems (see later).

Forward selection

The idea of sequential variable selection methods is to start with some initial model, and to then add, delete or change a variable in the initial model to improve the fit. We then add, delete or change a variable in the new model, and continue searching through the space of possible models in this way until some stopping criterion is satisfied. By starting with some initial model and making small changes to it we avoid the need to compute some criterion function for all possible models. The mechanisms for adding or deleting variables in the stepwise procedure are based on the hypothesis tests (partial t and F tests) which were discussed earlier in the course.

There are three basic sequential variable selection techniques. The first is forward selection. The idea of forward selection is to start with our best model involving one variable, and then add the variable at each step which most improves the fit until some stopping rule is satisfied. The hope with forward selection is that by taking our best simple model and improving it as much as we can at each step, we will arrive at a model that is good globally: this hope may not always be realized. We can summarize the steps in forward selection as follows:

1. Fit the model involving just an intercept

2. Until some stopping rule is satisfied, find the predictor not currently in the model (if there is one) for which the partial F-statistic is largest and add it to the model.

We haven't said anything yet about the stopping criterion in the method above. One approach is to stop when the model contains a certain fixed number of variables, p^* say. Another common stopping rule is to stop when none of the partial F statistics for the coefficients considered in the second step is significant at some preselected level. The default significance level used varies between computer packages. We give an example which will illustrate the idea of forward selection.

Example: cheddar cheese tastings

Consider the data on cheddar cheese tastings given in assignment three. The response here was the variable taste (a subjective measure of taste) and the predictors were measures of the concentrations of various chemicals in the samples (acetic, lactic and H2S).

Of all the one variable models, the largest partial F-statistic occurs for the model involving `H2S` (the F statistic for testing for the significance of the slope for H2S is 37.29, with a p-value of approximately 0).

Now we consider adding the variables not currently in the model (acetic and lactic) to the variable H2S. Fitting the model involving H2S and lactic, the partial F statistic for testing for the significance of lactic in the presence of H2S is approximately 6.25. If we fit the model involving H2S and acetic, the partial F statistic for testing for the significance of acetic in the presence of H2S is approximately 0.71. The largest of these is the F statistic for lactic, with a p-value of 0.019. Finally we consider adding acetic to the model involving H2S and lactic. The partial F statistic for acetic is approximately 0.005, with a p-value exceeding 0.9, clearly not significant at the 0.25 level. Since this is the only variable not already in the model our stopping rule is satisfied, and the result of our forward stepwise search is to choose the model involving H2S and lactic.

```
Stepwise Regression: Taste versus acetic,H2S,lactic

Forward Selection. Alpha-to-enter: 0.25

 Response is  taste   on  3 predictors, with N =   30

    Step          1         2
Constant     -9.787   -27.592

H2S            5.78      3.95
T-Value        6.11      3.47
P-Value       0.000     0.002

Lactic                   19.9
T-Value                  2.50
P-Value                 0.019

S              10.8      9.94
R-Sq          57.12     65.17
R-Sq(adj)     55.58     62.59
C-p             6.0       2.0
```

The "Step" columns labelled 1 and 2 here show the variables which are added at each stage. We see that H2S was added first, and for the simple linear regression model with H2S as predictor the estimated intercept was -9.787 and the estimated slope was 5.78. Also reported is the partial t statistic for testing significance of the slope which is 6.11 (we square this to get the partial F statistic) with associated p-value 0.000. The residual standard error is 10.8, the R^2 is 0.5712, the adjusted R^2 is 0.5558 and Mallow's C_p is 6.0. At the second stage, the variable lactic is added to the model: again we have coefficient estimates, the estimated residual standard error, the R^2, adjusted R^2, C_p and values of the partial t statistics. The model involving H2S and lactic is the final model here.

Backward elimination

The second main sequential variable selection algorithm is called backward elimination or backward selection. The idea of backward selection is to start with an initial model containing all predictors, and to then delete terms at each step which give the least degradation of fit until some stopping criterion is satisfied. More formally, we can describe the algorithm as follows:

1. Fit the model involving all the predictors.

2. Until some stopping criterion is satisfied, delete the variable currently in the model with the smallest partial F statistic.

The stopping rule for backward elimination is usually to stop when some fixed number of parameters is reached, or to stop when all of the partial F statistics are significant at a certain level (this is what R does, and the default significance level is 0.1).

Example: data on cheddar cheese tastings

If we fit the full model for the cheddar cheese tastings data, then we have that the partial F statistics for the three variables are approximately 9.80 for H2S, 0.005 for acetic and 5.20 for lactic. So at the first stage of backward elimination we delete the variable acetic. Now we consider the model involving H2S and lactic. In this model, the partial F statistics are approximately 12.04 for H2S and 6.25 for lactic. Since both of these are significant at the 0.1 level, our stopping criterion is satisfied and the final model is again the model involving H2S and lactic.

```
Stepwise Regression: Taste versus acetic,H2S,lactic

Backward elimination. Alpha-to-remove: 0.1

 Response is  taste   on  3 predictors, with N =   30

    Step         1        2
Constant    -28.88   -27.59

Acetic         0.3
T-Value       0.07
P-Value      0.942

H2S            3.9      3.9
T-Value       3.13     3.47
P-Value      0.004    0.002

Lactic        19.7     19.9
T-Value       2.28     2.50
P-Value      0.031    0.019
```

```
S             10.1    9.94
R-Sq         65.18   65.17
R-Sq(adj)    61.16   62.59
C-p            4.0     2.0
```

Again the "Step" columns labelled 1 and 2 show the variables in the model at each stage of the backward elimination procedure. We start with the full model, and in the second step move to the model involving just H2S and lactic. Also shown are coefficient estimates, the estimate of the error standard deviation, the R^2 value, adjusted R^2 and C_p values.

Stepwise methods

The final sequential variable selection algorithm we will discuss is called stepwise variable selection, and combines elements of the forward and backward elimination algorithms. There are many variations of the basic stepwise procedure, and we will just discuss the following algorithm. Repeat:

1. If there is at least one predictor in the current model, consider deleting the variable with the smallest partial F statistic provided this partial F statistic is not significant at some preset level α_{OUT}.

2. If no variable could be deleted in the first step, consider adding a variable to the model. Add the variable with the largest partial F statistic provided it is significant at some preset level α_{IN}.

3. If no variable could be added or deleted in the first two steps, then stop.

Default values for α_{IN} and α_{OUT} have to be supplied.
Example: data on cheddar cheese tastings

For the data on cheddar cheese tastings, we describe the steps involved in stepwise variable selection. The steps involved are actually the same as for forward selection in this case. The initial model contains just an intercept, and there is no variable to delete. Then we consider adding a variable: the variable to be added (with the largest partial F statistic of approximately 37.29) is H2S. This variable cannot be deleted, and so we consider adding another variable: the largest partial F is for lactic, with a partial F value of approximately 6.25 (which is significant at the 0.15 level). Now consider deleting a variable: the partial F statistics in the model involving Lactic and H2S are both significant at the 0.15 level, so there is no variable to delete. The partial F statistic for acetic is not significant at the 0.15 level, so there is no variable to add. So we stop, and the final model is the one involving Lactic and H2S.

```
Stepwise Regression: Taste versus acetic,H2S,lactic

 Alpha-to-Enter:      0.15    Alpha-to-Remove:      0.15

 Response is  taste   on  3 predictors, with N =   30

    Step         1        2
Constant    -9.787  -27.592

H2S           5.78     3.95
T-Value       6.11     3.47
P-Value      0.000    0.002

Lactic                 19.9
T-Value                2.50
P-Value               0.019

S             10.8     9.94
R-Sq         57.12    65.17
R-Sq(adj)    55.58    62.59
C-p            6.0      2.0
```

If you want to keep variables in the model regardless of their p-values, you can enter them as "Predictors to include in every model" after choosing `Stepwise...`. As we have mentioned there are many variations on the basic stepwise procedure: one common variation is to consider swaps of variables currently in the model with variables not in the model if this improves R^2, in addition to additions and deletions of variables.

4.6 Problems with forward selection, backward selection and stepwise algorithms

The use of sequential algorithms presents a number of statistical problems. Clearly the motivation for the forward selection, backward elimination and stepwise algorithms comes from hypothesis testing and in particular from the partial F tests for examining the significance of a predictor in the presence of the other terms in a linear model.

With hypothesis testing we make a comparison between two models, one of which is simpler than the other. In sequential variable selection procedures, we are comparing two models which differ by a single variable, and we might have good reason to believe that neither model is adequate. Consider the early stages of forward selection or the stepwise procedure for instance: here we might expect that there are many important omitted covariates, and the current model or any one variable addition to it cannot be reasonable. The residual variance is overestimated, and partial F statistics are "deflated" (since the residual variance appears in the denominator of the partial F statistic). Clearly there is also a problem of multiple testing in sequential algorithms, where decisions about ommission or inclusion of covariates are based on looking at a maximum or minimum of multiple partial F statistics.

For all the model selection procedures we have considered, there is also the problem of bias of parameter estimates and overstatement of significance levels because the same data are used both for model selection and estimation of parameters. We did show that the least squares estimator of β was unbiased if the assumed model was correct. However, there the model was known beforehand: if the final model chosen to be fitted is based on the data y, then the development we gave previously of distributions of estimators and test statistics is not really appropriate. The effect of selection has not been taken into account.

Example: regression for simulated data where the response and predictors are unrelated

To illustrate some of the problems with sequential algorithms we consider a simulated data set. Our discussion here is similar to that of Weisberg (1985), "Applied Linear Regression (Second Edition)," Wiley, New York, pp. 214–215.

Suppose we simulate 51 vectors of independent $N(0, 1)$ random variables of length 100 We take the first of these vectors as the response, and the last 50 vectors as the predictors. Now, of course, the response here is completely unrelated to all the predictors: but what happens when we apply a sequential variable selection procedure to this data set?

For a simulation I performed, the regression of the response on all fifty predictors (not doing any variable selection) gave an R^2 of 0.597. This is perhaps surprising: after all, the predictors are completely unrelated to the response. The high R^2 occurs simply because we have a large number of predictors, and some of them happen to be related to the response purely by chance. However, the F statistic for significance of the overall model indicates that the model with no predictors is to be preferred to the full model.

```
Regression Analysis

The regression equation is
C1 = - 0.126 + 0.277 C37 - 0.343 C45 - 0.219 C14 - 0.181 C4

Predictor        Coef       StDev          T         P
Constant     -0.12648     0.08919      -1.42     0.159
C37           0.27750     0.08253       3.36     0.001
C45          -0.34327     0.08991      -3.82     0.000
```

```
C14          -0.21942     0.08488      -2.58    0.011
C4           -0.18071     0.07892      -2.29    0.024

S = 0.8622      R-Sq = 25.4%     R-Sq(adj) = 22.2%

Analysis of Variance

Source            DF          SS          MS         F        P
Regression         4     24.0298      6.0074      8.08    0.000
Residual Error    95     70.6279      0.7435
Total             99     94.6576

Source       DF      Seq SS
C37           1      7.1434
C45           1      7.9000
C14           1      5.0881
C4            1      3.8982
```

We see here that the p value for overall significance of the model is less than 0.0005, and the p-values for the partial t tests are also less than 0.05 for all four of the predictors. By ignoring the fact that the data was used to select *which* model was fitted we compute inappropriate p values which indicate overall significance of the model and significance of the coefficients.

4.7 The MAXR procedure

A sequential variable selection procedure we have not yet considered is the so-called MAXR procedure. Unlike forward selection, backward selection and the stepwise algorithm, the final result of the MAXR procedure is not a single model but a list of models, one model for each possible subset size of the predictors.

MAXR begins in the same way as forward selection (that is, we find the best one variable model). The MAXR procedure then proceeds iteratively. We consider all the possible models obtained from the current model by adding a single variable. The variable which gives the largest increase in R^2 is chosen. Given this new model, we then consider each of the predictors in the model in turn, and we do a swap with the predictor not in the model producing the largest increase in R^2 (provided there is a swap that increases R^2). We repeatedly cycle through the predictors in this way making one on one swaps until no increase in R^2 can be made. When this happens, we store the current model as being the best model of the current size, and then repeat the cycle of adding a predictor and considering swaps. The procedure continues until all variables are in the model, and the result of the algorithm is a list of models (one for each subset size) which can be examined further.

More formally, we can write the algorithm as follows.

1. Set the initial model to be the best simple linear regression model (largest partial F-statistic or equivalently largest R^2).

2. Until all variables are in the model, do the following.

   (a) Add the variable not currently in the model which produces the largest increase in R^2. Repeatedly cycle through all the predictors in the new model, doing the following until R^2 does not increase in a cycle through all the predictors (and save the current model when this occurs).

      i. Swap the current predictor with the predictor not in the model producing the largest increase in R^2 (provided there is a swap which increases R^2).

Example: cheddar cheese tastings

We now discuss the implementation of the MAXR procedure for the data on cheddar cheese tastings. The MAXR procedure starts in the same way as forward selection: we simply choose the best one variable model by finding the predictor which has the highest absolute correlation with the response. As we saw in the last lecture, the best one variable model for the data on cheddar cheese tastings involved the predictor H2S. We save this model as being our best one variable model. Then we consider adding one of the variables acetic or lactic to the model involving H2S. The maximum increase in R^2 occurs when lactic is added (the R^2 for H2S and lactic is 0.652, and the R^2 for H2S and acetic is 0.582). Then for the model involving H2S and lactic, we consider a swap between H2S and acetic: does the model with acetic and lactic have a higher R^2? The answer here is no (the model with acetic and lactic has an R^2 of 0.520). Then we consider a swap between lactic and acetic: but we have already seen that the model with H2S and acetic is inferior to the one involving H2S and lactic. So there is no one on swap of acetic with one of the predictors H2S and lactic which can improve the model. So we store this model as our best two predictor model. Of course, there is only one three predictor model. This is the end of the MAXR procedure: we have a list of possible models for each subset size.

5 Residuals and diagnostics

So far we have talked about selection of a model and estimation of parameters under ideal conditions (that is, given that the assumed model is correct). We mentioned when we talked about simple linear regression the need for criticism of model assumptions, and we took a brief look at some methods for model criticism based on residuals. In this topic we take a more detailed look at various kinds of residuals in the context of the general linear model.

When we talked about simple linear regression, we mentioned that residuals could be used to help detect an incorrectly specified mean structure or to detect a violation of the constancy of error variance assumption. Residual plots can be helpful for detecting similar violations of assumptions in the general linear model, as well as for detecting outlying points and other problems with the model formulation. However, we should not expect the raw residuals to be the most helpful diagnostic for detecting all possible violations of model assumptions, and there are modifications of the raw residuals which sharpen their usefulness for investigating specific problems. A discussion of different kinds of residuals and related diagnostic plots is the subject of this topic.

5.1 Residual plots

Consider the general linear model

$$y = X\beta + \varepsilon$$

where $y = (y_1, ..., y_n)^T$ is an $n \times 1$ vector of responses, $\beta = (\beta_0, ..., \beta_k)^T$ is a $p \times 1$ vector of parameters, X is an $n \times p$ design matrix and $\varepsilon = (\varepsilon_1, ..., \varepsilon_n)$ is a vector of zero mean uncorrelated errors with common variance σ^2 say. As usual we write b for the least squares estimate

$$b = (X^T X)^{-1} X^T y$$

of β, and $\widehat{\sigma}^2$ for the usual unbiased estimator of σ^2. The fitted values are

$$\begin{aligned} \widehat{y} &= Xb \\ &= X(X^T X)^{-1} X^T y \\ &= Hy \end{aligned}$$

where H is the hat matrix introduced in our discussion of computation of the PRESS residuals. The vector of residuals is the vector of responses minus the vector of fitted values,

$$\begin{aligned} e &= y - Hy \\ &= (I - H)y \end{aligned} \qquad (35)$$

where I denotes the $n \times n$ identity matrix here. We can also write

$$\begin{aligned} e &= (I - H)y \\ &= (I - H)(X\beta + \varepsilon) \\ &= X\beta - HX\beta - H\varepsilon + \varepsilon. \end{aligned}$$

But $HX = X(X^T X)^{-1} X^T X = X$, so we have

$$e = (I - H)\varepsilon.$$

Hence in terms of the residuals,

$$e_i = (1 - h_{ii})\varepsilon_i - \sum_{j \neq i} h_{ij}\varepsilon_j.$$

In general, as the sample size n increases the elements of the hat matrix tend to zero and the residuals e_i are a good approximation to the errors ε_i. Note that $\mathbb{E}(e) = 0$ always holds, so that the residuals have mean zero. We considered plots of the raw residuals against predictors and fitted values in the context of the simple linear regression model for detecting an incorrectly specified mean structure or a violation of the assumption of constant error variance. Similar plots can be useful in the context of the general linear model, as the following examples show.

Example: Locating fast food restaurants

The following example is from Keller, Warrack and Bartel, "Statistics for Management and Economics: A Systematic Approach (Second Edition)," Wadsworth, Belmont, California, 1990, pp. 832–833.

In trying to find new locations for their restaurants, hamburger emporiums like McDonald's and Wendy's usually consider a number of factors. Suppose that a statistician working for a hamburger restaurant chain wants to construct a regression model that will help her make recommendations about new locations. She knows that this type of restaurant has as its primary market middle-income adults and their children (particularly children betwen the ages of 5 and 12). She collected the (fictitious) data in the table below showing annual sales in twenty five randomly chosen areas containing about five hundred households and exactly one of her employer's restaurants and exactly one competitor's restaurant. Consider fitting a linear model to these data in order to predict annual gross sales based on mean annual household income and mean age of children. Fitting the model results in the following.

```
The regression equation is
Sales = 668 + 11.4 Income + 16.8 Age
```

Annual Gross Sales ($1000s)	Mean Annual Household Income ($1000s)	Mean Age of Children
1128	23.5	10.5
1005	17.6	7.2
1212	26.3	7.6
893	16.5	5.9
1073	22.3	6.6
1179	26.1	6.3
1109	24.3	12.1
1019	20.9	14.9
1228	27.1	8.9
812	15.6	3.4
1193	25.7	10.5
983	3.5	6.0
1281	26.5	8.6
1156	25.7	11.6
1032	21.8	13.7
856	33.6	5.8
978	17.9	10.3
1017	18.3	5.3
1091	30.1	6.3
1048	29.8	5.3
1192	28.5	10.4
1256	27.5	8.7
1215	26.8	9.5
1233	24.3	8.3
950	17.8	6.1

```
Predictor        Coef       StDev          T        P
Constant        667.8       132.3       5.05    0.000
Income         11.430       4.677       2.44    0.023
Age            16.819       8.000       2.10    0.047

S = 111.6       R-Sq = 32.5%     R-Sq(adj) = 26.4%

Analysis of Variance

Source            DF          SS          MS          F        P
Regression         2      132071       66035       5.30    0.013
Residual Error    22      274025       12456
Total             24      406096

Source       DF       Seq SS
Income        1        77008
Age           1        55063
```

We see that the F test in the analysis of variance table indicates that this model is to be preferred to the model involving just an intercept term. A plot of residuals against the fitted values indicates some possible problems with the model formulation, however (Figure 5.11). It seems that for both small and large fitted values residuals tend to be negative, whereas away from the extreme fitted values the residuals tend to be positive. This plot does suggest that the mean structure of the model is not correctly specified: in particular, it may be that we need to include some additional terms in our linear model such as the square of age or square of income. We will look further at plots which can be used for assessing the need for nonlinear terms in the predictors in later lectures.

Example: risk assessment data

As a further example of the use of the raw residuals for detecting violations of model assumptions we consider the risk assessment data discussed in previous lectures. We fit the full model with mean risk assessment as the response, and all seven of the accounting determined measures of risk as predictors. A plot of the raw residuals against the fitted values is shown in Figure 5.12 In this example, we can see that the variability of the residuals seems to decrease for the most extreme fitted values, and so the constancy of variance assumption is doubtful here.

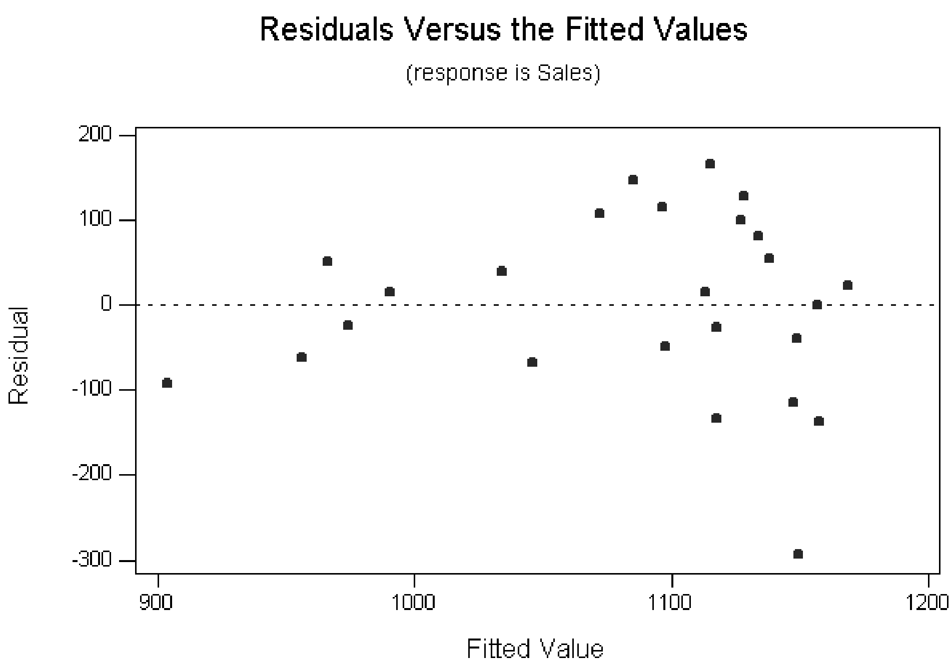

Figure 5.11: Scatter plot of residuals versus fitted values for fast food restaurants data

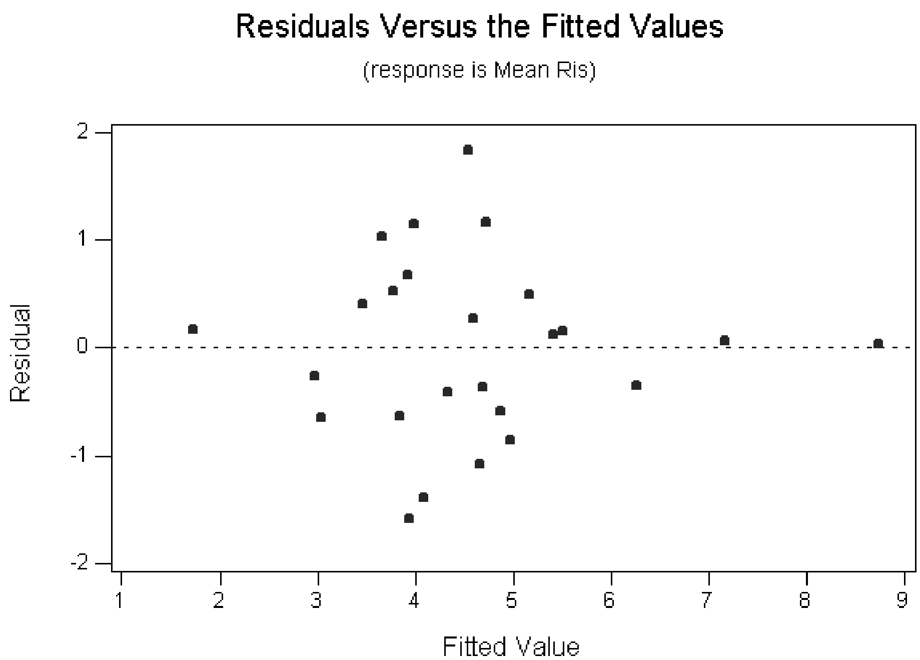

Figure 5.12: Scatter plot of residuals versus fitted values for risk assessment data

5.2 Use of residuals for detecting outlying points

There are many uses for residuals in diagnosing the fit of a model. We have discussed using the raw residuals for detecting if the mean structure of a model is not correctly specified, and for investigating whether or not the variance of the errors is constant. Another use for residuals is for detecting outlying points (single points which do not fit the overall pattern of the data). We will see in this section that the raw residuals are not necessarily the most useful quantities for detecting outlying points, or for many other purposes.

One problem with interpreting plots of the raw residuals such as the plot in the previous example is that the variance of the residuals is not constant, even if the errors ε_i do have constant variance. From (35) and since y has covariance matrix $\sigma^2 I$, and since $I - H$ is symmetric, we can write down the covariance matrix of the residuals as

$$\text{Var}(e) = \sigma^2 (I-H)^2.$$

But $I - H$ is idempotent (that is, $(I-H)^2 = I - H$) since

$$\begin{aligned}(I-H)^2 &= (I - X(X^TX)^{-1}X^T)^2 \\ &= I - X(X^TX)^{-1}X^T - X(X^TX)^{-1}X^T + X(X^TX)^{-1}X^TX(X^TX)^{-1}X^T \\ &= I - X(X^TX)^{-1}X^T \\ &= I - H\end{aligned}$$

Hence we have that

$$\text{Var}(e) = \sigma^2 (I-H). \tag{36}$$

We can learn a number of things from the above equation. Firstly, unlike the errors ε_i, $i = 1, ..., n$, the residuals e_i are not necessarily uncorrelated. In fact, for $i \neq j$,

$$Cov(e_i, e_j) = -\sigma^2 h_{ij}.$$

This expression is not necessarily zero, although in general the elements of the hat matrix will tend to zero as the sample size increases. Secondly, we have illustrated that the residuals e_i tend to be smaller in absolute value than the errors ε_i, even when the model assumptions hold: recall that the leverages h_{ii} (the diagonal elements of the hat matrix H) are all positive quantities, so that the variances

$$\text{Var}(e_i) = \sigma^2 (1 - h_{ii}),$$

are smaller than $\text{Var}(\varepsilon_i) = \sigma^2$. Furthermore, the potentially high influence points with large leverages h_{ii} are the ones corresponding to residuals with a small variance. In effect, an observation with a high leverage will tend to pull the fitted line towards itself, ensuring that the residual at that point will be small.

Example: capital asset pricing model

We illustrate some of the problems which can occur with plots of ordinary residuals by considering the capital asset pricing model data discussed in previous lectures. Here we had 21 measurements of a stock price (the response) as well as corresponding measurements of two predictors, the TSE 300 index and Price/Earnings ratio. For the purposes of this example I have deleted two of the observations and I will consider a simple linear regression model of Stock price on the TSE300 index. Figure 5.13 shows a scatter plot of stock price against the TSE300 index for my modified data set. For this modified data set, linearity of

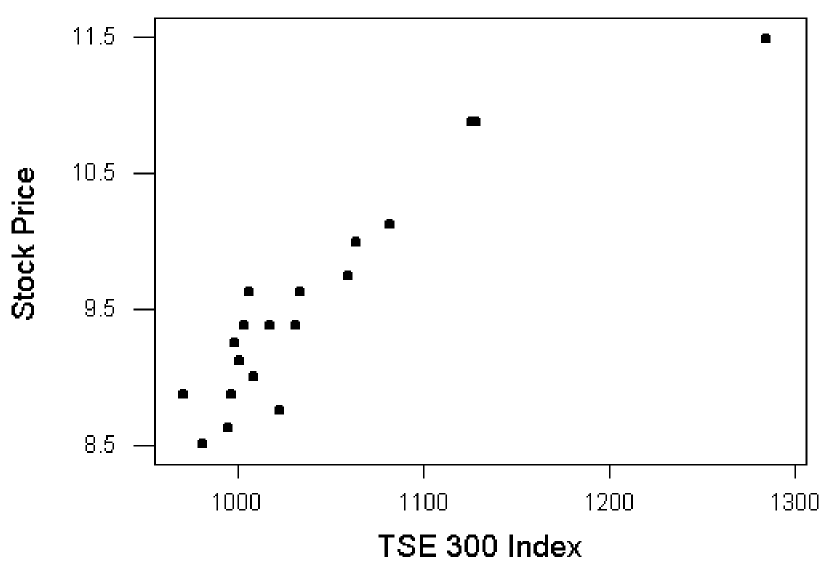

Figure 5.13: Scatter plot of stock price against TSE300 index for modified capital asset pricing model data.

the expected response in the predictor seems reasonable, apart from the extreme discrepant point. Of course, in a simple linear regression model we can display the data very easily with a scatter plot, and the residual plot is perhaps not very informative. But in multiple linear regression diagnostic residual plots may reveal patterns in the data that are not obvious. However, looking at the residual plot after fitting a simple linear regression model in this example is instructive for showing what can happen when there is a single outlier (Figure 5.14). We see

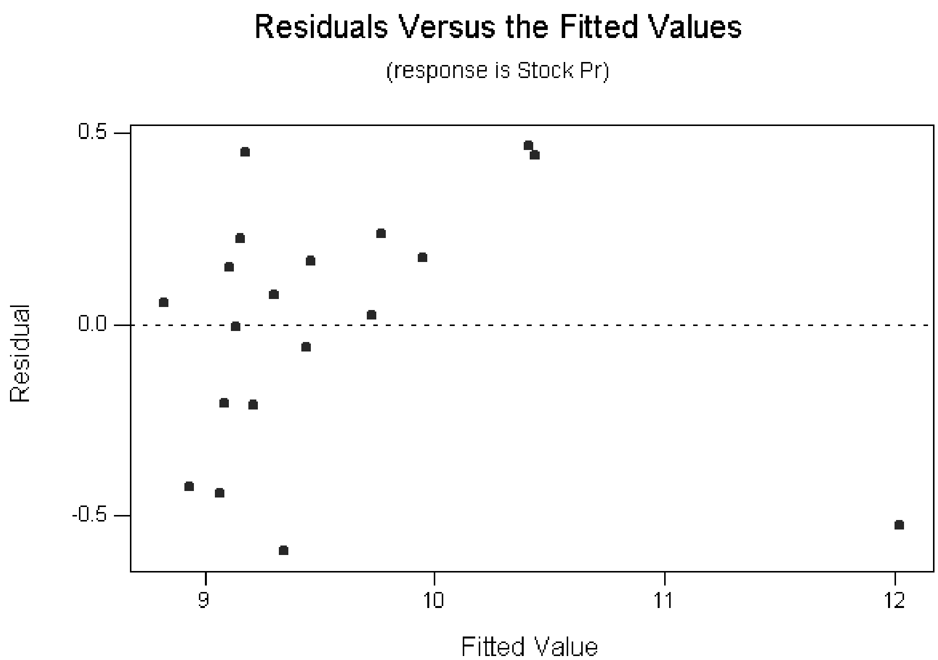

Figure 5.14: Residual plot for modified capital asset pricing model data.

that the residual corresponding to the discrepant point is no larger than many of the other residuals. This observation is a high influence point, and it has pulled the fitted line towards itself so that a plot of the raw residuals does not reflect the inconsistency of this observation with the rest of the data. The residual at this point has a smaller variance than the other residuals, making it difficult to see from the residual plot that there is an outlier.

The fact that the variance of the raw residuals is not constant can make it tricky to interpret plots of the raw residuals, as we have just seen. This has motivated the introduction of standardized and studentized residuals. The ith standardized residual is simply the ith raw residual e_i divided by its standard deviation

$$\frac{e_i}{\sigma\sqrt{1-h_{ii}}}.$$

Since this depends on σ (which is in general unknown) we define the *studentized residuals* (or more precisely the internally studentized residuals) in which σ is replaced by $\widehat{\sigma}$:

$$r_i \quad = \quad \frac{e_i}{\widehat{\sigma}\sqrt{1-h_{ii}}}.$$

For detection of outlying points you may be thinking that we could use the PRESS residuals which were defined when we looked at model selection. Recall that to define the PRESS residuals $e_{i,-i}$ we looked at the difference between the ith response and a prediction of that response obtained by fitting a model to the data with the ith response excluded. We gave a computational formula for $e_{i,-i}$, namely

$$e_{i,-i} = \frac{e_i}{1-h_{ii}}.$$

We might think that if we could standardize the PRESS residuals to have constant variance then these standardized PRESS residuals would be most appropriate for detecting a single outlying observation which does not follow the pattern of the rest of the data.

To standardize the PRESS residuals we first have to find their variances. We have

$$\begin{aligned}\mathrm{Var}(e_{i,-i}) &= \frac{\mathrm{Var}(e_i)}{(1-h_{ii})^2} \\ &= \frac{\sigma^2(1-h_{ii})}{(1-h_{ii})^2} \\ &= \frac{\sigma^2}{1-h_{ii}}.\end{aligned}$$

Hence the standardized PRESS residual is

$$\frac{\frac{e_i}{1-h_{ii}}}{\frac{\sigma}{\sqrt{1-h_{ii}}}} = \frac{e_i}{\sigma\sqrt{1-h_{ii}}}$$

which is simply r_i, the ordinary standardized residual! Hence it does seem appropriate to use the studentized residuals to look for single outlying points in the data.

Example: capital asset pricing model

As an illustration of the use of studentized residuals we return to our example of the capital asset pricing model in which we deleted two of the original observations and fitted a simple linear regression model with stock price as the response and TSE300 index as the predictor. Figure 5.14 shows the ordinary residuals plotted against the fitted values. The outlier here does not have the largest raw residual in absolute value. A plot of the studentized residuals for this data set is shown in Figure 5.15 In the plot of studentized residuals, the outlying point is more effectively highlighted. The outlier corresponds to a high leverage point (a leverage of approximately 0.65) so that the variance of the raw residual at this

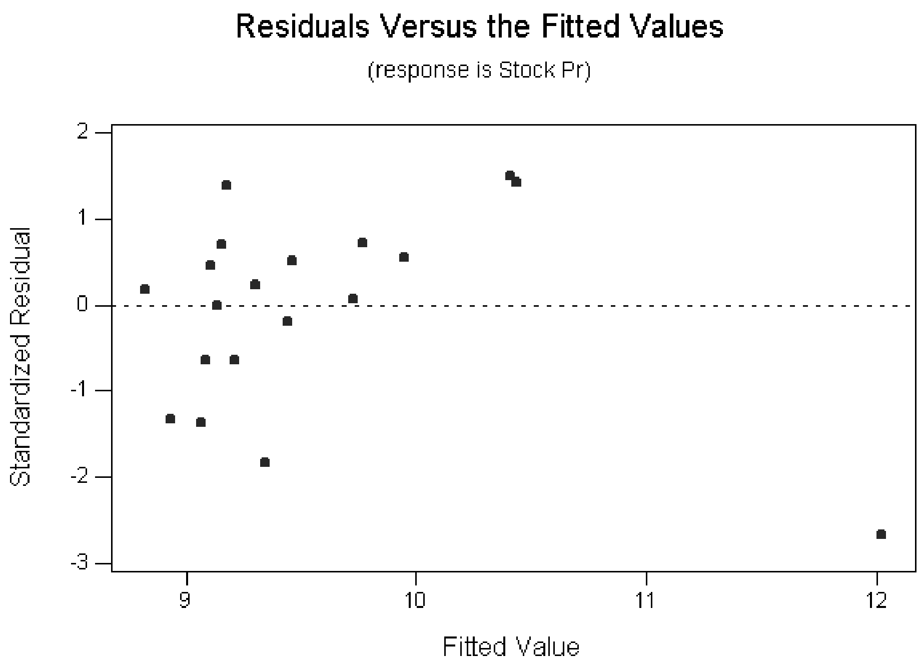

Figure 5.15: Plot of studentized residuals versus fitted values for modified capital asset pricing model data.

point differs substantially from the error variance σ^2.

Example: risk assessment data

For the risk assessment data, we previously plotted the raw residuals versus fitted values (Figure 5.12). For the same fitted model considered before, we can also consider plotting the studentized residuals against the fitted values. The basic pattern observed (Figure 5.16) is the same as before, but notice that there seem to be some differences in the relative magnitudes of the standardized residuals, with the ones corresponding to extreme fitted values being relatively larger compared to the others than before.

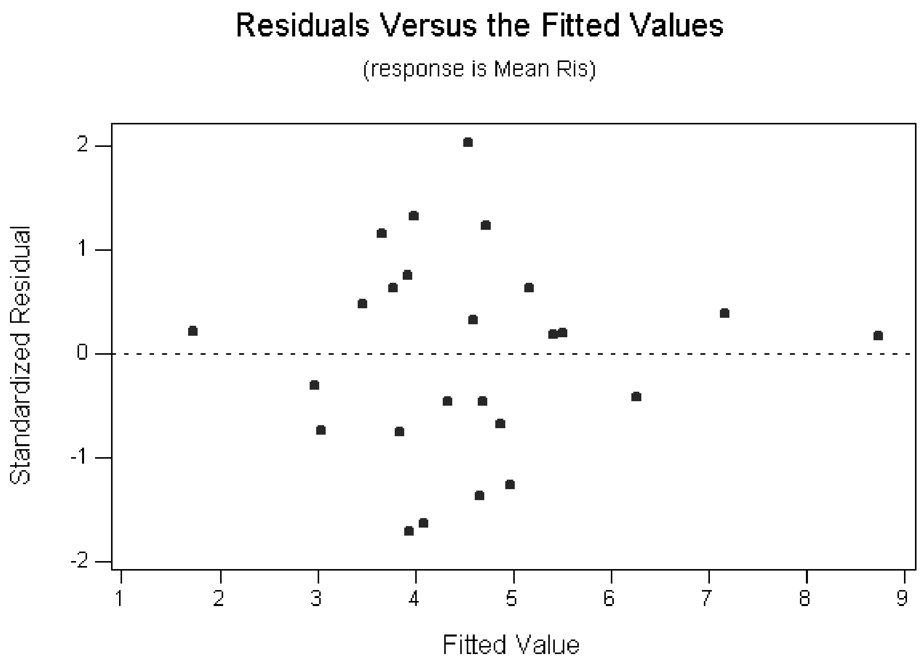

Figure 5.16: Plot of the studentized residuals against fitted values for risk assessment data.

Unfortunately, plots of residuals don't always help us to unambiguously diagnose a problem with the model formulation when it exists. The following example is given in Sanford Weisberg, "Applied Linear Regression (Second Edition)," Wiley, New York, 1985, pp. 131–132.

Example: technological innovation in jet fighters

Stanley and Miller, in a 1987 RAND corporation technical report, have attempted to build a descriptive model of the role of various design and performance factors in modelling technological innovation in jet fighter aircraft. Using data on American jet fighters built since 1940, they use the date of the first flight as a stand-in for a measure of technology; presumably, the level of technology is increasing with time. In some of their work, they considered the following variables:

$$\begin{aligned} FFD &= \text{first flight date, in months after January 1940} \\ SPR &= \text{specific power, proportional to power per unit weight} \\ RGF &= \text{flight range factor} \\ PLF &= \text{payload as a fraction of gross weight of aircraft} \\ SLF &= \text{sustained load factor} \\ CAR &= \text{1 if aircraft can land on a carrier; 0 otherwise.} \end{aligned}$$

Data on 22 planes flown between 1940 and 1979 were considered (we do not reproduce the full data set here). If we fit a model with FFD as response and the remaining variables as predictors, and if we plot the studentized residuals against fitted values, we obtain the graph in Figure 5.17. We see here that the plot of

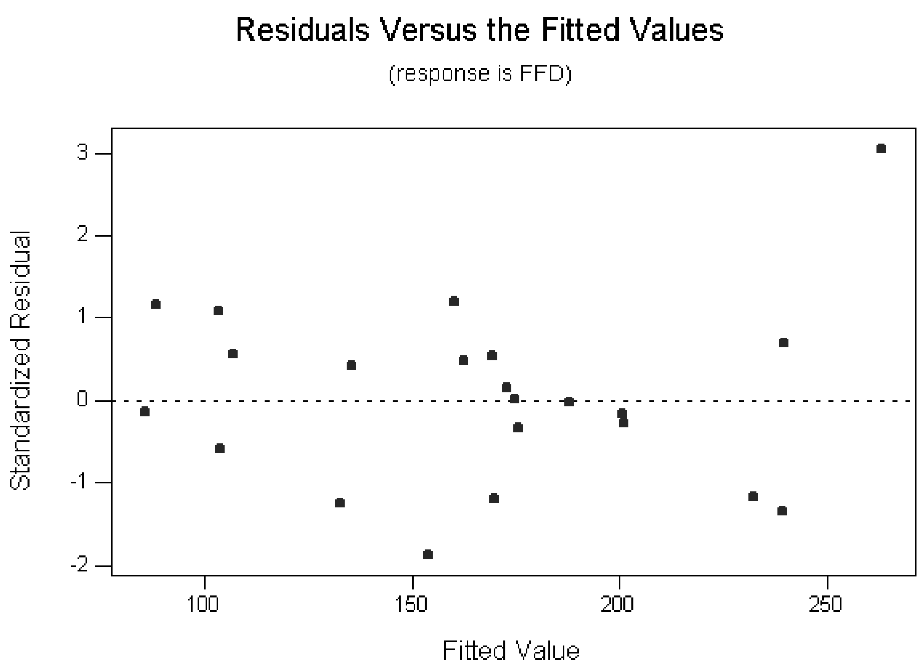

Figure 5.17: Plot of studentized residuals versus fitted values for jet fighter data.

studentized residuals is somewhat ambiguous. We might interpret the residual plot as indicating an outlying point on the extreme right, or as indicating an incorrectly specified mean structure, or perhaps as indicating that the constancy of variance assumption is violated. The plot of residuals against fitted values is not as helpful as we might hope.

We have pointed out that studentized residuals are useful for detection of single outlying observations. However, we have not yet formally discussed the question of how large the studentized residual needs to be to cause concern. Obviously the standardized residual has variance one, but when we estimate the unknown σ^2 in construction of the studentized residual there is some extra dispersion due to estimation of this parameter and we might expect that a distribution more heavy tailed than the normal such as a t distribution might be appropriate for assessing whether an observation is more extreme than would be expected under the model assumptions. We now discuss a statistic that can be used to formally test whether or not an observation seems to conform to the model. The test statistic used is the following:

$$t_i = \frac{e_i}{\widehat{\sigma}_{-i}\sqrt{1-h_{ii}}}$$

where $\widehat{\sigma}_{-i}$ is the estimate of the error variance obtained by fitting the model to all the data except the ith observation. Note that this statistic is exactly the same as the studentized residual r_i except that we use $\widehat{\sigma}_{-i}$ to estimate the error variance rather than $\widehat{\sigma}$. The quantities t_i are usually called the *externally studentized residuals* (externally studentized since we have not used the ith observation in estimation of σ^2.) Use of the quantities t_i for outlier detection does make good intuitive sense: if the ith observation really is an outlying observation, there is the danger that $\widehat{\sigma}^2$ overestimates σ^2 due to this extreme point and the residual r_i will be "deflated" because of this.

It can be shown that if the assumptions of the linear model hold, then t_i has a t-distribution with $n-p-1$ degees of freedom. We may test the hypothesis that the ith observation is an outlier by expanding the original model in one of two different ways. In the mean shift outlier model, we assume that the ith error ε_i does not have mean zero, but has mean Δ_i say. Alternatively, we may assume that ε_i has zero mean, but that the constancy of error variance assumption breaks down at the ith point, so that $\mathrm{Var}(\varepsilon_i) = \sigma^2 + \sigma_i^2$ say. In this second scenario an unusual observation occurs because of a large random fluctuation at the ith point consistent with the larger error variance. We can test the hypothesis

$$H_0 : \Delta_i = 0$$

in the mean shift outlier model against the alternative

$$H_1 : \Delta_i \neq 0$$

or test

$$H_0 : \sigma_i^2 = 0$$

against the alternative

$$H_1 : \sigma_i^2 \neq 0$$

in the model where constancy of error variance is violated using t_i as a test statistic with null distribution the t distribution with $n-p-1$ degrees of freedom. We will say more about testing for outliers with the externally studentized residuals, and about associated problems of multiple testing, in the next lecture.

5.3 Testing for outliers

At the end of the last lecture we mentioned that we could use the externally studentized residual t_i as a test statistic in a formal test of whether the ith observation conforms to the assumed model. In this subsection we develop this idea further.

We consider two kinds of departures from the assumed linear model. In the mean shift outlier model, we assume

$$\mathbb{E}(\varepsilon_i) = \Delta_i$$

where $\Delta_i \neq 0$. That is, in the mean shift outlier model it is assumed that the assumption of a zero mean for the ith error breaks down, possibly leading to an outlier for the ith observation.

The second kind of departure we consider for the linear model is that

$$\mathrm{Var}(\varepsilon_i) = \sigma^2 + \sigma_i^2$$

where $\sigma_i^2 > 0$. In this scenario, an outlier can occur for the ith observation due to a large random fluctuation in line with $\sigma_i^2 > 0$.

We can test either

$$H_0 : \Delta_i = 0$$

against the alternative

$$H_1 : \Delta_i \neq 0$$

in the mean shift outlier model or test

$$H_0 : \sigma_i^2 = 0$$

versus the alternative

$$H_1 : \sigma_i^2 > 0$$

in the model with inhomogeneous variance using

$$t_i = \frac{e_i}{\widehat{\sigma}_{-i}\sqrt{1-h_{ii}}}$$

as the test statistic. For both tests, under the null hypothesis t_i is a realization of a t random variable with $n-p-1$ degrees of freedom in the linear model with normal errors, so that if $T \sim t_{n-p-1}$ we compute the p-value for the tests above as

$$p = \mathbb{P}(|T| \geqslant |t_i|).$$

Example: Capital asset pricing model

Consider the modified capital asset pricing model data which we examined in the previous lecture. The data consisted of a response variable stock price and two predictor variables price/earnings ratio and TSE300 index. For illustrating some properties of raw and studentized residuals we deleted two observations from the data set. Figure 5.13 shows the scatter plot of stock price versus TSE300 index for the modified data set. Since the observation corresponding to the predictor value on the extreme right is so far away from the other predictor values we may have a prior interest in testing whether this observation follows the pattern of the rest of the data. We can apply the testing procedure above to this observation. The externally studentized residual for this point is approximately -3.37 here. To compute the p-value for testing zero Δ_i in the mean shift outlier model versus the alternative $\Delta_i \neq 0$, or for testing $\sigma_i^2 = 0$ versus the alternative $\sigma_i^2 > 0$ in the model with a possibly inhomogeneous variance, we let T be a t random variable with $n-p-1 = 19-2-1 = 16$ degrees of freedom and calculate

$$\mathbb{P}(|T| \geqslant 3.37)$$

which is approximately 0.0038. So we would reject the null hypothesis for either test at the 5 percent level, and we would conclude that this observation may need to be investigated further.

When we have a prior reason for believing that one of the observations may be unusual then the test we have described above is appropriate. However, if we were to conduct this test for all the observations in our data set, then we have a problem of multiple testing or simultaneous inference: although it is rare for an externally studentized residual to be as large as the upper 2.5 percentage point of a t_{n-p-1} distribution for a single observation, it is very likely that in a large data set at least some of the externally studentized residuals are this large even if all observations conform to the model. So we need to make an adjustment to the significance level of our outlier test when we apply it to all the observations in the data set. We can apply the idea of Bonferroni adjustment discussed in earlier lectures in this situation.

In particular, if we apply our outlier test for each observation in the data set, then we adjust the significance level of our test from α to α/n.

Example: capital asset pricing model data

Let us reconsider for the moment the modified capital asset pricing model data we discussed in the previous example. If we did not have prior knowledge that the last observation was unusual, then we might apply our outlier test to all nineteen observations in this data set. Since we were testing before at the 5 percent level, Bonferroni adjustment suggests we should use a significance level of 0.05/19 which is approximately 0.0026. Comparing this with the p-value we obtained previously of approximately 0.0038, we see that the observation would not be considered an outlier in this analysis. We should remember, however, that the Bonferroni adjustment is a conservative one, and we should not in any case treat formal tests too seriously in the context of model criticism: the purpose of the test is simply to bring to our attention individual observations that may require further investigation.

Example: jet fighter data set

Recall the data set on technological innovation in jet fighters discussed on page 156 of your notes. We showed a plot of the internally studentized residuals when we fitted a model with FFD as the response and the remaining five variables as predictors in this data set. A plot of the externally studentized residuals is shown in Figure 5.18. The value of the externally studentized residual on the extreme right here is approximately 4.63. If we compare with the plot of the internally studentized residuals given in your last set of lecture notes we see that this is an example where it may be desirable to use an external estimate of the error variance in the standardization: the internally studentized residual here is approximately 3.07, which is quite different to the value 4.63 when the possibly outlying point is excluded in compuation of the estimated standard deviation. There are 22 observations in this data set, so if we apply Bonferroni adjustment with an initial 5 percent significance level for testing for outliers, the Bonferroni significance level is 0.05/22 or approximately 0.0023. To compute the p-value for the outlier test, we let T be a t distributed random variable with $n-p-1 = 22-6-1 = 15$ degrees of freedom, and calculate

$$\mathbb{P}(|T| \geqslant 4.63)$$

which is approximately 0.0004. So we reject the null hypothesis, and this observation with an externally studentized residual of 4.63 might be considered to be an unusual one.

As we mentioned when discussing the capital asset pricing model data, the formal tests of hypotheses we have considered in this section are not to be taken too seriously in the context of model criticism. The purpose of the techniques we

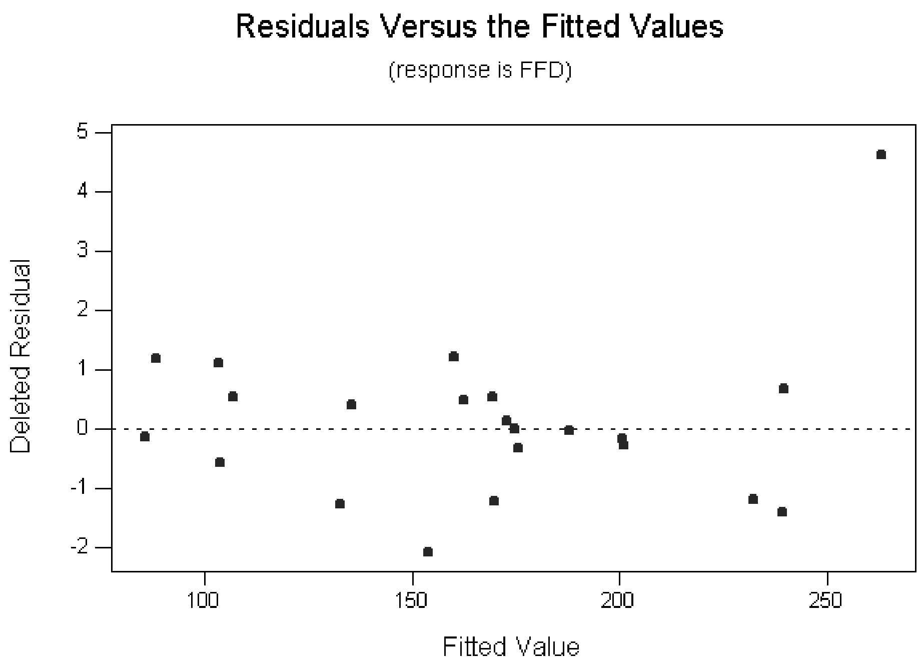

Figure 5.18: Plot of externally studentized residuals versus fitted values for jet fighter data set.

have discussed for criticism of the model is to highlight unusual observations that should be investigated further. We should ask ourselves why an outlier has occurred: was it measurement or recording error, or is it something which requires us to change the model? What we do with the outlier will usually depend on why we think the outlier occurred. Sometimes the right thing to do is to exclude the outlier from the analysis. In other situations the outlier may be the most interesting and informative observation in the data.

5.4 Use of indicator variables

In the previous subsection we described a way of testing a single observation thought to be suspect from prior knowledge, and a way of testing all of the observations to highlight possibly unusual cases. In this subsection we discuss a way of dealing with groups of observations which are thought from prior knowledge to be suspect.

Suppose that we can partition the observations y into a set of r observations which from prior knowledge may not conform to the model, and $n-r$ remaining observations. Without loss of generality we can reorder the observations, so we let $(y_1, ..., y_r)$ be the possible outliers, and we write $(y_{r+1}, ..., y_n)$ for the remaining observations. We will extend the mean shift outlier model to develop various tests for whether some or all of the first r observations might be considered unusual. If there is a common cause why the observations $y_1, ..., y_r$ might be considered outliers, then it might be reasonable to assume that $\mathbb{E}(\varepsilon_1) = ... = \mathbb{E}(\varepsilon_r) = \Delta$. That is, it might be assumed that the errors corresponding to the possibly unusual observations have a common mean which is not necessarily zero. We can develop a test of

$$H_0 : \Delta = 0$$

against the alternative

$$H_1 : \Delta \neq 0$$

as follows. Construct a new predictor variable which is 1 for the observations $y_1, ..., y_r$ and 0 for the remaining observations. If we refit our linear model with this new predictor added to the existing predictors, then the estimate of the coefficient corresponding to this new predictor is an estimate of Δ: furthermore, the partial t-test for this coefficient provides a test for $\Delta = 0$ in the context of this model. It can be shown that when $r = 1$ looking at this partial t-test is exactly the same as performing the test based on the externally studentized residual discussed in the previous subsection.

An alternative situation may occur in which the observations $y_1, ..., y_r$ might be thought to be suspcious, but there is no common cause why they are suspect. In this case, we can follow a similar approach to the above to examine these possible outliers: however, in this case, we can define r new predictors, with the first predictor being one for the first observation and zero otherwise, the second predictor being one for the second observation and zero otherwise, and so on. Again we can examine estimates of coefficients for these predictors and examine partial t-tests to highlight possibly unusual observations.

Example: capital asset pricing model data

We consider again the capital asset pricing model data. For the purposes of this example, consider the full data set and a simple linear regression of stock price on TSE300 index. The scatter plot of stock price against TSE300 index is shown in Figure 5.19. Suppose that the three observations corresponding to

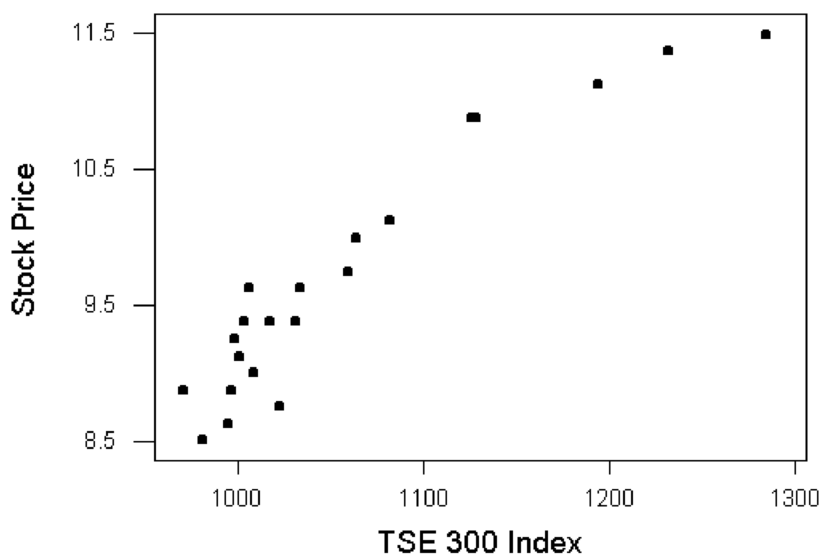

Figure 5.19: Scatter plot of stock price versus TSE300 index.

the predictors on the extreme right were thought to be suspicious on the basis of prior knowledge: the data were collected over time, and some event occurred which caused a shift in the stock price for the last three observations collected. We can set up an indicator variable to model the possible effect. The indicator variable is zero for the first eighteen observations, and is one for the last three.

```
The regression equation is
Stock Price = - 3.85 + 0.0129 TSE 300 Index - 0.795 Dummy
```

```
Predictor         Coef        StDev           T        P
Constant        -3.847        1.483       -2.59    0.018
TSE 300       0.012915     0.001439        8.97    0.000
Dummy          -0.7947       0.3487       -2.28    0.035

S = 0.2875      R-Sq = 91.4%     R-Sq(adj) = 90.5%

Analysis of Variance

Source            DF          SS          MS         F        P
Regression         2     15.8294      7.9147     95.77    0.000
Residual Error    18      1.4876      0.0826
Total             20     17.3170

Source       DF      Seq SS
TSE 300       1     15.4001
Dummy         1      0.4293
```

We see from the above output that the partial t-test for the coefficient of our dummy variable is significant at the 5 percent level: there does seem to be reason to further examine the last three observations and to possibly exclude them from our analysis.

5.5 Partial regression plots

In the simple linear regression model, a plot of the response against the predictor is a good way of summarizing the data. In the case of multiple linear regression, however, plots of the response against each of the predictors may not be a helpful way of summarizing the contribution of a predictor to the model. In particular, in multiple linear regression the existence of relationships between predictors can complicate interpretation of the role of a predictor. We also saw this problem when we talked about partial t tests and multicollinearity.

For a multiple linear regression model *partial regression plots* (sometimes called added variable plots) provide one useful way of displaying the relationship between a given predictor and the response after adjusting for the relationship between the predictor and the other covariates in the model. In our usual notation we write y for the response and $x_1, ..., x_k$ for a collection of k predictors. Suppose we are interested in the role that x_j will play in a multiple linear regression model with y as the response and $x_1, ..., x_k$ as covariates. To construct a partial regression plot, we first fit a multiple linear regression model of y on the predictors $x_1, ..., x_{j-1}, x_{j+1}, ..., x_k$ (that is, we consider a model involving all predictors except x_j). Write $e_{y,-x_j}$ for the vector of residuals from this fitted model. We then fit a multiple linear regression model with x_j as the response and with $x_1, .., x_{j-1}, x_{j+1}, ..., x_k$ as predictors (that is, we regress x_j on the remaining predictors). Write $e_{x_j,-x_j}$ for the residuals from this second fitted model. We can think of the vector of residuals $e_{x_j,-x_j}$ as the independent information in the predictor x_j, after removing linear dependence on the other covariates. In the partial regression plot we simply plot $e_{y,-x_j}$ against $e_{x_j,-x_j}$. This plot helps us to visualize the relationship between y and x_j after we have removed the linear dependence of both y and x_j on the remaining predictors.

It can be shown that the least squares slope of the partial regression plot is actually b_j, the least squares estimator of β_j in the multiple linear regression model involving all the predictors (see, for instance, p. 233 of Myers, "Classical and Modern Regression with Applications (Second Edition)", Duxbury, Belmont, California, 1990). The partial regression plot gives an effective visual representation of the contribution of a predictor in a multiple linear regression model, and it can be helpful for suggesting the need for an additional nonlinear term in the model (we might decide to add x_j^2 as a predictor for example). Partial regression plots may also be helfpul for identifying influential observations: the partial regression plots in which an unusual observation shows up most clearly correspond to predictors whose coefficients will be most affected by the unusual point.

Example: capital asset pricing model

We return to the example of the capital asset pricing model data. Recall that in this data set the response was stock price, and there were two predictors, price/earnings ratio and TSE300 index. We wish to consider the role of each predictor in a multiple linear regression model involving both predictors. The partial regression plots for price/earnings ratio and TSE300 index are shown below. The partial regression plot for TSE300 index may suggest the need to include an additional predictor such as the square of TSE300 index in the model.

If we do include the square of TSE300 index as an additional predictor then the partial t test for the coefficient of this predictor leads to a p-value of 0.004, so that this additional term does seem to contribute to the model.

Example: fast food restaurants

As an additional example we return to the data on fast food restaurants which was discussed in the last lecture. Recall that in this data set the response was annual sales and the predictors were a measure of income and the mean age of children. Below are the partial regression plots for the predictors income and age for this data set. The plots here suggest that the model might benefit from including a nonlinear term in income, or age, or possibly both.

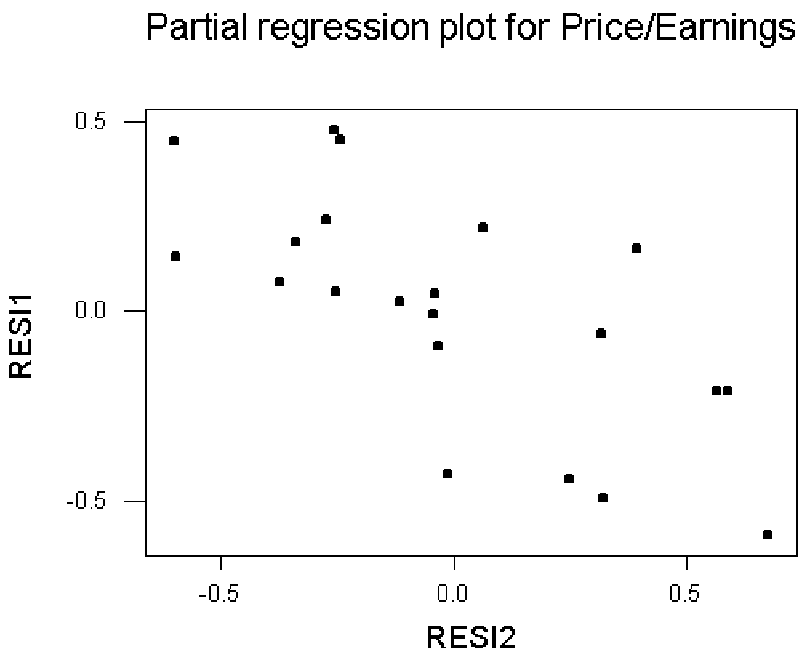

Figure 5.20: Partial regression plot for price/earnings ratio.

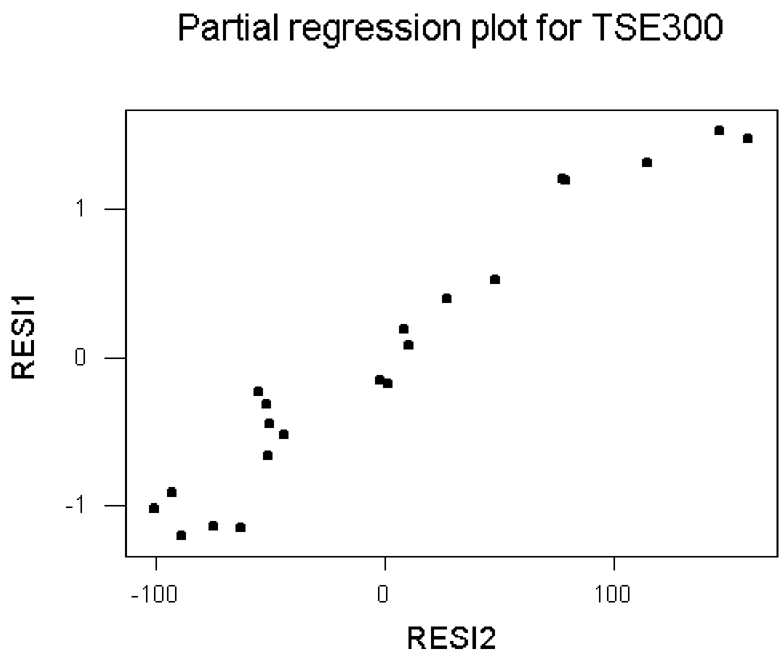

Figure 5.21: Partial regression plot for TSE300 index.

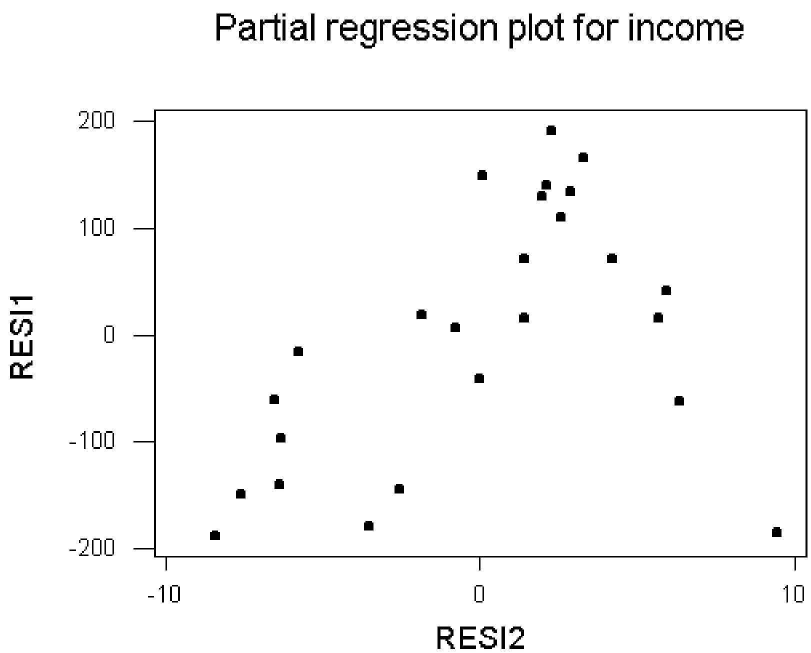

Figure 5.22: Partial regression plot for income.

Figure 5.23: Partial regression plot for age.

5.6 Partial residual plots

Another plot which is useful for describing the role of a predictor in a multiple linear regression model is the *partial residual plot*. Suppose as in the previous subsection that we have a response y and predictors $x_1, ..., x_k$. Write simply e for the vector of residuals which results from fitting a multiple linear regression model of y on all the predictors. Also, if as usual we write β for the vector of unknown parameters in the multiple linear regression model, write b_j for the least squares estimate of β_j. The partial residual plot is simply a plot of $e + b_j x_j$ against x_j. Again this plot shows the role of x_j in the mutliple linear regression involving all the predictors. The partial residual plot is probably more useful than the partial regression plot for indicating the need to include additional nonlinear terms in the predictors.

Example: capital asset pricing model.

Figures 5.24 and 5.25 show the partial residual plots for the predictors price/earnings ratio and TSE300 index for the capital asset pricing model data set. Again the

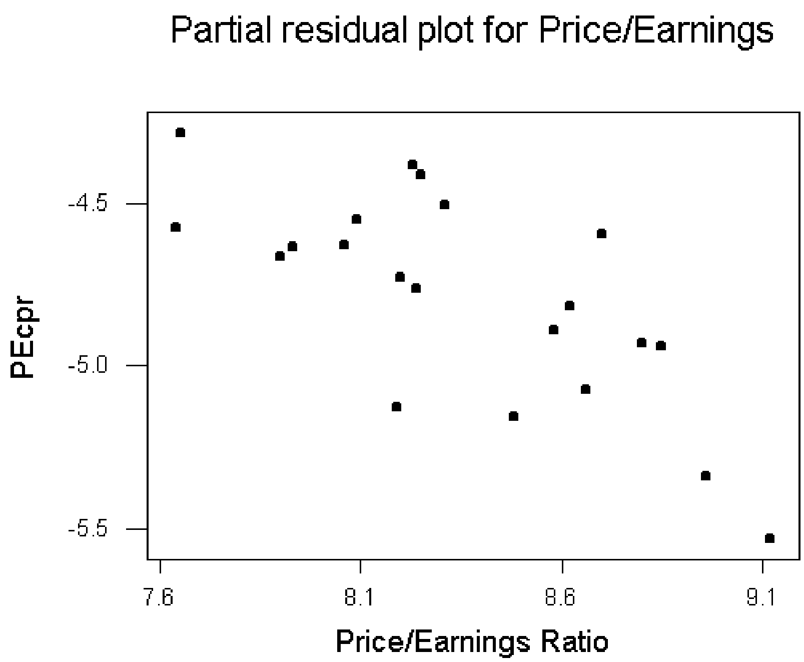

Figure 5.24: Partial residual plot for price/earnings ratio.

need for a nonlinear term in TSE300 index may be indicated.

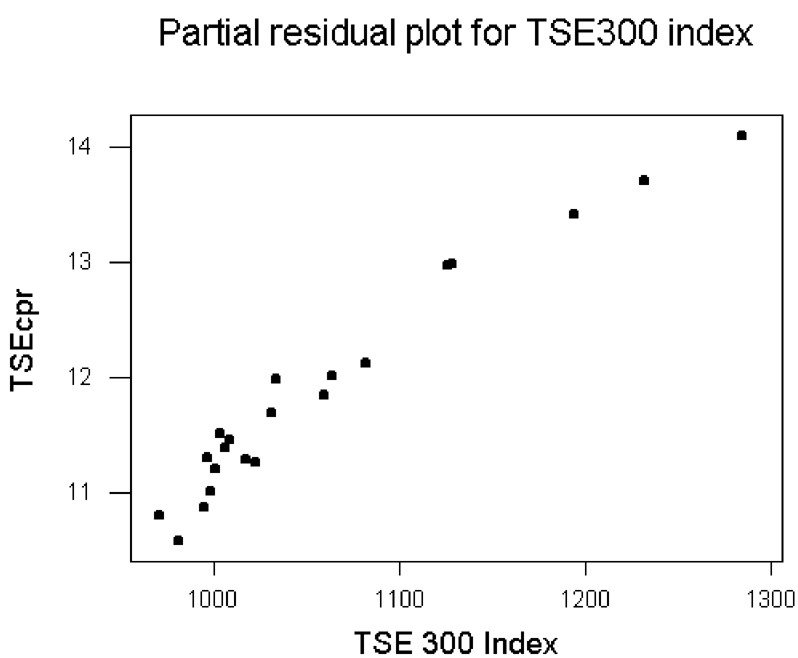

Figure 5.25: Partial residual plot for TSE300 index.

Example: fast food restaurants.

Figures 5.26 and 5.27 show the partial residual plots for the predictors income and age for the fast food restaurants data set. Again the possible need for

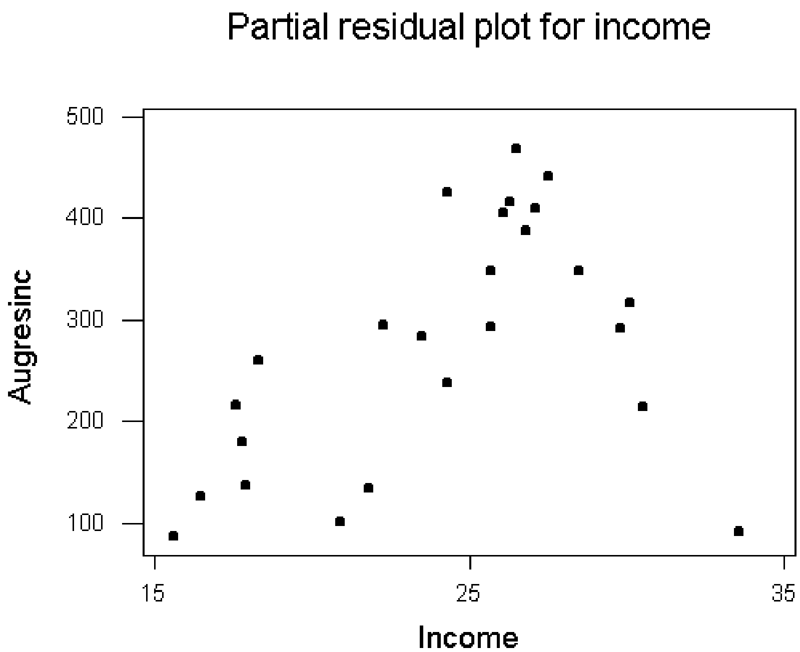

Figure 5.26: Partial residual plot for income.

nonlinear transformations of both predictors is indicated.

A further alternative to the partial regression and partial residual plots for detecting the need for nonlinear transformations of the predictors is the use of an augmented partial residual plot. In the augmented partial residual plot for the predictor x_j, we fit a linear model involving all the predictors with the additional predictor x_j^2: writing b_j for the least squares estimator of the coefficient for x_j and $b_j^{(2)}$ for the least squares estimator of the coefficient for x_j^2, and writing e for the vector of residuals, then in the augmented partial residual plot we plot $e + b_j x_j + b_j^{(2)} x_j^2$ against x_j. The motivation for the introduction of this plot was that if the contribution of x_j is really nonlinear, then the nonlinear contribution of x_j may be masked by its relationship with the other terms, and including x_j^2 as an additional predictor can help to alleviate this problem.

Figure 5.27: Partial residual plot for age.

5.7 Testing for normality

We have discussed in this topic so far some ways of detecting failures of assumptions through the analysis of residuals. In our development of the general linear model we saw that the assumption of normal errors was essential for hypothesis testing and computation of confidence intervals. However, normality was not required for derivation of parameter estimates or for computation of standard errors of estimators or predictions.

If the purpose of a regression analysis concerns inferences which depend crucially on the assumption of normality, it may be helpful to use a residual analysis to examine the validity of the normality assumption. It is unfortunately difficult to test for normality based on the residuals in small to moderate sized samples. Although formal tests for normality exist which can be applied to residuals for a fitted linear model, we will discuss in this subsection just one graphical method for assessing normality, and for diagnosing the nature of departures from normality. In particular, we discuss *normal probability plots*.

Normal probability plots

When we construct a normal probability plot we do assume that the residuals are identically distributed (at least approximately) so that before we construct a normal probability plot we should examine residual plots in order to detect any incorrect specification of the mean, inhomogeneity of the error variance or outliers: if one of these model failures occurs, then the residuals (of whatever kind) may not be close to identically distributed, and the normal probability plot will not be meaningful.

We now develop the idea of a normal probability plot. For a random sample $U_1, ..., U_n$ we define the ith order statistic of the sample to be the ith smallest value, and we write this quantity as $U_{(i)}$. Thus $U_{(1)}$ is the minimum of the sample and $U_{(n)}$ is the maximum for instance. Now suppose $Z_1, ..., Z_n$ is a random sample from a normal distribution with mean μ and variance σ^2. Then we can write $Z_i = \mu + \sigma W_i$ where $W_1, ..., W_n$ is a random sample from a normal distribution with mean zero and variance one. Now consider the order statistics of $Z_1, ..., Z_n$ and $W_1, ..., W_n$. Clearly we have

$$Z_{(i)} = \mu + \sigma W_{(i)}$$

since the ordering of the W_i's is not changed by the transformation. Taking expectations, we have

$$\mathbb{E}(Z_{(i)}) = \mu + \sigma \mathbb{E}(W_{(i)}).$$

The idea of the normal probability plot is as follows: if $Z_1, ..., Z_n$ is a random sample from a normal distribution with mean μ and variance σ^2, then if we plot the order statistics $Z_{(i)}$ against the expected order statistics $\mathbb{E}(W_{(i)})$, then this plot should be approximately a straight line with intercept μ and slope σ. Departure from linearity in this plot indicates that the random sample $Z_1, ..., Z_n$ may not be normally distributed. It should be noted that the idea of a probability plot is quite a general one: although we only consider normal probability plots, a similar idea can be used to graphically assess goodness of fit for other distributions.

We note before going on that in some statistical packages the order statistics of the sample are plotted on the x-axis, and the values $\mathbb{E}(W_{(i)})$ are plotted on the y-axis:

Example: cheddar cheese tastings

As an example of the use of normal probability plots for assessing normality in multiple linear regression we consider again the data on cheddar cheese tastings. Recall that for these data our interest was in predicting a measure of taste for cheddar cheese samples (taste) based on predictors which were related to the concentration of various chemicals in the cheese (H2S, lactic and acetic). In this example we fit a model for taste involving the predictors H2S and acetic, and use computer to generate a normal probability plot of the internally studentized residuals. The plot of internally studentized residuals against fitted values and the normal probability plot of internally studentized residuals are shown in Figures 5.28 and 5.29. There is probably no real reason to question the assumption of normality here, although the largest residual may be an outlier.

Example: relationship between income and education and job experience

As a further example we consider the data on the relationship between income and job experience and education discussed on one of your tutorial sheets. We fit a multiple linear regression model with income as the response and education and job experience as predictors. A plot of the internally studentized residuals against fitted values is shown in Figure 5.30 and a normal probability plot of the internally studentized residuals is shown in Figure 5.31. There seems to be no reason to question normality here.

We have stated that a departure from linearity in the normal probability plot indicates possible non-normality. We also stated at the beginning of this subsection that the nature of the departure from linearity in a normal probability plot can help us diagnose the way in which residuals differ from what is expected based on normality. We can learn a great deal about the characteristics of the errors by looking at the shape of a normal probability plot. For the moment consider a normal probability plot where the ordered data are plotted on the y-axis and the expected order statistics are plotted on the x-axis.

To help assess linearity in a normal probability plot, it is common to draw a

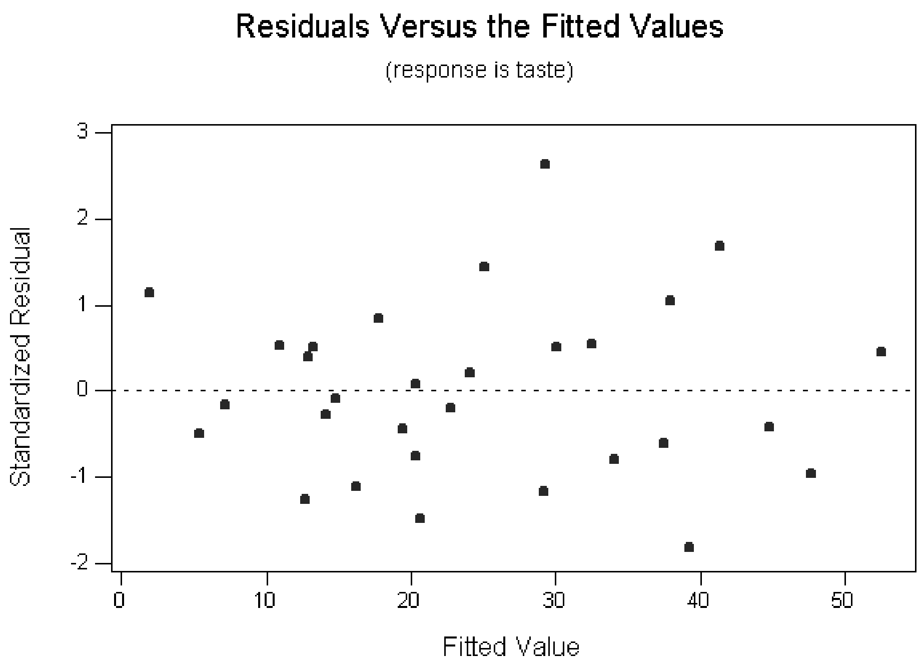

Figure 5.28: Plot of internally studentized residuals versus fitted values for cheddar cheese tastings data.

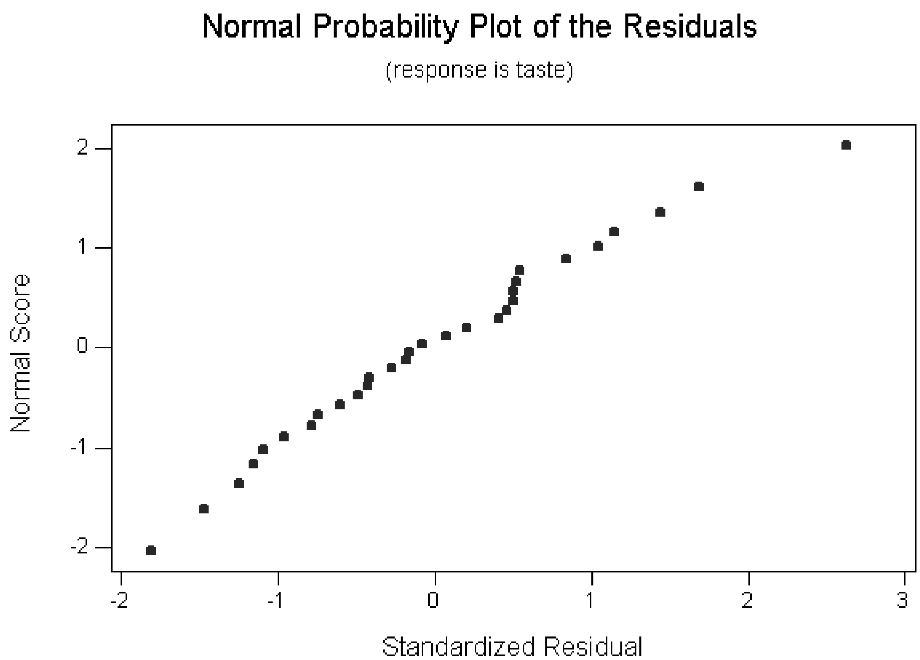

Figure 5.29: Normal probability plot of internally studentized residuals for fitted model to cheddar cheese tastings data.

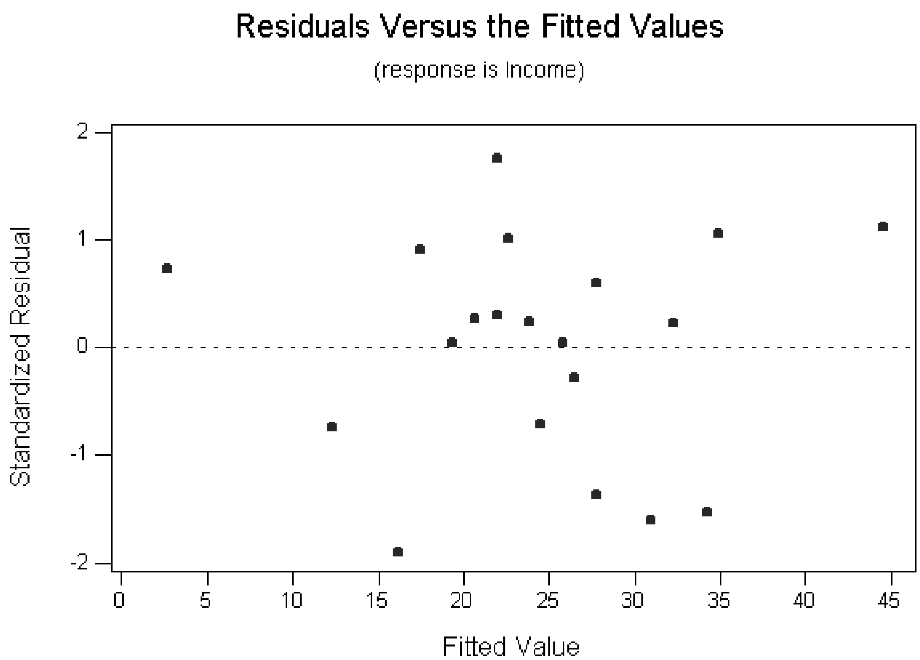

Figure 5.30: Plot of internally studentized residuals for data on income, education and job experience.

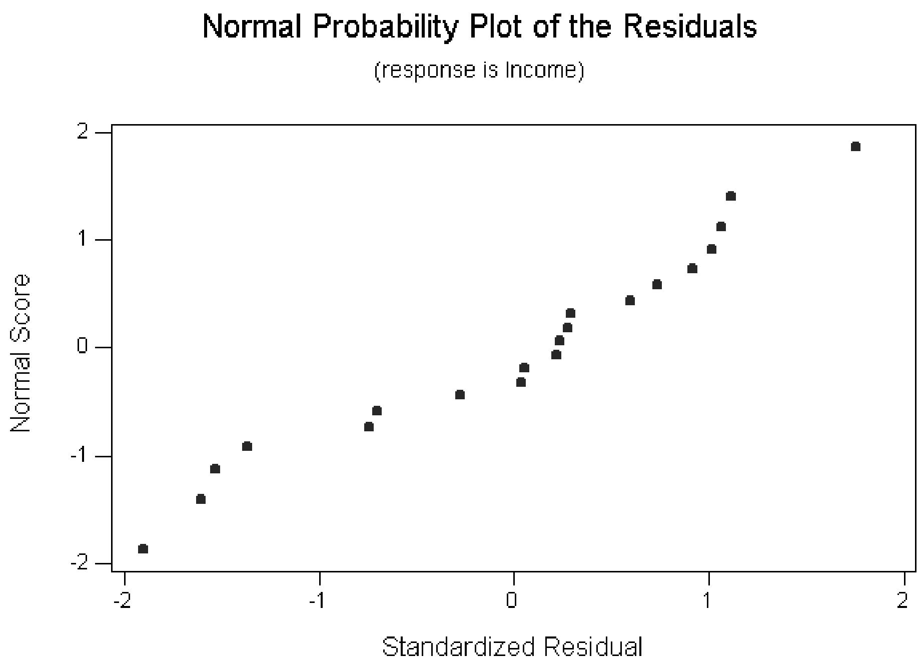

Figure 5.31: Normal probability plot for internally studentized residuals and data on income, education and job experience.

reference line on the plot. There are various ways of doing this. One method is to draw a straight line through the two points corresponding to the first and third quartiles of the sample (which are roughly the points which bound the middle half of the data). We can interpret characteristics of the residuals based on the way the probability plot deviates from the reference line. Points below the line correspond to points which are smaller than we would expect based on normality, and points above the line corresond to points which are larger than we expect.

The 'S' shape plot shown in Figure 5.32 indicates that the sample plotted has fewer extreme observations (both large and small) than we would expect under normality (and we say that we have a distribution which is light tailed with respect to the normal). The points above the line on the left indicate that the small values in the sample are larger than we would expect, whereas the values below the line on the right indicate that the large values in the sample are smaller than we would expect. If we reflect this shape through the line $y = x$,

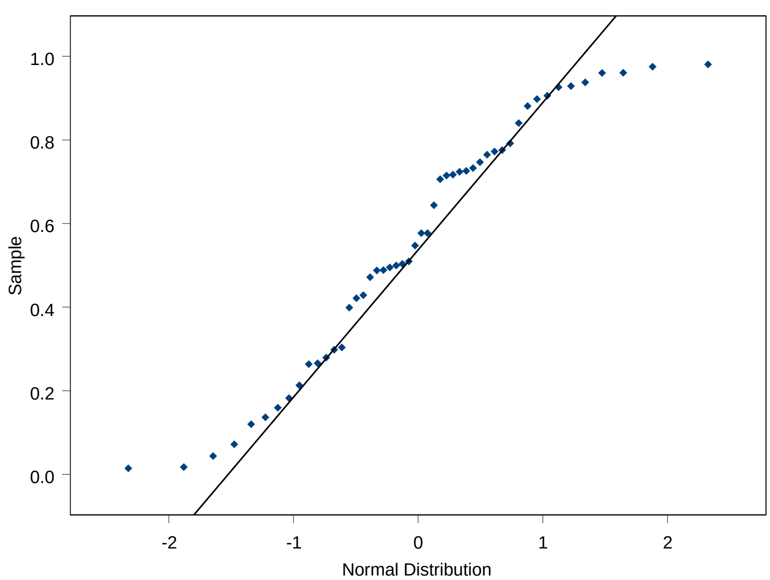

Figure 5.32: Normal probability plot for sample with fewer extreme values than we would expect based on the normal distribution (these data were simulated from a uniform distribution on $[0, 1]$). The data are plotted on the y-axis.

then we get a probability plot like the one shown in Figure 5.33, which indicates that the sample plotted has more extreme observations than we would expect under normality (and we say that we have a distribution which is heavy tailed with respect to the normal). Here the points below the line on the left indicate that the small values in the sample are smaller than we would expect based on normality, whereas the values above the line on the right indicate that the large values in the sample are larger than we expect. The plots of Figure 5.34

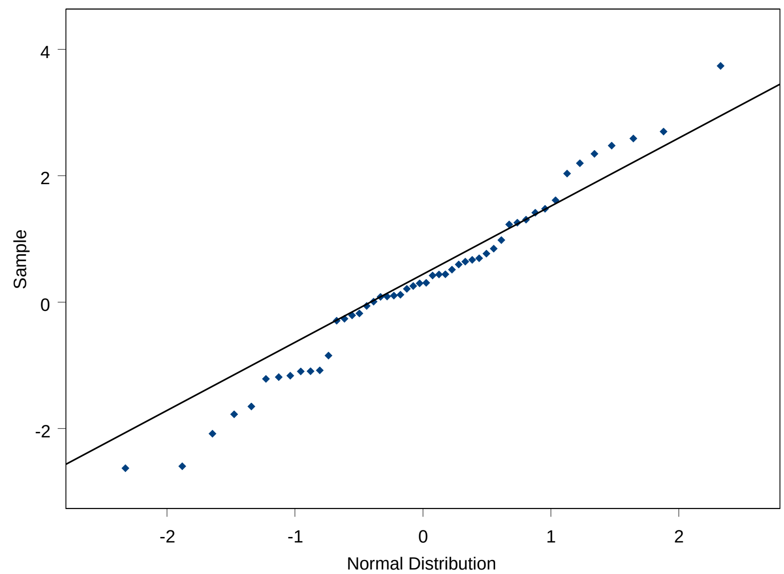

Figure 5.33: Normal probability plot for sample with more extreme values than we would expect based on the normal distribution (these data were simulated from a t distribution with 10 degrees of freedom). The data are plotted on the y-axis.

and Figure 5.35 show shapes which are characteristic of positive and negative skewness respectively.

We conclude this subsection with an example to illustrate that apparent deviations from normality in the normal probability plot may in fact just be due to a failure of other assumptions.

Example: inflation and central bank independence

We discuss the data given in Tutorial five on inflation rates and central bank independence for twenty three developed and developing countries. The predictors here are two measures of independence (QUES and LEGAL) as well as a binary variable DEV (which is 1 for developed economies and 0 for developing economies). In this example we fit a model for annual inflation rate with predictors QUES, LEGAL, DEV, QUES*DEV and LEGAL*DEV. A plot of internally

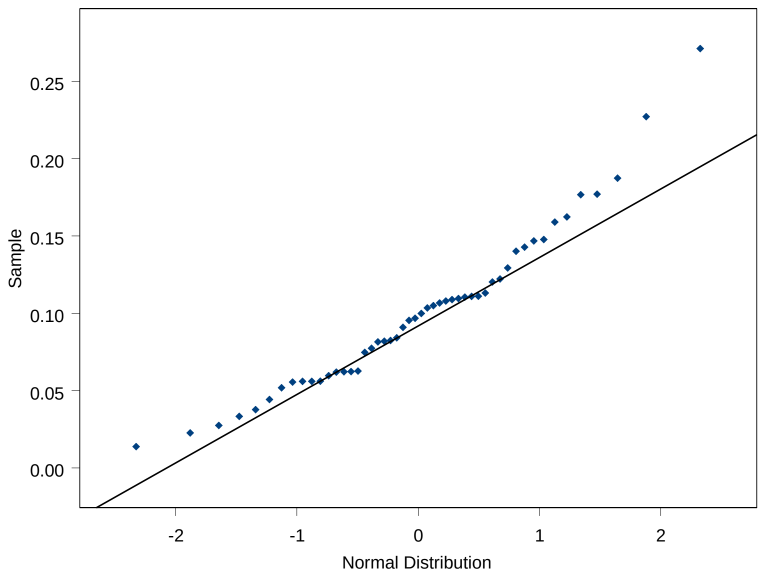

Figure 5.34: Normal probability plot for sample which is positively skewed (these data were simulated from a Beta(2,20) distribution). The data are plotted on the y-axis.

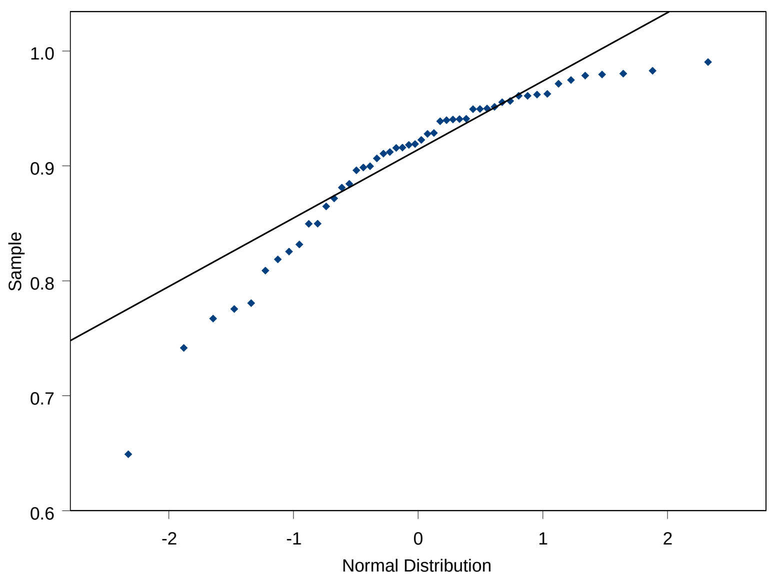

Figure 5.35: Normal probability plot for sample which is negatively skewed (these data were simulated from a Beta(20,2) distribution). The data are plotted on the y-axis.

studentized residuals against fitted values and the normal probability plot of these residuals is shown in Figures 5.36 and 5.37. The normal probability plot here

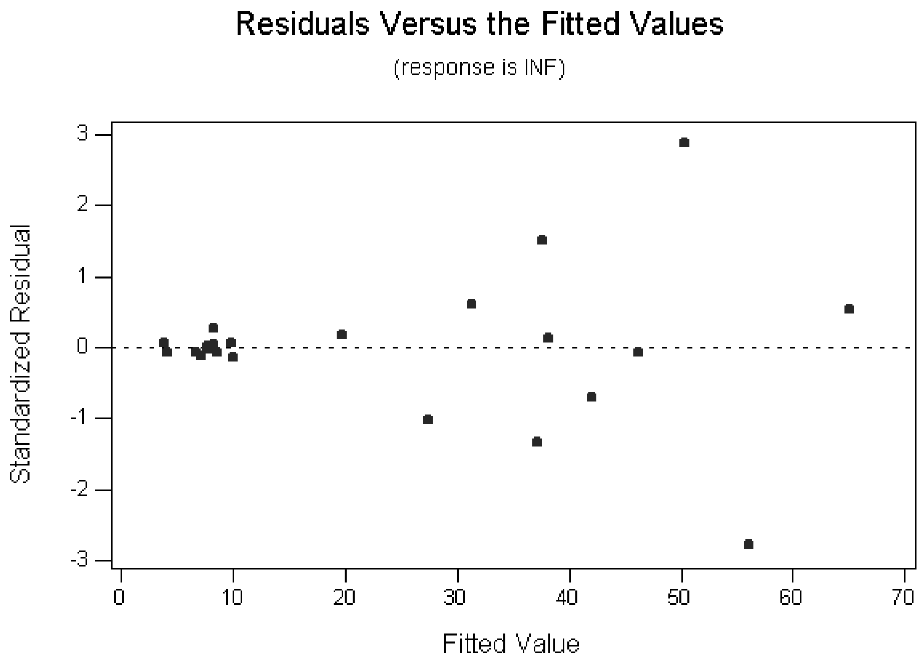

Figure 5.36: Plot of internally studentized residuals against fitted values for data on inflation and central bank independence.

indicates that the errors are heavy tailed with respect to the normal distribution: but the real problem here may be a violation of the constancy of variance assumption. Certainly from the residual plot it does not seem that the studentized residuals are identically distributed, and the normal probability plot is not very meaningful here.

Example: fast food restaurants

As a further example consider the fast food restaurants data set we have discussed in previous lectures. Here we are trying to predict a measure of sales for restaurants of a fast food chain in terms of the predictors income (a measure of income levels for households in the area) and age (the mean age of children for households in the area). For this example I have fitted a simple linear regression model with sales as the response and income as the predictor. A plot of internally studentized residuals against fitted values and the normal probability plot of these residuals are shown in Figures 5.38 and 5.39. The normal probability

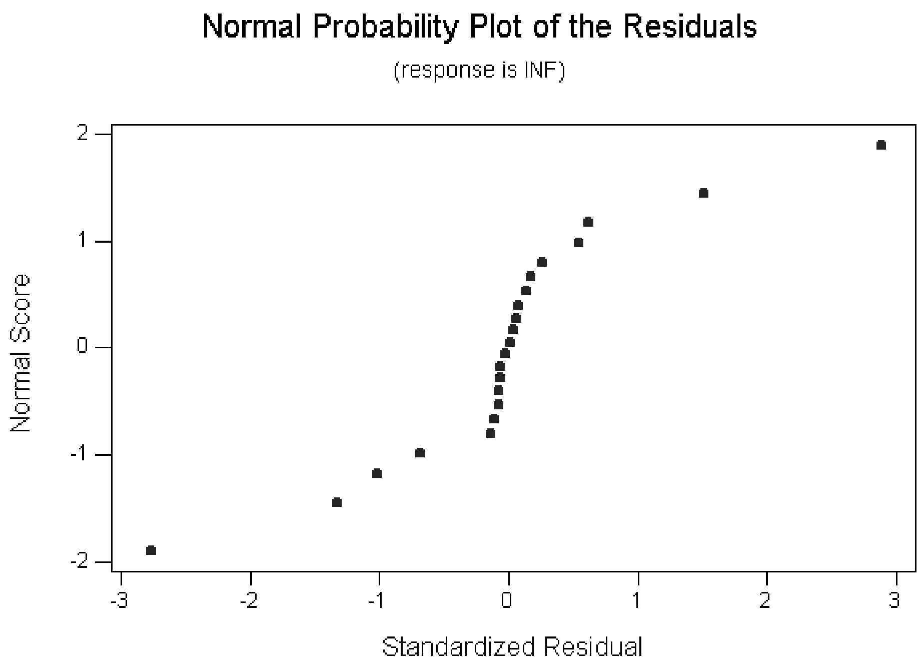

Figure 5.37: Normal probability plot for internally studentized residuals and data on inflation and central bank independence.

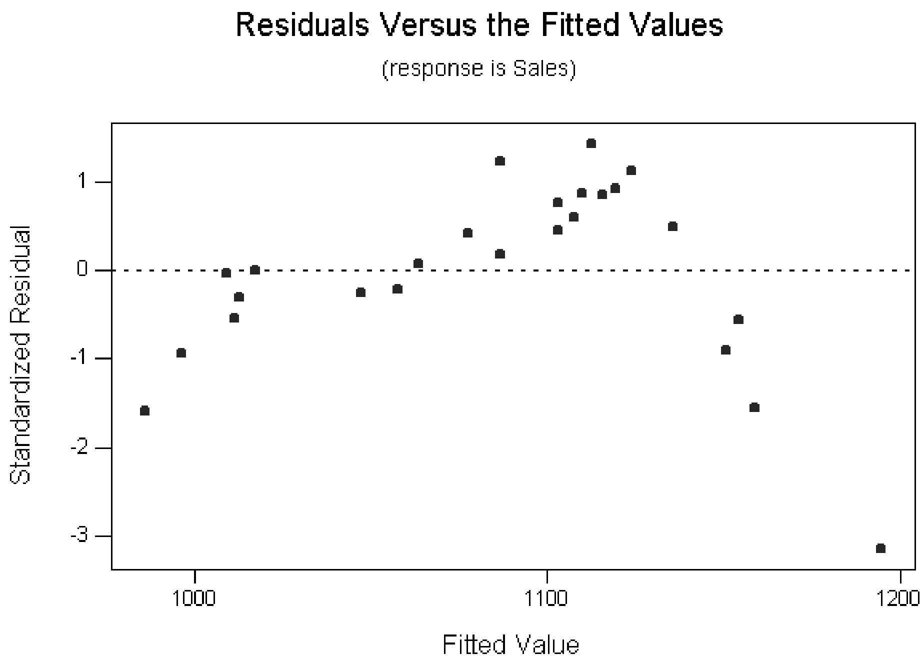

Figure 5.38: Plot of internally studentized residuals against fitted values for data on fast food restaurants.

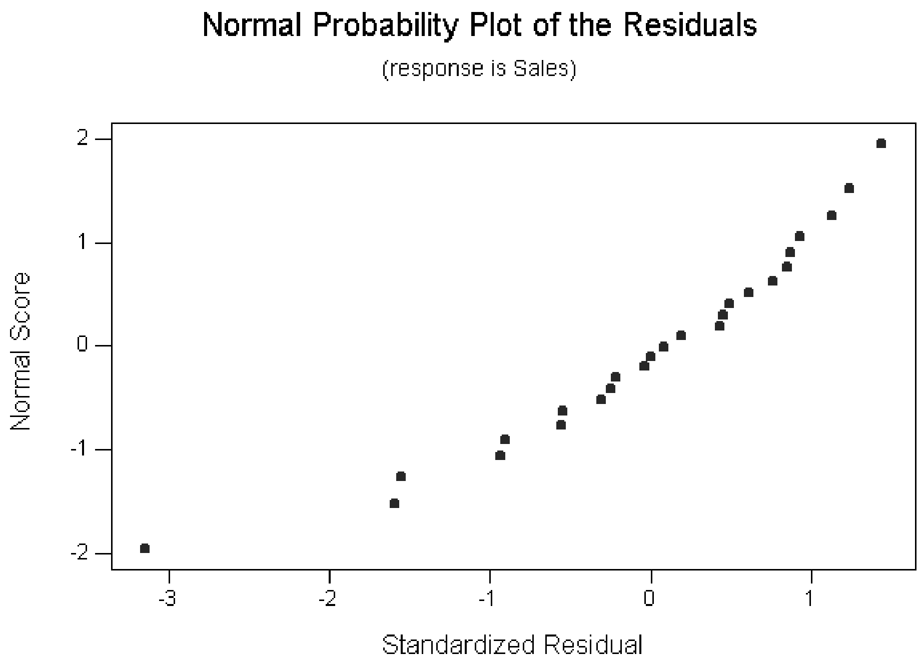

Figure 5.39: Normal probability plot for internally studentized residuals and data on fast food restaurants.

plot would indicate some negative skewness. However, if we look at the plot of studentized residuals against fitted values this indicates that the mean structure of the model is not correctly specified, so that again the normal probability plot is perhaps not terribly helpful.

5.8 Measures of influence

So far in this topic we have concentrated on the use of residuals for detecting an incorrectly specified mean, a failure of the constancy of variance assumption or the existence of outliers. We also considered assessment of normality in the general linear model using normal probability plots of residuals.

In this subsection we consider the notion of influence. We are interested here in whether a particular case (or cases) have a large impact on the inferences of interest in a study. Some of the measures we have studied in previous lectures (the externally studentized residuals and the leverages for instance) can help to identify potentially influential observations. Observations with a large externally studentized residual, for instance, correspond to unusual response values given the location in predictor space and these observations are often influential. Similarly, observations which correspond to unusual values for the predictors (high leverage observations) have the potential to exert a large amount of influence on inferences. However, high leverage observations which follow the pattern of the rest of the data may not be influential. Similarly, an outlier (a value with a large externally studentized residual) may not affect the inferences of interest in a statistical analysis.

Example: fish biomass data

As an example illustrating that a high leverage observation need not be influential, we consider the fish biomass data discussed on your last tutorial sheet. The response here is fish biomass, and there are four predictors which measure stream characteristics. Consider a simple linear regression model involving x_4, the fourth predictor (which is the area of the stream greater than 25cm in depth). The scatter plot of the response y against x_4 is shown in Figure 5.40. The two observations on the right are high leverage observations. One of them (the one with the smaller response value) clearly does not follow the pattern of the rest of the data, and this point would have a strong influence on the estimate of the slope in a simple linear regression model. However, if this point were omitted, the other high leverage observation would not affect the estimates of the regression parameters very much: it would not be an influential observation in this sense. So a large leverage value does not necessarily translate into high influence.

Example: Forbes data

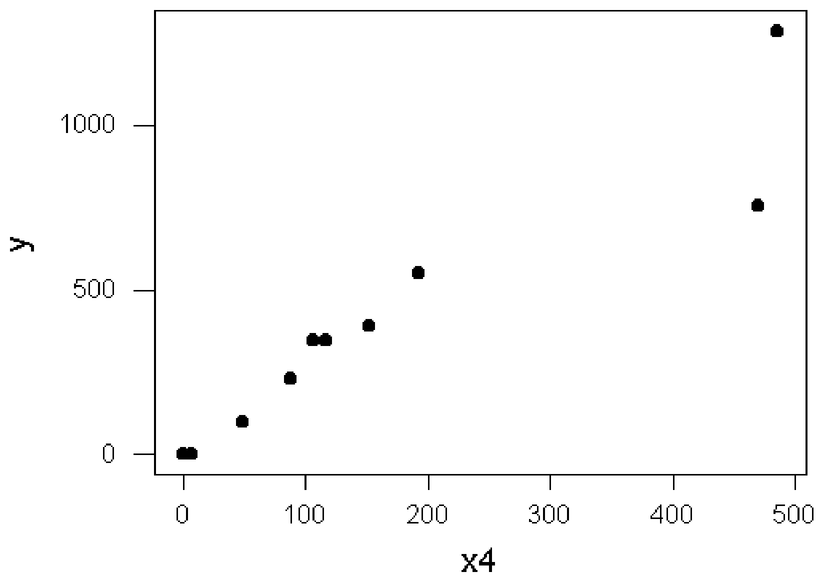

Figure 5.40: Scatter plot of y against x_4 for fish biomass data set.

The following example is discussed in some detail in Weisberg, "Applied Linear Regression (Second Edition)," Wiley, New York, 1985, p. 2. In the 1840s and 1850s a Scottish physicist, James D. Forbes, wanted to be able to estimate altitude above sea level from measurements of the boiling point of water. He knew that altitude could be determined from atmospheric pressure, measured with a barometer, with lower pressures corresponding to higher altitudes. He studied the relationship between pressure and boiling point. His interest in the problem was motivated by the difficulty of transporting the fragile barometers of the 1840s.

The scatter plot below (Figure 5.41) shows a measure of pressure as the response variable on the y-axis (pressure was measured in units of inches of mercury, and the pressures were then transformed by taking logs and multiplying by 100) and boiling point on the x-axis (measured in degrees Farenheit). There is an

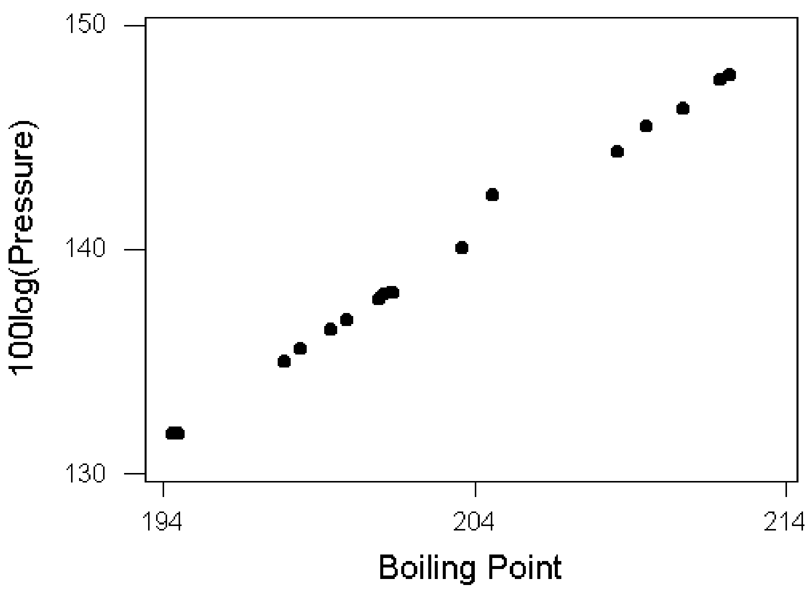

Figure 5.41: Scatterplot of transformed pressure (in units of 100 times log inches of mercury) against boiling point (in degrees Farenheit).

obvious outlier in this data set from the scatter plot: if we fit a simple linear regression model to these data, the externally studentized residual corresponding to this observation is 12.4037, which is extremely large. However, if we remove this outlying point and refit the model, we find that the estimate of the slope parameter in the simple linear regression model is little changed. So if this data set were observed in a situation where the inference of interest concerned the slope parameter in the model, the outlier would not correspond to an influential observation.

The need for detecting influential observations is obvious. If one observation has a substantial effect on the inferences of interest in a statistical analysis, then this observation must be examined carefully. But how do we detect influential observations? While a large externally studentized residual or leverage does not imply high influence as we have seen, it is certainly true that a high influence observation will have either a large externally studentized residual, high leverage or both. So a good place to start when attempting to identify potentially high influence observations is to examine the externally studentized residuals and leverages.

How do we assess when a residual or leverage value is large? We have already seen that the externally studentized residuals are t distributed if the model assumptions hold, and so this gives us a way of determining if an externally studentized residual is unusually large. But what about the leverages? It can be shown that the trace of the hat matrix is p, the number of elements in the vector β in the linear model. In other words,

$$\sum_{i=1}^{n} h_{ii} = p.$$

This implies that the average leverage value is p/n, and one guideline which is sometimes used for isolating large leverage values is to examine further observations with leverages larger than $2p/n$.

We give some examples now where we consider the externally studentized residuals and leverages. These examples also serve to illustrate how a single observation in a data set can substantially alter inferences of interest.

Example: rat data

The following example is discussed in Weisberg, "Applied Linear Regression (Second Edition)", Wiley, New York, 1985, p. 121 (data supplied by Dennis Cook). An experiment was conducted to investigate the amount of a particular drug present in the liver of a rat. Nineteen rats were randomly selected, weighed, placed under light ether anesthesia and given an oral dose of the drug. Because it was felt that large livers would absorb more of a given dose than smaller livers, the actual dose an animal received was approximately determined as 40mg of the drug per kilogram of body weight. (Liver weight is known to be strongly related to body weight). After a fixed length of time each rat was sacrificed, the liver weighed, and the percent of the dose in the liver determined.

The data are shown in the table. It was hypothesized that, for the method of

Body Weight	Liver Weight	Dose	y
176	6.5	0.88	0.42
176	9.5	0.88	0.25
190	9.0	1.00	0.56
176	8.9	0.88	0.23
200	7.2	1.00	0.23
167	8.9	0.83	0.32
188	8.0	0.94	0.37
195	10.0	0.98	0.41
176	8.0	0.88	0.33
165	7.9	0.84	0.38
158	6.9	0.80	0.27
148	7.3	0.74	0.36
149	5.2	0.75	0.21
163	8.4	0.81	0.28
170	7.2	0.85	0.34
186	6.8	0.94	0.28
146	7.3	0.73	0.30
181	9.0	0.90	0.37
149	6.4	0.75	0.46

determining the dose, there is no relationship between the percentage of the dose in the liver (y) and the body weight (x_1), liver weight (x_2) and relative dose (x_3). It can be shown that simple linear regressions of y on each of the predictors give insignificant slope terms, in line with the hypothesis. However, fitting a multiple linear regression for y involving predictors x_1, x_2 and x_3 results in the following output.

```
Regression Analysis

The regression equation is
y = 0.266 - 0.0212 Body Weight + 0.0143 Liver Weight + 4.18 Dose

Predictor        Coef       StDev          T        P
Constant       0.2659      0.1946       1.37    0.192
Body Wei    -0.021246    0.007974      -2.66    0.018
Liver We      0.01430     0.01722       0.83    0.419
Dose            4.178       1.523       2.74    0.015

S = 0.07729     R-Sq = 36.4%     R-Sq(adj) = 23.7%

Analysis of Variance

Source            DF          SS          MS         F        P
Regression         3    0.051265    0.017088      2.86    0.072
Residual Error    15    0.089609    0.005974
Total             18    0.140874

Source       DF      Seq SS
Body Wei      1    0.003216
Liver We      1    0.003067
Dose          1    0.044982
```

We see that the p-values for the partial t tests for the coefficients for body weight and dose are significant here. If we remove the term for liver weight, the coefficients for body weight and dose remain significant and the p value in the ANOVA table for overall significance of the model is 0.032. Given that the simple linear regressions indicate no relationship in line with our original hypothesis this seems like a strange result. The explanation of the problem can be revealed by examining externally studentized residuals and leverages. In the model involving body weight and dose, the leverage of the third observation is approximately 0.83, indicating that this is a potentially influential case because it has an unusual predictor vector (as a matter of fact, for this rat the usual rule for assigning dosage based on body weight was not used). If we omit this observation and refit the model we get the following output.

```
Regression Analysis

The regression equation is
y = 0.332 - 0.0044 Body Weight + 0.88 Dose

Predictor        Coef       StDev          T        P
Constant       0.3320      0.1954       1.70    0.110
Body Wei     -0.00444     0.01693      -0.26    0.797
Dose            0.875       3.400       0.26    0.800

S = 0.07622     R-Sq = 0.5%      R-Sq(adj) = 0.0%

Analysis of Variance

Source             DF          SS          MS         F         P
Regression          2    0.000423    0.000211      0.04     0.964
Residual Error     15    0.087138    0.005809
Total              17    0.087561

Source       DF      Seq SS
Body Wei      1    0.000038
Dose          1    0.000385
```

As we can see, no relationship between the predictors and the response in this multiple linear regression model is indicated, in line with our hypothesis. The apparent relationship was due to a single influential observation.

Example: BOQ data

We give a further example which illustrates how a single observation can be extremely influential in a regression analysis, and how potential influence can be detected by looking at the leverages and residuals. The data set we describe here is given in Myers, "Classical and Modern Regression (Second Edition)," Duxbury, Belmont, California, 1990, pp. 254–255. This data set describes manpower needs for bachelor officer's quarters in the US navy. We won't give the full data set here, but there are 25 observations in the data set. The response y is monthly man hours, and there are seven predictors, average daily occupancy (x_1), monthly average number of check-ins (x_2), weekly hours of service desk operation (x_3), square feet of common use area (x_4), number of building wings (x_5), operational berthing capacity (x_6) and number of rooms (x_7).

If we fit the model involving all the predictors, we find that one observation (number 23) has a very high leverage of 0.9885 and an externally studentized residual of -5.24. There are also a number of other observations which might need further investigation. Fitting the model without observation 23 drastically changes inferences about the coefficients: the coefficients for average daily occupancy and square feet of common use area become positive (which is important if the model is to have a sensible interpretation). Also, changes occur in the conclusions of the partial t-tests for average daily occupancy, square feet of common use area and operational berthing capacity. We will demonstrate this in the lecture.

While the leverages and residuals can help to expose potential influence, it would be nice to have measures which expose directly the influence of each observation. We will look at a number of different influence measures in this course, each of which reflects the way that an observation will effect a certain aspect of the fitted model. Perhaps the most commonly used measure of influence (and one that is computed in many statistical packages) is Cook's distance. The Cook's distance attempts to measure the change in the estimated parameters when an observation is deleted. Write b_{-i} for the estimate of the regression parameters β when we fit using all the data except the ith observation, and write b for the estimate of β based on the complete data set. If we write $\widehat{\sigma}^2$ for the usual unbiased estimator of σ^2, then the Cook's distance for the ith observation is simply

$$D_i = \frac{(b - b_{-i})^T X^T X (b - b_{-i})}{p\widehat{\sigma}^2}.$$

To understand the motivation for this influence measure, recall that a $100(1-\alpha)$ percentage confidence region for the vector of parameters β in the general linear model is given by

$$\left\{\beta^* : \frac{(b - \beta^*)^T X^T X (b - \beta^*)}{p\widehat{\sigma}^2} \leqslant F_{p,n-p;\alpha}\right\}$$

where $F_{p,n-p;\alpha}$ is the upper 100α percentage point of an F distribution with p and $n-p$ degrees of freedom. This follows from the fact that

$$\frac{(b - \beta)^T X^T X (b - \beta)}{p\widehat{\sigma}^2} \sim F_{p,n-p}.$$

So if we compare the Cook's distance to, say, the 50 percentage point of an F distribution with p and $n-p$ degrees of freedom, we are saying that deletion of the ith case moves the estimate of the parameters outside a 50 percent confidence region for β based on the fit to the full data. We point out that it is not necessary to refit the model with one observation deleted in order to compute the Cook's distance. It can be shown that

$$D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1 - h_{ii}}$$

where r_i is the internally studentized residual and h_{ii} is the leverage. So the Cook's distance gives an easily computed and interpreted measure of how much the parameter estimates are influenced by each observation.

Example: rat data

We return briefly to the rat data. We identified the third case here as being a potentially influential one, and confirmed this by deleting this case and reanalyzing the data. Examining the Cook's distance shows directly from the fit to the full model that this observation is an influential one: the Cook's distance for this observation is 1.82 (the next largest value for the Cook's distance is approximately 0.21).

Example: BOQ data

Consider once again the data on manpower needs for bachelor officer's quarters. In this data set, observation 23 was influential: does the Cook's distance reveal this as we expect? The Cook's distance for this observation is 115.041 (the next largest value is 5.89). We will illustrate in the lecture computation of percentage points of an F distribution for assessment of when the Cook's distance is large.

5.9 DFFITS, DFBETAS and other influence measures

Last lecture we discussed the use of studentized residuals and leverages to identify observations which potentially exert strong influence on a fitted linear regression model. At the end of the lecture we also discussed the Cook's distance, which is a measure of how much the estimate of β changes when we delete an observation from the model. The Cook's distance D_i was defined to be

$$D_i = \frac{(b_{-i}-b)^T X^T X (b_{-i}-b)}{p\widehat{\sigma}^2}$$

where b is the estimate of β obtained from a fit to the full model, b_{-i} is the estimate of β from a fit to all the data except the ith observation, X is the $n \times p$ design matrix and $\widehat{\sigma}^2$ is the usual unbiased estimator of σ^2. As a rule of thumb, we compare the Cook's distance to the upper 50 percentage point of an $F_{p,n-p}$ distribution: recall that if the Cook's distance is larger than this, then this means that b_{-i} lies outside a 50 percent confidence ellipsoid for β based on all the data. While consideration of confidence ellipsoids provides one interpretation for the Cook's distance, there is an alternative interpretation which leads to the introduction of a closely related influence measure.

Rearranging our expression for D_i gives

$$D_i = \frac{(Xb_{-i}-Xb)^T (Xb_{-i}-Xb)}{p\widehat{\sigma}^2}.$$

Now, note that Xb is simply $\widehat{y}$, the vector of fitted values for the complete data set. Also, Xb_{-i} is the vector of fitted values based on the fit with the ith case deleted, and we write $\widehat{y}_{-i} = Xb_{-i}$. So D_i becomes

$$D_i = \frac{(\widehat{y} - \widehat{y}_{-i})^T(\widehat{y} - \widehat{y}_{-i})}{p\widehat{\sigma}^2}.$$

So we can see that we can interpret D_i as representing the ordinary Euclidean distance between the vectors of fitted values for the full data set and the data set with the ith case deleted, scaled by a constant. You may have felt that looking at the change in the vector of fitted values would be the appropriate way of assessing influence, particularly if prediction is the goal of a regression analysis: we now see that this is closely related to looking at the Cook's distance, which measures the change in the estimate of β upon deletion of the ith case.

A statistic which is sometimes used to look at the change in the fitted value at the ith point upon deletion of the ith case is DFFITS_i, defined to be

$$\text{DFFITS}_i = \frac{\widehat{y}_i - \widehat{y}_{i,-i}}{\widehat{\sigma}_{-i}\sqrt{h_{ii}}}$$

where we have written $\widehat{y}_{i,-i}$ for the ith element of $\widehat{y}_{-i}$ and $\widehat{\sigma}_{-i}$ is our estimate of the residual standard deviation based on the fit with the ith case deleted. Note that the standard error of $\widehat{y}_i$ is just $\sigma\sqrt{h_{ii}}$, so DFFITS_i is estimating the number of standard errors by which the fit changes at the ith point when the ith case is deleted from the fit. One rule of thumb for assessing the magnitude of values of DFFITS is to say that an absolute value larger than 2 is large. The statistic DFFITS_i is closely related to the Cook's distance D_i: it can be shown that

$$\text{DFFITS}_i^2 = \frac{(b_{-i} - b)^T X^T X (b_{-i} - b)}{\widehat{\sigma}_{-i}^2},$$

so that the square of DFFITS_i is simply p times a statistic that is very similar to the Cook's distance. Evidently the Cook's distance and DFFITS give much the same information, although DFFITS does give information about the sign of the change in the fitted value upon deletion of the ith case. As for the Cook's distance, we can compute DFFITS without having to fit a new model when we delete each case: the computational formula for DFFITS_i is

$$\text{DFFITS}_i = t_i\sqrt{\frac{h_{ii}}{1 - h_{ii}}}$$

where t_i is the externally studentized residual and h_{ii} is the leverage. We return to some of the examples in the last lecture (the rat data and the BOQ data) to illustrate the use of DFFITS.

Example: rat data

Recall from last lecture the rat data, where we were interested in modelling the percentage of a drug in the livers of 19 rats after a period of time in terms of predictors body weight, liver weight and relative dose. When discussing this example in the last lecture we determined that the 3rd observation was very influential: it has a high leverage value, since the method for assigning doses for the other rats was not followed in this case, and so the vector of predictors is unusual here. The Cook's distance for this observation based on a model involving body weight and relative dose is 1.82, which may be compared to the upper 50 percentage point of an F distribution with 3 and 16 degrees of freedom (approximately 0.82). The DFFITS for the 3rd case and the model involving body weight and relative dose is approximately 2.35 here, which suggests that deletion of this case does substantially influence the fit at this point. Note that with $p = 3$ we have that p times the Cook's distance is approximately 5.46, which is roughly the square of DFFITS (2.35^2 or approximately 5.53). So the Cook's distance and DFFITS do give much the same kind of information.

Example: BOQ data

As a further example we consider the BOQ data. Recall that the goal of the analysis with this data set was to predict manpower needs for bachelor officer's quarters in the US Navy. The response is man hours (a measure of manpower needs), and there are seven predictors measured for 25 different naval installations. In the last lecture we considered fitting a model for the response including all seven predictors. We identified the 23rd case as being an unusual one: it has both a large externally studentized residual and a high leverage value. The Cook's distance for this observation is large (approximately 115.04) which is much larger than the upper 50 percentage point of an F distribution with 8 and 17 degrees of freedom (approximately 0.96). The DFFITS for the 23rd case is approximately -48.52, which shows that the fitted value at this point changes by approximately 48.52 times the standard error of the fitted value at this point when this case is deleted: this is certainly an influential case.

In defining the Cook's distance our motivation was to identify observations which have a substantial effect on the estimate of β. Once we have identified an observation which does appear to have an effect on estimation of β, it may be of interest to ask which components of the estimate b of β are influenced most. It may be that an unusual observation has a large influence on some of the coefficient estimates but not on others. An influence statistic which can help us to determine which of the coefficient estimates are most influenced by an unusual observation is the DFBETAS statistic. For a given observation i and the jth predictor, we define $\text{DFBETAS}_{j,i}$ to be

$$\text{DFBETAS}_{j,i} = \frac{b_j - b_{j,-i}}{\widehat{\sigma}_{-i}\sqrt{c_{jj}}}$$

where b_j is the jth element of b, $b_{j,-i}$ is the jth element of b_{-i}, $\widehat{\sigma}_{-i}$ is the estimate of σ obtained from the fit with the ith case deleted and c_{jj} is the jth diagonal element of $(X^TX)^{-1}$. Recall that the covariance matrix of b is simply $\sigma^2(X^TX)^{-1}$, so $\text{DFBETAS}_{j,i}$ estimates the number of standard errors by which the jth coefficient estimate changes when the ith case is deleted. As for DFFITS and the Cook's distance, there is a convenient computational formula for the DFBETAS statistic, but we do not discuss this here.

Example: BOQ data

As a further example we consider the BOQ data. From the Cook's distance observation 23 exerts substantial influence on the coefficient estimates: which coefficient estimates are most influenced? Again we can compute the DFBETAS to answer this question. For the intercept the DFBETAS for observation 23 is approximately -0.25, for x_1 it is -44.38, for x_2 it is 1.17, for x_3 it is 0.66, for x_4 it is -4.57, for x_5 it is 1.18, for x_6 it is -3.94 and for x_7 it is 7.28. Obviously the DFBETAS for x_4, x_6 and x_7 are all large here, and the DFBETAS for x_1 is very large at -44.38.

We discuss one more influence measure which is less commonly used than DFFITS and DFBETAS, and which gives an overall measure of how much the standard errors of parameter estimates are influenced by deletion of cases. In particular, we define COVRATIO_i to be

$$\text{COVRATIO}_i = \frac{\widehat{\sigma}_{-i}^{2p}}{\widehat{\sigma}^{2p}}\frac{1}{1-h_{ii}}.$$

We won't discuss the motivation for the definition of this statistic in too much detail. You should just think of this statistic as measuring in a global sense how much standard error estimates of the coefficients are influenced by deletion of the ith case. A rule of thumb for assessing the magnitude of this statistic is to further examine observations which have a COVRATIO larger than $1 + 3p/n$ or smaller than $1 - 3p/n$. We illustrate the use of the COVRATIO statistic by looking at the fish biomass data discussed last lecture.

Example: fish biomass data

Consider again the fish biomass data. For this data set the response y was fish biomass, and there were four predictors consisting of measurements of stream characteristics which were thought to relate to fish biomass. Recall that when we considered a simple linear regression model using the fourth predictor (area of stream with a depth greater than 25 centimetres) we saw that there were two observations with very high leverages (see Figure 5.40). We might expect that these observations exert a substantial influence on standard errors of coefficient estimates: computing the COVRATIO for these observations we get approximately 0.00589 for the third observation and approximately 0.492 for the fourth observation. These very small COVRATIO values reflect the fact that precision of estimation is substantially reduced by inclusion of these observations.

5.10 Dealing with violations of assumptions: transformations

In previous subsections we have looked at the use of residuals and other diagnostics to highlight possible violations of assumptions. However, we have not yet discussed how to fix the problems these diagnostics may detect. We now discuss the use of transformations (of either the response or the predictors) in an analysis where there is a possible violation of the model assumptions. We also discuss some of the implications of the use of transformations and their drawbacks. It is important to point out that applying a transformation to fix a violation of the model assumptions may cause another violation which did not occur on the original scale.

Variance Stabilizing Transformations

We will need to consider different kinds of transformations depending on which assumptions we believe are violated in the linear model. We consider first transformations which are appropriate when the assumption of a constant error variance may be violated (so-called variance stabilizing transformations). Weisberg, "Applied Linear Regression (Second Edition)", Wiley, New York, 1985, p. 134, gives a list of variance stabilizing transformations and describes situations in which these transformations are natural. It is actually quite common in a regression analysis for the variance of a response to depend on the mean level: in many situations, as the magnitude of the expected response grows, so does the variance of the response (see below for some examples). A transformation of the response may be useful in these situations so that the usual linear modelling and inferential techniques can be applied on the transformed scale.

A square root transformation $\sqrt{y}$ for the response y is appropriate when the error variance is proportional to the mean (of course the responses must be positive to take square roots). To see why the square root transformation is useful in this situation, consider a first order Taylor expansion of $\sqrt{y}$ about the expected value $\mathbb{E}(y)$ of y. We have that $\sqrt{y}$ is approximately

$$\sqrt{\mathbb{E}(y)} + \frac{1}{2\sqrt{\mathbb{E}(y)}}(y - \mathbb{E}(y)).$$

from this we have that the variance of the square root of y is approximately

$$\frac{1}{4E(y)}\mathrm{Var}(y)$$

and hence if the variance of y is proportional to its mean, the variance of the square root of y should be approximately constant. The square root transformation is often considered when a linear model is used for an approximate analysis of count data. If a Poisson distribution is appropriate for the counts, then the variance is proportional to the mean, and so the square root transformation has some theoretical rationale in this situation.

A log transformation is appropriate when the error standard deviation is proportional to the mean (that is, when the standard deviation is a percentage of the response). We can consider a Taylor expansion as in our discussion of the square root transformation to see why the log function will stabilize the variance in this situation. We have that $\log y$ is approximately

$$\log \mathbb{E}(y) + \frac{(y - \mathbb{E}(y))}{\mathbb{E}(y)}$$

so that the variance of $\log y$ is approximately

$$\frac{\mathrm{Var}(y)}{\mathbb{E}(y)^2}.$$

Again we require the responses to be positive for a log transformation to be applicable. If some of the responses are zero, the transformation $\log(y+1)$ is sometimes used. Another common variance stabilizing transformation is the inverse function, $1/y$. This transformation is applicable when the responses are positive and when the error standard deviation is proportional to the mean squared. If some of the responses are zero, the transformation $1/(y + 1)$ is sometimes used.

Evaluating transformations

Before we give some examples we discuss how to evaluate whether a transformation has helped to improve the fitted model. If we are interested in prediction, we saw when we discussed model selection that the PRESS statistic was one way of comparing models. We might think that we can use the PRESS statistic here for comparing models where we have transformed the response. However, we must remember that we are interested in predictions of the original response: that is, we must compare different models on the original untransformed scale.

One method for comparing models is the following. As usual write y for the response, and suppose that after fitting a multiple linear regression to y an examination of diagnostic statistics indicates the need for a variance stabilizing transformation. We apply the variance stabilizing transformation $z = f(y)$ (where f is invertible) and fit a multiple linear regression model to z. Is the model on the transformed scale to be preferred if our goal is to predict y? We can develop a statistic for the model on the transformed response which can be compared with the PRESS statistic for the model for y (or with the corresponding statistic for a model involving a different transformation of the response). As in our discussion of the PRESS statistic, write $\widehat{z}_{i,-i}$ for the fitted value for z obtained from fitting a model to all the data except the ith observation. The ith PRESS residual on the transformed scale is of course simply $z_i - \widehat{z}_{i,-i}$. However, as we are interested in prediction on the original scale we consider transforming $\widehat{z}_{i,-i}$ by $f^{-1}(\cdot)$ (where $f^{-1}(\cdot)$ denotes the inverse of $f(\cdot)$) to get a prediction of y_i, and then the analogue of the PRESS residual on the original scale is $y_i - f^{-1}(\widehat{z}_{i,-i})$. We can calculate

$$\sum_{i=1}^{n}(y_i - f^{-1}(\widehat{z}_{i,-i}))^2$$

and compare this to the PRESS statistic for the model for y or to a similar statistic for a different transformation to get an idea of whether predictive performance is improved on the original scale by using the variance stabilizing transformation. Alternatively, we could compare the sum of the absolute PRESS residuals for the model of the untransformed response with

$$\sum_{i=1}^{n}|y_i - f^{-1}(\widehat{z}_{i,-i})|.$$

It is important to stress that models must be compared on the original scale and that we cannot simply look at the R^2 or error standard deviation for models at different scales. We give some examples to illustrate the use of variance stabilizing transformations.

Example: snow geese data

The following example is from Weisberg, S., "Applied Linear Regression (Second Edition)," Wiley, New York, 1985, p. 102.

Aerial survey methods are regularly used to estimate the number of snow geese in their summer range areas west of Hudson Bay in Canada. To obtain estimates, small aircraft fly over the range and, when a flock of geese is spotted, an experienced person estimates the number of geese in the flock. To investigate the reliability of this method of counting, an experiment was conducted in which an airplane carrying two observers flew over 45 flocks, and each observer made

Photo	Observer	Photo	Observer
46	50	119	75
38	25	165	100
25	30	152	150
48	35	205	120
38	25	409	250
22	20	342	500
22	12	200	200
42	34	73	50
34	20	123	75
14	10	150	150
30	25	70	50
9	10	90	60
18	15	110	75
25	20	95	150
62	40	57	40
26	30	43	25
88	75	55	100
56	35	325	200
11	9	114	60
66	55	83	40
42	30	91	35
30	25	56	20
90	40		

an independent estimate of the number of birds in each flock. Also, a photograph of the flock was taken so that an exact count of the number of birds in the flock could be made.

We do not reproduce the full data set here, but in the table below the photo counts and counts for one of the observers are shown (see Weisberg (1985) for the full data set). A scatter plot of the photo count against observer count is shown in Figure 5.42 It is clear from the scatter plot that the variance increases with the mean response. For count data of this kind, a square root transformation is often appropriate. (One of the most commonly used models for count data is the Poisson distribution, for which the mean is equal to the variance). Figure 5.43 shows the scatter plot of the square root of the photo count against the square root of the observer count. An assumption of constant error variance in the simple linear regression model for $\sqrt{y}$ with $\sqrt{x}$ as the predictor would seem to be more nearly reasonable than in the model with y as the response and x as the

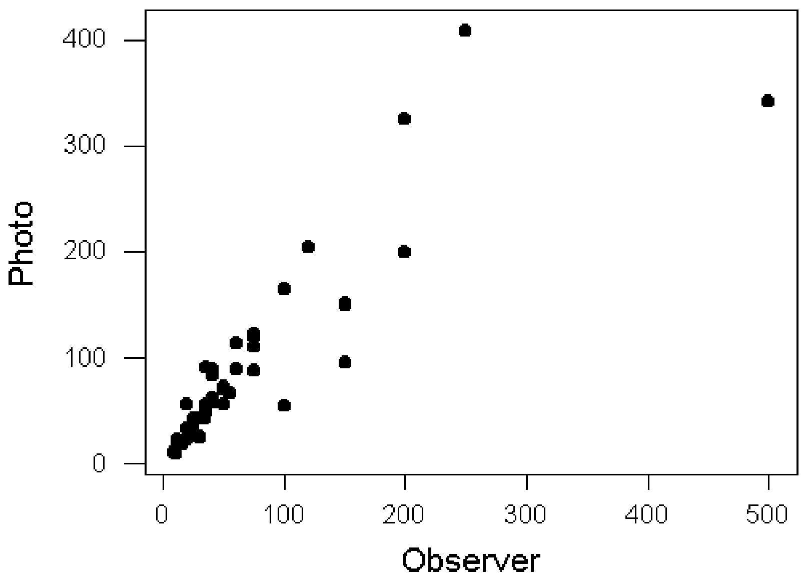

Figure 5.42: Scatter plot of photo count against observer count for data on snow geese.

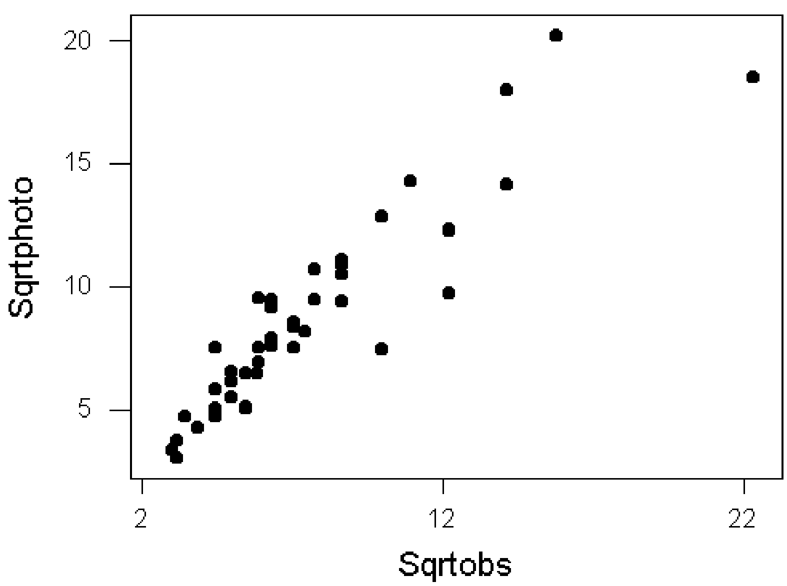

Figure 5.43: Scatter plot of square root of photo count against square root of observer count for data on snow geese.

predictor, although there is still a suggestion that the variance increases with the mean. Figure 5.44 shows the scatter plot of the log of photo count against the log of observer count. The log transformation would appear to be better than

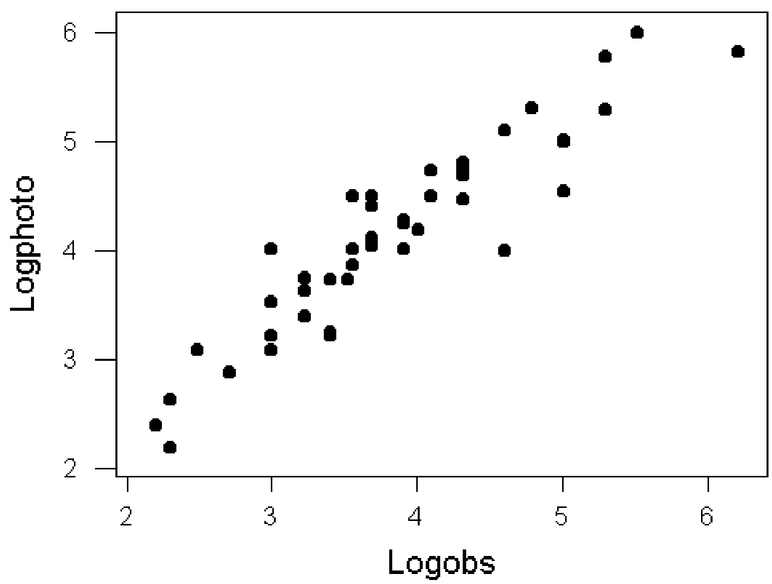

Figure 5.44: Scatter plot of log of photo count against log of observer count for data on snow geese.

the square root transformation for stabilizing the variance, although there are reasons for preferring the square root for interpretability (since the data consist of counts).

We can compare the models for y, $\sqrt{y}$ and $\log(y)$ in terms of predictive performance on the scale of y by using the analogue of the PRESS statistic that we have just developed. Computation of these statistics will be illustrated in lectures. The PRESS statistic for the simple linear regression model with y as response and x as predictor is approximately 172738. The analogue of this statistic for the model involving the square root of y is 137603, and for the log of y is 122740. The sum of the absolute PRESS residuals for the model involving y is approximately 1475.55, and the analogue of this statistic is 1295.89 and 1257.81 for the models involving the square root of y and log of y respectively. So for predictive performance on the untransformed scale it would seem that the model for $\log(y)$ is best.

Example: inflation data

We consider the data on inflation rates for 22 different countries which was considered in previous lectures. The response here was INF (inflation rate), and we were interested in predicting INF using measures of central bank independence (QUES, LEGAL) and an indicator variable DEV (one for developed countries, zero for developing countries). For the purposes of this example we consider just a simple linear regression model for INF with QUES as the predictor. The scatter plot of INF against QUES is given in Figure 5.45. Variation in the response seems

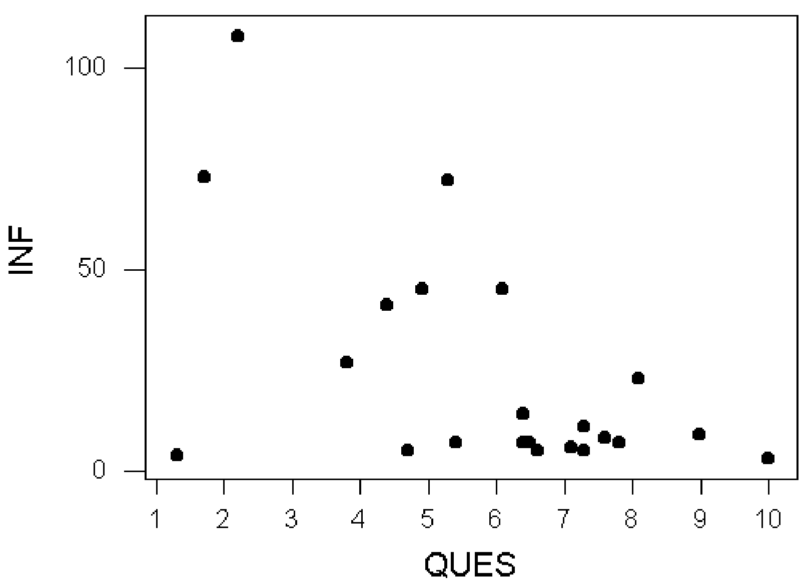

Figure 5.45: Scatter plot of INF against QUES for data on inflation rates and central bank independence.

to increase with the mean here, and a variance stabilizing transformation might be considered. Figure 5.46 shows a scatter plot of the log of INF against QUES, which seems to indicate that the assumption of constancy of variance would be more nearly reasonable if the response were modelled on this scale. There is a clear outlier in this scatter plot.

Again we can fit models at the different scales and compare predictive performance via the analogue of the PRESS statistic that we have developed. The PRESS statistic for the model for INF in terms of QUES is approximately

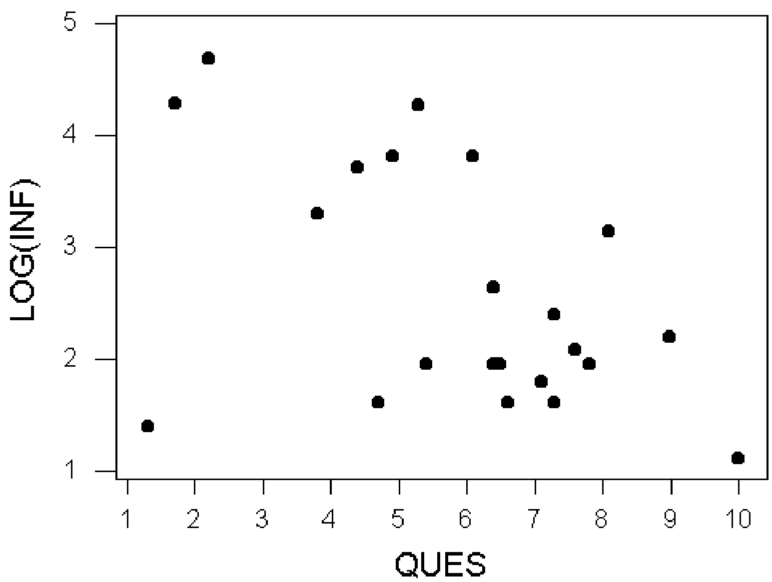

Figure 5.46: Scatter plot of log(INF) against QUES for data on inflation rates and central bank independence.

16070.9, and the analogue of this for the model with the log of INF as response is 21610.8. However, if we look at the sum of the absolute PRESS residuals for the model of INF we get 433.84, whereas for the model of log INF the corresponding statistic is 431.193. The conflict between the two measures of predictive performance is due to the outlier, and the model for the log of INF would be preferable on the untransformed scale if the outlier is removed.

5.11 Weighted Least Squares

We have considered the use of variance stabilizing transformations so that the usual theory of linear models can be applied in modelling the response on a suitable scale. However, there is an alternative to the use of variance stabilizing transformations. If we know the variances of the errors ε_i, $\mathrm{Var}(\varepsilon_i) = \sigma_i^2$ say, or if we know the variance of the errors up to an unknown constant, $\mathrm{Var}(\varepsilon_i) = \sigma^2 w_i$ where σ^2 is unknown and w_i, $i = 1, ..., n$ are known weights, then so-called *weighted least squares* can be applied. If we write V for the (diagonal) covariance matrix of the errors $(\varepsilon_1, ..., \varepsilon_n)$, then the maximum likelihood estimator $\widehat{\beta}$ of β assuming normality of the errors is

$$\widehat{\beta} = (X^\top V^{-1} X)^{-1}(X^\top V^{-1} y)$$

where X is the design matrix and y is the vector of responses. In the situation where $V = \sigma^2 W$ and W is the diagonal matrix with diagonal elements $w_1, ..., w_n$ we can write

$$\widehat{\beta} = (X^\top W^{-1} X)^{-1}(X^\top W^{-1} y)$$

and we see that $\widehat{\beta}$ doesn't depend on the (unknown) σ^2. The covariance matrix of $\widehat{\beta}$ can be shown to be

$$(X^\top V^{-1} X)^{-1}$$

which reduces to

$$\sigma^2 (X^\top W^{-1} X)^{-1}$$

when $V = \sigma^2 W$. It can be shown that $\widehat{\beta}$ minimizes

$$\sum_{i=1}^{n} w_i^{-1}(y_i - x_i^\top \beta)^2$$

which is a least squares type criterion in which observations with larger variances receive less weight. Much of the theory of linear models with a constant error variance can be carried over to the situation we have just described, and if we can specify the variances σ_i^2 of the errors or weights w_i in a natural way then modelling the inhomogeneity of the variances may be preferable to the application of a variance stabilizing transformation. Sometimes the variances σ_i^2 can be estimated: for instance, if we have many observations for each set of predictor values, then we could estimate σ_i^2 from the data. Another possiblity which is reasonable in some situations is to take the weights w_i to be given by the values of one of the predictors.

Example: transfer efficiency data

We illustrate the use of weighted regression with an example from Myers (1990), "Classical and Modern Regression Analysis with Applications (Second Edition)", Duxbury, Belmont, California, p. 281–282. The response y here is a measure of the efficiency of a particular electrostatic type of spray paint equipment. We are interested in modelling the response as a function of two predictors, air velocity and voltage. An experiment was conducted in which two levels of air velocity and two levels of voltage were considered, and ten observations were taken for each of the four possible air velocity/voltage combinations. The data are shown in the table below. Since we have ten replicates for each distinct combination of the

		Voltage			
		50		70	
Air Velocity	60	87.5	88.2	77.4	68.1
		88.1	87.3	70.7	65.3
		89.5	89.2	67.0	61.0
		86.2	85.9	71.7	81.7
		90.0	87.0	79.2	60.3
	120	82.5	81.3	61.2	50.7
		81.6	80.7	67.2	52.3
		77.4	79.3	55.9	68.6
		81.5	82.0	52.0	69.5
		79.7	79.2	63.5	70.1

predictor values, we can estimate the variance at each set of predictor values. We can use these estimated variances in a weighted regression analysis of the data.

```
Weighted analysis using weights in Weights

The regression equation is
Efficiency = 142 - 0.924 Voltage - 0.124 Air Velocity

Predictor        Coef       StDev          T         P
Constant      141.552       4.449      31.82     0.000
Voltage      -0.92402     0.08591     -10.76     0.000
Air Velo     -0.12389     0.01066     -11.62     0.000

S = 0.9839      R-Sq = 87.3%     R-Sq(adj) = 86.6%
```

It is important to note that while the ordinary least squares estimator will be consistent even in the case where the error variances are unequal in the linear model, an appropriate weighted least squares estimator will generally be much more efficient (have a smaller variance). The ordinary least squares estimator will be heavily influenced by unreliable high variance observations, whereas the role of these observations is donwplayed in the weighted analysis.

5.12 Nonlinear transformations of a single predictor

In the previous subsection we considered the use of a transformation of the response in order to stabilize the error variance. Another violation of model assumptions where transformations can be helpful is where there is evidence of non-linear relationships between the predictors and the response.

For simplicity we will consider just the case of a single predictor first. We have already encountered the idea in previous lectures of using transformations of the predictors in a linear model so that the mean structure is linear in the transformed predictors. In this subsection, we describe a number of different nonlinear relationships between a response and a predictor which can be written as linear relationships between a transformed response and a transformed predictor or predictors (see Myers, 1990, Section 7.3 for description of some additional nonlinear relationships and corresponding transformations). We write y for the response variable, and x for a predictor, and we consider various functional relationships between y and x (we ignore the random component of the model for the moment and make some remarks at the end of the subsection about the effect of transformations on the error structure).

Parabolic relationship

Suppose we do a scatter plot of y against x and we see a parabolic shape. If the relationship between y and x is parabolic (we ignore errors) then we have

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

so that by introducing the transformation x^2 of the original predictor x we have a relationship that is nonlinear in the predictor x but linear in the unknown parameters β_0, β_1 and β_2. So if we see a parabolic relationship between y and x in a scatter plot, it may be appropriate to fit a multiple linear regression model involving the predcitors x and x^2.

Exponential relationship

If the relationship between y and x is described by an exponential function, $y = \beta_0 \exp(\beta_1 x)$, then by taking logarithms of both sides we get

$$\log y = \log \beta_0 + \beta_1 x$$

or

$$\log y = \beta_0^* + \beta_1^* x.$$

So if we see an exponential pattern in a scatter plot of y against x, a transformation of y to $\log y$ might be considered.

Inverse exponential relationship

If the relationship between y and x is described by an inverse exponential function,

$$y = \beta_0 \exp(\beta_1 / x)$$

then by taking logarithms we get

$$\log y = \log \beta_0 + \beta_1 \frac{1}{x}$$

so an inverse exponential pattern in a scatter plot might indicate that a linear regression model with $\log(y)$ as the response and $1/x$ as the predictor may be appropriate.

There will be certain modelling situations in which some of these response/predictor relationships are natural: for instance, in modelling population growth in biology or ecology an exponential model might be appropriate. We give an example to illustrate the use of transformations when there is a nonlinear relationship between a response y and predictor x.

Example: surgical services data

This example is from your text book (Myers (1990), "Classical and Modern Regression with Applications (Second Edition)," Duxbury, Belmont, California, p. 299). These data were collected in an attempt to predict manpower requirements for surgical services at US naval hospitals. The response y is man hours per month measured for fifteen hospitals, and the predictor x is the number of surgical cases at these hospitals. The data are shown in the table below. A scatter plot of y against x is shown in Figure 5.47. The scatter plot certainly seems to show a nonlinear relationship between y and x. However, there are a number of different possibilities when it comes to modelling this nonlinear relationship. On the basis of the above discussion, we consider the simple linear regression model involving y and x, as well as a parabolic model and inverse exponential model.

For the simple linear regression model with y as response and x as the predictor, the PRESS statisic is $35,927,143$ and the sum of the absolute PRESS

y (Man Hours)	x (Surgical Cases)
1275	230
1350	235
1650	250
2000	277
3750	522
4222	545
5018	625
6125	713
6200	735
8150	820
9975	992
12200	1322
12750	1900
13014	2022
13275	2155

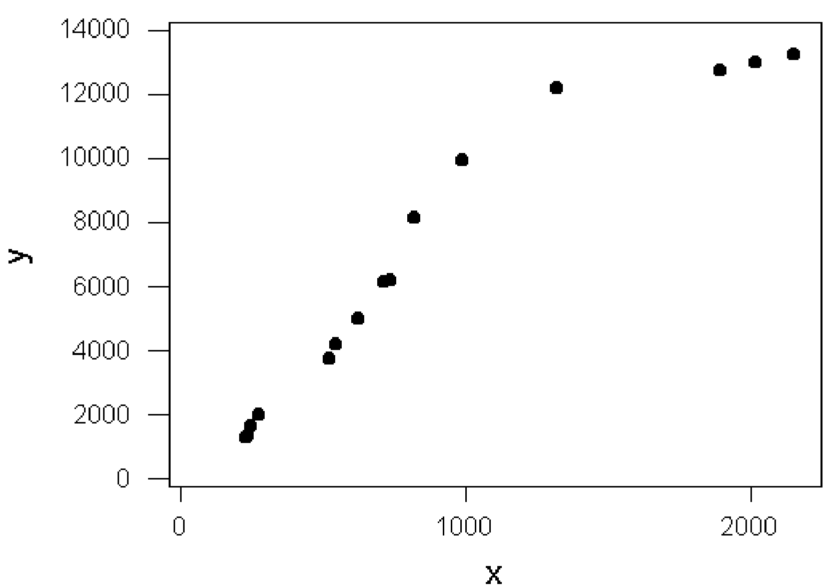

Figure 5.47: Scatter plot of y against x for surgical serices data.

residuals is 19064.7. For the parabolic model, (where y is the response and we use x and x^2 as predictors), the PRESS statisic is $7,057,692$ and the sum of the absolute PRESS residuals is 9348.06.

For the inverse exponential model, we fit a model with $\log y$ as the repsonse and $1/x$ as the predictor. We use the analogue of the PRESS residuals developed in the last lecture to derive statistics for comparison with the PRESS statistic and sum of absolute PRESS residuals. The sum of squared prediction errors on the original scale for this model is $20,684,140$, and the sum of absolute prediction errors is $14,780.7$.

From the above, it would appear that for predictive purposes the parabolic model is best. The inverse exponential model also seems better than the simple linear regression model with y as response and x as the predictor.

To conclude this subsection we make a cautionary remark about the effect of transformations of the response on the error structure of a model. Suppose that we have measured values of a response y, and corresponding values of a predictor x, and that there is some nonlinear relationship between the mean of y and x which can be linearized by a transformation. To be concrete, suppose we believe that the model

$$y_i = \beta_0 \exp(\beta_1 x_i) + \varepsilon_i$$

holds, where β_0 and β_1 are unknown parameters and where ε_i is a collection of zero mean errors with constant variance. Now, we know that by applying a log transformation to the mean of y_i we can obtain a function that is linear in x_i and linear in unknown parameters. But does this justify taking logarithms of the responses y_i and fitting a linear model? Taking logs of the above equation does not yield a model in which

$$\log y_i = \log \beta_0 + \beta_1 x_i + \eta_i$$

where the η_i are zero mean with a constant variance. In other words, if we have reasons for believing the first model stated above holds, then applying a transformation which linearizes the mean in unknown parameters does not justify the linear model above: we have to consider the effect on the errors of the transformation. It may be better to work with the original nonlinear model: we will not discuss nonlinear regression in this course, but you may see more on this if you do further statistics courses.

5.13 The Box-Tidwell Procedure

In the next two subsections we consider ways of choosing transformations of the predictors in multiple linear regression. Of course, simple scatter plots of the response against each of the predictors are not helpful for diagnosing the need to transform when we have two or more predictors. Relationships among the predictors may obscure the role that each individual predictor plays in a multiple linear regression model. We have already discussed ways of diagnosing the need to transform in a multiple regression. In particular, we have looked at partial residual and regression plots and added variable plots. However, it may not always be clear from these plots what kind of nonlinear transformation of a predictor or predictors may be best.

Weisberg (1985), "Applied Linear Regression (Second Edition)", Wiley, New York, distinguishes two situations in considering the need to transform the predictors in linear regression models. In the case where the expected response takes a minimum or maximum within the range of the predictors, it is natural to introduce powers of the original predictors and products of the original predictors (that is, we consider polynomial functions for approximating the response). In the case where the expected response is increasing or decreasing in the predictors, it may be more natural to consider so-called power transformations of the original predictors. For instance, if we wish to transform the first predictor x_1 in a multiple linear regression model, in the first situation we might introduce an additional term involving x_1^2 into the model. For the case where power transformations are considered, we replace the term involving x_1 with a term of the form $x_1^{\alpha_1}$ for some exponent α_1 (for instance, we might consider $\alpha_1 = 1/2$, the square root of x_1). In this subsection we discuss a procedure for estimating the exponent α_1 when a power transformation is considered, and in the next subsection we briefly consider polynomial regression.

Estimating a power transformation

The method we present for estimating an exponent in a power transformation is similar to one suggested by Box and Tidwell, and is developed in Weisberg ("Applied Linear Regression (Second Edition)", Wiley, New York, 1985, pp. 153–155). Suppose we have a response y and predictors $x_1, ..., x_k$ and consider the multiple linear regression model

$$y_i \quad = \quad \beta_0 + \beta_1 x_{i1} + ... + \beta_k x_{ik} + \varepsilon_i \tag{37}$$

where the ε_i are zero mean errors and the usual linear model assumptions hold. For simplicity suppose we wish to investigate the need to transform just the first predictor x_1 (although the discussion which follows may be generalized to the situation where we consider the need to transform more than one predictor simultaneously). We write the above model as

$$y_i = \beta_0 + \beta_1 x_{i1} + \sum_{j=2}^{k} \beta_j x_{ij} + \varepsilon_i.$$

Now consider the more general model

$$y_i \;=\; \beta_0 + \beta_1 x_{i1}^{\alpha_1} + \sum_{j=2}^{k} \beta_j x_{ij} + \varepsilon_i. \tag{38}$$

This model reduces to the original one if we set $\alpha_1 = 1$. When $\alpha_1 = 0$ we use the predictor $\log x_1$ instead of $x_1^{\alpha_1}$ (to see why this is natural, observe that if α_1 is known, fitting a linear model with $x_1^{\alpha_1}$ as a predictor is equivalent to fitting a model with

$$\frac{x_1^{\alpha_1} - 1}{\alpha_1}$$

as predictor, and that this last expression approaches $\log x_1$ as α_1 approaches zero.) The model (38) is nonlinear in the unknown parameters when α_1 is introduced. However, Box and Tidwell suggest a method based on linear modelling techniques that allows us to test for the need to transform x_1 and which allows us to get a crude estimate of α_1.

The method is based on performing a first order Taylor series expansion of $x_{i1}^{\alpha_1}$ about $\alpha_1 = 1$. We have that $x_{i1}^{\alpha_1}$ is approximately

$$x_{i1} + (\alpha_1 - 1) x_{i1} \log x_{i1}.$$

Substituting this approximation into (38) we obtain

$$y_i \;=\; \beta_0 + \beta_1 x_{i1} + \beta_1(\alpha_1 - 1) x_{i1} \log x_{i1} + \sum_{j=2}^{k} \beta_j x_{ij} + \varepsilon_i. \tag{39}$$

and if we write $\eta = \beta_1(\alpha_1 - 1)$ we have a model which is linear in unknown parameters $\beta_0, ..., \beta_k, \eta$ where we have introduced a new predictor $x_{i1} \log x_{i1}$. Examining an estimate of η can help us to decide whether there is a need to transform (since $\eta = 0$ when either $\alpha_1 = 1$ or $\beta_1 = 0$).

The crude method for estimating α_1 suggested by Weisberg (1985) is as follows. First we fit the linear model (37) where x_1 is untransformed. Write b_1 for the estimate of β_1 in this model. Then fit the model (39), and write $\widehat{\eta}$ for the estimate of η in this model. Since $\eta = \beta_1(\alpha_1 - 1)$, one natural estimate of α_1 is obtained by solving for α_1 in this equation and plugging in b_1 and $\widehat{\eta}$: we obtain an estimate

$$\widehat{\alpha_1} = \frac{\widehat{\eta}}{b_1} + 1.$$

Weisberg warns that this crude estimator may not perform very well in some situations, particularly if the partial t-statistic for b_1 is small (since in this case β_1 is possibly close to zero, and in the above expression where b_1 appears in the denominator we do not expect the estimate $\widehat{\alpha_1}$ to be reliable). We give some examples to illustrate the use of the Box-Tidwell procedure.

Example: size of Romanesque churches

The following example is discussed in Weisberg (1985) and concerns a data set supplied by Stephen J. Gould. The data set consists of two variables, which are the measured area (in hundreds of square metres) and perimeter (in hundreds of metres) for 25 Romanesque churches. The data are given in the table below. A

Perimeter	Area	Perimeter	Area
3.48	38.83	3.14	34.27
3.69	43.92	2.04	17.61
1.43	9.14	1.77	13.37
2.05	16.66	0.59	2.04
3.05	36.16	0.69	2.22
4.19	38.66	0.50	1.46
2.43	17.74	0.69	1.92
2.40	19.46	0.63	1.86
2.72	23.00	0.58	1.69
2.99	29.75	0.86	3.31
4.78	51.19	0.41	1.13
1.33	6.60	1.23	6.74
1.67	9.04		

scatter plot of area against perimeter is shown in Figure 5.48. From the scatter plot, evidently a simple linear regression model of area against perimeter does not seem to be reasonable here. Weisberg (1985) gives an argument which suggests using the square root of the area as the response. We will return to a discussion of transformation of the response in this example in later lectures. We investigate the need to transform the predictor perimeter in a model for the square root of area. We consider first the model

$$\sqrt{\text{Area}_i} = \beta_0 + \beta_1 \text{Perimeter}_i + \varepsilon_i.$$

The estimate of β_1 when this model is fitted is $b_1 = 1.544$. Now consider the model

$$\sqrt{\text{Area}_i} = \beta_0 + \beta_1 \text{Perimeter}_i + \eta \text{Perimeter}_i \log(\text{Perimeter}_i) + \varepsilon_i.$$

When we fit this model, we get $\widehat{\eta} = -0.6726$. Hence for a model involving a power transformation $\text{Perimeter}^{\alpha_1}$ a crude estimate of α_1 is approximately

$$\widehat{\alpha}_1 = \frac{-0.6726}{1.544} + 1 = 0.56.$$

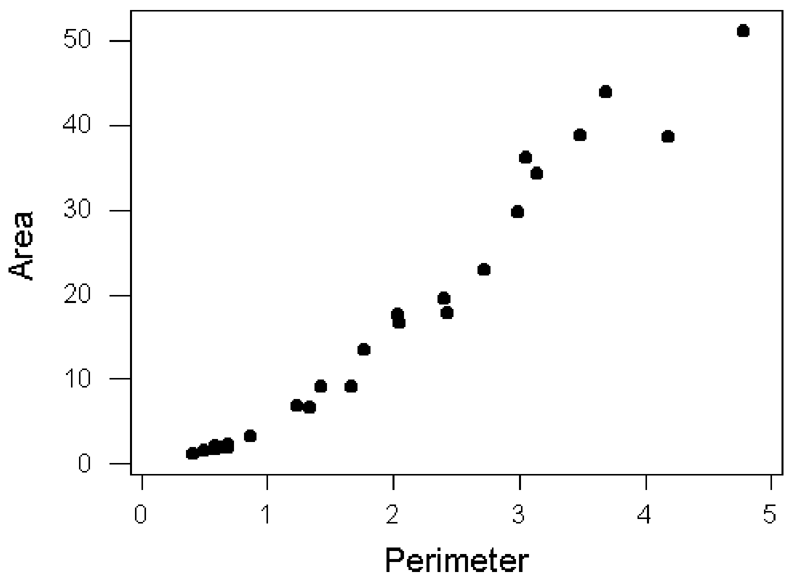

Figure 5.48: Scatter plot of area against perimeter for 25 Romanesque churches.

Since this value is close to 0.5, we consider a model involving a square root transformation. A scatter plot of the square root of area against the square root of perimeter is shown in Figure 5.49. We may still not be entirely happy with this

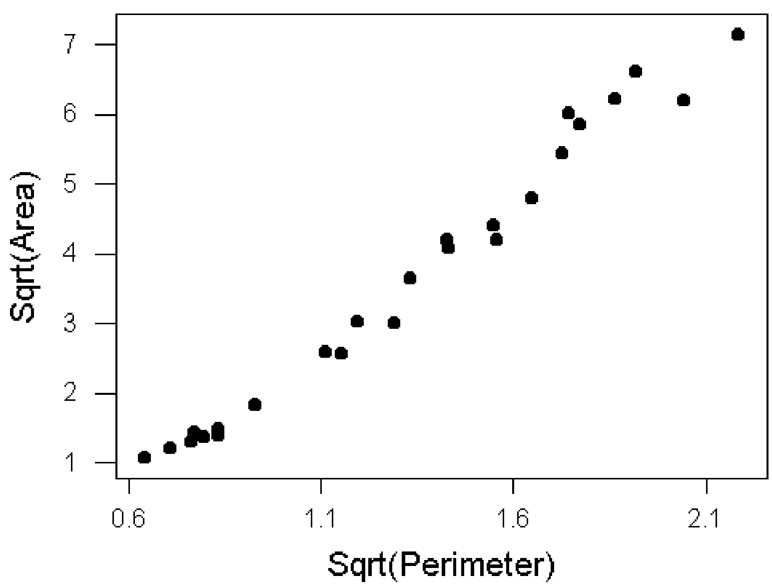

Figure 5.49: Scatter plot of square root of area against square root of perimeter for 25 Romanesque churches.

model. The relationship between the transformed response and transformed predictor does not seem to be quite linear (there seem to be some departures from linearity for large and small perimeter values) and possibly the variance increases as the mean response increases.

5.14 Polynomial regression

We mentioned in the last subsection that polynomial regression models are used by many analysts when an expected response takes a minimum or maximum value within the range of the predictors. In this subsection we discuss polynomial regression models, the meaning of interaction in polynomial regression models with multiple predictors, and give some examples.

Polynomials are of course a flexible class of models for a relationship between the expected response and one or more predictors in a regression problem (an argument involving a Taylor series expansion shows that we can approximate any suitably smooth function by a polynomial of sufficiently high order). However, there are difficulties with the use of high order polynomial models (both numerical and theoretical) as a general tool for function approximation in regression modelling. There are other (probably better) flexible ways of approximating a general smooth response function within the framework of the linear model. However, in many situations where a model which is linear in the original predictors is not appropriate we may wish to consider second order polynomial or other low order polynomial models for the expected response.

For simplicity, let us just consider a situation in which we have a measured response y and two predictors x_1 and x_2. Suppose we have fitted a linear model with x_1 and x_2 and added variable plots indicate the need for a nonlinear transformation of x_1 or x_2 or both. We might consider a second order polynomial model

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \varepsilon_i.$$

The term $\beta_5 x_1 x_2$ is called an interaction term. To see where this name comes from, suppose that $\beta_5 = 0$ (that is, assume there is no interaction term). Then observe that if we change x_1, the resulting change in the expected value of the response is the same regardless of the value of x_2. Similarly, if we change x_2, the change in the expected response is the same regardless of the value of x_1. However, if $\beta_5 \neq 0$, this is no longer the case: a change in one of the predictors does not produce the same change in the expected response regardless of the value of the other variable.

Example: fast food restaurants

As an example of the use of a polynomial regression model we consider the data on fast food restaurants discussed in previous lectures. For this data set we were interested in predicting annual gross sales for a store in a fast food restaurant chain based on mean annual household income and mean age of children for families in the area around the store. The variables are Sales (the response), Income (the mean annual household income) and Age (the mean age of children). We might expect that the expected response would achieve a maximum within the range of the predictors here: we expect that increasing income and age cause sales to increase up to a point, but this pattern would not persist indefinitely. We might expect that middle income families with teenage children would be most likely to consume fast food. The variables `Incomesq` and `Agesq` here are the square of `Income` and `Age`, and the variable `Income*Age` is the product of `Income` and `Age`.

```
Regression Analysis
```

```
The regression equation is
Sales = - 1134 + 173 Income + 23.5 Age - 3.73 Incomesq - 3.87 Agesq
           + 1.97 Income*Age

Predictor        Coef        StDev           T         P
Constant      -1134.0        320.0       -3.54     0.002
Income         173.20        28.20        6.14     0.000
Age             23.55        32.23        0.73     0.474
Incomesq      -3.7261       0.5422       -6.87     0.000
Agesq          -3.869        1.179       -3.28     0.004
Income*A       1.9673       0.9441        2.08     0.051

S = 44.70       R-Sq = 90.7%     R-Sq(adj) = 88.2%

Analysis of Variance

Source            DF           SS           MS          F         P
Regression         5       368140        73628      36.86     0.000
Residual Error    19        37956         1998
Total             24       406096

Source       DF      Seq SS
Income        1       77008
Age           1       55063
Incomesq      1      202845
Agesq         1       24551
Income*A      1        8674
```

The p-value for the partial t-statistic for `Income*Age` is 0.051, which suggests dropping the interaction term from the model at the five percent level in the presence of the other terms (although this is borderline). The p-value for the partial t-statistic for `Age` is large here also.

5.15 Box-Cox transformation of the response

In previous lectures we considered ways of transforming the predictors in multiple linear regression. In particular, we considered the Box-Tidwell procedure and polynomial regression models. These methods were possibly helpful when there was evidence of a nonlinear relationship between the expected response and the original predictors.

We have also considered the possibility of transforming the response in a simple or multiple regression model. So far we have only considered the use of a transformation of the response to achieve constancy of error variance when certain mean/variance relationships hold. However, a transformation of the response can also be helpful for achieving linearity of the mean response in the original predictors, and for making the assumption of normality of errors more reasonable. In this subsection we consider a class of transformations indexed by a single parameter, and a way of choosing a transformation within this class which attempts to make the specification of the mean structure, constancy of error variance assumption and normality of the errors reasonable on the transformed scale.

The class of transformations we consider is the class of *Box-Cox* transformations. We saw this class of transformations when we discussed the Box-Tidwell procedure. If y is the response (which we assume for the moment to be positive), then we consider

$$w = \begin{cases} \frac{y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(y) & \text{if } \lambda = 0. \end{cases}$$

Here λ is a real valued parameter. The reason for defining $w = \log(y)$ when $\lambda = 0$ is that $(y^\lambda - 1)/\lambda$ approaches $\log(y)$ as λ approaches zero (see our discussion of the Box-Tidwell procedure). If we find upon fitting a multiple linear regression model that the assumptions of the linear model are violated (either specification of the mean structure seems incorrect or the error variance does not seem to be constant for instance) then one thing that might be considered is to transform the response using the above transformation for a given value of λ. How do we choose λ? Box and Cox have suggested an automatic way of choosing λ based on the data, which we now describe.

Estimating $\boldsymbol{\lambda}$

The proposal by Box and Cox for estimating λ is to use maximum likelihood. If for a fixed λ the transformed responses w_i are normal and uncorrelated with common variance σ^2, then we can write down the joint density of $w_1, ..., w_n$. If $w = (w_1, ..., w_n)^\top$, we write

$$w = X\beta + \varepsilon$$

where as usual X is the design matrix ($n \times p$), β is a $p \times 1$ vector of parameters and ε is a vector of uncorrelated zero mean normal errors with variance σ^2. We can easily write down the density of w, which is of course a product of univariate normal densities. From this density function and from the relationship between w_i and the untransformed response y_i, we can write down the joint density of $y = (y_1, ..., y_n)^\top$. In terms of the parameters β, σ^2 and λ, this joint density is

$$\frac{1}{(2\pi)^{n/2}(\sigma^2)^{n/2}} \exp\left(-1/(2\sigma^2)(w - X\beta)^\top(w - X\beta)\right) \left(\prod_{i=1}^n y_i\right)^{\lambda-1}.$$

(If you are familiar with transformations of random variables, this is easily derived: if not, you may take this on trust).

This joint density function considered as a function of the parameters β, σ^2 and λ is the likelihood function. It can be shown that if we take logarithms and maximize with respect to β and σ^2, then ignoring additive constants and writing $RSS(\lambda)$ for the residual sum of squares when we use the value λ in the Box-Cox transformation, we obtain a function of λ, $L(\lambda)$:

$$L(\lambda) = -\frac{n}{2}\log(RSS(\lambda)) + n(\lambda - 1)\log\left((\prod_{i=1}^{n} y_i)^{1/n}\right).$$

To determine a rough value for λ, we can consider a grid of possible λ values between -2.0 and 2.0 say, compute the residual sum of squares $RSS(\lambda)$ by fitting the model with the appropriately transformed response and then plot $L(\lambda)$. We can read a crude value for λ off the graph (it is common to round off to an interpretable value, say to the nearest half so that we might have a log or square root transformation for instance). Generally the function $L(\lambda)$ will have a maximum in the interval $[-2, 2]$: if this is not the case, the Box-Cox transformation is probably not useful. We have assumed in this discussion that the responses are positive. If this is not the case, then we can add a constant to the responses so that they are, but the best choice for the constant to be added may not be clear. We give an example to illustrate the use of the Box-Cox transformation.

Example: size of Romanesque churches

As an illustration of the Box-Cox procedure we consider once more the data on the size of Romanesque churches considered in our discussion of the Box-Tidwell procedure. For this data set, there were measurements of area and perimeter for 25 Romanesque churches. We consider using a Box-Cox transformation of the response for this data set. A plot of $L(\lambda)$ against λ for the data on Romanesque churches is shown in Figure 5.50 This plot seems to indicate a Box-Cox transformation with a value for λ of approximately 0.5 would be appropriate.

Example: surgical services data

As a further example of the application of Box-Cox transformation, we consider the surgical services data (also discussed in previous lectures). Recall that for this data set we had a response y which was a measure of manpower and a predictor x which was the number of surgical cases at 15 US Naval hospitals. The goal of the study was to predict manpower needs. A plot of $L(\lambda)$ against λ for this data set is shown in Figure 5.51. The Box-Cox procedure does not seem to indicate a need to transform here.

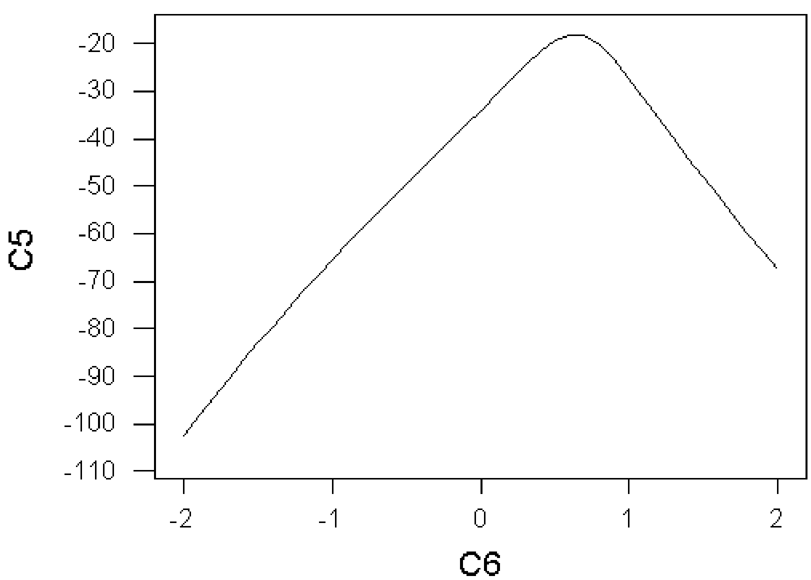

Figure 5.50: Plot of λ against $L(\lambda)$ for data on Romanesque churches.

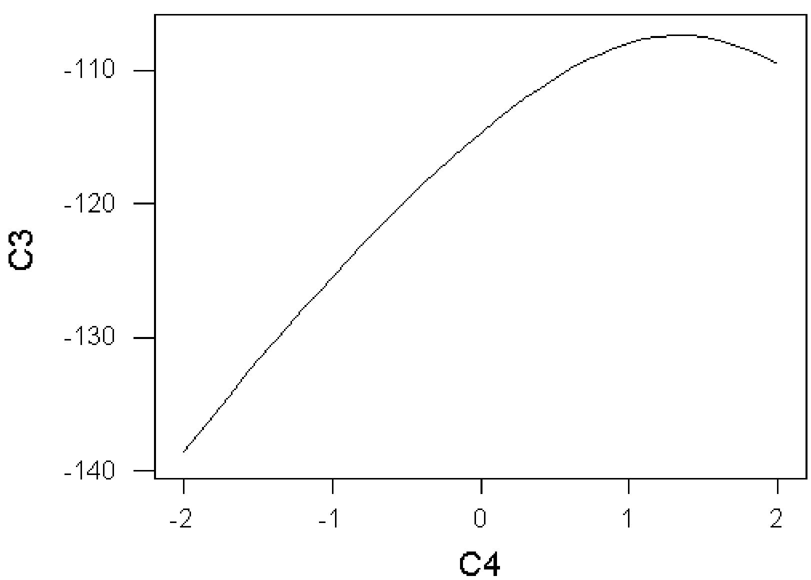

Figure 5.51: Plot of λ against $L(\lambda)$ for data on surgical services.

In the above examples we chose a rough value for λ by looking at a plot of $L(\lambda)$ against λ. However, in deciding on a suitable value for λ in the Box-Cox procedure it is helpful to have some measure of the uncertainty about the maximizer $\widehat{\lambda}$ of $L(\lambda)$. It can be shown that an approximate $100(1-\alpha)$ percentage confidence interval for λ based on $\widehat{\lambda}$ is given by

$$\{\lambda : L(\widehat{\lambda}) - L(\lambda) \leqslant \frac{1}{2}\chi^2_{1;\alpha}\}$$

where $\chi^2_{1;\alpha}$ is the upper 100α percentage point of a χ^2_1 distribution.

Example: Romanesque churches

Consider again the data on Romanesque churches. For this data set we said that a Box-Cox transformation with a value of λ of approximately 0.5 seemed to be indicated. To compute an approximate 95 percent confidence interval we need the upper 5 percentage point of a χ^2_1 distribution, which is approximately 3.8415. In generating the graph of Figure 5.50 I computed $L(\lambda)$ for values of λ 0.1 units apart between -2.0 and 2.0. The values at $\lambda = 0.4, 0.5, 0.6, 0.7$ and 0.8 are respectively $-21.554, -19.212, -17.925, -18.165$ and -20.038. If we assume that the maximizer is exactly at 0.6 (we can compute the maximizer of $L(\lambda)$ more precisely by using a finer grid) then the approximate 95 percent confidence interval for λ includes values of λ for which $L(\lambda)$ is within $0.5 \times 3.8415 = 1.92075$ of the maximum value. Based on this guideline, the value of 0.5 lies in an approximate 95 percent confidence interval, and a square root transformation of the response seems appropriate.

Example: Surgical services data

For the surgical services data set above, we stated that the need for a transformation of the response did not seem to be indicated by the Box-Cox procedure. Is this inference justified based on a confidence interval for λ? Again as a very crude guideline we can consider values of λ on our grid in the plot which lie within 1.92075 of the maximum value: the value $\lambda = 1.0$ satisfies this criterion, and so a transformation does not seem to be warranted.

In our discussion of transformations in this course we have presented a wide variety of techniques, and it may not be clear in any given situation which technique to apply or how to use these techniques in combination with each other. Weisberg, "Applied Linear Regression (Second Edition)," Wiley, New York, 1985, p. 156, suggests a general algorithm to apply when the data are strictly positive. He suggests first transforming any predictors which have a maximum to minimum ratio of larger than about ten to a log scale, then using the Box-Cox procedure and finally considering the Box-Tidwell procedure for predictors which have large t-values. It must be said that in many cases we may not wish to use a transformation of the response at all, and transformations can be overused.

6 Categorical predictors

In this section we discuss model building in multiple linear regression when some of our predictors are categorical variables. This discussion will lead on to consideration of the less than full rank linear model in the next section.

As an example of a categorical predictor in multiple linear regression consider the data set on inflation rates and central bank independence that we discussed in previous lectures and in tutorial five. For this data set we had for a number of countries the average annual inflation rate over $1980 - 1989$, as well as two measures of central bank independence (a questionaire measure of independence and a legal measure of independence). In addition to these two quantitative predictor variables, there was a predictor which took the value 1 for developed economies and 0 for developing economies. This last predictor variable is an example of a categorical predictor: the values 1 and 0 taken by this predictor are just arbitrary labels, numeric values that have no particular meaning except to distinguish two different groups within the observations. Instead of the values 1 and 0 we could equally well have used the labels A and B to distinguish the groups. We wish to develop methods for handling categorical variables like this one within the framework of the linear model.

To take another example, suppose a new drug has been developed for the treatment of high blood pressure. A study was undertaken with a group of patients having high blood pressure into the effectiveness of the new drug. Each patient was assigned at random to either a treatment group (patients in this group receive the new drug) or a control group (patients in this group receive the standard treatment for high blood pressure). After a month we measure the change in the blood pressure of each patient compared to their blood pressure at the beginning of the trial. We are interested in investigating the effect of the drug on the response (change in blood pressure), perhaps adjusting for the effect of other quantitative characteristics of the patients (such as age for instance). So for each patient we have the response (change in blood pressure) some quantitative predictors (such as age) as well as a predictor which records whether the patient was in the treatment or control group. This last predictor is an example of a categorical predictor variable with two levels: the two levels are "treatment" and "control".

6.1 Categorical predictors with two levels

We consider first the situation where we have a single quantitative predictor and a categorical predictor with just two levels. We write y for the response variable, x for the quantitative predictor, and z for the categorical predictor. Suppose that the categorical variable can take on values A and B.

Now define a binary quantitative variable w (sometimes called a dummy variable) from the predictor z as

$$w = \begin{cases} 1 & \text{if } z \text{ takes the value A} \\ 0 & \text{if } z \text{ takes the value B} \end{cases} \tag{40}$$

and consider what happens when we use w and x as predictors in a multiple linear regression with y as the response. That is, consider the model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 w_i + \varepsilon_i,$$

$i = 1, ..., n$, where β_0, β_1 and β_2 are unknown parameters and ε_i, $i = 1, ..., n$ is a collection of zero mean errors uncorrelated with a common variance σ^2.

If observation i has $w_i = 1$ (that is, if this observation belongs to the A group for the categorical predictor z) then we have

$$y_i = \beta_0 + \beta_2 + \beta_1 x_i + \varepsilon_i.$$

On the other hand, suppose that $w_i = 0$. In this case, we have

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

So the effect of introducing the dummy variable w into the regression is to have a shift in the intercept for the regression of y on x as we move between the two groups defined by z. This may be a sensible model. We can interpret the parameter β_2 as being the effect of being in group A compared to being in group B, and if we want to see if there is any difference between the two groups (for instance, if we want to see in our blood pressure example whether the new drug is more effective than the standard treatment after adjusting for the quantitative predictor age) then we can simply look at the partial t statistic for β_2.

The appropriateness of the above model involving the dummy variable w depends on the assumption that the slope of the regression relationship between y and x does not change for the groups A and B. The effect due to the group (A or B) does not depend on the level of the quantitative variable x and we say in this situation that there is *no interaction* between x and z. Effectively, the relationship between the expected value of y and x for the two groups is described by a pair of parallel lines, with one line for each group.

We can of course envisage a more general model in which the mean response varies linearly with x in the two groups, but with a different slope and intercept for the line in each group (that is, the relationship between the mean of y and x is described by a pair of lines which may not be parallel). Consider defining a new predictor variable as the product of the dummy variable w and the quantitative predictor x. Now suppose we fit the model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 w_i + \beta_3 x_i w_i + \varepsilon_i.$$

What happens in this situation for observations from the two groups A and B? If $w_i = 1$ (observation belongs to group A) then we have that

$$y_i = \beta_0 + \beta_2 + (\beta_1 + \beta_3)x_i + \varepsilon_i.$$

On the other hand, if $w_i = 0$ (observation belongs to group B) then we have that

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

So by fitting a model with predictors x_i, w_i and $x_i w_i$ we are allowing a linear relationship between the expected response and x in which there is a different intercept and slope within each of the groups A and B. We could test for interaction by looking at the partial t statistic for β_3 in the above model (if β_3 is zero, then we are back in the situation we had before with parallel lines for the two groups) and we could test whether there is any effect due to the categorical variable by testing

$$H_0 : \beta_2 = \beta_3 = 0$$

against the alternative that β_2 and β_3 are not both zero by doing an F-test.

In the above model it is important to note that we are assuming separate linear relationships between the expected value of y and x within the two groups A and B, but with a common error variance σ^2. If we believe that there is a different error variance for the two groups, then we can simply split the data and fit separate simple linear regressions of y on x within the two groups.

Example: inflation and central bank independence

We illustrate some of the ideas we have discussed by looking at the example on inflation and central bank independence. Here the categorical variable `DEV` already has levels which are either zero or one to distinguish the two groups, so we can use `DEV` as a quantitative predictor in the manner described above. The two quantitative predictors measuring central bank independence were `QUES` and `LEGAL` (the questionnaire and legal measures of independence) and in this example we consider only modelling inflation rate in terms of `QUES` and `DEV`. First we fit a multiple linear regression model with the predictors `QUES` and `DEV`. Recall that in this model we are postulating that there is a linear relationship between the expected inflation rate and `QUES` for both developed and developing economies with the same slope but a different intercept within the two groups.

```
Regression Analysis

The regression equation is
INF = 59.6 - 4.07 QUES - 22.8 DEV

Predictor          Coef       StDev            T        P
```

```
Constant         59.62        14.32        4.16    0.001
QUES            -4.072        2.753       -1.48    0.156
DEV             -22.79        12.13       -1.88    0.076

S = 22.65        R-Sq = 43.0%     R-Sq(adj) = 37.0%

Analysis of Variance

Source            DF          SS          MS          F         P
Regression         2      7345.2      3672.6       7.16     0.005
Residual Error    19      9750.0       513.2
Total             21     17095.3

Source       DF      Seq SS
QUES          1      5533.4
DEV           1      1811.8
```

Assuming that this model is adequate, we can test for whether being a developed economy is related to inflation after adjusting for `QUES` by looking at the partial t-statistic for DEV and its associated p-value. The conclusion of the test for $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$ here is uncertain: we would accept H_0 at the 5 percent level, but reject at the 10 percent level.

Now consider the model where we have a separate linear relationship between expected inflation rate and `QUES` within each `DEV` group. We fit a multiple linear regression model with inflation rate as the response and `QUES`, `DEV` and `QUES*DEV` as predictors.

```
Regression Analysis

The regression equation is
INF = 66.8 - 5.64 QUES - 57.1 DEV + 5.31 QUES*DEV

Predictor         Coef        StDev           T         P
Constant         66.80        16.58        4.03    0.001
QUES            -5.641        3.300       -1.71    0.105
DEV             -57.06        41.03       -1.39    0.181
QUES*DEV         5.314        6.074        0.87    0.393

S = 22.79        R-Sq = 45.3%     R-Sq(adj) = 36.2%

Analysis of Variance

Source            DF          SS          MS          F         P
```

```
Regression         3      7742.9      2581.0      4.97    0.011
Residual Error    18      9352.3       519.6
Total             21     17095.3

Source        DF      Seq SS
QUES           1      5533.4
DEV            1      1811.8
QUES*DEV       1       397.7
```

Assuming this model is an appropriate one, we can test for the presence of interaction between QUES and DEV by looking at the partial t statistic and p-value for the predictor QUES*DEV. The p-value here for testing $H_0 : \beta_3 = 0$ against the alternative $H_1 : \beta_3 \neq 0$ is 0.393, which would indicate acceptance of H_0 at the 5 percent level. That is, there seems to be no real evidence of any interaction.

We can test whether the level of DEV seems to have any relationship to inflation rate in the presence of QUES by testing

$$H_0 : \beta_2 = \beta_3 = 0$$

against the alternative

$$H_1 : \beta_2, \beta_3 \text{ not both zero.}$$

We use an F test. The value of the test statistic is the reduction of the residual sum of squares when DEV and QUES*DEV are added to the model, divided by the difference in the number of parameters for the larger and smaller models (which is two here), all divided by the mean square error. This is a realization of an F random variable with 2 and $n - 4 = 18$ degrees of freedom under H_0. The value of the test statistic is

$$\frac{(1811.8 + 397.7)/2}{519.6} = 2.13,$$

which can be compared with the upper 5 percentage point of an F distribution with 2 and 18 degrees of freedom (approximately 3.55). We accept the null hypothesis here at the 5 percent level (that is, DEV does not seem to be related to inflation rate in the presence of QUES according to this test).

In all these hypothesis tests we are assuming that the model we have fitted is appropriate. The appropriateness of all the models we have considered is in fact very doubtful here. Figures 6.52 and 6.53 show plots of the standardized residuals against fitted values and the predictor DEV respectively for the model involving QUES, DEV and QUES*DEV. Clearly here there seems to be a problem with the constancy of error variance assumption: in particular, the variability of the inflation rate would seem to depend on the level of DEV. A better model than the ones we have considered so far might be simply to fit separate simple linear regressions of INF on QUES within the two groups defined by levels of DEV.

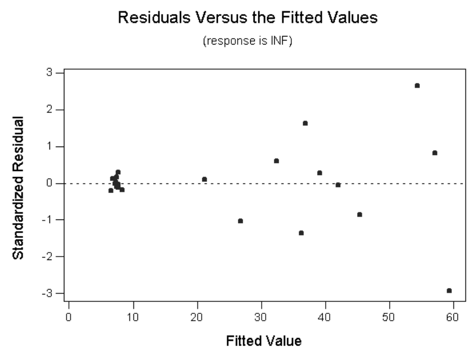

Figure 6.52: Scatter plot of standardized residuals against fitted values.

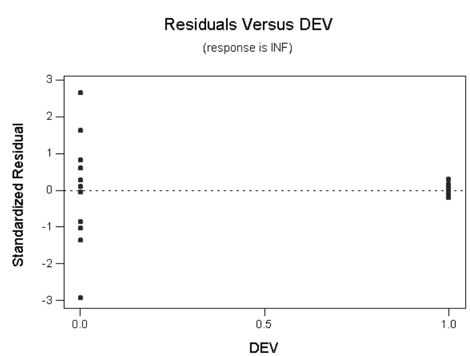

Figure 6.53: Scatter plot of standardized residuals against DEV.

Example: age and growth of mussel species.

The following example is from Myers, "Classical and Modern Regression with Applications (Second Edition)", Duxbury, Belmont, California, 1990, p. 154. In a project to study age and growth characteristics of selected mussel species from Southwest Virginia, the data below were taken from two distinct locations. It was desired to investigate whether location was a significant factor in the growth of the mussels.

Table 6.2: Age and weight of mussels: location 1.

Age	Weight (g)	Age	Weight (g)
3	0.44	11	3.96
3	0.50	11	3.84
3	0.66	12	5.58
3	0.78	12	5.64
4	1.20	12	4.26
4	1.18	13	6.00
4	1.08	13	2.54
6	1.12	13	3.82
6	1.72	14	4.50
7	1.04	14	5.18
8	2.62	14	4.04
9	1.88	15	6.38
10	2.26	15	4.08
11	4.10	16	4.56
11	2.12		

We can define a dummy variable `Location` which is 0 for location 1 and 1 for location 2. If we do a multiple linear regression of `Weight` on `Age`, `Location` and `Age*Location`, then we obtain the following.

```
Regression Analysis

The regression equation is
Weight = - 0.483 + 0.365 Age - 0.759 Location + 0.290 Age*Location

Predictor        Coef       StDev          T        P
Constant      -0.4828      0.3653      -1.32    0.191
Age           0.36494     0.03310      11.03    0.000
```

Table 6.3: Age and weight of mussels: location 2.

Age	Weight (g)	Age	Weight (g)
3	0.76	8	2.52
4	1.38	8	3.90
5	1.20	10	3.94
5	1.76	10	6.22
6	2.60	10	4.96
6	2.16	13	9.02
6	2.64	13	8.20
6	2.52	13	8.26
6	3.08	14	6.40
6	2.12	15	10.06
7	2.72	15	8.60
7	2.96	18	11.06
8	4.54	19	10.78
8	5.26	22	12.04
8	5.60	24	13.92

```
Location       -0.7592       0.5050       -1.50    0.138
Age*Loca       0.28998      0.04505        6.44    0.000

S = 0.8893      R-Sq = 91.7%     R-Sq(adj) = 91.3%

Analysis of Variance

Source              DF          SS          MS         F        P
Regression           3      536.45      178.82    226.09    0.000
Residual Error      61       48.25        0.79
Total               64      584.70

Source         DF      Seq SS
Age             1      428.11
Location        1       75.57
Age*Loca        1       32.77
```

We can do an F test for the significance of the `Location` term and its interaction with `Age`. The value of the appropriate test statistic for the F test is

$$\frac{(75.57 + 32.77)/2}{0.79} = 68.57.$$

This is to be compared with the upper 5 percentage point of an F distribution with 2 and 61 degrees of freedom (which is approximately 3.15). So we would reject the hypothesis that `Location` has no effect on mussel weight after adjusting for age.

6.2 Categorical variables with more than two levels

Categorical variables with more than two levels can be handled in much the same way as for the previous case. An example will help to illustrate the idea: the following example is from Keller, Warrack and Bartel, "Statistics for Management and Economics: A Systematic Approach (Second Edition)," Wadsworth, Belmont, California, 1990. Suppose that a real estate agent is interested in predicting the selling price of a house (y), and that the agent believes that the selling price of houses in the local area is related to the size of the house (x), and the kind of energy used for heating (measured as a categorical variable z with values "electricity", "heating oil" or "natural gas"). The agent wishes to develop a multiple linear regression model with selling price as the response and predictors x and z (size of house and type of heating).

Here z is a categorical variable with three levels. To handle this situation we proceed as follows. Define *two* dummy variables w_1 and w_2 to be

$$w_1 = \begin{cases} 1 & \text{if } z \text{ takes the value "electricity"} \\ 0 & \text{otherwise} \end{cases}$$

and

$$w_2 = \begin{cases} 1 & \text{if } z \text{ takes the value "heating oil"} \\ 0 & \text{otherwise.} \end{cases}$$

The dummy variables w_1 and w_2 together code the information in z, since w_1 is one only for those homes heated by electicity, w_2 is one only for those homes heated by heating oil, and w_1 and w_2 are both zero only for those homes heated by natural gas. As before we can consider fitting a multiple linear regression model with y as the response and with predictors x, w_1 and w_2. Fitting this model allows a linear relationship between expected selling price and size of house for each group defined by z, with a common slope within each group (the mean structure is defined by three parallel lines). If we want to allow separate linear relationships within the three groups then we can fit a model involving the predictors x, w_1, w_2, xw_1 and xw_2. Hypothesis testing can also proceed in a similar way to before.

6.3 Hypothesis testing for categorical variables with more than two levels

In the last subsection we started a discussion of what to do in the linear model when there is a categorical predictor taking on more than two values. We began to discuss an example on real estate prices. For this example the response y was the selling price of fifteen different houses in a certain area, and it was desired to predict selling price based on the size of the house and the method used for heating the house. The method of heating is a categorical predictor, with values "electricity", "heating oil" and "natural gas". The data are shown in Table 6.4. In the case of a categorical predictor variable taking on two values,

Table 6.4: Real estate prices data.

Price (y)	Size (x)	Heating (z)
89.5	20.0	electricity
79.9	14.8	electricity
83.1	20.5	heating oil
56.9	12.5	heating oil
66.6	18.0	heating oil
82.5	14.3	natural gas
126.3	27.5	electricity
79.3	16.5	electricity
119.9	24.3	natural gas
87.6	20.2	heating oil
112.6	22.0	natural gas
120.8	19.0	natural gas
78.5	12.3	natural gas
74.3	14.0	electricity
74.8	16.7	heating oil

our approach was to define a binary dummy variable which was then introduced as a quantitative predictor into the model. Using this dummy variable (and the products of this dummy variable with any quantitative predictors) we were able to test for the presence of an effect due to the categorical predictor, and to test for interaction between the categorical predictor and any quantitative predictors. A similar approach can be followed when dealing with a categorical variable with more than two levels. For the case of the real estate prices data, and denoting the selling price by y, the size of the house by x and the categorical predictor heating type by z, we define binary dummy variables w_1 and w_2 by

$$w_1 = \begin{cases} 1 & \text{if } z \text{ takes the value "electricity"} \\ 0 & \text{otherwise} \end{cases}$$

and

$$w_2 = \begin{cases} 1 & \text{if } z \text{ takes the value "heating oil"} \\ 0 & \text{otherwise.} \end{cases}$$

Now consider the model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 w_{i1} + \beta_3 w_{i2} + \beta_4 x_i w_{i1} + \beta_5 x_i w_{i2} + \varepsilon_i.$$

Again we can ask ourselves: what does this model reduce to for houses heated by electricity, heating oil and natural gas?

For electricity, $w_{i1} = 1$ and $w_{i2} = 0$, and we have

$$y_i = \beta_0 + \beta_2 + (\beta_1 + \beta_4)x_i.$$

For heating oil, $w_{i1} = 0$ and $w_{i2} = 1$ and we have

$$y_i = \beta_0 + \beta_3 + (\beta_1 + \beta_5)x_i.$$

Finally, for natural gas $w_{i1} = 0$ and $w_{i2} = 0$ and we have

$$y_i = \beta_0 + \beta_1 x_i.$$

So by the expedient of defining multiple binary dummy variables to code levels of a categorical predictor when that predictor can take on three or more distinct values, and by introducing these dummy variables into a multiple linear regression model along with the products of the dummy variables with the quantitative predictor x, we are in effect defining separate linear relationships between the mean of y and x within the groups.

Again there are some special cases of this general model worth considering. If β_4 and β_5 are both zero, then we have a model with no interaction (sometimes called an additive model). The meaning of interaction here is similar to before where we considered a categorical variable with two levels. If β_4 and β_5 are zero, then the three lines describing the relationship between the expected value of y and x for the the three heating method groups have a common slope but different intercepts. That is, the difference in the expected selling prices for two houses of the same size but different heating types would not depend on the size of house. Of course, we can test

$$H_0 : \beta_4 = \beta_5 = 0$$

against the alternative

$$H_1 : \beta_4,\ \beta_5 \text{ not both zero}$$

by using an F-test.

Another hypothesis test we might be interested in testing is whether there is any group effect. That is, does the heating system have any effect on selling price in the model involving size? In the model with interaction we could test

$$H_0 : \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

against the alternative

$$H_1 : \text{Not all } \beta_2, \beta_3, \beta_4 \text{ and } \beta_5 \text{ are zero}$$

with an appropriate F test. In the additive model (model with no interaction) we could test

$$H_0 : \beta_2 = \beta_3 = 0$$

against

$$H_1 : \beta_2, \beta_3 \text{ not both zero}$$

in a similar way. Partial t tests may be useful for testing for differences between pairs of groups. For instance, suppose we have fitted the model without interaction ($\beta_4 = \beta_5 = 0$). Then testing $H_0 : \beta_2 = 0$ against a two-sided alternative using the partial t statistic tests whether the effect for electicity and for natural gas is the same. Similarly, testing $H_0 : \beta_3 = 0$ against a two-sided alternative tests whether the effect for heating oil and for natural gas is the same. We now illlustrate some of these tests for the real estate data.

Example: hypothesis testing for real estate prices.

We begin by doing a scatter plot of the selling price against size, using different plotting symbols for the different heating type groups. This plot is shown in Figure 6.54. Of course, there are only a few observations in each group here but

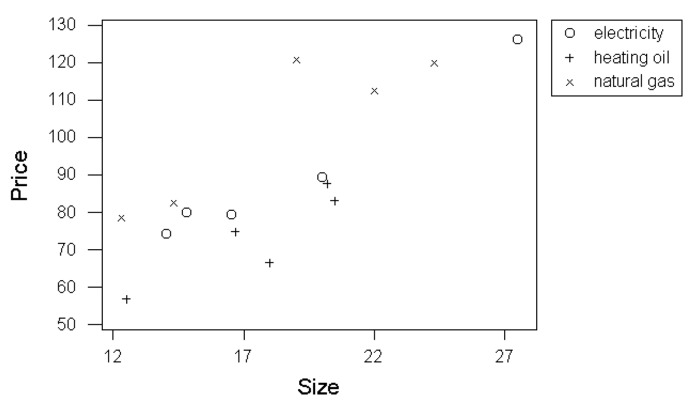

Figure 6.54: Scatter plot of selling price against size for real estate data using different plotting symbols for different heating type groups.

on the basis of this plot we might believe that there is no interaction between heating type and size (the relationship between expected selling price and size for the three groups seems to be well described by three parallel lines). We can test this by fitting a linear model with the dummy variables defined above as predictors, and with the products of these dummy variables and size as predictors.

We can then do an F test in the way outlined above to test for interaction. The computer output is shown below. The variable `dummy1` is w_1 and `dummy2` is w_2.

```
Regression Analysis

The regression equation is
Price = 34.8 + 3.70 Size - 14.7 Dummy1 - 21.6 Dummy2 + 0.054 Size*Dummy1
           - 0.26 Size*Dummy2

Predictor        Coef        StDev          T         P
Constant        34.77        13.92       2.50     0.034
Size           3.7048       0.7352       5.04     0.001
Dummy1         -14.66        19.02      -0.77     0.460
Dummy2         -21.55        24.69      -0.87     0.405
Size*Dum       0.0536       0.9981       0.05     0.958
Size*Dum       -0.258        1.360      -0.19     0.854

S = 7.431       R-Sq = 92.0%     R-Sq(adj) = 87.6%

Analysis of Variance

Source            DF          SS          MS         F         P
Regression         5      5733.3      1146.7     20.77     0.000
Residual Error     9       497.0        55.2
Total             14      6230.2

Source       DF      Seq SS
Size          1      4034.4
Dummy1        1         1.7
Dummy2        1      1694.0
Size*Dum      1         1.1
Size*Dum      1         2.0
```

From the table of sequential sums of squares and the analysis of variance table, the appropriate test statistic for testing for interaction is

$$\frac{(1.1 + 2.0)/2}{55.2} = 0.028.$$

We compare this with the upper 5 percentage point of an F distribution with 2 and 9 degrees of freedom (approximately 4.26). We see that there is no evidence for interaction here.

We can refit the model with no interaction (that is, fit the additive model).

```
Regression Analysis
```

```
The regression equation is
Price = 35.1 + 3.69 Size - 13.7 Dummy1 - 26.1 Dummy2

Predictor         Coef       StDev          T        P
Constant        35.070       8.182       4.29    0.001
Size            3.6882      0.4138       8.91    0.000
Dummy1         -13.664       4.265      -3.20    0.008
Dummy2         -26.109       4.277      -6.10    0.000

S = 6.742       R-Sq = 92.0%     R-Sq(adj) = 89.8%

Analysis of Variance

Source            DF          SS          MS        F        P
Regression         3      5730.2      1910.1    42.02    0.000
Residual Error    11       500.0        45.5
Total             14      6230.2

Source       DF      Seq SS
Size          1      4034.4
Dummy1        1         1.7
Dummy2        1      1694.0
```

We demonstrate testing for an effect for heating type in this model. The appropriate F statistic (from the sequential sums of squares and the analysis of variance table) is

$$\frac{(1.7 + 1694.0)/2}{45.5} = 18.63.$$

We compare this with the upper 5 percentage point of an F distribution with 2 and 11 degrees of freedom (approximately 3.98). In this case we reject the null hypothesis that there is no group effect. As mentioned above, the partial t statistics can tell us something about differences between pairs of groups. The partial t statistic for `dummy1` tests for a significant difference between electricity and natural gas, and the partial t statistic for `dummy2` tests for a singificant difference between heating oil and natural gas (from the p-values, both differences are significant here at the 5 percent level).

We end our discussion of categorical variables here for the moment. We have focused on the case of a single categorical variable and a single quantitative predictor. Situations involving multiple quantitative and categorical predictor variables may be handled in much the same way. We will say more about the general case in the next section.

7 Less than full rank linear models.

In our discussion of the linear model so far, we have always assumed that the design matrix X has full rank (that is, we have assumed that we can't express one column of X, or one predictor, as a linear combination of the others). However, a natural formulation of some linear statistical models may lead to a design matrix which does not have full rank. For these models, there is no unique solution to the normal equations, and so the vector of parameters β in the mean structure cannot be uniquely estimated. However, it may still be possible to estimate certain functions of the parameters and to formulate meaningful hypotheses to be tested in a less than full rank model. In this section we give an introduction to the analysis of less than full rank linear models.

We begin by giving a simple example of a less than full rank linear model. This example is from Heinz Kohler, "Statistics for Business and Economics (Second Edition)," Scott, Foresman and Co, Glenview, Illinois, 1988.

A doctor wants to test whether dieting men, on average, lose the same weight during a week, regardless of which of three diets are involved. The following sample data are available (in pounds of weight loss per week).

Diet A: 5.1, 6.7, 7.0, 5.3, 9.1, 4.3
Diet B: 6.1, 7.3, 8.1, 9.1, 10.3, 12.1
Diet C: 7.3, 8.9, 9.4, 12.0, 15.7, 9.5

A natural way to formulate a model for these data is as follows. Define μ to be a parameter representing the global mean of pounds lost for the population under study after dieting, and let α_1, α_2, α_3 denote effects on weight loss due to diet A, diet B and diet C respectively. In an extension of our previous notation, let y_{ij} $i = 1, 2, 3$, $j = 1, ..., 6$ denote the jth response value in the ith group (where the first group is the diet A group, the second group is the diet B group and the third is the diet C group). Then a sensible model for these data might postulate

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

where ε_{ij}, $i = 1, 2, 3$, $j = 1, ..., 6$ are a collection of zero mean errors which are uncorrelated with a common variance. Notice that this is a linear model. Just define a dummy variable x_1 which is one for the diet A group and zero otherwise, a dummy variable x_2 which is one for the diet B group and zero otherwise, and a dummy variable x_3 which is one for the diet C group and zero otherwise. Then if we do a multiple linear regression of the responses on the predictors x_1, x_2 and x_3 then we are simply fitting the model above. In this model, the coefficient for x_1 is what we have written as α_1, the coefficient for x_2 is α_2, the coefficient for x_3 is α_3 and the intercept term is μ. The model we have considered actually involves fitting a single categorical variable (diet group) and our discussion should remind you of what we did in the last section.

However, in the last section we used $k-1$ binary dummy variables to code the information in a categorical variable with k levels. The reason for this is that k binary variables are not required to code the information a k level categorical variable: in the example above, if x_1 and x_2 are both zero (so that the subject is not on diet A or B) then we know the subject is on diet C. If we looked at the design matrix for fitting the model involving x_1, x_2, x_3 and an intercept, we would see that the sum of the columns for x_1, x_2 and x_3 is equal to the first column (a column of ones). That is, $x_1+x_2+x_3=1$. So the design matrix here is not of full rank, and there is no unique solution to the normal equations. We can express the nature of the problem in a different way. Suppose we define a parameter μ' as $\mu'=\mu+c$ where c is a constant, and suppose we define parameters α_i', $i=1,2,3$ as $\alpha_i'=\alpha_i-c$. Then we see that $\mu+\alpha_i=\mu'+\alpha_i'$, and so these two sets of parameters result in the same model for the data. So we should not expect to be able to estimate all the parameters in the model (that is, we should not expect to be able to solve the normal equations). We have too many parameters in the model.

One solution to this problem is the one presented in the last section: just remove one of the binary dummy variables. Effectively, we set one of the parameters α_i to be zero. The reduced design matrix when one of the predictors is removed now has full rank, and analysis can proceed as before. This approach to dealing with the less than full rank model is called reparametrization.

However, reparameterization need not be used in dealing with a less than full rank linear model. We can in fact work with the original model, provided we only ask sensible questions about the parameters. We may not be able to estimate all the parameters, but we might be able to estimate functions of the parameters: for instance, in our example we could still estimate differences between the effects for different diets (that is, we can estimate quantities like $\alpha_1-\alpha_2$ for instance). It is also possible to formulate sensible hypotheses to be tested. There will be some discussion of the general approach to handling the less than full rank linear model in the coming weeks.

7.1 One way classification model with fixed effects

In the last subsection we discussed a simple example to introduce the idea of the less than full rank model. Our example was a special case of the so-called one way classification model with fixed effects. This model may be described in general as follows.

Suppose we collect random samples from k different populations (think of these k populations as being indexed by the k levels of some categorical variable). We write n_i, $i=1,...,k$ for the size of the sample from the ith population. Also, write y_{ij}, $j=1,...,n_i$ for the random sample collected from population i,

$i = 1, ..., k$. In the one way classification model with fixed effects it is assumed that

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

where μ and α_i, $i = 1, ..., k$ are unknown parameters, and the errors ε_{ij} have mean zero and are uncorrelated with a common variance σ^2. Normality of the errors needs to be assumed for hypothesis testing and interval estimation. The parameter μ is a global mean parameter, and the parameters α_i represent deviations from this mean for the k populations.

It is usually of interest in the one way classification model to test whether all the population means are the same, and to test for differences between pairs of population means. We will describe how to conduct these tests and describe the form of the ANOVA table for a one way classification model in the remainder of this subsection.

It should be evident from the discussion of last week that the one way classification model above is a less than full rank linear model. An example will help to show how we can write the one way classification model in the form of the general linear model. Suppose there are two populations, with a random sample of size two taken from each population. Now let

$$y = (y_{11}, y_{12}, y_{21}, y_{22})^\top$$

$$\varepsilon = (\varepsilon_{11}, \varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22})^\top$$

$$\beta = (\mu, \alpha_1, \alpha_2)^\top$$

and

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Then we can write

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

$i = 1, 2$, $j = 1, 2$, in matrix notation as

$$\begin{bmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{21} \\ \varepsilon_{22} \end{bmatrix}$$

or

$$y = X\beta + \varepsilon.$$

We can write the general one way classification model in a similar way. In the general case the vector β will be

$$\beta = (\mu, \alpha_1, ..., \alpha_k)^\top$$

and the design matrix X will have $k+1$ columns, with the first column being a column of ones, and the $(i+1)$th column consisting of entries which are one for observations in the ith population and zero otherwise. Observe that the design matrix X does not have full rank, since the sum of the last k columns of X is equal to the first column (see the above example for instance). Since X does not have full rank, the normal equations do not have a unique solution, and so there is no unique estimate of β. One approach to fixing this problem is to reparametrize the original model: we reparametrize so that for the new model the design matrix does have full rank, and then the parameters can be estimated.

There are in general many possible ways that we could reparametrize the one way classification model with fixed effects. If the mean of one of the populations can be considered to establish some reference level (without loss of generality say the reference population is the first one) then we could set $\alpha_1 = 0$ (this corresponds to deleting the second column of the original design matrix X). Then the parameter μ is the mean of the reference population, and the parameters $\alpha_2, ..., \alpha_k$ represent deviations from this mean for the remaining populations. This way of reparametrizing the model is often very natural in, say, medical trials where we are interested in comparing different methods of treatment with a control group.

The partial t-tests in the regression output for the parameters α_i $i = 2, ..., k$ in this reparametrized model test for a significant difference between the mean of population i and the mean of the first population (reference population). A test for equality of all population means amounts to testing

$$H_0 : \alpha_2 = ... = \alpha_k = 0$$

against the alternative

$$H_1 : \text{Not all } \alpha_2, ..., \alpha_k \text{ are zero.}$$

This hypothesis is tested using the usual F test. We will discuss another way of parametrizing the one way classification model in the next lecture. In setting $\alpha_1 = 0$ in the discussion above (removing a column of the design matrix) we have effectively added an additional equation to the normal equations for the less than full rank model, allowing a unique solution. Other arbitrary constraints could have been made on the original parameters: for instance, many statistical packages introduce the constraint $\sum_{i=1}^{k} \alpha_i = 0$ on the parameters in the original model. This will also allow a unique solution of the normal equations, where the effects α_i are required to be "centred" about the grand mean μ.

Example: completion times for tax returns

This example is from Keller, Warrack and Bartel, "Statistics for Management and Economics (Second Edition)," Wadsworth, Belmont, California 1990.

Three new formats have been developed recently for tax return forms (formats A, B and C say). To determine which, if any, are superior to the current form, 20 individuals were asked to participate in an experiment. Each of the three new forms and the old form were filled out by 5 different people. The amount of time taken by each person to complete the task is recorded in the accompanying table. At the 10 percent significance level, can we conclude that differences in the completion times exist among the four forms? Three binary dummy variables

Current Format	Format A	Format B	Format C
6.3	9.2	12.6	7.5
9.8	8.6	10.9	9.3
10.1	10.3	11.8	8.8
8.5	11.4	10.8	7.8
7.1	8.5	10.1	9.1

Table 7.5: Completion times for tax returns: old format, and three new formats (A, B and C).

were created (`Dummy1`, `Dummy2` and `Dummy3`). The first dummy variable is one for format A and zero otherwise, the second dummy variable is one for format B and zero otherwise, and the third dummy variable is one for format C and zero otherwise. Thus we have parametrized a one way classification model by the method discussed above, with the subjects filling out the current form acting as a control group. Fitting a multiple linear regression model with these dummy variables as predictors and the completion time (`y`) as the response gave the following output.

```
Regression Analysis

The regression equation is
y = 8.36 + 1.24 Dummy1 + 2.88 Dummy2 + 0.140 Dummy3

Predictor        Coef       StDev          T        P
Constant       8.3600      0.5409      15.46    0.000
Dummy1         1.2400      0.7649       1.62    0.125
Dummy2         2.8800      0.7649       3.77    0.002
Dummy3         0.1400      0.7649       0.18    0.857
```

```
S = 1.209       R-Sq = 53.2%     R-Sq(adj) = 44.4%

Analysis of Variance

Source            DF          SS          MS        F        P
Regression         3      26.573       8.858     6.06    0.006
Residual Error    16      23.404       1.463
Total             19      49.977

Source       DF       Seq SS
Dummy1        1        0.204
Dummy2        1       26.320
Dummy3        1        0.049
```

The F statistic in the ANOVA table is a realization of an $F_{3,16}$ random variable under the null hypothesis that

$$H_0 : \alpha_2 = \alpha_3 = \alpha_4 = 0$$

(that is, under the hypothesis that there is no difference in completion time among the four groups). The p-value for this F test is 0.006, so that we reject the null hypothesis that there is no difference among means in favour of the hypothesis that at least two of the population means differ. The partial t statistics in the above regression output for α_2, α_3 and α_4 relate to a test for a significant difference between mean completion time for the current form and mean completion time for formats A, B and C respectively. We see here that there seems to be a significant difference between mean completion time for the current form and for format B (the current form is better).

It is interesting to have a closer look at the analysis of variance table for the one way classification model. For the multiple linear regression model, if we write y_i $i = 1, ..., n$ for the responses, $\widehat{y}_i$ $i = 1, ..., n$ for the fitted values and $\bar{y}$ for the mean of the responses, then we know that

$$SS_{total} = SS_{reg} + SS_{res}$$

where

$$SS_{total} = \sum_{i=1}^{n} (y_i - \bar{y})^2,$$

$$SS_{reg} = \sum_{i=1}^{n} (\widehat{y}_i - \bar{y})^2$$

and

$$SS_{res} = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2$$

(this was stated in lectures and proved by the MATH2931 students on their third assignment). We can rewrite this analysis of variance identity in an interesting way for the one way classification model. However, first we introduce some notation. Write $\bar{y}_{i.}$ for the sample mean of observations in the ith population,

$$\bar{y}_{i.} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}.$$

Also, write $\bar{y}_{..}$ for the sample mean of all response values,

$$\bar{y}_{..} = \frac{1}{n}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}.$$

Thus the dots in the subscripts above mean that summation is taken with respect to that index in forming the mean.

It is easy to show that the fitted value for an observation from population i is $\bar{y}_{i.}$, and the analysis of variance identity becomes

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\bar{y}_{..})^2 = \sum_{i=1}^{k} n_i(\bar{y}_{i.}-\bar{y}_{..})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\bar{y}_{i.})^2.$$

Thus total variation can be partitioned into a component representing variation of group sample means from the grand mean ("between group variation") and a component representing variation of observations about their within group sample means ("within group variation"). Testing whether all population means are the same via the F test in the ANOVA table amounts to looking at whether between group variation is large with respect to within group variation, an intuitively sensible procedure. You will be asked to prove the above identity for the one way classification model in your final assignment. The ANOVA table in the regression output shows the components of variation in the above partition, and the degrees of freedom, mean squares and F ratio follow from our previous discussion of the ANOVA table for the multiple linear regression model.

In our previous discussion we have indicated how to test equality of all the population means, and how to test for equality of the mean in a reference population with one of the other population means via the partial t statistic when the one way classification model is parametrized in the way that we have described. However, how do we test, say, that $\alpha_i = \alpha_j$ for arbitrary i, j? That is, how do we test for equality of means for an arbitrary pair of populations in our experiment? It can be shown that the statistic

$$\frac{(\widehat{\alpha}_i - \widehat{\alpha}_j) - (\alpha_i - \alpha_j)}{\widehat{\sigma}\sqrt{\frac{1}{n_i}+\frac{1}{n_j}}}$$

(where $\widehat{\sigma}$ denotes the estimated residual standard deviation) has a t_{n-k} distribution. This result can be used to construct a confidence interval for $\alpha_i - \alpha_j$ or to do either a one-sided or two-sided test of equality of population means for the ith and jth populations. If a test like this one is to be conducted for every pair of populations in the experiment, then we obviously have a problem of multiple comparisons: one conservative way to deal with this is to use Bonferroni adjustment.

Example: completion times for tax returns

Suppose in the previous example that it was of particular interest to test for a difference between formats B and C. We wish to test

$$H_0 : \alpha_3 = \alpha_4$$

against the alternative

$$H_1 : \alpha_3 \neq \alpha_4.$$

From our previous numerical computations, we can evaluate the t-statistic given above (with $\alpha_3 - \alpha_4 = 0$ under H_0) as

$$\frac{2.88 - 0.14}{1.209\sqrt{\frac{1}{5} + \frac{1}{5}}} = 3.58.$$

For a two-sided test at the 5 percent level, we compare this with the upper 2.5 percentage point of a t distribution with 16 degrees of freedom (which is approximately 2.12). So we would reject the null hypothesis that there is no difference in completion times for formats B and C.

An alternative parametrization and the general linear hypothesis

In the last lecture we discussed the one way classification model. The one way classification model is of interest when we have random samples from k different populations, and are interested in comparing population means. With a random sample of size n_i from the ith population, $i = 1, ..., k$, we write the one way classification model as

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

where y_{ij} is the jth observation from the ith population, $i = 1, ..., k$, $j = 1, ..., n_i$, μ is a global mean, α_i is the deviation from the global mean for population i, $i = 1, .., k$ and the terms ε_{ij} are random errors. We showed that the one way classification model was an example of a less than full rank linear model, and in order to apply techniques from the full rank case we discussed ways of reparametrizing to full rank. In particular, we considered a situation where one of the k populations was considered a reference population (without loss of generality say it is the first population): it was then natural to set the deviation from the global mean for this population to zero ($\alpha_1 = 0$ say). In the resulting reparametrized model μ can be considered to be the mean of the reference population, and the parameters $\alpha_2, ..., \alpha_k$ describe deviations from the mean of the reference population for the other populations of interest in the experiment.

In this lecture we discuss an alternative parametrization of the one way classification model, and discuss hypothesis testing for this parametrization.

Instead of defining a reference population as in the previous lecture, an alternative way of parametrizing the one way classification model is to define

$$\mu_i = \mu + \alpha_i$$

(so that μ_i is the population mean for the ith population) and to write

$$y_{ij} = \mu_i + \varepsilon_{ij}.$$

We have thus reduced the number of parameters in the mean structure from the $k+1$ parameters $\mu, \alpha_1, ..., \alpha_k$ to the k parameters $\mu_1, ..., \mu_k$. The new parametrization gives a full rank linear model (you might like to check this for yourself). The parametrization above is attractive, since μ_i has a direct interpretation (the population mean for population i).

As for the parametrization considered in the last lecture, we are interested in testing whether all population means are equal, and in looking at differences in population means. To test equality of the population means we must test

$$H_0 : \mu_1 = ... = \mu_k$$

against the alternative

$$H_1 : \text{Not all } \mu_1, ..., \mu_k \text{ are equal.}$$

This test does not quite fit into the hypothesis testing framework we have discussed previously for the full rank linear model: testing $\mu_1 = ... = \mu_k$ does not correspond to testing whether a subset of $\mu_1, ..., \mu_k$ are zero. However, hypotheses like the one above can be tested within the framework of the *general linear hypothesis*, which we now discuss.

The general linear hypothesis

If we write the full rank linear model in the usual notation as

$$y = X\beta + \varepsilon$$

(where $\beta = (\beta_0, ..., \beta_k)$ so that there are $p = k + 1$ parameters in the mean structure) then for the general linear hypothesis we test

$$H_0 : C\beta = d$$

against the alternative

$$H_1 : C\beta \neq d$$

where C is an $m \times p$ matrix of rank m, $m \leqslant p$, and d is an $m \times 1$ vector.

It is worth considering a few special cases to illustrate the usefulness of the general linear hypothesis. Suppose we wish to test the hypothesis

$$H_0 : \beta_{r+1} = ... = \beta_k = 0$$

against the alternative

$$H_1 : \text{Not all } \beta_{r+1}, ..., \beta_k \text{ are zero.}$$

Then define C to be the $(k-r) \times p$ matrix consisting of the last $k-r$ rows of the $p \times p$ identity matrix I_p. Clearly as $I_p\beta = \beta$, we have that $C\beta$ is the vector

$$(\beta_{r+1}, ..., \beta_k)^\top.$$

Thus defining d to be a $(k-r) \times 1$ vector of zeros, the hypothesis

$$H_0 : \beta_{r+1} = ... = \beta_k = 0$$

can be written as $C\beta = d$. So any of the hypothesis tests we have considered previously may be considered as special cases of the general linear hypothesis.

The general linear hypothesis also allows us to test

$$H_0 : \mu_1 = ... = \mu_k$$

in the parametrization we have introduced for the one way classification model. We rewrite the above hypothesis as

$$H_0 : \mu_1 - \mu_2 = 0, \mu_1 - \mu_3 = 0, ..., \mu_1 - \mu_k = 0.$$

Clearly these two hypotheses are equivalent. Each of the equations $\mu_1 - \mu_2 = 0, ..., \mu_1 - \mu_k = 0$ can be expressed in the form $c^\top\beta = 0$ where c is a $k \times 1$ vector and $\beta = (\mu_1, ..., \mu_k)^\top$. For instance, if we let $c = (1, -1, 0, ..., 0)^\top$, then $c^\top\beta = 0$ is equivalent to $\mu_1 - \mu_2 = 0$. Hence by defining an appropriate $(k-1) \times k$ matrix C and by letting d be a $(k-1) \times 1$ vector of zeros, we can write the equations $\mu_1 - \mu_2 = 0, ..., \mu_1 - \mu_k = 0$ in the form $C\beta = d$. So the general linear hypothesis gives a very general framework for testing many hypotheses of interest in the general linear model.

What is the test statisic used for testing the general linear hypothesis? It can be shown that under $H_0 : C\beta = d$, and writing b for the least squares estimator of β, the statistic

$$\frac{(Cb-d)^\top(C(X^\top X)^{-1}C^\top)^{-1}(Cb-d)}{m\widehat{\sigma}^2}$$

has an $F_{m,n-p}$ distribution. For testing $\mu_1 = ... = \mu_k$ for the one way classification model, this statistic is computed and displayed in the analysis of variance table.
Example: tax return forms.

We continue our discussion of the data on completion times for different formats of tax return forms introduced in the last lecture. Recall that this data set was collected by dividing a group of twenty subjects into four groups, and giving each subject in the first group a tax return form in the format currently used, and members of the remaining groups forms in three new formats (A, B and C). The completion time for each of the subjects was recorded. This is a one way classification model with four groups. We showed in the last lecture that we could test if the mean completion time was the same for all four groups by introducing appropriate binary variables to fit the model where we adopt a parametrization with the mean completion time for the current form as a reference level.

```
One-way Analysis of Variance

Analysis of Variance for y
Source     DF        SS        MS        F        P
factor      3     26.57      8.86     6.06    0.006
Error      16     23.40      1.46
Total      19     49.98
                                   Individual 95% CIs For Mean
                                   Based on Pooled StDev
Level       N      Mean     StDev  --+---------+---------+---------+----
1           5     8.360     1.655  (-------*------)
2           5     9.600     1.235          (-------*-------)
3           5    11.240     0.971                     (-------*-------)
4           5     8.500     0.803   (-------*------)
                                   --+---------+---------+---------+----
Pooled StDev =    1.209            7.5       9.0      10.5      12.0

Fisher's pairwise comparisons

    Family error rate = 0.189
Individual error rate = 0.0500

Critical value = 2.120

Intervals for (column level mean) - (row level mean)

                   1           2           3
```

```
2       -2.862
         0.382

3       -4.502      -3.262
        -1.258      -0.018

4       -1.762      -0.522       1.118
         1.482       2.722       4.362
```

For the moment just consider the analysis of variance table. As we have mentioned, the hypothesis we wish to test (equality of population means) can be written in the form $C\beta = 0$ here. The test statistic in this case is

$$\frac{(Cb)^\top (C(X^\top X)^{-1}C^\top)^{-1}(Cb)}{(k-1)\widehat{\sigma}^2}$$

for an appropriate matrix C. In the analysis of variance table in the sum of squares (SS) column and the factor row, 26.57 is the value of

$$(Cb)^\top (C(X^\top X)^{-1}C^\top)^{-1}(Cb).$$

The degrees of freedom in the factor row is 3 (which is $k-1$). The sum of squares for the error row is as usual the residual sum of squares. In the mean square (MS) column we have in the error row the estimated residual variance $\widehat{\sigma}^2$ (the residual sum of squares divided by its degrees of freedom), and in the factor row we have

$$\frac{(Cb)^\top (C(X^\top X)^{-1}C^\top)^{-1}(Cb)}{(k-1)}.$$

The F statistic is the ratio of the two entries in the mean square column, which is the test statistic for the general linear hypothesis for testing equality of population means. The p-value is 0.006, indicating we reject equality of population means. You should compare the ANOVA table here with the one we obtained in the last lecture in the alternative parametrization we have considered.

Below the analysis of variance table is a table giving the number of observations in each of the four groups, the estimated coefficients in the model (that is, the estimated population means) and the estimated standard errors of the coefficients. Also, 95 percent confidence intervals for the population means are depicted graphically. These confidence intervals are just the usual ones for the coefficients in a full rank linear model based on the t distribution. Below this is the estimated residual standard deviation (which is 1.209 here). Below this under the section labelled “Fisher’s pairwise comparisons” are 95% confidence intervals for differences in pairs of population means. If $\widehat{\mu}_i$, $i = 1, ..., k$ are the estimated means for the k populations, it can be shown that for $i \neq j$

$$\frac{(\widehat{\mu}_i - \widehat{\mu}_j) - (\mu_i - \mu_j)}{\widehat{\sigma}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}}}$$

has a t_{n-k} distribution. This result is the basis for computing the intervals in the table.

7.2 Randomized complete block designs

In the last two lectures we have been discussing the one way classification model with fixed effects (sometimes called a completely randomized experiment). In the completely randomized experiment we have random samples from k populations, and we are interested in comparing the means of these populations.

Today we will discuss an important extension of the completely randomized design where we are interested in comparing population means for k populations of interest, but where the members of the k populations can be divided into groups based on a second classifying variable (we talk about blocks for the groupings induced by the classification). We will consider the so-called *randomized complete block experiment* where we attempt to reduce unexplained variation and improve precision of comparisons between population means by taking account of variation between blocks.

An example will help to illustrate the idea. This example is from Keller, Warrack and Bartel, “Statistics for Management and Economics: A Systematic Approach (Second Edition),” Wadsworth, Belmont, California. It was desired to compare the annual incomes of business students with different majors two years after graduation. A random sample of six students was taken of graduates in accounting, marketing and finance and their income was determined. The data are shown in the table below (income is given in thousands of dollars). We

Accounting	Marketing	Finance
27	23	48
22	36	35
33	27	46
25	44	36
38	39	28
29	32	29

are interested in making comparisons between mean income levels for students majoring in different disciplines. One way to test for any differences between income levels is to analyze the data as a one way classification model with fixed effects. We have seen how to do this in previous lectures. Fitting a one way classification model results in the following output.

```
One-way Analysis of Variance
```

```
Analysis of Variance
Source     DF        SS         MS        F        P
Factor      2     193.0       96.5     1.77    0.205
Error      15     819.5       54.6
Total      17    1012.5
                                   Individual 95% CIs For Mean
                                   Based on Pooled StDev
Level       N      Mean      StDev ---+---------+---------+---------+---
Accounti    6    29.000      5.762  (---------*----------)
Marketin    6    33.500      7.765         (----------*----------)
Finance     6    37.000      8.390               (----------*---------)
                                   ---+---------+---------+---------+---
Pooled StDev =    7.391            24.0      30.0      36.0      42.0
```

From the p-value in the analysis of variance table, no significant difference between mean annual income levels seems to be indicated. We can see that only a small fraction of total variation is explained by the model here. With a sample of size six from each group, it will be difficult to detect differences in population means which are small compared to the residual standard deviation (which is about 7000 dollars). Now, suppose it was thought that annual income following graduation was dependent on performance at university. We can classify the members of each of the populations of interest (accounting majors, marketing majors and finance majors) according to their average grade (A+, A, B+, B, C+ or C) during the course of their studies. A new experiment is conducted in which we again take a sample of size six from the three populations, but in each of the three groups of six students we take one student with an A+ average, one with an A average, one with a B+ average, one with a B average, one with a C+ average and one with a C average: we say that students in different grade levels form different *blocks* in this new experiment. So we have three populations, and six blocks, and one observation for each population/block combination (eighteen observations in all). The table below shows the data collected for this new experiment. The idea of introducing the blocking variable here (average grade) is to reduce residual variation to allow more precise estimates of differences between population means. A key assumption here is that there is no interaction between the blocking variable, average grade, and the major of the student: it is assumed that the difference between mean salary for, say, an accounting and marketing graduate with the same average grade does not depend of whether that grade is an A+, a C or anything else. We allow the grade to have an additive effect on mean annual income, but the amount of the effect must not depend on what the student studied.

For notational convenience we consider first the situation where there is precisely one observation for each population/block combination. We write y_{ij} for

Average	Accounting	Marketing	Finance
A+	41	45	51
A	36	38	45
B+	27	33	31
B	32	29	35
C+	26	31	32
C	23	25	27

the observation for population i, $i = 1, .., k$ and block j, $j = 1, ..., b$. The model we are considering in a randomized complete block experiment is

$$y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}.$$

Here μ is a global mean, α_i is an effect for population i, β_j is an effect for block j and the ε_{ij} are zero mean random errors, $i = 1, ..., k$, $j = 1, ..., b$. We make the usual distributional assumptions about the errors.

We want to develop a test for equality of the population means (that is, we want to test if $\alpha_i = 0$ for $i = 1, ..., k$). You have probably already realized that the model we have written down above is a less than full rank linear model. There are $k + b + 1$ parameters in the mean structure, so that the design matrix X has dimensions $n \times (k + b + 1)$. The rank of X is $k + b - 1$ here: effectively there are two redundant parameters. As for the one way classification model, there are a number of approaches to developing a hypothesis test for equality of population means. One approach is to reparametrize to a full rank model and to apply methods for the full rank case. For instance, we could set $\alpha_1 = \beta_1 = 0$, and then test

$$H_0 : \alpha_2 = ... = \alpha_k = 0$$

against the alternative

$$H_1 : \text{Not all } \alpha_2, ..., \alpha_k \text{ are zero}$$

in this reparametrized model to test equality of population means. There are other parametrizations that could be considered in which the general linear hypothesis can be employed to test equality of means. Also, it is possible to work directly with the less than full rank linear model, although we have not developed the necessary matrix concepts to do this. Whichever method is adopted, the analysis of variance table will be the same, and the same conclusion will be reached for the test for equality of population means.

It is worth discussing the general form of the analysis of variance table for the randomized complete block design. In a randomized complete block design we can partition up total variation into a component of variation for the population mean differences (sometimes called the sum of squares due to treatments), a component for block differences (sum of squares due to blocks) and residual variation. The sum of squares due to treatments is simply the regression sum of squares you would get if you ignored blocking and used a one way classification model for the samples from the different populations. The sum of squares due to blocks is simply the regression sum of squares you would get if you ignored population mean differences and fitted a one way classification model treating the different blocks as the populations of interest. It is a special feature of the randomized complete block design where you have equal numbers of observations for each population/block combination that these two sums of squares add to the regression sum of squares you get for fitting the full model.

Income levels for business students when analyzing the data as a randomized complete block experiment is given below.

```
Two-way Analysis of Variance

Analysis of Variance for y
Source         DF         SS          MS         F         P
Major           2     108.44       54.22     10.38     0.004
Grade           5     854.94      170.99     32.74     0.000
Error          10      52.22        5.22
Total          17    1015.61
```

Our treatment sum of squares here is 108.44, block sum of squares is 854.94, and residual sum of squares is 52.22. The estimated residual variance is 5.22, and we see that blocking has considerably reduced the proportion of total variation unexplained by the model: this may allow us to detect differences between population means which we could not have detected previously for the same expenditure of resources. The F statistic for testing equality of mean income for the different majors is 10.38 here (the mean square values in the 'MS' column are the sums of squares divided by the degrees of freedom, and the F statistic 10.38 is the mean square due to treatments divided by the mean square error: we look at whether the variation between different major types is large compared to unexplained variation). The p-value for this F-test is 0.004, indicating a significant difference between means for different majors at the 5 percent level. The p-value for the blocking variable should be ignored here, although the F statistic for blocking does give us a rough indication of whether blocking was effective in reducing residual variation to allow more precise comparisons among the populations of interest.

7.3 Factorial Experiments

Often experiments are conducted where one or more observations of a response variable are made for each combination of possible levels of a number of categorical predictors (factors). These kind of experiments are called factorial experiments.

Example: insanity and phases of the moon

The data described here are given by Blackman and Catalina (1993) in the article "The Moon and the Emergency Room," *Perceptual and Motor Skills*, 37, pp. 624–626. The admission rate (in patients per day) to the emergency room of a Virginia mental health clinic was measured before, during and after the 12 full moons from August 1971 to July 1972. This is a factorial experiment: for each level of the factor `month` (twelve levels) and each level of the factor `moon` (three levels, before, during or after the full moon) we have an observation of the response.

Example: paper plane experiment

The data described here was collected by Stewart Fischer and David Tippetts, statistics students at the Queensland University of Technology in a subject taught by Dr Margaret Mackisack.

In the experiment conducted by these students, the relationship between weight, design and angle of release for a paper aeroplane and the distance travelled by the aeroplane was investigated. Two different weights were considered (factor `Paper` describing the weight of the sheet of paper used with levels 1=80 grams and 2=50 grams) as well as two different designs (1=sophisticated design, 2=simple design) and two different angles of release (1=horizontal, 2=45 degrees). For each distinct combination of levels of the three factors ($2.2.2 = 8$ different combinations) there are two observations of the response - 16 observations in all.

Factorial experiments arise frequently in experiments designed to investigate process improvement in manufacturing. In a statistical approach to process improvement, typically a sequence of experiments is conducted. In the initial stages of an investigation, an attempt will be made to identify any variables (factors) which affect the process being studied. Then a factorial experiment will be conducted where a response (some characteristic of the process being studied) is measured for each combination of a small number of levels for the factors identified as possibly influential. Analysis of the factorial experiment identifies which factors are most influential in explaining process variation. Further experiments are then conducted to attempt to optimize settings for the factors found to be important in the initial experiment (response surface methodology is the name given to these kind of sequential experimentation methods which aim to optimize some process).

When the levels of categorical variables or factors are under the control of an experimenter, there are many advantages in designing a study in the form of a factorial experiment. These advantages include the ability to detect interactions between factors (we discuss what this means later) as well as the need for fewer observations of the response to achieve the same precision of estimation of effects of interest. You will discuss these advantages more in later courses when you look at experimental design.

Analysis of factorial experiments

For simplicity, we consider the analysis of an experiment with two factors (A and B say). Factor A has a levels, and factor B has b levels. For each of the ab combinations of possible levels of the two factors, we observe the response m times. Write y_{ijk} for the kth observation of the response at the ith level of factor A and jth level of factor B, $i = 1, ..., a$, $j = 1, ..., b$, $k = 1, ..., m$. The model we consider is

$$y_{ijk} \quad = \quad \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk} \tag{41}$$

where μ is an overall mean, α_i is the effect for the ith level of factor A, β_j is the effect for the jth level of factor B, γ_{ij} is a term representing an interaction between the ith level of factor A and the jth level of factor B (more on this in a moment), and the ε_{ijk} are zero mean uncorrelated normal errors with constant variance.

Write $\mu_{ij} = \mathbb{E}(y_{ijk})$ for the mean of observations at level i of factor A and at level j of factor B. The main effect at the ith level of factor A is defined to be

$$\frac{1}{b}\sum_{j'} \mu_{ij'} - \frac{1}{ab}\sum_{i',j'} \mu_{i'j'}.$$

This is the difference between the average of means with factor A fixed at level i and the average of all means.

The interaction between factor A at level i and factor B at level j is defined to be

$$\left(\mu_{ij} - \frac{1}{a}\sum_{i'} \mu_{i'j}\right) - \left(\frac{1}{b}\sum_{j'} \mu_{ij'} - \frac{1}{ab}\sum_{i',j'} \mu_{i'j'}\right).$$

The first bracketed term above is the main effect you would get for factor A at level i in an experiment with factor B fixed at level j. The second bracketed term is the main effect for factor A at level i in the full experiment. If the interaction is nonzero, it means that the effect of factor A at level i depends on the level j of factor B. This is the meaning of interaction.

When there is interaction between factors, it may be hard to interpret the main effects (when the way that one factor affects the response depends on the level of another factor, it may not be meaningful to talk about an average effect for the factor). You may check that if all the terms γ_{ij} are zero in the model (41) then the interactions as we have defined them above are zero.

In experiments with more than two factors, higher order interactions (that is, interactions between more than two factors) can be defined, but we leave this for later statistics courses.

We illustrate the idea of interaction with the paper plane experiment.

Example: paper plane experiment

For simplicity we analyse the paper plane experiment as a two factor experiment, ignoring the angle of release factor. So we consider the two factor experiment with factors `Paper` (weight of paper used in the plane) and `Plane` (design of plane, sophisticated or simple).

Below is shown the so-called main effects plot for the paper plane experiment. This plot shows response means for different levels of the factors. As we can see, lighter planes seem to travel further and the planes with a sophisticated design seem to travel further than planes with a simple design. The differences between the points plotted on the graph and the horizontal lines on the graphs (the overall mean of responses) give empirical estimates of the main effects as we've defined them above.

Figure 7.56 shows a so-called interaction plot. Plotted on the y-axis are response means, and plotted on the x-axis here are the levels of the factor `Plane` (plane design, 1=sophisticated and 2=simple). The two lines (dashed and solid) are for the light paper and heavy paper respectively. If there were no interaction, then the lines in the plot would be parallel - for no interaction, the way that the mean response should change as we change the level of `Plane` should not depend on the level of `Paper`.

The model (41) is overparametrized: it is a less than full rank linear model where the normal equations have no unique solution. We can impose some constraints on the parameters, and adding these constraint equations to the normal equations ensures that the normal equations have a unique solution.

The usual constraints are

$$\sum_i \alpha_i = 0, \ \ \sum_j \beta_j = 0$$

$$\sum_{i=1}^{a} \gamma_{ij} = 0, \ \ j = 1, ..., b$$

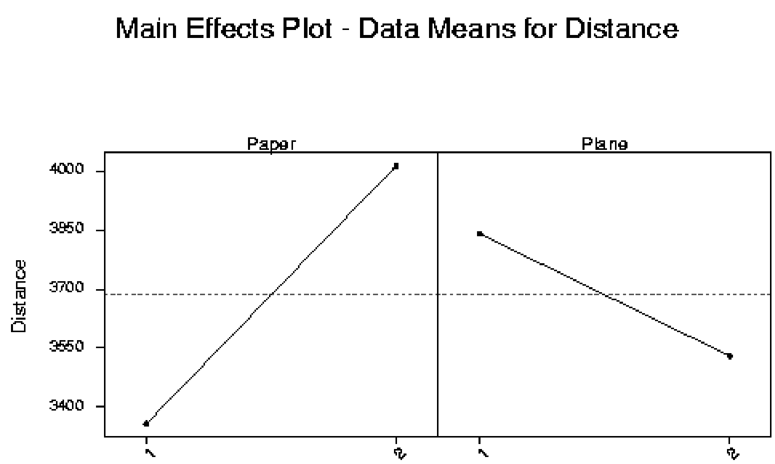

Figure 7.55: Main effects plot for data on paper planes.

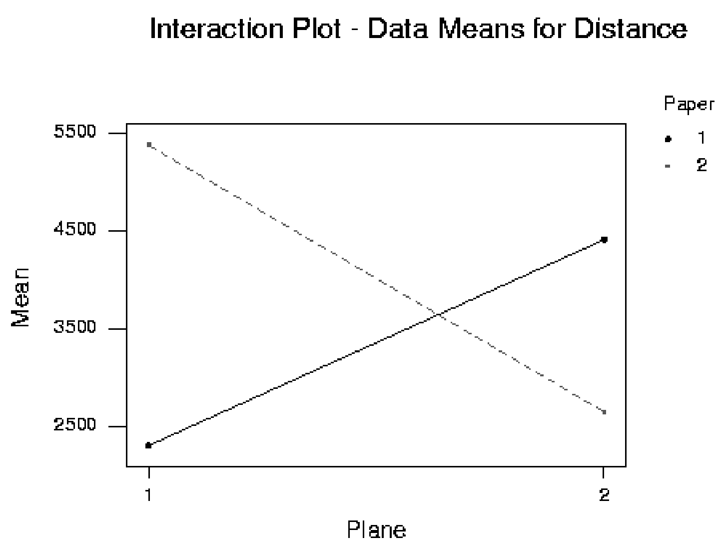

Figure 7.56: Interaction plot for data on paper planes.

$$\sum_{j=1}^{b} \gamma_{ij} = 0, \;\; i = 1, ..., a.$$

An alternative to adding constraints is to reparametrize the model. As in our previous discussion, we can code the information in factor A through a set of $a-1$ dummy variables, and the information in factor B by a set of $b-1$ dummy variables. The interaction effects can be represented by including the products of the dummy variables for the two factors as additional predictors. Reparametrization in this way results in a full rank linear model, and we can then estimate parameters and test hypotheses in the usual way. The interpretation of parameters will depend on the way that the factors have been coded as dummy variables.

Often a factorial experiment will be conducted in which $m = 1$ (that is, there is only one observation for each combination of levels of the factors). In this case, the residual sum of squares in fitting the model is zero (the number of parameters in the mean for the reparametrized model is equal to the total number of observations) and in order to estimate the error variance we need to assume that at least some of the interaction parameters are zero. If we assume all interaction parameters are zero, then we have the additive model

$$y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}.$$

Here there is assumed to be no interaction between the factors.

Example: insanity and phases of the moon

We return to the example on admissions to the emergency room of a hospital and phases of the moon. In this experiment the daily rate of admissions to the emergency room of a mental hospital was measured for each of twelve months. For each of the twleve months there are measurements for the periods before, during and after the full moon. This is a two factor factorial experiment with factors month of the year and phase of the moon (a three level factor with levels before, during and after).

Below is the main effects plot and shows for each of the factors the average of all responses for each level of the factor. As we can see, there seems to be some seasonal pattern to the level of admissions, and perhaps also admissions are higher during the full moon. Figure 7.58 shows the interaction plot. In the figure, mean responses are plotted on the y-axis, the x-axis shows the month for each mean and the different lines are for the three levels of the phase of the moon factor. As mentioned above, in an additive model (no interaction) the lines should be roughly parallel - if there is interaction, this means that the effect of the phase of the moon factor depends on the level of the month factor and the lines would not be parallel. Since we only have one observation for each combination of levels of the factors, we cannot fit the model with interactions.

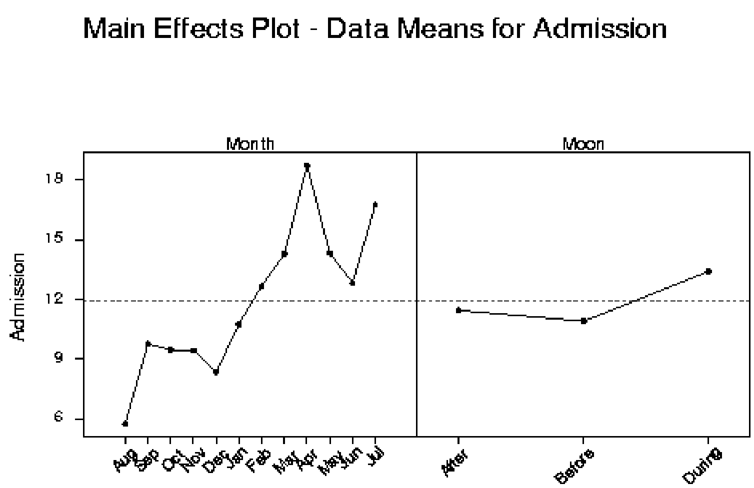

Figure 7.57: Main effects plot for data on insanity and phases of the moon.

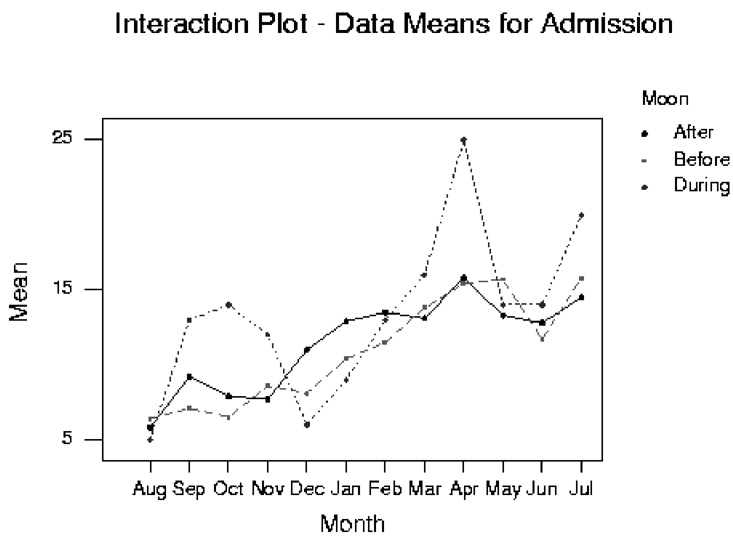

Figure 7.58: Interaction plot for data on insanity and phases of the moon.

```
Two-way Analysis of Variance


Analysis of Variance for Admission
Source        DF        SS         MS        F        P
Moon           2     41.51      20.76     3.57    0.045
Month         11    455.58      41.42     7.13    0.000
Error         22    127.82       5.81
Total         35    624.92
```

From the p-values in the table, it can be seen that there is a significant effect for both month of year and phase of the moon - it does seem as though there is strong evidence in the additive model that phase of the moon has an effect on mental hospital admissions.

When we have more than one observation for each distinct combination of levels of the factors in a factorial experiment, then we are able to estimate interaction terms in our model.

Example: paper plane experiment

In the paper plane experiment, we can fit a model including interactions since we have replicate observations for each distinct combination of levels of the factors. Again we consider a two factor experiment including only the factors `Plane` (design of plane) and `Paper` (paper weight). The analysis of variance table is given below. A significant interaction effect is indicated, as was shown in the interaction plot we discussed earlier.

```
Two-way Analysis of Variance


Analysis of Variance for Distance
Source        DF        SS         MS        F        P
Paper          1   1718721    1718721     2.16    0.168
Plane          1    385641     385641     0.48    0.500
Interaction    1  23386896   23386896    29.35    0.000
Error         12   9561029     796752
Total         15  35052287
```

8 Logistic Regression

This course has been concerned with the general linear model, and we have seen how flexible the linear model is for interpreting relationships in data. However, the assumptions of the linear model can be restrictive in some situations. Much of the theory of the linear model we have discussed has relied on the assumption of normal errors: this often may not be reasonable, and if the assumption of normality is to hold the responses must also be continuous. Constancy of variance for the errors is another assumption of the linear model which is often violated.

In this subsection we discuss how we can do regression when the responses are binary (zero or one). The model we will consider is perhaps the simplest example of a *generalized linear model*. Generalized linear models are a flexible class of models (including normal linear models as a special case) which are often used for analyzing discrete and other kinds of non-normal data. Generalized linear models will be discussed in much greater depth in some of your later statistics courses. For now, we will consider only logistic regression models for describing the dependence of a binary response on one or more predictors.

There are many situations in regression modelling where we are interested in a binary response. For instance, for a number of banks over a period of time we may have a collection of predictor variables describing the financial viability of the banks as well as a response variable which is one if the bank fails over the period of observation and zero otherwise. We are interested in estimating the probability of failure based on the predictors describing the financial viability of the bank.

The use of regression models for a binary response is also common in medical statistics. For instance, we might be interested in estimating the probability that someone will suffer a heart attack based on lifestyle factors. We could take a random sample of subjects, measure predictors which are thought to relate to heart attack risk (such as weight, blood pressure, cholesterol, etc.) and follow these subjects over a period of time. We could then record a response variable indicating whether the subject suffered a heart attack during the period of investigation. Again we have a binary response which we are interested in relating to predictor variables: what is the probability of heart attack for someone with given weight, blood pressure, cholesterol, etc.?

Clearly the linear model is inappropriate for a regression analysis when the response is binary. Normality doesn't hold, and the assumption of a constant variance cannot hold in general either. One method of analysis appropriate for a binary response variable is based on the logistic regression model.

As before we write $y = (y_1, ..., y_n)$ for a vector of n responses, but where now each value y_i is either zero or one. Regarding each response value y_i as random, we write

$$p_i = Pr(y_i = 1).$$

Note also that

$$E(y_i) = 0.Pr(y_i = 0) + 1.Pr(y_i = 1) = Pr(y_i = 1)$$

so that $p_i = E(y_i)$. In the linear model we assumed that the expected response was linear in a vector of unknown parameters β of length p, say: that is,

$$E(y_i) = x_i^T \beta$$

where we have written $x_i = (x_{i1}, ..., x_{ip})^T$ for a vector of p predictors (and where $x_{i1} = 1$). Can we assume the same thing here, that

$$p_i = x_i^T \beta?$$

Clearly there are some problems with this, since p_i is a probability (and hence must lie between zero and one) but the term $x_i^T \beta$ can potentially be negative or greater than one. One solution is to assume that some transformation of p_i is a linear function of the parameters β: we can use a transformation which maps a number between zero and one (a probability) to the whole real line, and then model the probability on this transformed scale.

One appropriate transformation of the probability p_i might be

$$l(p_i) = \log\left(\frac{p_i}{1-p_i}\right).$$

Note that as $p_i \to 0$, $l(p_i) \to -\infty$, whereas as $p_i \to 1$, $l(p_i) \to \infty$. Furthermore, $l(p_i)$ increases as p_i increases. In a logistic regression model it is assumed that

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \beta. \tag{42}$$

How are parameters estimated in the logistic regression model? An obvious approach is to use maximum likelihood to estimate the parameters β. Rearranging (42), we have that

$$p_i = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}. \tag{43}$$

If we assume that the responses y_i are independent, the likelihood function is the product of the probability functions for the observed responses y_i. If $y_i = 1$, the contribution to the likelihood from the ith response is the probability that $y_i = 1$, or p_i. If $y_i = 0$, the contribution to the likelihood is the probability that $y_i = 0$, or $1 - p_i$. Noting that the function

$$p_i^{y_i}(1-p_i)^{1-y_i}$$

takes the value p_i when $y_i = 1$ and the value $1 - p_i$ when $y_i = 0$, we can write the likelihood function as

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{1-y_i}$$

where p_i is given by (43). We must maximize this likelihood with respect to β, or equivalently maximize the log-likelihood

$$\log L(\beta) = \sum_{i=1}^{n} y_i \log\left(\frac{p_i}{1-p_i}\right) + \sum_{i=1}^{n} \log(1-p_i).$$

After differentiating with respect to β, we find that the maximum likelihood estimator $\widehat{\beta}$ of β satisfies

$$\sum_{i=1}^{n} \frac{\exp(x_i^T\widehat{\beta})}{1+\exp(x_i^T\widehat{\beta})} x_i = \sum_{i=1}^{n} y_i x_i \tag{44}$$

(you might like to prove this as an exercise). Now, for the linear model the equations we needed to solve to obtain maximum likelihood estimates of β (the normal equations) were linear equations, and we know from linear algebra when a unique solution will exist and how to find it. However, the equations which need to be solved to find the maximum likelihood estimate of β in the case of logistic regression are nonlinear equations, and in general an iterative technique is needed to solve them. We do not discuss the algorithm used in this course.

So far we have discussed the use of logistic regression for modelling a binary response. It is easy to extend the idea of logistic regression to deal with responses which are binomially distributed. Recall that a binomial random variable can be regarded as the number of successes in n independent trials with a probability p of success on each trial (or the number of heads in n tosses of a coin where the coin has probability p of a head on each flip). If we have responses y_i which are the number of successes in n_i trials with probability p_i of success on each trial, and if we are interested in modelling the probability p_i in terms of predictor variables (with n_i known) then logistic regression can be applied: we just regard each binomial response y_i as being equivalent to n_i binary responses, with y_i of these binary responses being one, and $n_i - y_i$ of them being zero. We now illustrate the fitting of logistic regression models in MINITAB.

Example: response of cows to electric shocks

This example is from Weisberg, "Applied Linear Regression (Second Edition)," Wiley, New York, 1985. An experiment was conducted into the effect of small electrical currents on farm animals, with the eventual goal of understanding the effects of high voltage power lines on livestock. Seven cows were subejcted to six different shock intensities ($0, 1, 2, 3, 4$ and 5 milliamps). Each cow was given thirty shocks, five at each intensity, in a random order. The experiment was then repeated. For each shock, the response is mouth movement, which is either present or absent. At each shock level, there are 70 observations of the response. We ignore any effects due to the individual cows in our analysis (although we could employ the idea of blocking here). We can model the number of positive responses at each shock level by considering these to be binomial random variables with $n = 70$ and a probability of success depending on the shock level.

Fitting the logistic regression model in MINITAB gives the following output.

```
Binary Logistic Regression

Link Function:  Logit

Response  Information

Variable  Value         Count
Number o  Success         200
          Failure         220
Number o  Total           420

Logistic Regression Table
                                                 Odds        95% CI
Predictor       Coef       StDev         Z      P    Ratio    Lower    Upper
Constant     -3.3010      0.3238    -10.20 0.000
Current       1.2459      0.1119     11.13 0.000     3.48     2.79     4.33

Log-Likelihood = -170.078
Test that all slopes are zero: G = 241.134, DF = 1, P-Value = 0.000
```

From the MINITAB output, we have that the fitted probability of mouth movement at shock level i is

$$\frac{\exp(-3.3010 + (1.2459).\texttt{Current}_i)}{1 + \exp(-3.3010 + (1.2459).\texttt{Current}_i)}.$$

Also listed in the output are estimated standard errors of the coeficients, and p-values for testing the hypothesis that the various coefficients in the model are zero (these are analogous to the partial t-tests we looked at in the linear model). We will discuss in more depth what some of the output given here means in the next lecture.

8.1 Hypothesis testing in logistic regression

When building a logistic regression model for a binary response variable we may wish to test hypotheses in the same way that we did for the linear model. For instance, we might wish to test whether the predictors are at all useful in a logistic regression for explaining variation in the response, or we may wish to test whether a subset of the predictors can be deleted from the model. In this subsection we extend the framework we have developed for hypothesis testing in linear models to logistic regression.

We will start by rewriting some of the things we know about the linear model in a different way. Consider the situation where we have a set of continuous responses $(y_1, ..., y_n)^T$. We can obtain a linear model which gives a "perfect fit" by having a parameter for every observation: we write

$$y_i = \mu_i + \varepsilon_i$$

where the terms ε_i are zero mean errors which are here independent and normal with a common variance σ^2. Now, of course the model we have just suggested is probably not a very helpful one for answering the questions motivating any real experiment, but the perfect fit model (usually called a saturated model) can help us to assess the fit of models with fewer parameters.

What are the maximum likelihood estimators of the parameters μ_i in the above model? If we write $\mu = (\mu_1, ..., \mu_n)^T$ for the vector of the unknown mean parameters, and if we assume that σ^2 is known, we can write the likelihood in terms of the unknown parameters μ as

$$\begin{aligned} L(\mu) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu_i)^2\right). \end{aligned}$$

Clearly maximizing $L(\mu)$ with respect to μ is equivalent to maximizing

$$-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu_i)^2.$$

The above expression is clearly never positive, and the maximum value it achieves is zero when $\mu_i = y_i$. So the maximum likelihood estimator of μ_i is $\widehat{\mu}_i = y_i$. The maximum value attained by the likelihood is thus

$$L(\widehat{\mu}) = (2\pi\sigma^2)^{-n/2}.$$

Now consider the model

$$y_i = x_i^T\beta + \varepsilon_i \qquad (45)$$

where β is a $p \times 1$ vector of parameters, and $x_i = (x_{i1}, ..., x_{ip})^T$ is a vector of predictors associated with the ith response. Writing $\widehat{\beta}$ for the maximum likelihood estimator of β in this model, the maximum value achieved by the likelihood function is

$$L(\widehat{\beta}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x_i^T\widehat{\beta})^2\right)$$

Clearly the maximized value of the likelihood for the saturated model is always larger than the maximized value of the likelihood for a model with fewer parameters. Looking at the ratio

$$\frac{L(\widehat{\beta})}{L(\widehat{\mu})}$$

tells us something about how close the smaller model comes to attaining the ideal of a perfect fit established by the saturated model. We have here that

$$\frac{L(\widehat{\beta})}{L(\widehat{\mu})} = \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x_i^T\widehat{\beta})^2\right).$$

Hence

$$-2\log\left(\frac{L(\widehat{\beta})}{L(\widehat{\mu})}\right) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i - x_i^T\widehat{\beta})^2.$$

We say that

$$\lambda(\beta) = -2\log\left(\frac{L(\widehat{\beta})}{L(\widehat{\mu})}\right)$$

is the *scaled deviance* of the model (45). The scaled deviance for a normal linear model is simply the residual sum of squares for the model divided by σ^2. The *deviance* of the model (45) is the scaled deviance multiplied by the variance parameter σ^2: the deviance for a normal linear model is thus simply the residual sum of squares (note: you will sometimes find different definitions of the deviance in different textbooks).

We have rewritten our usual measure of lack of fit for a normal linear model, the residual sum of squares, by considering the maximum value of the likelihood for the model and comparing this to the maximimum value of the likelihood for a saturated model. Rewriting the residual sum of squares in this way enables us to generalize to new situations. In particular, we can define the deviance for a logistic regression model: we can consider a saturated model (with a parameter for each observation) and define the scaled deviance as -2 times the log of the ratio of the maximized likelihood for the model to the maximized likelihood for the saturated model. In logistic regression the distributions of the responses are determined by their means, and there is no additional scale or dispersion parameter (like σ^2 in the normal linear case). Thus for logistic regression, the deviance and the scaled deviance have the same definition. As before, the deviance tells us something about how close our model comes to achieving the perfect fit established by the saturated model: it is a measure of lack of fit, analogous to the residual sum of squares.

Looking at differences of scaled deviances is the key idea in hypothesis testing for logistic regression. Again an analogy with the case of the normal linear model is helpful. Let $\beta = (\beta_0, ..., \beta_k)^T$ denote the vector of parameters in the mean structure for a normal linear model containing k predictors. Partition β into two subvectors, $\beta^{(1)} = (\beta_0, ..., \beta_r)^T$ and $\beta^{(2)} = (\beta_{r+1}, ..., \beta_k)^T$. Suppose that we wish to test the hypothesis

$$H_0 : \beta^{(2)} = 0$$

against the alternative

$$H_1 : \beta^{(2)} \neq 0.$$

By testing this hypothesis we are comparing two different models: we compare a model containing all k predictors with a smaller model in which the last $k - r$ predictors in the original model are excluded. Recall that our procedure for testing in this situation involved fitting the smaller model, and then looking at the reduction in the residual sum of squares when the larger model was fitted. We could divide this reduction in the residual sum of squares by the number of additional parameters in the larger model ($k - r$ here), divide all this by the estimated residual error variance (the mean square error) and then assuming that the smaller model was adequate this was a realization of an F random variable with $k-r$ and $n-k-1$ degrees of freedom (where n is the number of observations).

The derivation of the distribution of the F statistic used for testing $\beta^{(2)} = 0$ involved showing that the difference in the scaled deviances for the larger and smaller models (that is, the difference in the residual sum of squares divided by σ^2) had a χ^2_{k-r} distribution. If σ^2 was known, this result would enable testing $\beta^{(2)} = 0$ via a χ^2 statistic (the reason for using the F statistic was that σ^2 is not known in general, and so to come up with a statistic with a distribution free of unknown parameters which can be used for hypothesis testing we need to divide by the mean square error). So we have seen that for testing $\beta^{(2)} = 0$ in the normal linear model with σ^2 known we can simply refer the difference of scaled deviances for the larger and smaller model divided by the difference in the number of parameters for the larger and smaller model to a χ^2 distribution.

This result is exact for a normal linear model. Fortunately for us, this result also holds approximately in large samples (that is, as $n \to \infty$) for comparing two fixed models in logistic regression. For logistic regression there is no variance or dispersion parameter (like σ^2) to estimate, and carrying over our previous notation we can test whether $\beta^{(2)}$ is zero by comparing the difference in scaled deviances for the larger and smaller models divided by the difference in the number of parameters for the larger and smaller models ($k - r$ here) to a percentage point of a χ^2_{k-r} distribution.

This result may be used to test the hypothesis that $\beta_1 = ... = \beta_k = 0$ (that is, that none of the predictors in a logistic regression model are helpful for explaining variation in the response) against the alternative that not all these parameters are zero. Also, of course we can compare the full model with a model in which one predictor is excluded to get a test analogous to a two-sided partial t-test for significance of coefficients in the normal linear model (there is also another way of testing significance of individual coefficients in the logistic regression based on large sample normality results for the maximum likelihood estimator). We will discuss some of these tests when we look at the MINITAB output for some examples during the lecture.